

Gradation Compression

The data except this specification conform to those of each product.

1. Product Type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function of Gradation Compression which makes the output data be 8 bit is described in this document.
When this function is used, the data is compressed and reduced.

3. Features

◆ Support the number of conversion bit

Conversion from 12 bit to 8 bit

Conversion from 10 bit to 8 bit

◆ Position of the compression

The 1 or 2 position can be set by the register.

◆ Compression gain setting

The compression gain can be set by the register.

◆ Supported mode

Normal mode (XVS / XHS sync)

Sequential Trigger mode

Fast Trigger mode

◆ Supported readout drive mode

All-pixel scan mode

Vertical / Horizontal 1 / 2 Subsampling mode

2 × 2 FD Binning mode^{*1}

ROI mode

^{*1} monochrome sensors only

◆ Supported interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

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4. Register Map

The register map for Gradation Compression is shown below.

Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
30h	3430h	0	ODBIT [1:0]	Fixed to 2h	0h	00h	S
		1					—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
B8h	35B8h	0	CMPEN [0]	Set Gradation Compression enable 0: Normal mode 1: Gradation Compression mode (according to CCMP1 / 2)	0	00h	S
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
B9h	35B9h	0	CCMP1 [3:0]	Set the first start point of gradation compression (data position) 0000: Gradation compression off (both of the first and second) 0001: 000h 0010: 010h 0011: 020h 0100: 040h 0101: 080h 0110: 100h 0111: 200h 1000: 400h 1001: 800h Others: prohibited	0h	00h	S
		1					
		2					
		3					
		4	CCMP2 [3:0]	Set the second start point of gradation compression (data position) 0000: The second Gradation compression off 0001: prohibited 0010: 010h 0011: 020h 0100: 040h 0101: 080h 0110: 100h 0111: 200h 1000: 400h 1001: 800h Others: prohibited *In case of setting more than 0010, to set CCMP2 < CCMP1 is prohibited.	0h	00h	S
		5					
		6					
		7					

Address		bit	Register Name	Description	Default value after reset		Reflection timing	
4-wire	I ² C				By register	By address		
BAh	35BAh	0	ACMP1 [3:0]	Set the gain of the first gradation compression 0000: 1 / 1 0001: 1 / 2 0010: 1 / 4 0011: 1 / 8 0100: 1 / 16 0101: 1 / 32 0110: 1 / 64 0111: 1 / 128 1000: 1 / 256 1001: 1 / 512 1010: 1 / 1024 1011: 1 / 2048 Others: prohibited	0h	00h	S	
		1						
		2						
		3						
		4		ACMP2 [3:0]	Set the gain of the second gradation compression 0000: 1 / 1 0001: 1 / 2 0010: 1 / 4 0011: 1 / 8 0100: 1 / 16 0101: 1 / 32 0110: 1 / 64 0111: 1 / 128 1000: 1 / 256 1001: 1 / 512 1010: 1 / 1024 1011: 1 / 2048 Others: prohibited		0h	S
		5						
		6						
		7						

5. Readout Drive Mode

5.1. IMX530-AAMJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	21.9	5320	4600	358.0	716.0	520.8
	4	594	(8)	11.2			698.0	1396.0	1015.3
	8	594	12	21.9			358.0	716.0	520.8
	4	594	(8)	11.2			698.0	1396.0	1015.3
Vertical / Horizontal 1/2 subsampling	8	594	10	79.4	2660	2300	193.5	387.0	281.5
	4	594	(8)	41.7			369.0	738.0	536.8
	8	594	12	79.4			193.5	387.0	281.5
	4	594	(8)	41.7			369.0	738.0	536.8
2 × 2 FD binning mode	8	594	10	79.4	2660	2300	193.5	387.0	281.5
	4	594	(8)	41.7			369.0	738.0	536.8
	8	594	12	79.4			193.5	387.0	281.5
	4	594	(8)	41.7			369.0	738.0	536.8
ROI	8	594	10	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594	(8)	^{*2}			698.0	1396.0	1015.3
	8	594	12	^{*2}			358.0	716.0	520.8
	4	594	(8)	^{*2}			698.0	1396.0	1015.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	11.1	5320	4600	706.0	1412.0	1027.0
	4	297	(8)	5.6			1382.5	2765.0	2011.0
	8	297	12	11.1			706.0	1412.0	1027.0
	4	297	(8)	5.6			1382.5	2765.0	2011.0
Vertical / Horizontal 1/2 subsampling	8	297	10	41.1	2660	2300	375.0	750.0	545.5
	4	297	(8)	21.4			720.5	1441.0	1048.0
	8	297	12	41.1			375.0	750.0	545.5
	4	297	(8)	21.4			720.5	1441.0	1048.0
2 × 2 FD binning mode	8	297	10	41.1	2660	2300	375.0	750.0	545.5
	4	297	(8)	21.4			720.5	1441.0	1048.0
	8	297	12	41.1			375.0	750.0	545.5
	4	297	(8)	21.4			720.5	1441.0	1048.0
ROI	8	297	10	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297	(8)	^{*2}			1382.5	2765.0	2011.0
	8	297	12	^{*2}			706.0	1412.0	1027.0
	4	297	(8)	^{*2}			1382.5	2765.0	2011.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	30.1	5320	4600	260.0	520.0	378.2
	4	891	(8)	15.8			495.0	990.0	720.0
	8	891	12	30.1			260.0	520.0	378.2
	4	891	(8)	15.8			495.0	990.0	720.0
Vertical / Horizontal 1/2 subsampling	8	891	10	100.1	2660	2300	153.0	306.0	222.6
	4	891	(8)	54.8			281.0	562.0	408.8
	8	891	12	100.1			153.0	306.0	222.6
	4	891	(8)	54.8			281.0	562.0	408.8
2 × 2 FD binning mode	8	891	10	100.1	2660	2300	153.0	306.0	222.6
	4	891	(8)	54.8			281.0	562.0	408.8
	8	891	12	100.1			153.0	306.0	222.6
	4	891	(8)	54.8			281.0	562.0	408.8
ROI	8	891	10	*2	*3	*3	260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0
	8	891	12	*2			260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	15.4	5320	4600	508.0	1016.0	739.0
	4	445.5	(8)	8.0			972.5	1945.0	1414.6
	8	445.5	12	15.4			508.0	1016.0	739.0
	4	445.5	(8)	8.0			972.5	1945.0	1414.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	53.1	2660	2300	290.0	580.0	421.9
	4	445.5	(8)	28.8			536.5	1073.0	780.4
	8	445.5	12	53.1			290.0	580.0	421.9
	4	445.5	(8)	28.8			536.5	1073.0	780.4
2 × 2 FD binning mode	8	445.5	10	53.1	2660	2300	290.0	580.0	421.9
	4	445.5	(8)	28.8			536.5	1073.0	780.4
	8	445.5	12	53.1			290.0	580.0	421.9
	4	445.5	(8)	28.8			536.5	1073.0	780.4
ROI	8	445.5	10	*2	*3	*3	508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6
	8	445.5	12	*2			508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 30.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	102.6	5320	4600	75.5	151.0	109.9
	4	4752		67.4			115.5	231.0	168.0
	2	4752		35.0			223.5	447.0	325.1
	1	4752		17.8			440.0	880.0	640.0
	8	4752	12 (8)	74.4			104.5	209.0	152.0
	4	4752		67.4			115.5	231.0	168.0
	2	4752		35.0			223.5	447.0	325.1
	1	4752		17.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	385.2	2660	2300	38.0	76.0	55.3
	4	4752		240.6			62.0	124.0	90.2
	2	4752		132.3			115.0	230.0	167.3
	1	4752		68.2			225.0	450.0	327.3
	8	4752	12 (8)	283.3			52.5	105.0	76.4
	4	4752		240.6			62.0	124.0	90.2
	2	4752		132.3			115.0	230.0	167.3
	1	4752		68.2			225.0	450.0	327.3
2 × 2 FD binning mode	8	4752	10 (8)	385.2	2660	2300	38.0	76.0	55.3
	4	4752		240.6			62.0	124.0	90.2
	2	4752		132.3			115.0	230.0	167.3
	1	4752		68.2			225.0	450.0	327.3
	8	4752	12 (8)	283.3			52.5	105.0	76.4
	4	4752		240.6			62.0	124.0	90.2
	2	4752		132.3			115.0	230.0	167.3
	1	4752		68.2			225.0	450.0	327.3
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	75.5	151.0	109.9
	4	4752		^{*2}			115.5	231.0	168.0
	2	4752		^{*2}			223.5	447.0	325.1
	1	4752		^{*2}			440.0	880.0	640.0
	8	4752	12 (8)	^{*2}			104.5	209.0	152.0
	4	4752		^{*2}			115.5	231.0	168.0
	2	4752		^{*2}			223.5	447.0	325.1
	1	4752		^{*2}			440.0	880.0	640.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 102.6 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	64.1	5320	4600	121.5	243.0	176.8
	4	2376		34.4			227.5	455.0	331.0
	2	2376		17.7			443.5	887.0	645.1
	1	2376		8.9			880.0	1760.0	1280.0
	8	2376	12 (8)	44.4			176.0	352.0	256.0
	4	2376		23.7			330.5	661.0	480.8
	2	2376		11.9			655.0	1310.0	952.8
	1	2376		6.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	227.5	2660	2300	66.0	132.0	96.0
	4	2376		126.7			120.0	240.0	174.6
	2	2376		67.6			227.0	454.0	330.2
	1	2376		34.3			450.0	900.0	654.6
	8	2376	12 (8)	162.5			93.0	186.0	135.3
	4	2376		90.0			170.0	340.0	247.3
	2	2376		45.5			338.5	677.0	492.4
	1	2376		23.0			672.0	1344.0	977.5
2 × 2 FD binning mode	8	2376	10 (8)	227.5	2660	2300	66.0	132.0	96.0
	4	2376		126.7			120.0	240.0	174.6
	2	2376		67.6			227.0	454.0	330.2
	1	2376		34.3			450.0	900.0	654.6
	8	2376	12 (8)	162.5			93.0	186.0	135.3
	4	2376		90.0			170.0	340.0	247.3
	2	2376		45.5			338.5	677.0	492.4
	1	2376		23.0			672.0	1344.0	977.5
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	121.5	243.0	176.8
	4	2376		^{*2}			227.5	455.0	331.0
	2	2376		^{*2}			443.5	887.0	645.1
	1	2376		^{*2}			880.0	1760.0	1280.0
	8	2376	12 (8)	^{*2}			176.0	352.0	256.0
	4	2376		^{*2}			330.5	661.0	480.8
	2	2376		^{*2}			655.0	1310.0	952.8
	1	2376		^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 64.1 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	32.6	5320	4600	240.0	480.0	349.1
	4	1188		17.3			452.5	905.0	658.2
	2	1188		8.8			887.0	1774.0	1290.2
	1	1188		4.4			1759.5	3519.0	2559.3
	8	1188	12 (8)	32.6			240.0	480.0	349.1
	4	1188		17.3			452.5	905.0	658.2
	2	1188		8.8			887.0	1774.0	1290.2
	1	1188		4.4			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	118.7	2660	2300	129.0	258.0	187.7
	4	1188		64.7			237.5	475.0	345.5
	2	1188		34.0			454.0	908.0	660.4
	1	1188		17.1			899.5	1799.0	1308.4
	8	1188	12 (8)	118.7			129.0	258.0	187.7
	4	1188		64.7			237.5	475.0	345.5
	2	1188		34.0			454.0	908.0	660.4
	1	1188		17.1			899.5	1799.0	1308.4
2 × 2 FD binning mode	8	1188	10 (8)	118.7	2660	2300	129.0	258.0	187.7
	4	1188		64.7			237.5	475.0	345.5
	2	1188		34.0			454.0	908.0	660.4
	1	1188		17.1			899.5	1799.0	1308.4
	8	1188	12 (8)	118.7			129.0	258.0	187.7
	4	1188		64.7			237.5	475.0	345.5
	2	1188		34.0			454.0	908.0	660.4
	1	1188		17.1			899.5	1799.0	1308.4
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	240.0	480.0	349.1
	4	1188		^{*2}			452.5	905.0	658.2
	2	1188		^{*2}			887.0	1774.0	1290.2
	1	1188		^{*2}			1759.5	3519.0	2559.3
	8	1188	12 (8)	^{*2}			240.0	480.0	349.1
	4	1188		^{*2}			452.5	905.0	658.2
	2	1188		^{*2}			887.0	1774.0	1290.2
	1	1188		^{*2}			1759.5	3519.0	2559.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 32.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

5.2. IMX530-AAQJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	21.9	5320	4600	358.0	716.0	520.8
	4	594	(8)	11.2			698.0	1396.0	1015.3
	8	594	12	21.9			358.0	716.0	520.8
	4	594	(8)	11.2			698.0	1396.0	1015.3
Vertical / Horizontal 1/2 subsampling	8	594	10	79.9	2660	2300	193.5	387.0	281.5
	4	594	(8)	42.1			369.0	738.0	536.8
	8	594	12	79.9			193.5	387.0	281.5
	4	594	(8)	42.1			369.0	738.0	536.8
ROI	8	594	10	*2	*3	*3	358.0	716.0	520.8
	4	594	(8)	*2			698.0	1396.0	1015.3
	8	594	12	*2			358.0	716.0	520.8
	4	594	(8)	*2			698.0	1396.0	1015.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	11.1	5320	4600	706.0	1412.0	1027.0
	4	297	(8)	5.6			1382.5	2765.0	2011.0
	8	297	12	11.1			706.0	1412.0	1027.0
	4	297	(8)	5.6			1382.5	2765.0	2011.0
Vertical / Horizontal 1/2 subsampling	8	297	10	41.4	2660	2300	375.0	750.0	545.5
	4	297	(8)	21.6			720.5	1441.0	1048.0
	8	297	12	41.4			375.0	750.0	545.5
	4	297	(8)	21.6			720.5	1441.0	1048.0
ROI	8	297	10	*2	*3	*3	706.0	1412.0	1027.0
	4	297	(8)	*2			1382.5	2765.0	2011.0
	8	297	12	*2			706.0	1412.0	1027.0
	4	297	(8)	*2			1382.5	2765.0	2011.0

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	30.1	5320	4600	260.0	520.0	378.2
	4	891	(8)	15.8			495.0	990.0	720.0
	8	891	12	30.1			260.0	520.0	378.2
	4	891	(8)	15.8			495.0	990.0	720.0
Vertical / Horizontal 1/2 subsampling	8	891	10	100.7	2660	2300	153.0	306.0	222.6
	4	891	(8)	55.3			281.0	562.0	408.8
	8	891	12	100.7			153.0	306.0	222.6
	4	891	(8)	55.3			281.0	562.0	408.8
ROI	8	891	10	*2	*3	*3	260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0
	8	891	12	*2			260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	15.4	5320	4600	508.0	1016.0	739.0
	4	445.5	(8)	8.0			972.5	1945.0	1414.6
	8	445.5	12	15.4			508.0	1016.0	739.0
	4	445.5	(8)	8.0			972.5	1945.0	1414.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	53.6	2660	2300	290.0	580.0	421.9
	4	445.5	(8)	29.0			536.5	1073.0	780.4
	8	445.5	12	53.6			290.0	580.0	421.9
	4	445.5	(8)	29.0			536.5	1073.0	780.4
ROI	8	445.5	10	*2	*3	*3	508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6
	8	445.5	12	*2			508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6

- *1 The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 30.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]
*2 See the section of "ROI mode"
*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	102.6	5320	4600	75.5	151.0	109.9
	4	4752		67.4			115.5	231.0	168.0
	2	4752		35.0			223.5	447.0	325.1
	1	4752		17.8			440.0	880.0	640.0
	8	4752	12 (8)	74.4			104.5	209.0	152.0
	4	4752		67.4			115.5	231.0	168.0
	2	4752		35.0			223.5	447.0	325.1
	1	4752		17.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	200.8	2660	2300	75.5	151.0	109.9
	4	4752		200.8			75.5	151.0	109.9
	2	4752		133.3			115.0	230.0	167.3
	1	4752		68.7			225.0	450.0	327.3
	8	4752	12 (8)	146.3			104.5	209.0	152.0
	4	4752		146.3			104.5	209.0	152.0
	2	4752		133.3			115.0	230.0	167.3
	1	4752		68.8			225.0	450.0	327.3
ROI	8	4752	10 (8)	*2	*3	*3	75.5	151.0	109.9
	4	4752		*2			115.5	231.0	168.0
	2	4752		*2			223.5	447.0	325.1
	1	4752		*2			440.0	880.0	640.0
	8	4752	12 (8)	*2			104.5	209.0	152.0
	4	4752		*2			115.5	231.0	168.0
	2	4752		*2			223.5	447.0	325.1
	1	4752		*2			440.0	880.0	640.0

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 102.6 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	64.1	5320	4600	121.5	243.0	176.8
	4	2376		34.4			227.5	455.0	331.0
	2	2376		17.7			443.5	887.0	645.1
	1	2376		8.9			880.0	1760.0	1280.0
	8	2376	12 (8)	44.4			176.0	352.0	256.0
	4	2376		23.7			330.5	661.0	480.8
	2	2376		11.9			655.0	1310.0	952.8
	1	2376		6.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	200.8	2660	2300	75.5	151.0	109.9
	4	2376		127.8			120.0	240.0	174.6
	2	2376		68.2			227.0	454.0	330.2
	1	2376		34.6			450.0	900.0	654.6
	8	2376	12 (8)	146.3			104.5	209.0	152.0
	4	2376		90.8			170.0	340.0	247.3
	2	2376		45.9			338.5	677.0	492.4
	1	2376		23.2			672.0	1344.0	977.5
ROI	8	2376	10 (8)	*2	*3	*3	121.5	243.0	176.8
	4	2376		*2			227.5	455.0	331.0
	2	2376		*2			443.5	887.0	645.1
	1	2376		*2			880.0	1760.0	1280.0
	8	2376	12 (8)	*2			176.0	352.0	256.0
	4	2376		*2			330.5	661.0	480.8
	2	2376		*2			655.0	1310.0	952.8
	1	2376		*2			1304.5	2609.0	1897.5

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 64.1 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	32.6	5320	4600	240.0	480.0	349.1
	4	1188		17.3			452.5	905.0	658.2
	2	1188		8.8			887.0	1774.0	1290.2
	1	1188		4.4			1759.5	3519.0	2559.3
	8	1188	12 (8)	32.6			240.0	480.0	349.1
	4	1188		17.3			452.5	905.0	658.2
	2	1188		8.8			887.0	1774.0	1290.2
	1	1188		4.4			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	119.3	2660	2300	129.0	258.0	187.7
	4	1188		65.2			237.5	475.0	345.5
	2	1188		34.3			454.0	908.0	660.4
	1	1188		17.3			899.5	1799.0	1308.4
	8	1188	12 (8)	119.3			129.0	258.0	187.7
	4	1188		65.2			237.5	475.0	345.5
	2	1188		34.3			454.0	908.0	660.4
	1	1188		17.3			899.5	1799.0	1308.4
ROI	8	1188	10 (8)	*2	*3	*3	240.0	480.0	349.1
	4	1188		*2			452.5	905.0	658.2
	2	1188		*2			887.0	1774.0	1290.2
	1	1188		*2			1759.5	3519.0	2559.3
	8	1188	12 (8)	*2			240.0	480.0	349.1
	4	1188		*2			452.5	905.0	658.2
	2	1188		*2			887.0	1774.0	1290.2
	1	1188		*2			1759.5	3519.0	2559.3

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 32.6 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

5.3. IMX531-AAMJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	26.0	4504	4504	307.0	614.0	446.6
	4	594	(8)	13.4			596.0	1192.0	867.0
	8	594	12	26.0			307.0	614.0	446.6
	4	594	(8)	13.4			596.0	1192.0	867.0
Vertical / Horizontal 1/2 subsampling	8	594	10	94.1	2252	2252	166.0	332.0	241.5
	4	594	(8)	50.0			314.0	628.0	456.8
	8	594	12	94.1			166.0	332.0	241.5
	4	594	(8)	50.0			314.0	628.0	456.8
2 × 2 FD binning mode	8	594	10	94.1	2252	2252	166.0	332.0	241.5
	4	594	(8)	50.0			314.0	628.0	456.8
	8	594	12	94.1			166.0	332.0	241.5
	4	594	(8)	50.0			314.0	628.0	456.8
ROI	8	594	10	^{*2}	^{*3}	^{*3}	307.0	614.0	446.6
	4	594	(8)	^{*2}			596.0	1192.0	867.0
	8	594	12	^{*2}			307.0	614.0	446.6
	4	594	(8)	^{*2}			596.0	1192.0	867.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	13.2	4504	4504	604.0	1208.0	878.6
	4	297	(8)	6.8			1178.5	2357.0	1714.2
	8	297	12	13.2			604.0	1208.0	878.6
	4	297	(8)	6.8			1178.5	2357.0	1714.2
Vertical / Horizontal 1/2 subsampling	8	297	10	48.8	2252	2252	322.0	644.0	468.4
	4	297	(8)	25.6			614.5	1229.0	893.9
	8	297	12	48.8			322.0	644.0	468.4
	4	297	(8)	25.6			614.5	1229.0	893.9
2 × 2 FD binning mode	8	297	10	48.8	2252	2252	322.0	644.0	468.4
	4	297	(8)	25.6			614.5	1229.0	893.9
	8	297	12	48.8			322.0	644.0	468.4
	4	297	(8)	25.6			614.5	1229.0	893.9
ROI	8	297	10	^{*2}	^{*3}	^{*3}	604.0	1208.0	878.6
	4	297	(8)	^{*2}			1178.5	2357.0	1714.2
	8	297	12	^{*2}			604.0	1208.0	878.6
	4	297	(8)	^{*2}			1178.5	2357.0	1714.2

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 26.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	35.3	4504	4504	226.0	452.0	328.8
	4	891	(8)	18.7			427.0	854.0	621.1
	8	891	12	35.3			226.0	452.0	328.8
	4	891	(8)	18.7			427.0	854.0	621.1
Vertical / Horizontal 1/2 subsampling	8	891	10	118.3	2252	2252	132.0	264.0	192.0
	4	891	(8)	65.5			239.0	478.0	347.7
	8	891	12	118.3			132.0	264.0	192.0
	4	891	(8)	65.5			239.0	478.0	347.7
2 × 2 FD binning mode	8	891	10	118.3	2252	2252	132.0	264.0	192.0
	4	891	(8)	65.5			239.0	478.0	347.7
	8	891	12	118.3			132.0	264.0	192.0
	4	891	(8)	65.5			239.0	478.0	347.7
ROI	8	891	10	^{*2}	^{*3}	^{*3}	226.0	452.0	328.8
	4	891	(8)	^{*2}			427.0	854.0	621.1
	8	891	12	^{*2}			226.0	452.0	328.8
	4	891	(8)	^{*2}			427.0	854.0	621.1

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	18.2	4504	4504	440.0	880.0	640.0
	4	445.5	(8)	9.5			836.5	1673.0	1216.8
	8	445.5	12	18.2			440.0	880.0	640.0
	4	445.5	(8)	9.5			836.5	1673.0	1216.8
Vertical / Horizontal 1/2 subsampling	8	445.5	10	62.2	2252	2252	252.0	504.0	366.6
	4	445.5	(8)	34.2			460.5	921.0	669.9
	8	445.5	12	62.2			252.0	504.0	366.6
	4	445.5	(8)	34.2			460.5	921.0	669.9
2 × 2 FD binning mode	8	445.5	10	62.2	2252	2252	252.0	504.0	366.6
	4	445.5	(8)	34.2			460.5	921.0	669.9
	8	445.5	12	62.2			252.0	504.0	366.6
	4	445.5	(8)	34.2			460.5	921.0	669.9
ROI	8	445.5	10	^{*2}	^{*3}	^{*3}	440.0	880.0	640.0
	4	445.5	(8)	^{*2}			836.5	1673.0	1216.8
	8	445.5	12	^{*2}			440.0	880.0	640.0
	4	445.5	(8)	^{*2}			836.5	1673.0	1216.8

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 35.3 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	104.7	4504	4504	75.5	151.0	109.9
	4	4752		80.1			99.0	198.0	144.0
	2	4752		42.5			187.5	375.0	272.8
	1	4752		21.4			373.5	747.0	543.3
	8	4752	12 (8)	75.9			104.5	209.0	152.0
	4	4752		75.9			104.5	209.0	152.0
	2	4752		42.5			187.5	375.0	272.8
	1	4752		21.4			373.5	747.0	543.3
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	392.6	2252	2252	38.0	76.0	55.3
	4	4752		282.5			53.5	107.0	77.9
	2	4752		155.4			99.5	199.0	144.8
	1	4752		80.8			194.0	388.0	282.2
	8	4752	12 (8)	288.8			52.5	105.0	76.4
	4	4752		283.4			53.5	107.0	77.9
	2	4752		155.4			99.5	199.0	144.8
	1	4752		80.8			194.0	388.0	282.2
2 × 2 FD binning mode	8	4752	10 (8)	392.6	2252	2252	38.0	76.0	55.3
	4	4752		282.5			53.5	107.0	77.9
	2	4752		155.4			99.5	199.0	144.8
	1	4752		80.8			194.0	388.0	282.2
	8	4752	12 (8)	288.8			52.5	105.0	76.4
	4	4752		283.4			53.5	107.0	77.9
	2	4752		155.4			99.5	199.0	144.8
	1	4752		80.8			194.0	388.0	282.2
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	75.5	151.0	109.9
	4	4752		^{*2}			99.0	198.0	144.0
	2	4752		^{*2}			187.5	375.0	272.8
	1	4752		^{*2}			373.5	747.0	543.3
	8	4752	12 (8)	^{*2}			104.5	209.0	152.0
	4	4752		^{*2}			104.5	209.0	152.0
	2	4752		^{*2}			187.5	375.0	272.8
	1	4752		^{*2}			373.5	747.0	543.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 104.7 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	75.9	4504	4504	104.5	209.0	152.0
	4	2376		40.9			195.0	390.0	283.7
	2	2376		21.3			374.5	749.0	544.8
	1	2376		10.7			747.0	1494.0	1086.6
	8	2376	12 (8)	52.9			150.5	301.0	219.0
	4	2376		28.6			279.5	559.0	406.6
	2	2376		14.4			557.0	1114.0	810.2
	1	2376		7.2			1109.0	2218.0	1613.1
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	264.6	2252	2252	57.5	115.0	83.7
	4	2376		148.7			104.0	208.0	151.3
	2	2376		80.1			195.5	391.0	284.4
	1	2376		40.6			388.0	776.0	564.4
	8	2376	12 (8)	192.0			80.0	160.0	116.4
	4	2376		106.6			146.5	293.0	213.1
	2	2376		54.8			287.0	574.0	417.5
	1	2376		27.7			568.5	1137.0	827.0
2 × 2 FD binning mode	8	2376	10 (8)	264.6	2252	2252	57.5	115.0	83.7
	4	2376		148.7			104.0	208.0	151.3
	2	2376		80.1			195.5	391.0	284.4
	1	2376		40.6			388.0	776.0	564.4
	8	2376	12 (8)	192.0			80.0	160.0	116.4
	4	2376		106.6			146.5	293.0	213.1
	2	2376		54.8			287.0	574.0	417.5
	1	2376		27.7			568.5	1137.0	827.0
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	104.5	209.0	152.0
	4	2376		^{*2}			195.0	390.0	283.7
	2	2376		^{*2}			374.5	749.0	544.8
	1	2376		^{*2}			747.0	1494.0	1086.6
	8	2376	12 (8)	^{*2}			150.5	301.0	219.0
	4	2376		^{*2}			279.5	559.0	406.6
	2	2376		^{*2}			557.0	1114.0	810.2
	1	2376		^{*2}			1109.0	2218.0	1613.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 75.9 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	38.7	4504	4504	206.0	412.0	299.7
	4	1188		20.7			387.0	774.0	563.0
	2	1188		10.7			749.0	1498.0	1089.5
	1	1188		5.3			1493.5	2987.0	2172.4
	8	1188	12 (8)	38.7			206.0	412.0	299.7
	4	1188		20.7			387.0	774.0	563.0
	2	1188		10.7			749.0	1498.0	1089.5
	1	1188		5.3			1493.5	2987.0	2172.4
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	138.1	2252	2252	112.0	224.0	163.0
	4	1188		76.6			204.5	409.0	297.5
	2	1188		40.3			391.0	782.0	568.8
	1	1188		20.3			775.5	1551.0	1128.0
	8	1188	12 (8)	138.5			112.0	224.0	163.0
	4	1188		76.6			204.5	409.0	297.5
	2	1188		40.3			391.0	782.0	568.8
	1	1188		20.3			775.5	1551.0	1128.0
2 × 2 FD binning mode	8	1188	10 (8)	138.1	2252	2252	112.0	224.0	163.0
	4	1188		76.6			204.5	409.0	297.5
	2	1188		40.3			391.0	782.0	568.8
	1	1188		20.3			775.5	1551.0	1128.0
	8	1188	12 (8)	138.5			112.0	224.0	163.0
	4	1188		76.6			204.5	409.0	297.5
	2	1188		40.3			391.0	782.0	568.8
	1	1188		20.3			775.5	1551.0	1128.0
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	206.0	412.0	299.7
	4	1188		^{*2}			387.0	774.0	563.0
	2	1188		^{*2}			749.0	1498.0	1089.5
	1	1188		^{*2}			1493.5	2987.0	2172.4
	8	1188	12 (8)	^{*2}			206.0	412.0	299.7
	4	1188		^{*2}			387.0	774.0	563.0
	2	1188		^{*2}			749.0	1498.0	1089.5
	1	1188		^{*2}			1493.5	2987.0	2172.4

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 38.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	26.0	4504	4504	307.0	614.0	446.6
	4	594	(8)	13.4			596.0	1192.0	867.0
	8	594	12	26.0			307.0	614.0	446.6
	4	594	(8)	13.4			596.0	1192.0	867.0
Vertical / Horizontal 1/2 subsampling	8	594	10	94.7	2252	2252	166.0	332.0	241.5
	4	594	(8)	50.5			314.0	628.0	456.8
	8	594	12	94.7			166.0	332.0	241.5
	4	594	(8)	50.5			314.0	628.0	456.8
ROI	8	594	10	*2	*3	*3	307.0	614.0	446.6
	4	594	(8)	*2			596.0	1192.0	867.0
	8	594	12	*2			307.0	614.0	446.6
	4	594	(8)	*2			596.0	1192.0	867.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	13.2	4504	4504	604.0	1208.0	878.6
	4	297	(8)	6.8			1178.5	2357.0	1714.2
	8	297	12	13.2			604.0	1208.0	878.6
	4	297	(8)	6.8			1178.5	2357.0	1714.2
Vertical / Horizontal 1/2 subsampling	8	297	10	49.2	2252	2252	322.0	644.0	468.4
	4	297	(8)	25.9			614.5	1229.0	893.9
	8	297	12	49.2			322.0	644.0	468.4
	4	297	(8)	25.9			614.5	1229.0	893.9
ROI	8	297	10	*2	*3	*3	604.0	1208.0	878.6
	4	297	(8)	*2			1178.5	2357.0	1714.2
	8	297	12	*2			604.0	1208.0	878.6
	4	297	(8)	*2			1178.5	2357.0	1714.2

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 26.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	35.3	4504	4504	226.0	452.0	328.8
	4	891	(8)	18.7			427.0	854.0	621.1
	8	891	12	35.3			226.0	452.0	328.8
	4	891	(8)	18.7			427.0	854.0	621.1
Vertical / Horizontal 1/2 subsampling	8	891	10	118.9	2252	2252	132.0	264.0	192.0
	4	891	(8)	66.1			239.0	478.0	347.7
	8	891	12	118.9			132.0	264.0	192.0
	4	891	(8)	66.1			239.0	478.0	347.7
ROI	8	891	10	^{*2}	^{*3}	^{*3}	226.0	452.0	328.8
	4	891	(8)	^{*2}			427.0	854.0	621.1
	8	891	12	^{*2}			226.0	452.0	328.8
	4	891	(8)	^{*2}			427.0	854.0	621.1

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	18.2	4504	4504	440.0	880.0	640.0
	4	445.5	(8)	9.5			836.5	1673.0	1216.8
	8	445.5	12	18.2			440.0	880.0	640.0
	4	445.5	(8)	9.5			836.5	1673.0	1216.8
Vertical / Horizontal 1/2 subsampling	8	445.5	10	62.7	2252	2252	252.0	504.0	366.6
	4	445.5	(8)	34.5			460.5	921.0	669.9
	8	445.5	12	62.7			252.0	504.0	366.6
	4	445.5	(8)	34.5			460.5	921.0	669.9
ROI	8	445.5	10	^{*2}	^{*3}	^{*3}	440.0	880.0	640.0
	4	445.5	(8)	^{*2}			836.5	1673.0	1216.8
	8	445.5	12	^{*2}			440.0	880.0	640.0
	4	445.5	(8)	^{*2}			836.5	1673.0	1216.8

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 35.3 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	104.7	4504	4504	75.5	151.0	109.9
	4	4752		80.1			99.0	198.0	144.0
	2	4752		42.5			187.5	375.0	272.8
	1	4752		21.4			373.5	747.0	543.3
	8	4752	12 (8)	75.9			104.5	209.0	152.0
	4	4752		75.9			104.5	209.0	152.0
	2	4752		42.5			187.5	375.0	272.8
	1	4752		21.4			373.5	747.0	543.3
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	204.8	2252	2252	75.5	151.0	109.9
	4	4752		204.8			75.5	151.0	109.9
	2	4752		156.5			99.5	199.0	144.8
	1	4752		81.3			194.0	388.0	282.2
	8	4752	12 (8)	149.2			104.5	209.0	152.0
	4	4752		149.2			104.5	209.0	152.0
	2	4752		149.2			104.5	209.0	152.0
	1	4752		81.3			194.0	388.0	282.2
ROI	8	4752	10 (8)	*2	*3	*3	75.5	151.0	109.9
	4	4752		*2			99.0	198.0	144.0
	2	4752		*2			187.5	375.0	272.8
	1	4752		*2			373.5	747.0	543.3
	8	4752	12 (8)	*2			104.5	209.0	152.0
	4	4752		*2			104.5	209.0	152.0
	2	4752		*2			187.5	375.0	272.8
	1	4752		*2			373.5	747.0	543.3

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 104.7 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	75.9	4504	4504	104.5	209.0	152.0
	4	2376		40.9			195.0	390.0	283.7
	2	2376		21.3			374.5	749.0	544.8
	1	2376		10.7			747.0	1494.0	1086.6
	8	2376	12 (8)	52.9			150.5	301.0	219.0
	4	2376		28.6			279.5	559.0	406.6
	2	2376		14.4			557.0	1114.0	810.2
	1	2376		7.2			1109.0	2218.0	1613.1
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	204.8	2252	2252	75.5	151.0	109.9
	4	2376		149.9			104.0	208.0	151.3
	2	2376		80.7			195.5	391.0	284.4
	1	2376		40.9			388.0	776.0	564.4
	8	2376	12 (8)	149.2			104.5	209.0	152.0
	4	2376		107.1			146.5	293.0	213.1
	2	2376		55.2			287.0	574.0	417.5
	1	2376		28.0			568.5	1137.0	827.0
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	104.5	209.0	152.0
	4	2376		^{*2}			195.0	390.0	283.7
	2	2376		^{*2}			374.5	749.0	544.8
	1	2376		^{*2}			747.0	1494.0	1086.6
	8	2376	12 (8)	^{*2}			150.5	301.0	219.0
	4	2376		^{*2}			279.5	559.0	406.6
	2	2376		^{*2}			557.0	1114.0	810.2
	1	2376		^{*2}			1109.0	2218.0	1613.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 75.9 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	38.7	4504	4504	206.0	412.0	299.7
	4	1188		20.7			387.0	774.0	563.0
	2	1188		10.7			749.0	1498.0	1089.5
	1	1188		5.3			1493.5	2987.0	2172.4
	8	1188	12 (8)	38.7			206.0	412.0	299.7
	4	1188		20.7			387.0	774.0	563.0
	2	1188		10.7			749.0	1498.0	1089.5
	1	1188		5.3			1493.5	2987.0	2172.4
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	139.5	2252	2252	112.0	224.0	163.0
	4	1188		77.1			204.5	409.0	297.5
	2	1188		40.6			391.0	782.0	568.8
	1	1188		20.5			775.5	1551.0	1128.0
	8	1188	12 (8)	139.7			112.0	224.0	163.0
	4	1188		77.1			204.5	409.0	297.5
	2	1188		40.6			391.0	782.0	568.8
	1	1188		20.5			775.5	1551.0	1128.0
ROI	8	1188	10 (8)	*2	*3	*3	206.0	412.0	299.7
	4	1188		*2			387.0	774.0	563.0
	2	1188		*2			749.0	1498.0	1089.5
	1	1188		*2			1493.5	2987.0	2172.4
	8	1188	12 (8)	*2			206.0	412.0	299.7
	4	1188		*2			387.0	774.0	563.0
	2	1188		*2			749.0	1498.0	1089.5
	1	1188		*2			1493.5	2987.0	2172.4

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 38.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

5.5. IMX532-AAMJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	32.7	5320	3032	358.0	716.0	520.8
	4	594	(8)	16.8			698.0	1396.0	1015.3
	8	594	12	32.7			358.0	716.0	520.8
	4	594	(8)	16.8			698.0	1396.0	1015.3
Vertical / Horizontal 1/2 subsampling	8	594	10	117.5	2660	1516	193.5	387.0	281.5
	4	594	(8)	61.9			369.0	738.0	536.8
	8	594	12	117.5			193.5	387.0	281.5
	4	594	(8)	61.9			369.0	738.0	536.8
2 × 2 FD binning mode	8	594	10	117.5	2660	1516	193.5	387.0	281.5
	4	594	(8)	61.9			369.0	738.0	536.8
	8	594	12	117.5			193.5	387.0	281.5
	4	594	(8)	61.9			369.0	738.0	536.8
ROI	8	594	10	*2	*3	*3	358.0	716.0	520.8
	4	594	(8)	*2			698.0	1396.0	1015.3
	8	594	12	*2			358.0	716.0	520.8
	4	594	(8)	*2			698.0	1396.0	1015.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	16.6	5320	3032	706.0	1412.0	1027.0
	4	297	(8)	8.5			1382.5	2765.0	2011.0
	8	297	12	16.6			706.0	1412.0	1027.0
	4	297	(8)	8.5			1382.5	2765.0	2011.0
Vertical / Horizontal 1/2 subsampling	8	297	10	60.9	2660	1516	375.0	750.0	545.5
	4	297	(8)	31.8			720.5	1441.0	1048.0
	8	297	12	60.9			375.0	750.0	545.5
	4	297	(8)	31.8			720.5	1441.0	1048.0
2 × 2 FD binning mode	8	297	10	60.9	2660	1516	375.0	750.0	545.5
	4	297	(8)	31.8			720.5	1441.0	1048.0
	8	297	12	60.9			375.0	750.0	545.5
	4	297	(8)	31.8			720.5	1441.0	1048.0
ROI	8	297	10	*2	*3	*3	706.0	1412.0	1027.0
	4	297	(8)	*2			1382.5	2765.0	2011.0
	8	297	12	*2			706.0	1412.0	1027.0
	4	297	(8)	*2			1382.5	2765.0	2011.0

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.7 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	45.1	5320	3032	260.0	520.0	378.2
	4	891	(8)	23.7			495.0	990.0	720.0
	8	891	12	45.1			260.0	520.0	378.2
	4	891	(8)	23.7			495.0	990.0	720.0
Vertical / Horizontal 1/2 subsampling	8	891	10	147.9	2660	1516	153.0	306.0	222.6
	4	891	(8)	81.3			281.0	562.0	408.8
	8	891	12	147.9			153.0	306.0	222.6
	4	891	(8)	81.3			281.0	562.0	408.8
2 × 2 FD binning mode	8	891	10	147.9	2660	1516	153.0	306.0	222.6
	4	891	(8)	81.3			281.0	562.0	408.8
	8	891	12	147.9			153.0	306.0	222.6
	4	891	(8)	81.3			281.0	562.0	408.8
ROI	8	891	10	*2	*3	*3	260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0
	8	891	12	*2			260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	23.1	5320	3032	508.0	1016.0	739.0
	4	445.5	(8)	12.1			972.5	1945.0	1414.6
	8	445.5	12	23.1			508.0	1016.0	739.0
	4	445.5	(8)	12.1			972.5	1945.0	1414.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	78.8	2660	1516	290.0	580.0	421.9
	4	445.5	(8)	42.8			536.5	1073.0	780.4
	8	445.5	12	78.8			290.0	580.0	421.9
	4	445.5	(8)	42.8			536.5	1073.0	780.4
2 × 2 FD binning mode	8	445.5	10	78.8	2660	1516	290.0	580.0	421.9
	4	445.5	(8)	42.8			536.5	1073.0	780.4
	8	445.5	12	78.8			290.0	580.0	421.9
	4	445.5	(8)	42.8			536.5	1073.0	780.4
ROI	8	445.5	10	*2	*3	*3	508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6
	8	445.5	12	*2			508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6

*1 The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 45.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	152.5	5320	3032	75.5	151.0	109.9
	4	4752		100.5			115.5	231.0	168.0
	2	4752		52.3			223.5	447.0	325.1
	1	4752		26.7			440.0	880.0	640.0
	8	4752	12 (8)	110.8			104.5	209.0	152.0
	4	4752		100.5			115.5	231.0	168.0
	2	4752		52.3			223.5	447.0	325.1
	1	4752		26.7			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	557.6	2660	1516	38.0	76.0	55.3
	4	4752		351.4			62.0	124.0	90.2
	2	4752		194.9			115.0	230.0	167.3
	1	4752		101.1			225.0	450.0	327.3
	8	4752	12 (8)	413.0			52.5	105.0	76.4
	4	4752		351.4			62.0	124.0	90.2
	2	4752		194.9			115.0	230.0	167.3
	1	4752		101.1			225.0	450.0	327.3
2 × 2 FD binning mode	8	4752	10 (8)	557.6	2660	1516	38.0	76.0	55.3
	4	4752		351.4			62.0	124.0	90.2
	2	4752		194.9			115.0	230.0	167.3
	1	4752		101.1			225.0	450.0	327.3
	8	4752	12 (8)	413.0			52.5	105.0	76.4
	4	4752		351.4			62.0	124.0	90.2
	2	4752		194.9			115.0	230.0	167.3
	1	4752		101.1			225.0	450.0	327.3
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	75.5	151.0	109.9
	4	4752		^{*2}			115.5	231.0	168.0
	2	4752		^{*2}			223.5	447.0	325.1
	1	4752		^{*2}			440.0	880.0	640.0
	8	4752	12 (8)	^{*2}			104.5	209.0	152.0
	4	4752		^{*2}			115.5	231.0	168.0
	2	4752		^{*2}			223.5	447.0	325.1
	1	4752		^{*2}			440.0	880.0	640.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 152.5 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	95.6	5320	3032	121.5	243.0	176.8
	4	2376		51.4			227.5	455.0	331.0
	2	2376		26.4			443.5	887.0	645.1
	1	2376		13.3			880.0	1760.0	1280.0
	8	2376	12 (8)	66.3			176.0	352.0	256.0
	4	2376		35.5			330.5	661.0	480.8
	2	2376		17.9			655.0	1310.0	952.8
	1	2376		9.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	333.2	2660	1516	66.0	132.0	96.0
	4	2376		186.8			120.0	240.0	174.6
	2	2376		100.2			227.0	454.0	330.2
	1	2376		51.0			450.0	900.0	654.6
	8	2376	12 (8)	238.7			93.0	186.0	135.3
	4	2376		133.1			170.0	340.0	247.3
	2	2376		67.5			338.5	677.0	492.4
	1	2376		34.1			672.0	1344.0	977.5
2 × 2 FD binning mode	8	2376	10 (8)	333.2	2660	1516	66.0	132.0	96.0
	4	2376		186.8			120.0	240.0	174.6
	2	2376		100.2			227.0	454.0	330.2
	1	2376		51.0			450.0	900.0	654.6
	8	2376	12 (8)	238.7			93.0	186.0	135.3
	4	2376		133.1			170.0	340.0	247.3
	2	2376		67.5			338.5	677.0	492.4
	1	2376		34.1			672.0	1344.0	977.5
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	121.5	243.0	176.8
	4	2376		^{*2}			227.5	455.0	331.0
	2	2376		^{*2}			443.5	887.0	645.1
	1	2376		^{*2}			880.0	1760.0	1280.0
	8	2376	12 (8)	^{*2}			176.0	352.0	256.0
	4	2376		^{*2}			330.5	661.0	480.8
	2	2376		^{*2}			655.0	1310.0	952.8
	1	2376		^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 95.6 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	48.7	5320	3032	240.0	480.0	349.1
	4	1188		25.9			452.5	905.0	658.2
	2	1188		13.2			887.0	1774.0	1290.2
	1	1188		6.6			1759.5	3519.0	2559.3
	8	1188	12 (8)	48.7			240.0	480.0	349.1
	4	1188		25.9			452.5	905.0	658.2
	2	1188		13.2			887.0	1774.0	1290.2
	1	1188		6.6			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	175.4	2660	1516	129.0	258.0	187.7
	4	1188		95.7			237.5	475.0	345.5
	2	1188		50.6			454.0	908.0	660.4
	1	1188		25.5			899.5	1799.0	1308.4
	8	1188	12 (8)	175.4			129.0	258.0	187.7
	4	1188		95.7			237.5	475.0	345.5
	2	1188		50.6			454.0	908.0	660.4
	1	1188		25.5			899.5	1799.0	1308.4
2 × 2 FD binning mode	8	1188	10 (8)	175.4	2660	1516	129.0	258.0	187.7
	4	1188		95.7			237.5	475.0	345.5
	2	1188		50.6			454.0	908.0	660.4
	1	1188		25.5			899.5	1799.0	1308.4
	8	1188	12 (8)	175.4			129.0	258.0	187.7
	4	1188		95.7			237.5	475.0	345.5
	2	1188		50.6			454.0	908.0	660.4
	1	1188		25.5			899.5	1799.0	1308.4
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	240.0	480.0	349.1
	4	1188		^{*2}			452.5	905.0	658.2
	2	1188		^{*2}			887.0	1774.0	1290.2
	1	1188		^{*2}			1759.5	3519.0	2559.3
	8	1188	12 (8)	^{*2}			240.0	480.0	349.1
	4	1188		^{*2}			452.5	905.0	658.2
	2	1188		^{*2}			887.0	1774.0	1290.2
	1	1188		^{*2}			1759.5	3519.0	2559.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 48.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

5.6. IMX532-AAQJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	32.7	5320	3032	358.0	716.0	520.8
	4	594	(8)	16.8			698.0	1396.0	1015.3
	8	594	12	32.7			358.0	716.0	520.8
	4	594	(8)	16.8			698.0	1396.0	1015.3
Vertical / Horizontal 1/2 subsampling	8	594	10	118.7	2660	1516	193.5	387.0	281.5
	4	594	(8)	62.7			369.0	738.0	536.8
	8	594	12	118.7			193.5	387.0	281.5
	4	594	(8)	62.7			369.0	738.0	536.8
ROI	8	594	10	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594	(8)	^{*2}			698.0	1396.0	1015.3
	8	594	12	^{*2}			358.0	716.0	520.8
	4	594	(8)	^{*2}			698.0	1396.0	1015.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	16.6	5320	3032	706.0	1412.0	1027.0
	4	297	(8)	8.5			1382.5	2765.0	2011.0
	8	297	12	16.6			706.0	1412.0	1027.0
	4	297	(8)	8.5			1382.5	2765.0	2011.0
Vertical / Horizontal 1/2 subsampling	8	297	10	61.7	2660	1516	375.0	750.0	545.5
	4	297	(8)	32.2			720.5	1441.0	1048.0
	8	297	12	61.7			375.0	750.0	545.5
	4	297	(8)	32.2			720.5	1441.0	1048.0
ROI	8	297	10	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297	(8)	^{*2}			1382.5	2765.0	2011.0
	8	297	12	^{*2}			706.0	1412.0	1027.0
	4	297	(8)	^{*2}			1382.5	2765.0	2011.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.7 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	45.1	5320	3032	260.0	520.0	378.2
	4	891	(8)	23.7			495.0	990.0	720.0
	8	891	12	45.1			260.0	520.0	378.2
	4	891	(8)	23.7			495.0	990.0	720.0
Vertical / Horizontal 1/2 subsampling	8	891	10	149.4	2660	1516	153.0	306.0	222.6
	4	891	(8)	82.3			281.0	562.0	408.8
	8	891	12	149.4			153.0	306.0	222.6
	4	891	(8)	82.3			281.0	562.0	408.8
ROI	8	891	10	*2	*3	*3	260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0
	8	891	12	*2			260.0	520.0	378.2
	4	891	(8)	*2			495.0	990.0	720.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	23.1	5320	3032	508.0	1016.0	739.0
	4	445.5	(8)	12.1			972.5	1945.0	1414.6
	8	445.5	12	23.1			508.0	1016.0	739.0
	4	445.5	(8)	12.1			972.5	1945.0	1414.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	79.8	2660	1516	290.0	580.0	421.9
	4	445.5	(8)	43.3			536.5	1073.0	780.4
	8	445.5	12	79.8			290.0	580.0	421.9
	4	445.5	(8)	43.3			536.5	1073.0	780.4
ROI	8	445.5	10	*2	*3	*3	508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6
	8	445.5	12	*2			508.0	1016.0	739.0
	4	445.5	(8)	*2			972.5	1945.0	1414.6

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 45.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	152.5	5320	3032	75.5	151.0	109.9
	4	4752		100.5			115.5	231.0	168.0
	2	4752		52.3			223.5	447.0	325.1
	1	4752		26.7			440.0	880.0	640.0
	8	4752	12 (8)	110.8			104.5	209.0	152.0
	4	4752		100.5			115.5	231.0	168.0
	2	4752		52.3			223.5	447.0	325.1
	1	4752		26.7			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	295.5	2660	1516	75.5	151.0	109.9
	4	4752		295.5			75.5	151.0	109.9
	2	4752		197.3			115.0	230.0	167.3
	1	4752		102.1			225.0	450.0	327.3
	8	4752	12 (8)	216.0			104.5	209.0	152.0
	4	4752		216.0			104.5	209.0	152.0
	2	4752		197.3			115.0	230.0	167.3
	1	4752		102.3			225.0	450.0	327.3
ROI	8	4752	10 (8)	*2	*3	*3	75.5	151.0	109.9
	4	4752		*2			115.5	231.0	168.0
	2	4752		*2			223.5	447.0	325.1
	1	4752		*2			440.0	880.0	640.0
	8	4752	12 (8)	*2			104.5	209.0	152.0
	4	4752		*2			115.5	231.0	168.0
	2	4752		*2			223.5	447.0	325.1
	1	4752		*2			440.0	880.0	640.0

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 152.5 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	95.6	5320	3032	121.5	243.0	176.8
	4	2376		51.4			227.5	455.0	331.0
	2	2376		26.4			443.5	887.0	645.1
	1	2376		13.3			880.0	1760.0	1280.0
	8	2376	12 (8)	66.3			176.0	352.0	256.0
	4	2376		35.5			330.5	661.0	480.8
	2	2376		17.9			655.0	1310.0	952.8
	1	2376		9.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	295.5	2660	1516	75.5	151.0	109.9
	4	2376		189.1			120.0	240.0	174.6
	2	2376		101.4			227.0	454.0	330.2
	1	2376		51.5			450.0	900.0	654.6
	8	2376	12 (8)	216.0			104.5	209.0	152.0
	4	2376		134.8			170.0	340.0	247.3
	2	2376		68.3			338.5	677.0	492.4
	1	2376		34.6			672.0	1344.0	977.5
ROI	8	2376	10 (8)	*2	*3	*3	121.5	243.0	176.8
	4	2376		*2			227.5	455.0	331.0
	2	2376		*2			443.5	887.0	645.1
	1	2376		*2			880.0	1760.0	1280.0
	8	2376	12 (8)	*2			176.0	352.0	256.0
	4	2376		*2			330.5	661.0	480.8
	2	2376		*2			655.0	1310.0	952.8
	1	2376		*2			1304.5	2609.0	1897.5

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 95.6 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	48.7	5320	3032	240.0	480.0	349.1
	4	1188		25.9			452.5	905.0	658.2
	2	1188		13.2			887.0	1774.0	1290.2
	1	1188		6.6			1759.5	3519.0	2559.3
	8	1188	12 (8)	48.7			240.0	480.0	349.1
	4	1188		25.9			452.5	905.0	658.2
	2	1188		13.2			887.0	1774.0	1290.2
	1	1188		6.6			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	176.7	2660	1516	129.0	258.0	187.7
	4	1188		96.9			237.5	475.0	345.5
	2	1188		51.1			454.0	908.0	660.4
	1	1188		25.9			899.5	1799.0	1308.4
	8	1188	12 (8)	176.7			129.0	258.0	187.7
	4	1188		96.9			237.5	475.0	345.5
	2	1188		51.1			454.0	908.0	660.4
	1	1188		25.9			899.5	1799.0	1308.4
ROI	8	1188	10 (8)	*2	*3	*3	240.0	480.0	349.1
	4	1188		*2			452.5	905.0	658.2
	2	1188		*2			887.0	1774.0	1290.2
	1	1188		*2			1759.5	3519.0	2559.3
	8	1188	12 (8)	*2			240.0	480.0	349.1
	4	1188		*2			452.5	905.0	658.2
	2	1188		*2			887.0	1774.0	1290.2
	1	1188		*2			1759.5	3519.0	2559.3

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 48.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	41.5	4096	3000	283.0	566.0	411.7
	4	594	(8)	21.5			548.0	1096.0	797.1
	8	594	12	41.5			283.0	566.0	411.7
	4	594	(8)	21.5			548.0	1096.0	797.1
Vertical / Horizontal 1/2 subsampling	8	594	10	145.5	2048	1500	154.0	308.0	224.0
	4	594	(8)	78.4			290.0	580.0	421.9
	8	594	12	145.5			154.0	308.0	224.0
	4	594	(8)	78.4			290.0	580.0	421.9
2 × 2 FD binning mode	8	594	10	145.5	2048	1500	154.0	308.0	224.0
	4	594	(8)	78.4			290.0	580.0	421.9
	8	594	12	145.5			154.0	308.0	224.0
	4	594	(8)	78.4			290.0	580.0	421.9
ROI	8	594	10	^{*2}	^{*3}	^{*3}	283.0	566.0	411.7
	4	594	(8)	^{*2}			548.0	1096.0	797.1
	8	594	12	^{*2}			283.0	566.0	411.7
	4	594	(8)	^{*2}			548.0	1096.0	797.1

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	21.2	4096	3000	556.0	1112.0	808.8
	4	297	(8)	10.9			1082.5	2165.0	1574.6
	8	297	12	21.2			556.0	1112.0	808.8
	4	297	(8)	10.9			1082.5	2165.0	1574.6
Vertical / Horizontal 1/2 subsampling	8	297	10	76.7	2048	1500	298.0	596.0	433.5
	4	297	(8)	40.7			566.5	1133.0	824.0
	8	297	12	76.7			298.0	596.0	433.5
	4	297	(8)	40.7			566.5	1133.0	824.0
2 × 2 FD binning mode	8	297	10	76.7	2048	1500	298.0	596.0	433.5
	4	297	(8)	40.7			566.5	1133.0	824.0
	8	297	12	76.7			298.0	596.0	433.5
	4	297	(8)	40.7			566.5	1133.0	824.0
ROI	8	297	10	^{*2}	^{*3}	^{*3}	556.0	1112.0	808.8
	4	297	(8)	^{*2}			1082.5	2165.0	1574.6
	8	297	12	^{*2}			556.0	1112.0	808.8
	4	297	(8)	^{*2}			1082.5	2165.0	1574.6

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.5 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	60.7	4096	3000	192.5	385.0	280.0
	4	891	(8)	31.9			369.5	739.0	537.5
	8	891	12	60.7			192.5	385.0	280.0
	4	891	(8)	31.9			369.5	739.0	537.5
Vertical / Horizontal 1/2 subsampling	8	891	10	205.5	2048	1500	106.5	213.0	155.0
	4	891	(8)	114.6			197.5	395.0	287.3
	8	891	12	205.5			106.5	213.0	155.0
	4	891	(8)	114.6			197.5	395.0	287.3
2 × 2 FD binning mode	8	891	10	205.5	2048	1500	106.5	213.0	155.0
	4	891	(8)	114.6			197.5	395.0	287.3
	8	891	12	205.5			106.5	213.0	155.0
	4	891	(8)	114.6			197.5	395.0	287.3
ROI	8	891	10	^{*2}	^{*3}	^{*3}	192.5	385.0	280.0
	4	891	(8)	^{*2}			369.5	739.0	537.5
	8	891	12	^{*2}			192.5	385.0	280.0
	4	891	(8)	^{*2}			369.5	739.0	537.5

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	31.5	4096	3000	374.0	748.0	544.0
	4	445.5	(8)	16.3			725.5	1451.0	1055.3
	8	445.5	12	31.5			374.0	748.0	544.0
	4	445.5	(8)	16.3			725.5	1451.0	1055.3
Vertical / Horizontal 1/2 subsampling	8	445.5	10	112.0	2048	1500	202.0	404.0	293.9
	4	445.5	(8)	60.2			381.5	763.0	555.0
	8	445.5	12	112.0			202.0	404.0	293.9
	4	445.5	(8)	60.2			381.5	763.0	555.0
2 × 2 FD binning mode	8	445.5	10	112.0	2048	1500	202.0	404.0	293.9
	4	445.5	(8)	60.2			381.5	763.0	555.0
	8	445.5	12	112.0			202.0	404.0	293.9
	4	445.5	(8)	60.2			381.5	763.0	555.0
ROI	8	445.5	10	^{*2}	^{*3}	^{*3}	374.0	748.0	544.0
	4	445.5	(8)	^{*2}			725.5	1451.0	1055.3
	8	445.5	12	^{*2}			374.0	748.0	544.0
	4	445.5	(8)	^{*2}			725.5	1451.0	1055.3

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	175.0	4096	3000	64.5	129.0	93.9
	4	4752		125.9			91.0	182.0	132.4
	2	4752		67.7			172.0	344.0	250.2
	1	4752		34.3			343.0	686.0	499.0
	8	4752	12 (8)	120.9			95.0	190.0	138.2
	4	4752		120.9			95.0	190.0	138.2
	2	4752		67.7			172.0	344.0	250.2
	1	4752		34.3			343.0	686.0	499.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	592.4	2048	1500	32.5	65.0	47.3
	4	4752		417.2			49.0	98.0	71.3
	2	4752		236.9			91.5	183.0	133.1
	1	4752		126.2			178.5	357.0	259.7
	8	4752	12 (8)	430.3			47.5	95.0	69.1
	4	4752		417.2			49.0	98.0	71.3
	2	4752		236.9			91.5	183.0	133.1
	1	4752		126.2			178.5	357.0	259.7
2 × 2 FD binning mode	8	4752	10 (8)	592.4	2048	1500	32.5	65.0	47.3
	4	4752		417.2			49.0	98.0	71.3
	2	4752		236.9			91.5	183.0	133.1
	1	4752		126.2			178.5	357.0	259.7
	8	4752	12 (8)	430.3			47.5	95.0	69.1
	4	4752		417.2			49.0	98.0	71.3
	2	4752		236.9			91.5	183.0	133.1
	1	4752		126.2			178.5	357.0	259.7
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	64.5	129.0	93.9
	4	4752		^{*2}			91.0	182.0	132.4
	2	4752		^{*2}			172.0	344.0	250.2
	1	4752		^{*2}			343.0	686.0	499.0
	8	4752	12 (8)	^{*2}			95.0	190.0	138.2
	4	4752		^{*2}			95.0	190.0	138.2
	2	4752		^{*2}			172.0	344.0	250.2
	1	4752		^{*2}			343.0	686.0	499.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 175.0 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	119.0	4096	3000	96.5	193.0	140.4
	4	2376		65.4			178.5	357.0	259.7
	2	2376		34.2			343.5	687.0	499.7
	1	2376		17.2			685.5	1371.0	997.1
	8	2376	12 (8)	119.0			96.5	193.0	140.4
	4	2376		65.4			178.5	357.0	259.7
	2	2376		34.2			343.5	687.0	499.7
	1	2376		17.2			685.5	1371.0	997.1
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	385.5	2048	1500	53.5	107.0	77.9
	4	2376		230.6			94.0	188.0	136.8
	2	2376		125.1			180.0	360.0	261.9
	1	2376		64.0			357.0	714.0	519.3
	8	2376	12 (8)	385.5			53.5	107.0	77.9
	4	2376		230.6			94.0	188.0	136.8
	2	2376		125.1			180.0	360.0	261.9
	1	2376		64.0			357.0	714.0	519.3
2 × 2 FD binning mode	8	2376	10 (8)	385.5	2048	1500	53.5	107.0	77.9
	4	2376		230.6			94.0	188.0	136.8
	2	2376		125.1			180.0	360.0	261.9
	1	2376		64.0			357.0	714.0	519.3
	8	2376	12 (8)	385.5			53.5	107.0	77.9
	4	2376		230.6			94.0	188.0	136.8
	2	2376		125.1			180.0	360.0	261.9
	1	2376		64.0			357.0	714.0	519.3
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	96.5	193.0	140.4
	4	2376		^{*2}			178.5	357.0	259.7
	2	2376		^{*2}			343.5	687.0	499.7
	1	2376		^{*2}			685.5	1371.0	997.1
	8	2376	12 (8)	^{*2}			96.5	193.0	140.4
	4	2376		^{*2}			178.5	357.0	259.7
	2	2376		^{*2}			343.5	687.0	499.7
	1	2376		^{*2}			685.5	1371.0	997.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 119.0 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	61.5	4096	3000	190.0	380.0	276.4
	4	1188		33.2			355.0	710.0	516.4
	2	1188		17.2			687.0	1374.0	999.3
	1	1188		8.6			1370.5	2741.0	1993.5
	8	1188	12 (8)	61.5			190.0	380.0	276.4
	4	1188		33.2			355.0	710.0	516.4
	2	1188		17.2			687.0	1374.0	999.3
	1	1188		8.6			1370.5	2741.0	1993.5
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	210.4	2048	1500	104.0	208.0	151.3
	4	1188		120.7			186.5	373.0	271.3
	2	1188		63.5			360.0	720.0	523.7
	1	1188		32.3			714.0	1428.0	1038.6
	8	1188	12 (8)	210.4			104.0	208.0	151.3
	4	1188		120.7			186.5	373.0	271.3
	2	1188		63.5			360.0	720.0	523.7
	1	1188		32.3			714.0	1428.0	1038.6
2 × 2 FD binning mode	8	1188	10 (8)	210.4	2048	1500	104.0	208.0	151.3
	4	1188		120.7			186.5	373.0	271.3
	2	1188		63.5			360.0	720.0	523.7
	1	1188		32.3			714.0	1428.0	1038.6
	8	1188	12 (8)	210.4			104.0	208.0	151.3
	4	1188		120.7			186.5	373.0	271.3
	2	1188		63.5			360.0	720.0	523.7
	1	1188		32.3			714.0	1428.0	1038.6
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	190.0	380.0	276.4
	4	1188		^{*2}			355.0	710.0	516.4
	2	1188		^{*2}			687.0	1374.0	999.3
	1	1188		^{*2}			1370.5	2741.0	1993.5
	8	1188	12 (8)	^{*2}			190.0	380.0	276.4
	4	1188		^{*2}			355.0	710.0	516.4
	2	1188		^{*2}			687.0	1374.0	999.3
	1	1188		^{*2}			1370.5	2741.0	1993.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

5.8. IMX535-AAQJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	41.6	4096	3000	283.0	566.0	411.7
	4	594	(8)	21.6			548.0	1096.0	797.1
	8	594	12	41.6			283.0	566.0	411.7
	4	594	(8)	21.6			548.0	1096.0	797.1
Vertical / Horizontal 1/2 subsampling	8	594	10	146.9	2048	1500	154.0	308.0	224.0
	4	594	(8)	79.6			290.0	580.0	421.9
	8	594	12	146.9			154.0	308.0	224.0
	4	594	(8)	79.6			290.0	580.0	421.9
ROI	8	594	10	*2	*3	*3	283.0	566.0	411.7
	4	594	(8)	*2			548.0	1096.0	797.1
	8	594	12	*2			283.0	566.0	411.7
	4	594	(8)	*2			548.0	1096.0	797.1

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	21.2	4096	3000	556.0	1112.0	808.8
	4	297	(8)	10.9			1082.5	2165.0	1574.6
	8	297	12	21.2			556.0	1112.0	808.8
	4	297	(8)	10.9			1082.5	2165.0	1574.6
Vertical / Horizontal 1/2 subsampling	8	297	10	77.6	2048	1500	298.0	596.0	433.5
	4	297	(8)	41.1			566.5	1133.0	824.0
	8	297	12	77.6			298.0	596.0	433.5
	4	297	(8)	41.1			566.5	1133.0	824.0
ROI	8	297	10	*2	*3	*3	556.0	1112.0	808.8
	4	297	(8)	*2			1082.5	2165.0	1574.6
	8	297	12	*2			556.0	1112.0	808.8
	4	297	(8)	*2			1082.5	2165.0	1574.6

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.6 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	60.7	4096	3000	192.5	385.0	280.0
	4	891	(8)	31.9			369.5	739.0	537.5
	8	891	12	60.7			192.5	385.0	280.0
	4	891	(8)	31.9			369.5	739.0	537.5
Vertical / Horizontal 1/2 subsampling	8	891	10	207.9	2048	1500	106.5	213.0	155.0
	4	891	(8)	115.7			197.5	395.0	287.3
	8	891	12	207.9			106.5	213.0	155.0
	4	891	(8)	115.7			197.5	395.0	287.3
ROI	8	891	10	*2	*3	*3	192.5	385.0	280.0
	4	891	(8)	*2			369.5	739.0	537.5
	8	891	12	*2			192.5	385.0	280.0
	4	891	(8)	*2			369.5	739.0	537.5

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	31.5	4096	3000	374.0	748.0	544.0
	4	445.5	(8)	16.3			725.5	1451.0	1055.3
	8	445.5	12	31.5			374.0	748.0	544.0
	4	445.5	(8)	16.3			725.5	1451.0	1055.3
Vertical / Horizontal 1/2 subsampling	8	445.5	10	113.1	2048	1500	202.0	404.0	293.9
	4	445.5	(8)	60.9			381.5	763.0	555.0
	8	445.5	12	113.1			202.0	404.0	293.9
	4	445.5	(8)	60.9			381.5	763.0	555.0
ROI	8	445.5	10	*2	*3	*3	374.0	748.0	544.0
	4	445.5	(8)	*2			725.5	1451.0	1055.3
	8	445.5	12	*2			374.0	748.0	544.0
	4	445.5	(8)	*2			725.5	1451.0	1055.3

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	175.2	4096	3000	64.5	129.0	93.9
	4	4752		126.0			91.0	182.0	132.4
	2	4752		67.8			172.0	344.0	250.2
	1	4752		34.3			343.0	686.0	499.0
	8	4752	12 (8)	121.0			95.0	190.0	138.2
	4	4752		121.0			95.0	190.0	138.2
	2	4752		67.8			172.0	344.0	250.2
	1	4752		34.3			343.0	686.0	499.0
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	330.7	2048	1500	64.5	129.0	93.9
	4	4752		330.7			64.5	129.0	93.9
	2	4752		239.7			91.5	183.0	133.1
	1	4752		127.4			178.5	357.0	259.7
	8	4752	12 (8)	232.0			95.0	190.0	138.2
	4	4752		232.0			95.0	190.0	138.2
	2	4752		232.0			95.0	190.0	138.2
	1	4752		127.4			178.5	357.0	259.7
ROI	8	4752	10 (8)	*2	*3	*3	64.5	129.0	93.9
	4	4752		*2			91.0	182.0	132.4
	2	4752		*2			172.0	344.0	250.2
	1	4752		*2			343.0	686.0	499.0
	8	4752	12 (8)	*2			95.0	190.0	138.2
	4	4752		*2			95.0	190.0	138.2
	2	4752		*2			172.0	344.0	250.2
	1	4752		*2			343.0	686.0	499.0

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 175.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	119.1	4096	3000	96.5	193.0	140.4
	4	2376		65.4			178.5	357.0	259.7
	2	2376		34.3			343.5	687.0	499.7
	1	2376		17.2			685.5	1371.0	997.1
	8	2376	12 (8)	119.1			96.5	193.0	140.4
	4	2376		65.4			178.5	357.0	259.7
	2	2376		34.3			343.5	687.0	499.7
	1	2376		17.2			685.5	1371.0	997.1
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	330.7	2048	1500	64.5	129.0	93.9
	4	2376		233.9			94.0	188.0	136.8
	2	2376		126.3			180.0	360.0	261.9
	1	2376		64.9			357.0	714.0	519.3
	8	2376	12 (8)	232.0			95.0	190.0	138.2
	4	2376		232.0			95.0	190.0	138.2
	2	2376		126.3			180.0	360.0	261.9
	1	2376		64.9			357.0	714.0	519.3
ROI	8	2376	10 (8)	*2	*3	*3	96.5	193.0	140.4
	4	2376		*2			178.5	357.0	259.7
	2	2376		*2			343.5	687.0	499.7
	1	2376		*2			685.5	1371.0	997.1
	8	2376	12 (8)	*2			96.5	193.0	140.4
	4	2376		*2			178.5	357.0	259.7
	2	2376		*2			343.5	687.0	499.7
	1	2376		*2			685.5	1371.0	997.1

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 119.1 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	61.5	4096	3000	190.0	380.0	276.4
	4	1188		33.2			355.0	710.0	516.4
	2	1188		17.2			687.0	1374.0	999.3
	1	1188		8.6			1370.5	2741.0	1993.5
	8	1188	12 (8)	61.5			190.0	380.0	276.4
	4	1188		33.2			355.0	710.0	516.4
	2	1188		17.2			687.0	1374.0	999.3
	1	1188		8.6			1370.5	2741.0	1993.5
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	212.9	2048	1500	104.0	208.0	151.3
	4	1188		121.9			186.5	373.0	271.3
	2	1188		64.4			360.0	720.0	523.7
	1	1188		32.7			714.0	1428.0	1038.6
	8	1188	12 (8)	212.9			104.0	208.0	151.3
	4	1188		121.9			186.5	373.0	271.3
	2	1188		64.4			360.0	720.0	523.7
	1	1188		32.7			714.0	1428.0	1038.6
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	190.0	380.0	276.4
	4	1188		^{*2}			355.0	710.0	516.4
	2	1188		^{*2}			687.0	1374.0	999.3
	1	1188		^{*2}			1370.5	2741.0	1993.5
	8	1188	12 (8)	^{*2}			190.0	380.0	276.4
	4	1188		^{*2}			355.0	710.0	516.4
	2	1188		^{*2}			687.0	1374.0	999.3
	1	1188		^{*2}			1370.5	2741.0	1993.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 61.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

5.9. IMX536-AAMJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	59.9	2840	2840	205.5	411.0	299.0
	4	594	(8)	31.6			393.0	786.0	571.7
	8	594	12	59.9			205.5	411.0	299.0
	4	594	(8)	31.6			393.0	786.0	571.7
Vertical / Horizontal 1/2 subsampling	8	594	10	198.1	1420	1420	116.5	233.0	169.5
	4	594	(8)	110.6			215.0	430.0	312.8
	8	594	12	198.1			116.5	233.0	169.5
	4	594	(8)	110.6			215.0	430.0	312.8
2 × 2 FD binning mode	8	594	10	198.1	1420	1420	116.5	233.0	169.5
	4	594	(8)	110.6			215.0	430.0	312.8
	8	594	12	198.1			116.5	233.0	169.5
	4	594	(8)	110.6			215.0	430.0	312.8
ROI	8	594	10	^{*2}	^{*3}	^{*3}	205.5	411.0	299.0
	4	594	(8)	^{*2}			393.0	786.0	571.7
	8	594	12	^{*2}			205.5	411.0	299.0
	4	594	(8)	^{*2}			393.0	786.0	571.7

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	31.1	2840	2840	399.0	798.0	580.4
	4	297	(8)	16.2			768.5	1537.0	1117.9
	8	297	12	31.1			399.0	798.0	580.4
	4	297	(8)	16.2			768.5	1537.0	1117.9
Vertical / Horizontal 1/2 subsampling	8	297	10	108.2	1420	1420	221.0	442.0	321.5
	4	297	(8)	58.5			412.5	825.0	600.0
	8	297	12	108.2			221.0	442.0	321.5
	4	297	(8)	58.5			412.5	825.0	600.0
2 × 2 FD binning mode	8	297	10	108.2	1420	1420	221.0	442.0	321.5
	4	297	(8)	58.5			412.5	825.0	600.0
	8	297	12	108.2			221.0	442.0	321.5
	4	297	(8)	58.5			412.5	825.0	600.0
ROI	8	297	10	^{*2}	^{*3}	^{*3}	399.0	798.0	580.4
	4	297	(8)	^{*2}			768.5	1537.0	1117.9
	8	297	12	^{*2}			399.0	798.0	580.4
	4	297	(8)	^{*2}			768.5	1537.0	1117.9

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 59.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	87.0	2840	2840	140.5	281.0	204.4
	4	891	(8)	46.5			266.0	532.0	387.0
	8	891	12	87.0			140.5	281.0	204.4
	4	891	(8)	46.5			266.0	532.0	387.0
Vertical / Horizontal 1/2 subsampling	8	891	10	277.7	1420	1420	81.5	163.0	118.6
	4	891	(8)	159.7			147.5	295.0	214.6
	8	891	12	277.7			81.5	163.0	118.6
	4	891	(8)	159.7			147.5	295.0	214.6
2 × 2 FD binning mode	8	891	10	277.7	1420	1420	81.5	163.0	118.6
	4	891	(8)	159.7			147.5	295.0	214.6
	8	891	12	277.7			81.5	163.0	118.6
	4	891	(8)	159.7			147.5	295.0	214.6
ROI	8	891	10	^{*2}	^{*3}	^{*3}	140.5	281.0	204.4
	4	891	(8)	^{*2}			266.0	532.0	387.0
	8	891	12	^{*2}			140.5	281.0	204.4
	4	891	(8)	^{*2}			266.0	532.0	387.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	45.9	2840	2840	269.5	539.0	392.0
	4	445.5	(8)	24.1			516.0	1032.0	750.6
	8	445.5	12	45.9			269.5	539.0	392.0
	4	445.5	(8)	24.1			516.0	1032.0	750.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	156.0	1420	1420	151.0	302.0	219.7
	4	445.5	(8)	85.8			278.5	557.0	405.1
	8	445.5	12	156.0			151.0	302.0	219.7
	4	445.5	(8)	85.8			278.5	557.0	405.1
2 × 2 FD binning mode	8	445.5	10	156.0	1420	1420	151.0	302.0	219.7
	4	445.5	(8)	85.8			278.5	557.0	405.1
	8	445.5	12	156.0			151.0	302.0	219.7
	4	445.5	(8)	85.8			278.5	557.0	405.1
ROI	8	445.5	10	^{*2}	^{*3}	^{*3}	269.5	539.0	392.0
	4	445.5	(8)	^{*2}			516.0	1032.0	750.6
	8	445.5	12	^{*2}			269.5	539.0	392.0
	4	445.5	(8)	^{*2}			516.0	1032.0	750.6

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 87.0 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	184.0	2840	2840	64.5	129.0	93.9
	4	4752		182.5			65.0	130.0	94.6
	2	4752		99.4			122.5	245.0	178.2
	1	4752		51.3			240.5	481.0	349.9
	8	4752	12 (8)	127.2			95.0	190.0	138.2
	4	4752		127.2			95.0	190.0	138.2
	2	4752		99.4			122.5	245.0	178.2
	1	4752		51.3			240.5	481.0	349.9
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	618.1	1420	1420	32.5	65.0	47.3
	4	4752		575.8			35.5	71.0	51.7
	2	4752		349.6			63.5	127.0	92.4
	1	4752		183.8			127.5	255.0	185.5
	8	4752	12 (8)	450.2			47.5	95.0	69.1
	4	4752		450.2			47.5	95.0	69.1
	2	4752		349.6			63.5	127.0	92.4
	1	4752		183.8			127.5	255.0	185.5
2 × 2 FD binning mode	8	4752	10 (8)	618.1	1420	1420	32.5	65.0	47.3
	4	4752		575.8			35.5	71.0	51.7
	2	4752		349.6			63.5	127.0	92.4
	1	4752		183.8			127.5	255.0	185.5
	8	4752	12 (8)	450.2			47.5	95.0	69.1
	4	4752		450.2			47.5	95.0	69.1
	2	4752		349.6			63.5	127.0	92.4
	1	4752		183.8			127.5	255.0	185.5
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	64.5	129.0	93.9
	4	4752		^{*2}			65.0	130.0	94.6
	2	4752		^{*2}			122.5	245.0	178.2
	1	4752		^{*2}			240.5	481.0	349.9
	8	4752	12 (8)	^{*2}			95.0	190.0	138.2
	4	4752		^{*2}			95.0	190.0	138.2
	2	4752		^{*2}			122.5	245.0	178.2
	1	4752		^{*2}			240.5	481.0	349.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 184.0 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	170.2	2840	2840	70.0	140.0	101.9
	4	2376		96.5			126.5	253.0	184.0
	2	2376		50.9			242.5	485.0	352.8
	1	2376		25.9			480.5	961.0	699.0
	8	2376	12 (8)	127.2			95.0	190.0	138.2
	4	2376		96.5			126.5	253.0	184.0
	2	2376		50.9			242.5	485.0	352.8
	1	2376		25.9			480.5	961.0	699.0
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	516.1	1420	1420	40.5	81.0	59.0
	4	2376		328.9			67.5	135.0	98.2
	2	2376		185.2			126.5	253.0	184.0
	1	2376		93.8			255.0	510.0	371.0
	8	2376	12 (8)	450.2			47.5	95.0	69.1
	4	2376		328.9			67.5	135.0	98.2
	2	2376		185.2			126.5	253.0	184.0
	1	2376		93.8			255.0	510.0	371.0
2 × 2 FD binning mode	8	2376	10 (8)	516.1	1420	1420	40.5	81.0	59.0
	4	2376		328.9			67.5	135.0	98.2
	2	2376		185.2			126.5	253.0	184.0
	1	2376		93.8			255.0	510.0	371.0
	8	2376	12 (8)	450.2			47.5	95.0	69.1
	4	2376		328.9			67.5	135.0	98.2
	2	2376		185.2			126.5	253.0	184.0
	1	2376		93.8			255.0	510.0	371.0
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	70.0	140.0	101.9
	4	2376		^{*2}			126.5	253.0	184.0
	2	2376		^{*2}			242.5	485.0	352.8
	1	2376		^{*2}			480.5	961.0	699.0
	8	2376	12 (8)	^{*2}			95.0	190.0	138.2
	4	2376		^{*2}			126.5	253.0	184.0
	2	2376		^{*2}			242.5	485.0	352.8
	1	2376		^{*2}			480.5	961.0	699.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 170.2 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	89.2	2840	2840	137.0	274.0	199.3
	4	1188		49.1			251.5	503.0	365.9
	2	1188		25.6			485.0	970.0	705.5
	1	1188		13.0			961.0	1922.0	1397.9
	8	1188	12 (8)	89.2			137.0	274.0	199.3
	4	1188		49.1			251.5	503.0	365.9
	2	1188		25.6			485.0	970.0	705.5
	1	1188		13.0			961.0	1922.0	1397.9
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	288.8	1420	1420	78.0	156.0	113.5
	4	1188		173.6			135.0	270.0	196.4
	2	1188		94.5			253.0	506.0	368.0
	1	1188		47.6			509.5	1019.0	741.1
	8	1188	12 (8)	288.8			78.0	156.0	113.5
	4	1188		173.6			135.0	270.0	196.4
	2	1188		94.5			253.0	506.0	368.0
	1	1188		47.6			509.5	1019.0	741.1
2 × 2 FD binning mode	8	1188	10 (8)	288.8	1420	1420	78.0	156.0	113.5
	4	1188		173.6			135.0	270.0	196.4
	2	1188		94.5			253.0	506.0	368.0
	1	1188		47.6			509.5	1019.0	741.1
	8	1188	12 (8)	288.8			78.0	156.0	113.5
	4	1188		173.6			135.0	270.0	196.4
	2	1188		94.5			253.0	506.0	368.0
	1	1188		47.6			509.5	1019.0	741.1
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	137.0	274.0	199.3
	4	1188		^{*2}			251.5	503.0	365.9
	2	1188		^{*2}			485.0	970.0	705.5
	1	1188		^{*2}			961.0	1922.0	1397.9
	8	1188	12 (8)	^{*2}			137.0	274.0	199.3
	4	1188		^{*2}			251.5	503.0	365.9
	2	1188		^{*2}			485.0	970.0	705.5
	1	1188		^{*2}			961.0	1922.0	1397.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 89.2 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	60.0	2840	2840	205.5	411.0	299.0
	4	594	(8)	31.6			393.0	786.0	571.7
	8	594	12	60.0			205.5	411.0	299.0
	4	594	(8)	31.6			393.0	786.0	571.7
Vertical / Horizontal 1/2 subsampling	8	594	10	200.6	1420	1420	116.5	233.0	169.5
	4	594	(8)	111.8			215.0	430.0	312.8
	8	594	12	200.6			116.5	233.0	169.5
	4	594	(8)	111.8			215.0	430.0	312.8
ROI	8	594	10	*2	*3	*3	205.5	411.0	299.0
	4	594	(8)	*2			393.0	786.0	571.7
	8	594	12	*2			205.5	411.0	299.0
	4	594	(8)	*2			393.0	786.0	571.7

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	31.2	2840	2840	399.0	798.0	580.4
	4	297	(8)	16.2			768.5	1537.0	1117.9
	8	297	12	31.2			399.0	798.0	580.4
	4	297	(8)	16.2			768.5	1537.0	1117.9
Vertical / Horizontal 1/2 subsampling	8	297	10	109.0	1420	1420	221.0	442.0	321.5
	4	297	(8)	59.3			412.5	825.0	600.0
	8	297	12	109.0			221.0	442.0	321.5
	4	297	(8)	59.3			412.5	825.0	600.0
ROI	8	297	10	*2	*3	*3	399.0	798.0	580.4
	4	297	(8)	*2			768.5	1537.0	1117.9
	8	297	12	*2			399.0	798.0	580.4
	4	297	(8)	*2			768.5	1537.0	1117.9

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.0 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	87.1	2840	2840	140.5	281.0	204.4
	4	891	(8)	46.6			266.0	532.0	387.0
	8	891	12	87.1			140.5	281.0	204.4
	4	891	(8)	46.6			266.0	532.0	387.0
Vertical / Horizontal 1/2 subsampling	8	891	10	280.4	1420	1420	81.5	163.0	118.6
	4	891	(8)	160.9			147.5	295.0	214.6
	8	891	12	243.6			95.0	190.0	138.2
	4	891	(8)	160.9			147.5	295.0	214.6
ROI	8	891	10	*2	*3	*3	140.5	281.0	204.4
	4	891	(8)	*2			266.0	532.0	387.0
	8	891	12	*2			140.5	281.0	204.4
	4	891	(8)	*2			266.0	532.0	387.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	46.0	2840	2840	269.5	539.0	392.0
	4	445.5	(8)	24.1			516.0	1032.0	750.6
	8	445.5	12	46.0			269.5	539.0	392.0
	4	445.5	(8)	24.1			516.0	1032.0	750.6
Vertical / Horizontal 1/2 subsampling	8	445.5	10	157.2	1420	1420	151.0	302.0	219.7
	4	445.5	(8)	87.2			278.5	557.0	405.1
	8	445.5	12	157.2			151.0	302.0	219.7
	4	445.5	(8)	87.2			278.5	557.0	405.1
ROI	8	445.5	10	*2	*3	*3	269.5	539.0	392.0
	4	445.5	(8)	*2			516.0	1032.0	750.6
	8	445.5	12	*2			269.5	539.0	392.0
	4	445.5	(8)	*2			516.0	1032.0	750.6

- *1 The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 87.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]
*2 See the section of "ROI mode"
*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	184.2	2840	2840	64.5	129.0	93.9
	4	4752		182.8			65.0	130.0	94.6
	2	4752		99.5			122.5	245.0	178.2
	1	4752		51.4			240.5	481.0	349.9
	8	4752	12 (8)	127.3			95.0	190.0	138.2
	4	4752		127.3			95.0	190.0	138.2
	2	4752		99.5			122.5	245.0	178.2
	1	4752		51.4			240.5	481.0	349.9
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	346.7	1420	1420	64.5	129.0	93.9
	4	4752		346.7			64.5	129.0	93.9
	2	4752		346.7			64.5	129.0	93.9
	1	4752		185.2			127.5	255.0	185.5
	8	4752	12 (8)	243.6			95.0	190.0	138.2
	4	4752		243.6			95.0	190.0	138.2
	2	4752		243.6			95.0	190.0	138.2
	1	4752		185.2			127.5	255.0	185.5
ROI	8	4752	10 (8)	*2	*3	*3	64.5	129.0	93.9
	4	4752		*2			65.0	130.0	94.6
	2	4752		*2			122.5	245.0	178.2
	1	4752		*2			240.5	481.0	349.9
	8	4752	12 (8)	*2			95.0	190.0	138.2
	4	4752		*2			95.0	190.0	138.2
	2	4752		*2			122.5	245.0	178.2
	1	4752		*2			240.5	481.0	349.9

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 184.2 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	170.4	2840	2840	70.0	140.0	101.9
	4	2376		96.6			126.5	253.0	184.0
	2	2376		51.0			242.5	485.0	352.8
	1	2376		25.9			480.5	961.0	699.0
	8	2376	12 (8)	127.3			95.0	190.0	138.2
	4	2376		96.6			126.5	253.0	184.0
	2	2376		51.0			242.5	485.0	352.8
	1	2376		25.9			480.5	961.0	699.0
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	346.7	1420	1420	64.5	129.0	93.9
	4	2376		332.1			67.5	135.0	98.2
	2	2376		186.6			126.5	253.0	184.0
	1	2376		95.2			255.0	510.0	371.0
	8	2376	12 (8)	243.6			95.0	190.0	138.2
	4	2376		243.6			95.0	190.0	138.2
	2	2376		186.6			126.5	253.0	184.0
	1	2376		95.2			255.0	510.0	371.0
ROI	8	2376	10 (8)	*2	*3	*3	70.0	140.0	101.9
	4	2376		*2			126.5	253.0	184.0
	2	2376		*2			242.5	485.0	352.8
	1	2376		*2			480.5	961.0	699.0
	8	2376	12 (8)	*2			95.0	190.0	138.2
	4	2376		*2			126.5	253.0	184.0
	2	2376		*2			242.5	485.0	352.8
	1	2376		*2			480.5	961.0	699.0

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 170.4 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	89.3	2840	2840	137.0	274.0	199.3
	4	1188		49.2			251.5	503.0	365.9
	2	1188		25.7			485.0	970.0	705.5
	1	1188		13.0			961.0	1922.0	1397.9
	8	1188	12 (8)	89.3			137.0	274.0	199.3
	4	1188		49.2			251.5	503.0	365.9
	2	1188		25.7			485.0	970.0	705.5
	1	1188		13.0			961.0	1922.0	1397.9
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	291.6	1420	1420	78.0	156.0	113.5
	4	1188		174.9			135.0	270.0	196.4
	2	1188		96.0			253.0	506.0	368.0
	1	1188		48.1			509.5	1019.0	741.1
	8	1188	12 (8)	243.6			95.0	190.0	138.2
	4	1188		174.9			135.0	270.0	196.4
	2	1188		96.0			253.0	506.0	368.0
	1	1188		48.1			509.5	1019.0	741.1
ROI	8	1188	10 (8)	*2	*3	*3	137.0	274.0	199.3
	4	1188		*2			251.5	503.0	365.9
	2	1188		*2			485.0	970.0	705.5
	1	1188		*2			961.0	1922.0	1397.9
	8	1188	12 (8)	*2			137.0	274.0	199.3
	4	1188		*2			251.5	503.0	365.9
	2	1188		*2			485.0	970.0	705.5
	1	1188		*2			961.0	1922.0	1397.9

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 89.3 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

5.11. IMX537-AAMJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	91.4	2448	2048	181.5	363.0	264.0
	4	594	(8)	48.7			345.0	690.0	501.9
	8	594	12	91.4			181.5	363.0	264.0
	4	594	(8)	48.7			345.0	690.0	501.9
Vertical / Horizontal 1/2 subsampling	8	594	10	290.2	1224	1024	104.5	209.0	152.0
	4	594	(8)	166.4			191.0	382.0	277.9
	8	594	12	290.2			104.5	209.0	152.0
	4	594	(8)	166.4			191.0	382.0	277.9
2 × 2 FD binning mode	8	594	10	290.2	1224	1024	104.5	209.0	152.0
	4	594	(8)	166.4			191.0	382.0	277.9
	8	594	12	290.2			104.5	209.0	152.0
	4	594	(8)	166.4			191.0	382.0	277.9
ROI	8	594	10	^{*2}	^{*3}	^{*3}	181.5	363.0	264.0
	4	594	(8)	^{*2}			345.0	690.0	501.9
	8	594	12	^{*2}			181.5	363.0	264.0
	4	594	(8)	^{*2}			345.0	690.0	501.9

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	47.9	2448	2048	351.0	702.0	510.6
	4	297	(8)	25.1			672.5	1345.0	978.2
	8	297	12	47.9			351.0	702.0	510.6
	4	297	(8)	25.1			672.5	1345.0	978.2
Vertical / Horizontal 1/2 subsampling	8	297	10	161.3	1224	1024	197.0	394.0	286.6
	4	297	(8)	88.4			364.5	729.0	530.2
	8	297	12	161.3			197.0	394.0	286.6
	4	297	(8)	88.4			364.5	729.0	530.2
2 × 2 FD binning mode	8	297	10	161.3	1224	1024	197.0	394.0	286.6
	4	297	(8)	88.4			364.5	729.0	530.2
	8	297	12	161.3			197.0	394.0	286.6
	4	297	(8)	88.4			364.5	729.0	530.2
ROI	8	297	10	^{*2}	^{*3}	^{*3}	351.0	702.0	510.6
	4	297	(8)	^{*2}			672.5	1345.0	978.2
	8	297	12	^{*2}			351.0	702.0	510.6
	4	297	(8)	^{*2}			672.5	1345.0	978.2

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 91.4 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	131.7	2448	2048	124.5	249.0	181.1
	4	891	(8)	71.3			234.0	468.0	340.4
	8	891	12	131.7			124.5	249.0	181.1
	4	891	(8)	71.3			234.0	468.0	340.4
Vertical / Horizontal 1/2 subsampling	8	891	10	397.0	1224	1024	73.5	147.0	107.0
	4	891	(8)	236.8			131.5	263.0	191.3
	8	891	12	397.0			73.5	147.0	107.0
	4	891	(8)	236.8			131.5	263.0	191.3
2 × 2 FD binning mode	8	891	10	397.0	1224	1024	73.5	147.0	107.0
	4	891	(8)	236.8			131.5	263.0	191.3
	8	891	12	397.0			73.5	147.0	107.0
	4	891	(8)	236.8			131.5	263.0	191.3
ROI	8	891	10	*2	*3	*3	124.5	249.0	181.1
	4	891	(8)	*2			234.0	468.0	340.4
	8	891	12	*2			124.5	249.0	181.1
	4	891	(8)	*2			234.0	468.0	340.4

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	70.2	2448	2048	237.5	475.0	345.5
	4	445.5	(8)	37.4			452.0	904.0	657.5
	8	445.5	12	70.2			237.5	475.0	345.5
	4	445.5	(8)	37.4			452.0	904.0	657.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10	230.7	1224	1024	135.0	270.0	196.4
	4	445.5	(8)	129.8			246.5	493.0	358.6
	8	445.5	12	230.7			135.0	270.0	196.4
	4	445.5	(8)	129.8			246.5	493.0	358.6
2 × 2 FD binning mode	8	445.5	10	230.7	1224	1024	135.0	270.0	196.4
	4	445.5	(8)	129.8			246.5	493.0	358.6
	8	445.5	12	230.7			135.0	270.0	196.4
	4	445.5	(8)	129.8			246.5	493.0	358.6
ROI	8	445.5	10	*2	*3	*3	237.5	475.0	345.5
	4	445.5	(8)	*2			452.0	904.0	657.5
	8	445.5	12	*2			237.5	475.0	345.5
	4	445.5	(8)	*2			452.0	904.0	657.5

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 131.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	245.5	2448	2048	64.5	129.0	93.9
	4	4752		245.5			64.5	129.0	93.9
	2	4752		152.1			107.0	214.0	155.7
	1	4752		79.5			209.5	419.0	304.8
	8	4752	12 (8)	170.7			95.0	190.0	138.2
	4	4752		170.7			95.0	190.0	138.2
	2	4752		152.1			107.0	214.0	155.7
	1	4752		79.5			209.5	419.0	304.8
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	784.5	1224	1024	32.5	65.0	47.3
	4	4752		784.5			32.5	65.0	47.3
	2	4752		509.8			55.5	111.0	80.8
	1	4752		272.5			112.0	224.0	163.0
	8	4752	12 (8)	581.5			47.5	95.0	69.1
	4	4752		581.5			47.5	95.0	69.1
	2	4752		509.8			55.5	111.0	80.8
	1	4752		272.5			112.0	224.0	163.0
2 × 2 FD binning mode	8	4752	10 (8)	784.5	1224	1024	32.5	65.0	47.3
	4	4752		784.5			32.5	65.0	47.3
	2	4752		509.8			55.5	111.0	80.8
	1	4752		272.5			112.0	224.0	163.0
	8	4752	12 (8)	581.5			47.5	95.0	69.1
	4	4752		581.5			47.5	95.0	69.1
	2	4752		509.8			55.5	111.0	80.8
	1	4752		272.5			112.0	224.0	163.0
ROI	8	4752	10 (8)	^{*2}	^{*3}	^{*3}	64.5	129.0	93.9
	4	4752		^{*2}			64.5	129.0	93.9
	2	4752		^{*2}			107.0	214.0	155.7
	1	4752		^{*2}			209.5	419.0	304.8
	8	4752	12 (8)	^{*2}			95.0	190.0	138.2
	4	4752		^{*2}			95.0	190.0	138.2
	2	4752		^{*2}			107.0	214.0	155.7
	1	4752		^{*2}			209.5	419.0	304.8

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 245.5 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	245.5	2448	2048	64.5	129.0	93.9
	4	2376		148.2			110.0	220.0	160.0
	2	2376		78.7			211.5	423.0	307.7
	1	2376		40.2			419.0	838.0	609.5
	8	2376	12 (8)	170.7			95.0	190.0	138.2
	4	2376		148.2			110.0	220.0	160.0
	2	2376		78.7			211.5	423.0	307.7
	1	2376		40.2			419.0	838.0	609.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	714.2	1224	1024	36.5	73.0	53.1
	4	2376		465.7			61.5	123.0	89.5
	2	2376		275.0			111.0	222.0	161.5
	1	2376		142.8			224.0	448.0	325.9
	8	2376	12 (8)	581.5			47.5	95.0	69.1
	4	2376		465.7			61.5	123.0	89.5
	2	2376		275.0			111.0	222.0	161.5
	1	2376		142.8			224.0	448.0	325.9
2 × 2 FD binning mode	8	2376	10 (8)	714.2	1224	1024	36.5	73.0	53.1
	4	2376		465.7			61.5	123.0	89.5
	2	2376		275.0			111.0	222.0	161.5
	1	2376		142.8			224.0	448.0	325.9
	8	2376	12 (8)	581.5			47.5	95.0	69.1
	4	2376		465.7			61.5	123.0	89.5
	2	2376		275.0			111.0	222.0	161.5
	1	2376		142.8			224.0	448.0	325.9
ROI	8	2376	10 (8)	^{*2}	^{*3}	^{*3}	64.5	129.0	93.9
	4	2376		^{*2}			110.0	220.0	160.0
	2	2376		^{*2}			211.5	423.0	307.7
	1	2376		^{*2}			419.0	838.0	609.5
	8	2376	12 (8)	^{*2}			95.0	190.0	138.2
	4	2376		^{*2}			110.0	220.0	160.0
	2	2376		^{*2}			211.5	423.0	307.7
	1	2376		^{*2}			419.0	838.0	609.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 245.5 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	135.5	2448	2048	121.0	242.0	176.0
	4	1188		75.9			219.5	439.0	319.3
	2	1188		39.8			423.0	846.0	615.3
	1	1188		20.2			838.0	1676.0	1219.0
	8	1188	12 (8)	135.5			121.0	242.0	176.0
	4	1188		75.9			219.5	439.0	319.3
	2	1188		39.8			423.0	846.0	615.3
	1	1188		20.2			838.0	1676.0	1219.0
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	416.9	1224	1024	70.0	140.0	101.9
	4	1188		253.9			121.0	242.0	176.0
	2	1188		144.1			222.0	444.0	323.0
	1	1188		72.9			448.0	896.0	651.7
	8	1188	12 (8)	416.9			70.0	140.0	101.9
	4	1188		253.9			121.0	242.0	176.0
	2	1188		144.1			222.0	444.0	323.0
	1	1188		72.9			448.0	896.0	651.7
2 × 2 FD binning mode	8	1188	10 (8)	416.9	1224	1024	70.0	140.0	101.9
	4	1188		253.9			121.0	242.0	176.0
	2	1188		144.1			222.0	444.0	323.0
	1	1188		72.9			448.0	896.0	651.7
	8	1188	12 (8)	416.9			70.0	140.0	101.9
	4	1188		253.9			121.0	242.0	176.0
	2	1188		144.1			222.0	444.0	323.0
	1	1188		72.9			448.0	896.0	651.7
ROI	8	1188	10 (8)	^{*2}	^{*3}	^{*3}	121.0	242.0	176.0
	4	1188		^{*2}			219.5	439.0	319.3
	2	1188		^{*2}			423.0	846.0	615.3
	1	1188		^{*2}			838.0	1676.0	1219.0
	8	1188	12 (8)	^{*2}			121.0	242.0	176.0
	4	1188		^{*2}			219.5	439.0	319.3
	2	1188		^{*2}			423.0	846.0	615.3
	1	1188		^{*2}			838.0	1676.0	1219.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 135.5 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

5.12. IMX537-AAQJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10	91.6	2448	2048	181.5	363.0	264.0
	4	594	(8)	48.8			345.0	690.0	501.9
	8	594	12	91.6			181.5	363.0	264.0
	4	594	(8)	48.8			345.0	690.0	501.9
Vertical / Horizontal 1/2 subsampling	8	594	10	295.0	1224	1024	104.5	209.0	152.0
	4	594	(8)	168.1			191.0	382.0	277.9
	8	594	12	295.0			104.5	209.0	152.0
	4	594	(8)	168.1			191.0	382.0	277.9
ROI	8	594	10	*2	*3	*3	181.5	363.0	264.0
	4	594	(8)	*2			345.0	690.0	501.9
	8	594	12	*2			181.5	363.0	264.0
	4	594	(8)	*2			345.0	690.0	501.9

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10	47.9	2448	2048	351.0	702.0	510.6
	4	297	(8)	25.2			672.5	1345.0	978.2
	8	297	12	47.9			351.0	702.0	510.6
	4	297	(8)	25.2			672.5	1345.0	978.2
Vertical / Horizontal 1/2 subsampling	8	297	10	163.5	1224	1024	197.0	394.0	286.6
	4	297	(8)	90.2			364.5	729.0	530.2
	8	297	12	163.5			197.0	394.0	286.6
	4	297	(8)	90.2			364.5	729.0	530.2
ROI	8	297	10	*2	*3	*3	351.0	702.0	510.6
	4	297	(8)	*2			672.5	1345.0	978.2
	8	297	12	*2			351.0	702.0	510.6
	4	297	(8)	*2			672.5	1345.0	978.2

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 91.6 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10	131.9	2448	2048	124.5	249.0	181.1
	4	891	(8)	71.4			234.0	468.0	340.4
	8	891	12	131.9			124.5	249.0	181.1
	4	891	(8)	71.4			234.0	468.0	340.4
Vertical / Horizontal 1/2 subsampling	8	891	10	403.4	1224	1024	73.5	147.0	107.0
	4	891	(8)	239.2			131.5	263.0	191.3
	8	891	12	322.4			95.0	190.0	138.2
	4	891	(8)	239.2			131.5	263.0	191.3
ROI	8	891	10	*2	*3	*3	124.5	249.0	181.1
	4	891	(8)	*2			234.0	468.0	340.4
	8	891	12	*2			124.5	249.0	181.1
	4	891	(8)	*2			234.0	468.0	340.4

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10	70.4	2448	2048	237.5	475.0	345.5
	4	445.5	(8)	37.4			452.0	904.0	657.5
	8	445.5	12	70.4			237.5	475.0	345.5
	4	445.5	(8)	37.4			452.0	904.0	657.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10	233.0	1224	1024	135.0	270.0	196.4
	4	445.5	(8)	131.6			246.5	493.0	358.6
	8	445.5	12	233.0			135.0	270.0	196.4
	4	445.5	(8)	131.6			246.5	493.0	358.6
ROI	8	445.5	10	*2	*3	*3	237.5	475.0	345.5
	4	445.5	(8)	*2			452.0	904.0	657.5
	8	445.5	12	*2			237.5	475.0	345.5
	4	445.5	(8)	*2			452.0	904.0	657.5

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 131.9 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	4752	10 (8)	245.9	2448	2048	64.5	129.0	93.9
	4	4752		245.9			64.5	129.0	93.9
	2	4752		152.4			107.0	214.0	155.7
	1	4752		79.6			209.5	419.0	304.8
	8	4752	12 (8)	171.0			95.0	190.0	138.2
	4	4752		171.0			95.0	190.0	138.2
	2	4752		152.4			107.0	214.0	155.7
	1	4752		79.6			209.5	419.0	304.8
Vertical / Horizontal 1/2 subsampling	8	4752	10 (8)	453.9	1224	1024	64.5	129.0	93.9
	4	4752		453.9			64.5	129.0	93.9
	2	4752		453.9			64.5	129.0	93.9
	1	4752		277.1			112.0	224.0	163.0
	8	4752	12 (8)	322.4			95.0	190.0	138.2
	4	4752		322.4			95.0	190.0	138.2
	2	4752		322.4			95.0	190.0	138.2
	1	4752		277.1			112.0	224.0	163.0
ROI	8	4752	10 (8)	*2	*3	*3	64.5	129.0	93.9
	4	4752		*2			64.5	129.0	93.9
	2	4752		*2			107.0	214.0	155.7
	1	4752		*2			209.5	419.0	304.8
	8	4752	12 (8)	*2			95.0	190.0	138.2
	4	4752		*2			95.0	190.0	138.2
	2	4752		*2			107.0	214.0	155.7
	1	4752		*2			209.5	419.0	304.8

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 245.9 [frame/s] mode: 4752 [Mbps/Lane] × 8 = 38.016 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	2376	10 (8)	245.9	2448	2048	64.5	129.0	93.9
	4	2376		148.5			110.0	220.0	160.0
	2	2376		78.9			211.5	423.0	307.7
	1	2376		40.3			419.0	838.0	609.5
	8	2376	12 (8)	171.0			95.0	190.0	138.2
	4	2376		148.5			110.0	220.0	160.0
	2	2376		78.9			211.5	423.0	307.7
	1	2376		40.3			419.0	838.0	609.5
Vertical / Horizontal 1/2 subsampling	8	2376	10 (8)	453.9	1224	1024	64.5	129.0	93.9
	4	2376		453.9			64.5	129.0	93.9
	2	2376		279.6			111.0	222.0	161.5
	1	2376		144.3			224.0	448.0	325.9
	8	2376	12 (8)	322.4			95.0	190.0	138.2
	4	2376		322.4			95.0	190.0	138.2
	2	2376		279.6			111.0	222.0	161.5
	1	2376		144.3			224.0	448.0	325.9
ROI	8	2376	10 (8)	*2	*3	*3	64.5	129.0	93.9
	4	2376		*2			110.0	220.0	160.0
	2	2376		*2			211.5	423.0	307.7
	1	2376		*2			419.0	838.0	609.5
	8	2376	12 (8)	*2			95.0	190.0	138.2
	4	2376		*2			110.0	220.0	160.0
	2	2376		*2			211.5	423.0	307.7
	1	2376		*2			419.0	838.0	609.5

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 245.9 [frame/s] mode: 2376 [Mbps/Lane] × 8 = 19.008 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	1188	10 (8)	135.7	2448	2048	121.0	242.0	176.0
	4	1188		76.0			219.5	439.0	319.3
	2	1188		39.9			423.0	846.0	615.3
	1	1188		20.2			838.0	1676.0	1219.0
	8	1188	12 (8)	135.7			121.0	242.0	176.0
	4	1188		76.0			219.5	439.0	319.3
	2	1188		39.9			423.0	846.0	615.3
	1	1188		20.2			838.0	1676.0	1219.0
Vertical / Horizontal 1/2 subsampling	8	1188	10 (8)	422.2	1224	1024	70.0	140.0	101.9
	4	1188		258.2			121.0	242.0	176.0
	2	1188		145.6			222.0	444.0	323.0
	1	1188		73.9			448.0	896.0	651.7
	8	1188	12 (8)	322.4			95.0	190.0	138.2
	4	1188		258.2			121.0	242.0	176.0
	2	1188		145.6			222.0	444.0	323.0
	1	1188		73.9			448.0	896.0	651.7
ROI	8	1188	10 (8)	*2	*3	*3	121.0	242.0	176.0
	4	1188		*2			219.5	439.0	319.3
	2	1188		*2			423.0	846.0	615.3
	1	1188		*2			838.0	1676.0	1219.0
	8	1188	12 (8)	*2			121.0	242.0	176.0
	4	1188		*2			219.5	439.0	319.3
	2	1188		*2			423.0	846.0	615.3
	1	1188		*2			838.0	1676.0	1219.0

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 135.7 [frame/s] mode: 1188 [Mbps/Lane] × 8 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	21.9	5320	4600	358.0	716.0	520.8
	4	594		11.2			698.0	1396.0	1015.3
	2	594		5.7			1372.0	2744.0	1995.7
	8	594	12 (8)	21.9			358.0	716.0	520.8
	4	594		11.2			698.0	1396.0	1015.3
	2	594		5.7			1372.0	2744.0	1995.7
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	80.0	2660	2300	193.5	387.0	281.5
	4	594		42.2			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
	8	594	12 (8)	80.0			193.5	387.0	281.5
	4	594		42.2			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
2 × 2 FD binning mode	8	594	10 (8)	80.0	2660	2300	193.5	387.0	281.5
	4	594		42.2			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
	8	594	12 (8)	80.0			193.5	387.0	281.5
	4	594		42.2			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7
	8	594	12 (8)	^{*2}			358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	11.1	5320	4600	706.0	1412.0	1027.0
	4	297		5.7			1382.5	2765.0	2011.0
	2	297		2.8			2735.5	5471.0	3979.0
	8	297	12 (8)	11.1			706.0	1412.0	1027.0
	4	297		5.7			1382.5	2765.0	2011.0
	2	297		2.8			2735.5	5471.0	3979.0
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	41.5	2660	2300	375.0	750.0	545.5
	4	297		21.6			720.5	1441.0	1048.0
	2	297		11.0			1411.5	2823.0	2053.1
	8	297	12 (8)	41.5			375.0	750.0	545.5
	4	297		21.6			720.5	1441.0	1048.0
	2	297		11.0			1411.5	2823.0	2053.1
2 × 2 FD binning mode	8	297	10 (8)	41.5	2660	2300	375.0	750.0	545.5
	4	297		21.6			720.5	1441.0	1048.0
	2	297		11.0			1411.5	2823.0	2053.1
	8	297	12 (8)	41.5			375.0	750.0	545.5
	4	297		21.6			720.5	1441.0	1048.0
	2	297		11.0			1411.5	2823.0	2053.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0
	8	297	12 (8)	^{*2}			706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 11.1 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	30.2	5320	4600	260.0	520.0	378.2
	4	891		15.9			495.0	990.0	720.0
	2	891		8.2			955.0	1910.0	1389.1
	8	891	12 (8)	30.2			260.0	520.0	378.2
	4	891		15.9			495.0	990.0	720.0
	2	891		8.2			955.0	1910.0	1389.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	100.9	2660	2300	153.0	306.0	222.6
	4	891		55.4			281.0	562.0	408.8
	2	891		30.1			519.0	1038.0	755.0
	8	891	12 (8)	100.9			153.0	306.0	222.6
	4	891		55.4			281.0	562.0	408.8
	2	891		30.1			519.0	1038.0	755.0
2 × 2 FD binning mode	8	891	10 (8)	100.9	2660	2300	153.0	306.0	222.6
	4	891		55.4			281.0	562.0	408.8
	2	891		30.1			519.0	1038.0	755.0
	8	891	12 (8)	100.9			153.0	306.0	222.6
	4	891		55.4			281.0	562.0	408.8
	2	891		30.1			519.0	1038.0	755.0
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1
	8	891	12 (8)	^{*2}			260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 30.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	15.5	5320	4600	508.0	1016.0	739.0
	4	445.5		8.1			972.5	1945.0	1414.6
	2	445.5		4.1			1901.5	3803.0	2765.9
	8	445.5	12 (8)	15.5			508.0	1016.0	739.0
	4	445.5		8.1			972.5	1945.0	1414.6
	2	445.5		4.1			1901.5	3803.0	2765.9
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	53.6	2660	2300	290.0	580.0	421.9
	4	445.5		29.1			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
	8	445.5	12 (8)	53.6			290.0	580.0	421.9
	4	445.5		29.1			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
2 × 2 FD binning mode	8	445.5	10 (8)	53.6	2660	2300	290.0	580.0	421.9
	4	445.5		29.1			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
	8	445.5	12 (8)	53.6			290.0	580.0	421.9
	4	445.5		29.1			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9
	8	445.5	12 (8)	^{*2}			508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 15.5 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	35.1	5320	4600	223.5	447.0	325.1
	1	4752	(8)	17.8			440.0	880.0	640.0
	2	4752	12	35.1			223.5	447.0	325.1
	1	4752	(8)	17.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	133.6	2660	2300	115.0	230.0	167.3
	1	4752	(8)	68.8			225.0	450.0	327.3
	2	4752	12	133.6			115.0	230.0	167.3
	1	4752	(8)	68.8			225.0	450.0	327.3
2 × 2 FD binning mode	2	4752	10	133.6	2660	2300	115.0	230.0	167.3
	1	4752	(8)	68.8			225.0	450.0	327.3
	2	4752	12	133.6			115.0	230.0	167.3
	1	4752	(8)	68.8			225.0	450.0	327.3
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0
	2	4752	12	^{*2}			223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	17.7	5320	4600	443.5	887.0	645.1
	1	2376	(8)	8.9			880.0	1760.0	1280.0
	2	2376	12	12.0			655.0	1310.0	952.8
	1	2376	(8)	6.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	68.3	2660	2300	227.0	454.0	330.2
	1	2376	(8)	34.6			450.0	900.0	654.6
	2	2376	12	46.0			338.5	677.0	492.4
	1	2376	(8)	23.2			672.0	1344.0	977.5
2 × 2 FD binning mode	2	2376	10	68.3	2660	2300	227.0	454.0	330.2
	1	2376	(8)	34.6			450.0	900.0	654.6
	2	2376	12	46.0			338.5	677.0	492.4
	1	2376	(8)	23.2			672.0	1344.0	977.5
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	443.5	887.0	645.1
	1	2376	(8)	^{*2}			880.0	1760.0	1280.0
	2	2376	12	^{*2}			655.0	1310.0	952.8
	1	2376	(8)	^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 35.1 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of “ROI mode”

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	8.8	5320	4600	887.0	1774.0	1290.2
	1	1188	(8)	4.4			1759.5	3519.0	2559.3
	2	1188	12	8.8			887.0	1774.0	1290.2
	1	1188	(8)	4.4			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	2	1188	10	34.3	2660	2300	454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
	2	1188	12	34.3			454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
2 × 2 FD binning mode	2	1188	10	34.3	2660	2300	454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
	2	1188	12	34.3			454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3
	2	1188	12	^{*2}			887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 8.8 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	21.9	5320	4600	358.0	716.0	520.8
	4	594		11.2			698.0	1396.0	1015.3
	2	594		5.7			1372.0	2744.0	1995.7
	8	594	12 (8)	21.9			358.0	716.0	520.8
	4	594		11.2			698.0	1396.0	1015.3
	2	594		5.7			1372.0	2744.0	1995.7
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	80.2	2660	2300	193.5	387.0	281.5
	4	594		42.3			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
	8	594	12 (8)	74.3			209.0	418.0	304.0
	4	594		42.3			369.0	738.0	536.8
	2	594		22.0			710.0	1420.0	1032.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7
	8	594	12 (8)	^{*2}			358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	11.1	5320	4600	706.0	1412.0	1027.0
	4	297		5.7			1382.5	2765.0	2011.0
	2	297		2.8			2735.5	5471.0	3979.0
	8	297	12 (8)	11.1			706.0	1412.0	1027.0
	4	297		5.7			1382.5	2765.0	2011.0
	2	297		2.8			2735.5	5471.0	3979.0
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	41.6	2660	2300	375.0	750.0	545.5
	4	297		21.7			720.5	1441.0	1048.0
	2	297		11.1			1411.5	2823.0	2053.1
	8	297	12 (8)	41.6			375.0	750.0	545.5
	4	297		21.7			720.5	1441.0	1048.0
	2	297		11.1			1411.5	2823.0	2053.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0
	8	297	12 (8)	^{*2}			706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 11.1 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	30.2	5320	4600	260.0	520.0	378.2
	4	891		15.9			495.0	990.0	720.0
	2	891		8.2			955.0	1910.0	1389.1
	8	891	12 (8)	30.2			260.0	520.0	378.2
	4	891		15.9			495.0	990.0	720.0
	2	891		8.2			955.0	1910.0	1389.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	101.2	2660	2300	153.0	306.0	222.6
	4	891		55.5			281.0	562.0	408.8
	2	891		30.2			519.0	1038.0	755.0
	8	891	12 (8)	74.3			209.0	418.0	304.0
	4	891		55.5			281.0	562.0	408.8
	2	891		30.2			519.0	1038.0	755.0
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1
	8	891	12 (8)	^{*2}			260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 30.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	15.5	5320	4600	508.0	1016.0	739.0
	4	445.5		8.1			972.5	1945.0	1414.6
	2	445.5		4.1			1901.5	3803.0	2765.9
	8	445.5	12 (8)	15.5			508.0	1016.0	739.0
	4	445.5		8.1			972.5	1945.0	1414.6
	2	445.5		4.1			1901.5	3803.0	2765.9
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	53.7	2660	2300	290.0	580.0	421.9
	4	445.5		29.2			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
	8	445.5	12 (8)	53.7			290.0	580.0	421.9
	4	445.5		29.2			536.5	1073.0	780.4
	2	445.5		15.2			1029.5	2059.0	1497.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9
	8	445.5	12 (8)	^{*2}			508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 15.5 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	35.1	5320	4600	223.5	447.0	325.1
	1	4752	(8)	17.8			440.0	880.0	640.0
	2	4752	12	35.1			223.5	447.0	325.1
	1	4752	(8)	17.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	102.5	2660	2300	151.0	302.0	219.7
	1	4752	(8)	69.1			225.0	450.0	327.3
	2	4752	12	74.3			209.0	418.0	304.0
	1	4752	(8)	69.1			225.0	450.0	327.3
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0
	2	4752	12	^{*2}			223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	17.7	5320	4600	443.5	887.0	645.1
	1	2376	(8)	8.9			880.0	1760.0	1280.0
	2	2376	12	12.0			655.0	1310.0	952.8
	1	2376	(8)	6.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	68.6	2660	2300	227.0	454.0	330.2
	1	2376	(8)	34.7			450.0	900.0	654.6
	2	2376	12	46.1			338.5	677.0	492.4
	1	2376	(8)	23.3			672.0	1344.0	977.5
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	443.5	887.0	645.1
	1	2376	(8)	^{*2}			880.0	1760.0	1280.0
	2	2376	12	^{*2}			655.0	1310.0	952.8
	1	2376	(8)	^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 35.1 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	8.8	5320	4600	887.0	1774.0	1290.2
	1	1188	(8)	4.4			1759.5	3519.0	2559.3
	2	1188	12	8.8			887.0	1774.0	1290.2
	1	1188	(8)	4.4			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	2	1188	10	34.4	2660	2300	454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
	2	1188	12	34.4			454.0	908.0	660.4
	1	1188	(8)	17.4			899.5	1799.0	1308.4
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3
	2	1188	12	^{*2}			887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 8.8 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of “ROI mode”

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	26.1	4504	4504	307.0	614.0	446.6
	4	594		13.5			596.0	1192.0	867.0
	2	594		6.8			1168.0	2336.0	1699.0
	8	594	12 (8)	26.1			307.0	614.0	446.6
	4	594		13.5			596.0	1192.0	867.0
	2	594		6.8			1168.0	2336.0	1699.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	94.9	2252	2252	166.0	332.0	241.5
	4	594		50.6			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
	8	594	12 (8)	94.9			166.0	332.0	241.5
	4	594		50.6			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
2 × 2 FD binning mode	8	594	10 (8)	94.9	2252	2252	166.0	332.0	241.5
	4	594		50.6			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
	8	594	12 (8)	94.9			166.0	332.0	241.5
	4	594		50.6			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	307.0	614.0	446.6
	4	594		^{*2}			596.0	1192.0	867.0
	2	594		^{*2}			1168.0	2336.0	1699.0
	8	594	12 (8)	^{*2}			307.0	614.0	446.6
	4	594		^{*2}			596.0	1192.0	867.0
	2	594		^{*2}			1168.0	2336.0	1699.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 26.1 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	13.3	4504	4504	604.0	1208.0	878.6
	4	297		6.8			1178.5	2357.0	1714.2
	2	297		3.4			2327.5	4655.0	3385.5
	8	297	12 (8)	13.3			604.0	1208.0	878.6
	4	297		6.8			1178.5	2357.0	1714.2
	2	297		3.4			2327.5	4655.0	3385.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	49.3	2252	2252	322.0	644.0	468.4
	4	297		25.9			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
	8	297	12 (8)	49.3			322.0	644.0	468.4
	4	297		25.9			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
2 × 2 FD binning mode	8	297	10 (8)	49.3	2252	2252	322.0	644.0	468.4
	4	297		25.9			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
	8	297	12 (8)	49.3			322.0	644.0	468.4
	4	297		25.9			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	604.0	1208.0	878.6
	4	297		^{*2}			1178.5	2357.0	1714.2
	2	297		^{*2}			2327.5	4655.0	3385.5
	8	297	12 (8)	^{*2}			604.0	1208.0	878.6
	4	297		^{*2}			1178.5	2357.0	1714.2
	2	297		^{*2}			2327.5	4655.0	3385.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 13.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	35.4	4504	4504	226.0	452.0	328.8
	4	891		18.8			427.0	854.0	621.1
	2	891		9.8			819.0	1638.0	1191.3
	8	891	12 (8)	35.4			226.0	452.0	328.8
	4	891		18.8			427.0	854.0	621.1
	2	891		9.8			819.0	1638.0	1191.3
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	119.1	2252	2252	132.0	264.0	192.0
	4	891		66.2			239.0	478.0	347.7
	2	891		35.9			443.0	886.0	644.4
	8	891	12 (8)	119.1			132.0	264.0	192.0
	4	891		66.2			239.0	478.0	347.7
	2	891		35.9			443.0	886.0	644.4
2 × 2 FD binning mode	8	891	10 (8)	119.1	2252	2252	132.0	264.0	192.0
	4	891		66.2			239.0	478.0	347.7
	2	891		35.9			443.0	886.0	644.4
	8	891	12 (8)	119.1			132.0	264.0	192.0
	4	891		66.2			239.0	478.0	347.7
	2	891		35.9			443.0	886.0	644.4
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	226.0	452.0	328.8
	4	891		^{*2}			427.0	854.0	621.1
	2	891		^{*2}			819.0	1638.0	1191.3
	8	891	12 (8)	^{*2}			226.0	452.0	328.8
	4	891		^{*2}			427.0	854.0	621.1
	2	891		^{*2}			819.0	1638.0	1191.3

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 35.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	18.2	4504	4504	440.0	880.0	640.0
	4	445.5		9.6			836.5	1673.0	1216.8
	2	445.5		4.9			1629.5	3259.0	2370.2
	8	445.5	12 (8)	18.2			440.0	880.0	640.0
	4	445.5		9.6			836.5	1673.0	1216.8
	2	445.5		4.9			1629.5	3259.0	2370.2
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	62.8	2252	2252	252.0	504.0	366.6
	4	445.5		34.5			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
	8	445.5	12 (8)	62.8			252.0	504.0	366.6
	4	445.5		34.5			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
2 × 2 FD binning mode	8	445.5	10 (8)	62.8	2252	2252	252.0	504.0	366.6
	4	445.5		34.5			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
	8	445.5	12 (8)	62.8			252.0	504.0	366.6
	4	445.5		34.5			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	440.0	880.0	640.0
	4	445.5		^{*2}			836.5	1673.0	1216.8
	2	445.5		^{*2}			1629.5	3259.0	2370.2
	8	445.5	12 (8)	^{*2}			440.0	880.0	640.0
	4	445.5		^{*2}			836.5	1673.0	1216.8
	2	445.5		^{*2}			1629.5	3259.0	2370.2

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 18.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	42.7	4504	4504	187.5	375.0	272.8
	1	4752	(8)	21.5			373.5	747.0	543.3
	2	4752	12	38.3			209.0	418.0	304.0
	1	4752	(8)	21.5			373.5	747.0	543.3
Vertical / Horizontal 1/2 subsampling	2	4752	10	156.7	2252	2252	99.5	199.0	144.8
	1	4752	(8)	81.5			194.0	388.0	282.2
	2	4752	12	149.5			104.5	209.0	152.0
	1	4752	(8)	81.5			194.0	388.0	282.2
2 × 2 FD binning mode	2	4752	10	156.7	2252	2252	99.5	199.0	144.8
	1	4752	(8)	81.5			194.0	388.0	282.2
	2	4752	12	149.5			104.5	209.0	152.0
	1	4752	(8)	81.5			194.0	388.0	282.2
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	187.5	375.0	272.8
	1	4752	(8)	^{*2}			373.5	747.0	543.3
	2	4752	12	^{*2}			209.0	418.0	304.0
	1	4752	(8)	^{*2}			373.5	747.0	543.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	21.4	4504	4504	374.5	749.0	544.8
	1	2376	(8)	10.7			747.0	1494.0	1086.6
	2	2376	12	14.4			557.0	1114.0	810.2
	1	2376	(8)	7.2			1109.0	2218.0	1613.1
Vertical / Horizontal 1/2 subsampling	2	2376	10	80.8	2252	2252	195.5	391.0	284.4
	1	2376	(8)	41.0			388.0	776.0	564.4
	2	2376	12	55.3			287.0	574.0	417.5
	1	2376	(8)	28.0			568.5	1137.0	827.0
2 × 2 FD binning mode	2	2376	10	80.8	2252	2252	195.5	391.0	284.4
	1	2376	(8)	41.0			388.0	776.0	564.4
	2	2376	12	55.3			287.0	574.0	417.5
	1	2376	(8)	28.0			568.5	1137.0	827.0
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	374.5	749.0	544.8
	1	2376	(8)	^{*2}			747.0	1494.0	1086.6
	2	2376	12	^{*2}			557.0	1114.0	810.2
	1	2376	(8)	^{*2}			1109.0	2218.0	1613.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.7 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	10.7	4504	4504	749.0	1498.0	1089.5
	1	1188	(8)	5.3			1493.5	2987.0	2172.4
	2	1188	12	10.7			749.0	1498.0	1089.5
	1	1188	(8)	5.3			1493.5	2987.0	2172.4
Vertical / Horizontal 1/2 subsampling	2	1188	10	40.7	2252	2252	391.0	782.0	568.8
	1	1188	(8)	20.5			775.5	1551.0	1128.0
	2	1188	12	40.7			391.0	782.0	568.8
	1	1188	(8)	20.5			775.5	1551.0	1128.0
2 × 2 FD binning mode	2	1188	10	40.7	2252	2252	391.0	782.0	568.8
	1	1188	(8)	20.5			775.5	1551.0	1128.0
	2	1188	12	40.7			391.0	782.0	568.8
	1	1188	(8)	20.5			775.5	1551.0	1128.0
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	749.0	1498.0	1089.5
	1	1188	(8)	^{*2}			1493.5	2987.0	2172.4
	2	1188	12	^{*2}			749.0	1498.0	1089.5
	1	1188	(8)	^{*2}			1493.5	2987.0	2172.4

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 10.7 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	26.1	4504	4504	307.0	614.0	446.6
	4	594		13.5			596.0	1192.0	867.0
	2	594		6.8			1168.0	2336.0	1699.0
	8	594	12 (8)	26.1			307.0	614.0	446.6
	4	594		13.5			596.0	1192.0	867.0
	2	594		6.8			1168.0	2336.0	1699.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	95.2	2252	2252	166.0	332.0	241.5
	4	594		50.7			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
	8	594	12 (8)	75.9			209.0	418.0	304.0
	4	594		50.7			314.0	628.0	456.8
	2	594		26.4			604.0	1208.0	878.6
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	307.0	614.0	446.6
	4	594		^{*2}			596.0	1192.0	867.0
	2	594		^{*2}			1168.0	2336.0	1699.0
	8	594	12 (8)	^{*2}			307.0	614.0	446.6
	4	594		^{*2}			596.0	1192.0	867.0
	2	594		^{*2}			1168.0	2336.0	1699.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 26.1 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	13.3	4504	4504	604.0	1208.0	878.6
	4	297		6.8			1178.5	2357.0	1714.2
	2	297		3.4			2327.5	4655.0	3385.5
	8	297	12 (8)	13.3			604.0	1208.0	878.6
	4	297		6.8			1178.5	2357.0	1714.2
	2	297		3.4			2327.5	4655.0	3385.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	49.4	2252	2252	322.0	644.0	468.4
	4	297		26.0			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
	8	297	12 (8)	49.4			322.0	644.0	468.4
	4	297		26.0			614.5	1229.0	893.9
	2	297		13.3			1199.5	2399.0	1744.8
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	604.0	1208.0	878.6
	4	297		^{*2}			1178.5	2357.0	1714.2
	2	297		^{*2}			2327.5	4655.0	3385.5
	8	297	12 (8)	^{*2}			604.0	1208.0	878.6
	4	297		^{*2}			1178.5	2357.0	1714.2
	2	297		^{*2}			2327.5	4655.0	3385.5

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 13.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cotal ropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	35.4	4504	4504	226.0	452.0	328.8
	4	891		18.8			427.0	854.0	621.1
	2	891		9.8			819.0	1638.0	1191.3
	8	891	12 (8)	35.4			226.0	452.0	328.8
	4	891		18.8			427.0	854.0	621.1
	2	891		9.8			819.0	1638.0	1191.3
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	104.6	2252	2252	151.0	302.0	219.7
	4	891		66.4			239.0	478.0	347.7
	2	891		36.0			443.0	886.0	644.4
	8	891	12 (8)	75.9			209.0	418.0	304.0
	4	891		66.4			239.0	478.0	347.7
	2	891		36.0			443.0	886.0	644.4
ROI	8	891	10 (8)	*2	*3	*3	226.0	452.0	328.8
	4	891		*2			427.0	854.0	621.1
	2	891		*2			819.0	1638.0	1191.3
	8	891	12 (8)	*2			226.0	452.0	328.8
	4	891		*2			427.0	854.0	621.1
	2	891		*2			819.0	1638.0	1191.3

*1 The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 35.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	18.2	4504	4504	440.0	880.0	640.0
	4	445.5		9.6			836.5	1673.0	1216.8
	2	445.5		4.9			1629.5	3259.0	2370.2
	8	445.5	12 (8)	18.2			440.0	880.0	640.0
	4	445.5		9.6			836.5	1673.0	1216.8
	2	445.5		4.9			1629.5	3259.0	2370.2
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	63.0	2252	2252	252.0	504.0	366.6
	4	445.5		34.6			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
	8	445.5	12 (8)	63.0			252.0	504.0	366.6
	4	445.5		34.6			460.5	921.0	669.9
	2	445.5		18.2			877.5	1755.0	1276.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	440.0	880.0	640.0
	4	445.5		^{*2}			836.5	1673.0	1216.8
	2	445.5		^{*2}			1629.5	3259.0	2370.2
	8	445.5	12 (8)	^{*2}			440.0	880.0	640.0
	4	445.5		^{*2}			836.5	1673.0	1216.8
	2	445.5		^{*2}			1629.5	3259.0	2370.2

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 18.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	42.7	4504	4504	187.5	375.0	272.8
	1	4752	(8)	21.5			373.5	747.0	543.3
	2	4752	12	38.3			209.0	418.0	304.0
	1	4752	(8)	21.5			373.5	747.0	543.3
Vertical / Horizontal 1/2 subsampling	2	4752	10	104.6	2252	2252	151.0	302.0	219.7
	1	4752	(8)	81.7			194.0	388.0	282.2
	2	4752	12	75.9			209.0	418.0	304.0
	1	4752	(8)	75.9			209.0	418.0	304.0
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	187.5	375.0	272.8
	1	4752	(8)	^{*2}			373.5	747.0	543.3
	2	4752	12	^{*2}			209.0	418.0	304.0
	1	4752	(8)	^{*2}			373.5	747.0	543.3

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	21.4	4504	4504	374.5	749.0	544.8
	1	2376	(8)	10.7			747.0	1494.0	1086.6
	2	2376	12	14.4			557.0	1114.0	810.2
	1	2376	(8)	7.2			1109.0	2218.0	1613.1
Vertical / Horizontal 1/2 subsampling	2	2376	10	81.0	2252	2252	195.5	391.0	284.4
	1	2376	(8)	41.1			388.0	776.0	564.4
	2	2376	12	55.4			287.0	574.0	417.5
	1	2376	(8)	28.1			568.5	1137.0	827.0
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	374.5	749.0	544.8
	1	2376	(8)	^{*2}			747.0	1494.0	1086.6
	2	2376	12	^{*2}			557.0	1114.0	810.2
	1	2376	(8)	^{*2}			1109.0	2218.0	1613.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.7 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	10.7	4504	4504	749.0	1498.0	1089.5
	1	1188	(8)	5.3			1493.5	2987.0	2172.4
	2	1188	12	10.7			749.0	1498.0	1089.5
	1	1188	(8)	5.3			1493.5	2987.0	2172.4
Vertical / Horizontal 1/2 subsampling	2	1188	10	40.8	2252	2252	391.0	782.0	568.8
	1	1188	(8)	20.6			775.5	1551.0	1128.0
	2	1188	12	40.8			391.0	782.0	568.8
	1	1188	(8)	20.6			775.5	1551.0	1128.0
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	749.0	1498.0	1089.5
	1	1188	(8)	^{*2}			1493.5	2987.0	2172.4
	2	1188	12	^{*2}			749.0	1498.0	1089.5
	1	1188	(8)	^{*2}			1493.5	2987.0	2172.4

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 10.7 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of “ROI mode”

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	32.9	5320	3032	358.0	716.0	520.8
	4	594		16.9			698.0	1396.0	1015.3
	2	594		8.6			1372.0	2744.0	1995.7
	8	594	12 (8)	32.9			358.0	716.0	520.8
	4	594		16.9			698.0	1396.0	1015.3
	2	594		8.6			1372.0	2744.0	1995.7
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	119.0	2660	1516	193.5	387.0	281.5
	4	594		62.8			369.0	738.0	536.8
	2	594		32.8			710.0	1420.0	1032.8
	8	594	12 (8)	119.0			193.5	387.0	281.5
	4	594		62.8			369.0	738.0	536.8
	2	594		32.8			710.0	1420.0	1032.8
2 × 2 FD binning mode	8	594	10 (8)	119.0	2660	1516	193.5	387.0	281.5
	4	594		62.8			369.0	738.0	536.8
	2	594		32.8			710.0	1420.0	1032.8
	8	594	12 (8)	119.0			193.5	387.0	281.5
	4	594		62.8			369.0	738.0	536.8
	2	594		32.8			710.0	1420.0	1032.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7
	8	594	12 (8)	^{*2}			358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	16.7	5320	3032	706.0	1412.0	1027.0
	4	297		8.5			1382.5	2765.0	2011.0
	2	297		4.3			2735.5	5471.0	3979.0
	8	297	12 (8)	16.7			706.0	1412.0	1027.0
	4	297		8.5			1382.5	2765.0	2011.0
	2	297		4.3			2735.5	5471.0	3979.0
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	61.8	2660	1516	375.0	750.0	545.5
	4	297		32.3			720.5	1441.0	1048.0
	2	297		16.5			1411.5	2823.0	2053.1
	8	297	12 (8)	61.8			375.0	750.0	545.5
	4	297		32.3			720.5	1441.0	1048.0
	2	297		16.5			1411.5	2823.0	2053.1
2 × 2 FD binning mode	8	297	10 (8)	61.8	2660	1516	375.0	750.0	545.5
	4	297		32.3			720.5	1441.0	1048.0
	2	297		16.5			1411.5	2823.0	2053.1
	8	297	12 (8)	61.8			375.0	750.0	545.5
	4	297		32.3			720.5	1441.0	1048.0
	2	297		16.5			1411.5	2823.0	2053.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0
	8	297	12 (8)	^{*2}			706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 16.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	45.3	5320	3032	260.0	520.0	378.2
	4	891		23.8			495.0	990.0	720.0
	2	891		12.3			955.0	1910.0	1389.1
	8	891	12 (8)	45.3			260.0	520.0	378.2
	4	891		23.8			495.0	990.0	720.0
	2	891		12.3			955.0	1910.0	1389.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	149.7	2660	1516	153.0	306.0	222.6
	4	891		82.5			281.0	562.0	408.8
	2	891		44.9			519.0	1038.0	755.0
	8	891	12 (8)	149.7			153.0	306.0	222.6
	4	891		82.5			281.0	562.0	408.8
	2	891		44.9			519.0	1038.0	755.0
2 × 2 FD binning mode	8	891	10 (8)	149.7	2660	1516	153.0	306.0	222.6
	4	891		82.5			281.0	562.0	408.8
	2	891		44.9			519.0	1038.0	755.0
	8	891	12 (8)	149.7			153.0	306.0	222.6
	4	891		82.5			281.0	562.0	408.8
	2	891		44.9			519.0	1038.0	755.0
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1
	8	891	12 (8)	^{*2}			260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 45.3 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	23.2	5320	3032	508.0	1016.0	739.0
	4	445.5		12.1			972.5	1945.0	1414.6
	2	445.5		6.2			1901.5	3803.0	2765.9
	8	445.5	12 (8)	23.2			508.0	1016.0	739.0
	4	445.5		12.1			972.5	1945.0	1414.6
	2	445.5		6.2			1901.5	3803.0	2765.9
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	80.0	2660	1516	290.0	580.0	421.9
	4	445.5		43.4			536.5	1073.0	780.4
	2	445.5		22.7			1029.5	2059.0	1497.5
	8	445.5	12 (8)	80.0			290.0	580.0	421.9
	4	445.5		43.4			536.5	1073.0	780.4
	2	445.5		22.7			1029.5	2059.0	1497.5
2 × 2 FD binning mode	8	445.5	10 (8)	80.0	2660	1516	290.0	580.0	421.9
	4	445.5		43.4			536.5	1073.0	780.4
	2	445.5		22.7			1029.5	2059.0	1497.5
	8	445.5	12 (8)	80.0			290.0	580.0	421.9
	4	445.5		43.4			536.5	1073.0	780.4
	2	445.5		22.7			1029.5	2059.0	1497.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9
	8	445.5	12 (8)	^{*2}			508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 23.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	52.5	5320	3032	223.5	447.0	325.1
	1	4752	(8)	26.8			440.0	880.0	640.0
	2	4752	12	52.5			223.5	447.0	325.1
	1	4752	(8)	26.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	197.8	2660	1516	115.0	230.0	167.3
	1	4752	(8)	102.3			225.0	450.0	327.3
	2	4752	12	197.8			115.0	230.0	167.3
	1	4752	(8)	102.3			225.0	450.0	327.3
2 × 2 FD binning mode	2	4752	10	197.8	2660	1516	115.0	230.0	167.3
	1	4752	(8)	102.3			225.0	450.0	327.3
	2	4752	12	197.8			115.0	230.0	167.3
	1	4752	(8)	102.3			225.0	450.0	327.3
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0
	2	4752	12	^{*2}			223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	26.6	5320	3032	443.5	887.0	645.1
	1	2376	(8)	13.4			880.0	1760.0	1280.0
	2	2376	12	18.0			655.0	1310.0	952.8
	1	2376	(8)	9.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	101.7	2660	1516	227.0	454.0	330.2
	1	2376	(8)	51.6			450.0	900.0	654.6
	2	2376	12	68.5			338.5	677.0	492.4
	1	2376	(8)	34.7			672.0	1344.0	977.5
2 × 2 FD binning mode	2	2376	10	101.7	2660	1516	227.0	454.0	330.2
	1	2376	(8)	51.6			450.0	900.0	654.6
	2	2376	12	68.5			338.5	677.0	492.4
	1	2376	(8)	34.7			672.0	1344.0	977.5
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	443.5	887.0	645.1
	1	2376	(8)	^{*2}			880.0	1760.0	1280.0
	2	2376	12	^{*2}			655.0	1310.0	952.8
	1	2376	(8)	^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 52.5 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	13.3	5320	3032	887.0	1774.0	1290.2
	1	1188	(8)	6.7			1759.5	3519.0	2559.3
	2	1188	12	13.3			887.0	1774.0	1290.2
	1	1188	(8)	6.7			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	2	1188	10	51.2	2660	1516	454.0	908.0	660.4
	1	1188	(8)	25.9			899.5	1799.0	1308.4
	2	1188	12	51.2			454.0	908.0	660.4
	1	1188	(8)	25.9			899.5	1799.0	1308.4
2 × 2 FD binning mode	2	1188	10	51.2	2660	1516	454.0	908.0	660.4
	1	1188	(8)	25.9			899.5	1799.0	1308.4
	2	1188	12	51.2			454.0	908.0	660.4
	1	1188	(8)	25.9			899.5	1799.0	1308.4
ROI	2	1188	10	*2	*3	*3	887.0	1774.0	1290.2
	1	1188	(8)	*2			1759.5	3519.0	2559.3
	2	1188	12	*2			887.0	1774.0	1290.2
	1	1188	(8)	*2			1759.5	3519.0	2559.3

*1 The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 13.3 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	32.9	5320	3032	358.0	716.0	520.8
	4	594		16.9			698.0	1396.0	1015.3
	2	594		8.6			1372.0	2744.0	1995.7
	8	594	12 (8)	32.9			358.0	716.0	520.8
	4	594		16.9			698.0	1396.0	1015.3
	2	594		8.6			1372.0	2744.0	1995.7
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	119.4	2660	1516	193.5	387.0	281.5
	4	594		63.1			369.0	738.0	536.8
	2	594		33.0			710.0	1420.0	1032.8
	8	594	12 (8)	110.7			209.0	418.0	304.0
	4	594		63.1			369.0	738.0	536.8
	2	594		33.0			710.0	1420.0	1032.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7
	8	594	12 (8)	^{*2}			358.0	716.0	520.8
	4	594		^{*2}			698.0	1396.0	1015.3
	2	594		^{*2}			1372.0	2744.0	1995.7

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 32.9 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	16.7	5320	3032	706.0	1412.0	1027.0
	4	297		8.5			1382.5	2765.0	2011.0
	2	297		4.3			2735.5	5471.0	3979.0
	8	297	12 (8)	16.7			706.0	1412.0	1027.0
	4	297		8.5			1382.5	2765.0	2011.0
	2	297		4.3			2735.5	5471.0	3979.0
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	62.1	2660	1516	375.0	750.0	545.5
	4	297		32.5			720.5	1441.0	1048.0
	2	297		16.6			1411.5	2823.0	2053.1
	8	297	12 (8)	62.1			375.0	750.0	545.5
	4	297		32.5			720.5	1441.0	1048.0
	2	297		16.6			1411.5	2823.0	2053.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0
	8	297	12 (8)	^{*2}			706.0	1412.0	1027.0
	4	297		^{*2}			1382.5	2765.0	2011.0
	2	297		^{*2}			2735.5	5471.0	3979.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 16.7 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	45.3	5320	3032	260.0	520.0	378.2
	4	891		23.8			495.0	990.0	720.0
	2	891		12.3			955.0	1910.0	1389.1
	8	891	12 (8)	45.3			260.0	520.0	378.2
	4	891		23.8			495.0	990.0	720.0
	2	891		12.3			955.0	1910.0	1389.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	150.5	2660	1516	153.0	306.0	222.6
	4	891		82.7			281.0	562.0	408.8
	2	891		45.1			519.0	1038.0	755.0
	8	891	12 (8)	110.7			209.0	418.0	304.0
	4	891		82.7			281.0	562.0	408.8
	2	891		45.1			519.0	1038.0	755.0
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1
	8	891	12 (8)	^{*2}			260.0	520.0	378.2
	4	891		^{*2}			495.0	990.0	720.0
	2	891		^{*2}			955.0	1910.0	1389.1

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 45.3 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	23.2	5320	3032	508.0	1016.0	739.0
	4	445.5		12.1			972.5	1945.0	1414.6
	2	445.5		6.2			1901.5	3803.0	2765.9
	8	445.5	12 (8)	23.2			508.0	1016.0	739.0
	4	445.5		12.1			972.5	1945.0	1414.6
	2	445.5		6.2			1901.5	3803.0	2765.9
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	80.2	2660	1516	290.0	580.0	421.9
	4	445.5		43.6			536.5	1073.0	780.4
	2	445.5		22.8			1029.5	2059.0	1497.5
	8	445.5	12 (8)	80.2			290.0	580.0	421.9
	4	445.5		43.6			536.5	1073.0	780.4
	2	445.5		22.8			1029.5	2059.0	1497.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9
	8	445.5	12 (8)	^{*2}			508.0	1016.0	739.0
	4	445.5		^{*2}			972.5	1945.0	1414.6
	2	445.5		^{*2}			1901.5	3803.0	2765.9

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 23.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	52.5	5320	3032	223.5	447.0	325.1
	1	4752	(8)	26.8			440.0	880.0	640.0
	2	4752	12	52.5			223.5	447.0	325.1
	1	4752	(8)	26.8			440.0	880.0	640.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	152.3	2660	1516	151.0	302.0	219.7
	1	4752	(8)	102.9			225.0	450.0	327.3
	2	4752	12	110.7			209.0	418.0	304.0
	1	4752	(8)	102.9			225.0	450.0	327.3
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0
	2	4752	12	^{*2}			223.5	447.0	325.1
	1	4752	(8)	^{*2}			440.0	880.0	640.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	26.6	5320	3032	443.5	887.0	645.1
	1	2376	(8)	13.4			880.0	1760.0	1280.0
	2	2376	12	18.0			655.0	1310.0	952.8
	1	2376	(8)	9.0			1304.5	2609.0	1897.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	102.2	2660	1516	227.0	454.0	330.2
	1	2376	(8)	51.8			450.0	900.0	654.6
	2	2376	12	68.8			338.5	677.0	492.4
	1	2376	(8)	34.8			672.0	1344.0	977.5
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	443.5	887.0	645.1
	1	2376	(8)	^{*2}			880.0	1760.0	1280.0
	2	2376	12	^{*2}			655.0	1310.0	952.8
	1	2376	(8)	^{*2}			1304.5	2609.0	1897.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 52.5 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	13.3	5320	3032	887.0	1774.0	1290.2
	1	1188	(8)	6.7			1759.5	3519.0	2559.3
	2	1188	12	13.3			887.0	1774.0	1290.2
	1	1188	(8)	6.7			1759.5	3519.0	2559.3
Vertical / Horizontal 1/2 subsampling	2	1188	10	51.4	2660	1516	454.0	908.0	660.4
	1	1188	(8)	26.0			899.5	1799.0	1308.4
	2	1188	12	51.4			454.0	908.0	660.4
	1	1188	(8)	26.0			899.5	1799.0	1308.4
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3
	2	1188	12	^{*2}			887.0	1774.0	1290.2
	1	1188	(8)	^{*2}			1759.5	3519.0	2559.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 13.3 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of “ROI mode”

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	41.8	4096	3000	283.0	566.0	411.7
	4	594		21.7			548.0	1096.0	797.1
	2	594		11.1			1072.0	2144.0	1559.3
	8	594	12 (8)	41.8			283.0	566.0	411.7
	4	594		21.7			548.0	1096.0	797.1
	2	594		11.1			1072.0	2144.0	1559.3
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	147.3	2048	1500	154.0	308.0	224.0
	4	594		79.8			290.0	580.0	421.9
	2	594		42.0			556.0	1112.0	808.8
	8	594	12 (8)	147.3			154.0	308.0	224.0
	4	594		79.8			290.0	580.0	421.9
	2	594		42.0			556.0	1112.0	808.8
2 × 2 FD binning mode	8	594	10 (8)	147.3	2048	1500	154.0	308.0	224.0
	4	594		79.8			290.0	580.0	421.9
	2	594		42.0			556.0	1112.0	808.8
	8	594	12 (8)	147.3			154.0	308.0	224.0
	4	594		79.8			290.0	580.0	421.9
	2	594		42.0			556.0	1112.0	808.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	283.0	566.0	411.7
	4	594		^{*2}			548.0	1096.0	797.1
	2	594		^{*2}			1072.0	2144.0	1559.3
	8	594	12 (8)	^{*2}			283.0	566.0	411.7
	4	594		^{*2}			548.0	1096.0	797.1
	2	594		^{*2}			1072.0	2144.0	1559.3

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.8 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	21.4	4096	3000	556.0	1112.0	808.8
	4	297		11.0			1082.5	2165.0	1574.6
	2	297		5.6			2135.5	4271.0	3106.2
	8	297	12 (8)	21.4			556.0	1112.0	808.8
	4	297		11.0			1082.5	2165.0	1574.6
	2	297		5.6			2135.5	4271.0	3106.2
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	77.8	2048	1500	298.0	596.0	433.5
	4	297		41.2			566.5	1133.0	824.0
	2	297		21.3			1103.5	2207.0	1605.1
	8	297	12 (8)	77.8			298.0	596.0	433.5
	4	297		41.2			566.5	1133.0	824.0
	2	297		21.3			1103.5	2207.0	1605.1
2 × 2 FD binning mode	8	297	10 (8)	77.8	2048	1500	298.0	596.0	433.5
	4	297		41.2			566.5	1133.0	824.0
	2	297		21.3			1103.5	2207.0	1605.1
	8	297	12 (8)	77.8			298.0	596.0	433.5
	4	297		41.2			566.5	1133.0	824.0
	2	297		21.3			1103.5	2207.0	1605.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	556.0	1112.0	808.8
	4	297		^{*2}			1082.5	2165.0	1574.6
	2	297		^{*2}			2135.5	4271.0	3106.2
	8	297	12 (8)	^{*2}			556.0	1112.0	808.8
	4	297		^{*2}			1082.5	2165.0	1574.6
	2	297		^{*2}			2135.5	4271.0	3106.2

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 21.4 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	61.1	4096	3000	192.5	385.0	280.0
	4	891		32.1			369.5	739.0	537.5
	2	891		16.6			718.5	1437.0	1045.1
	8	891	12 (8)	57.7			204.0	408.0	296.8
	4	891		32.1			369.5	739.0	537.5
	2	891		16.6			718.5	1437.0	1045.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	208.4	2048	1500	106.5	213.0	155.0
	4	891		116.0			197.5	395.0	287.3
	2	891		62.1			374.5	749.0	544.8
	8	891	12 (8)	208.4			106.5	213.0	155.0
	4	891		116.0			197.5	395.0	287.3
	2	891		62.1			374.5	749.0	544.8
2 × 2 FD binning mode	8	891	10 (8)	208.4	2048	1500	106.5	213.0	155.0
	4	891		116.0			197.5	395.0	287.3
	2	891		62.1			374.5	749.0	544.8
	8	891	12 (8)	208.4			106.5	213.0	155.0
	4	891		116.0			197.5	395.0	287.3
	2	891		62.1			374.5	749.0	544.8
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	192.5	385.0	280.0
	4	891		^{*2}			369.5	739.0	537.5
	2	891		^{*2}			718.5	1437.0	1045.1
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			369.5	739.0	537.5
	2	891		^{*2}			718.5	1437.0	1045.1

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 61.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	31.7	4096	3000	374.0	748.0	544.0
	4	445.5		16.4			725.5	1451.0	1055.3
	2	445.5		8.3			1427.5	2855.0	2076.4
	8	445.5	12 (8)	31.7			374.0	748.0	544.0
	4	445.5		16.4			725.5	1451.0	1055.3
	2	445.5		8.3			1427.5	2855.0	2076.4
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	113.4	2048	1500	202.0	404.0	293.9
	4	445.5		61.1			381.5	763.0	555.0
	2	445.5		31.6			739.5	1479.0	1075.7
	8	445.5	12 (8)	113.4			202.0	404.0	293.9
	4	445.5		61.1			381.5	763.0	555.0
	2	445.5		31.6			739.5	1479.0	1075.7
2 × 2 FD binning mode	8	445.5	10 (8)	113.4	2048	1500	202.0	404.0	293.9
	4	445.5		61.1			381.5	763.0	555.0
	2	445.5		31.6			739.5	1479.0	1075.7
	8	445.5	12 (8)	113.4			202.0	404.0	293.9
	4	445.5		61.1			381.5	763.0	555.0
	2	445.5		31.6			739.5	1479.0	1075.7
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	374.0	748.0	544.0
	4	445.5		^{*2}			725.5	1451.0	1055.3
	2	445.5		^{*2}			1427.5	2855.0	2076.4
	8	445.5	12 (8)	^{*2}			374.0	748.0	544.0
	4	445.5		^{*2}			725.5	1451.0	1055.3
	2	445.5		^{*2}			1427.5	2855.0	2076.4

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 31.7 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	68.3	4096	3000	172.0	344.0	250.2
	1	4752	(8)	34.5			343.0	686.0	499.0
	2	4752	12	57.7			204.0	408.0	296.8
	1	4752	(8)	34.5			343.0	686.0	499.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	240.3	2048	1500	91.5	183.0	133.1
	1	4752	(8)	127.7			178.5	357.0	259.7
	2	4752	12	217.1			102.0	204.0	148.4
	1	4752	(8)	127.7			178.5	357.0	259.7
2 × 2 FD binning mode	2	4752	10	240.3	2048	1500	91.5	183.0	133.1
	1	4752	(8)	127.7			178.5	357.0	259.7
	2	4752	12	217.1			102.0	204.0	148.4
	1	4752	(8)	127.7			178.5	357.0	259.7
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	172.0	344.0	250.2
	1	4752	(8)	^{*2}			343.0	686.0	499.0
	2	4752	12	^{*2}			204.0	408.0	296.8
	1	4752	(8)	^{*2}			343.0	686.0	499.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	34.5	4096	3000	343.5	687.0	499.7
	1	2376	(8)	17.3			685.5	1371.0	997.1
	2	2376	12	34.5			343.5	687.0	499.7
	1	2376	(8)	17.3			685.5	1371.0	997.1
Vertical / Horizontal 1/2 subsampling	2	2376	10	126.6	2048	1500	180.0	360.0	261.9
	1	2376	(8)	65.1			357.0	714.0	519.3
	2	2376	12	126.6			180.0	360.0	261.9
	1	2376	(8)	65.1			357.0	714.0	519.3
2 × 2 FD binning mode	2	2376	10	126.6	2048	1500	180.0	360.0	261.9
	1	2376	(8)	65.1			357.0	714.0	519.3
	2	2376	12	126.6			180.0	360.0	261.9
	1	2376	(8)	65.1			357.0	714.0	519.3
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	343.5	687.0	499.7
	1	2376	(8)	^{*2}			685.5	1371.0	997.1
	2	2376	12	^{*2}			343.5	687.0	499.7
	1	2376	(8)	^{*2}			685.5	1371.0	997.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 68.3 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	17.3	4096	3000	687.0	1374.0	999.3
	1	1188	(8)	8.7			1370.5	2741.0	1993.5
	2	1188	12	17.3			687.0	1374.0	999.3
	1	1188	(8)	8.7			1370.5	2741.0	1993.5
Vertical / Horizontal 1/2 subsampling	2	1188	10	64.6	2048	1500	360.0	720.0	523.7
	1	1188	(8)	32.8			714.0	1428.0	1038.6
	2	1188	12	64.6			360.0	720.0	523.7
	1	1188	(8)	32.8			714.0	1428.0	1038.6
2 × 2 FD binning mode	2	1188	10	64.6	2048	1500	360.0	720.0	523.7
	1	1188	(8)	32.8			714.0	1428.0	1038.6
	2	1188	12	64.6			360.0	720.0	523.7
	1	1188	(8)	32.8			714.0	1428.0	1038.6
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	687.0	1374.0	999.3
	1	1188	(8)	^{*2}			1370.5	2741.0	1993.5
	2	1188	12	^{*2}			687.0	1374.0	999.3
	1	1188	(8)	^{*2}			1370.5	2741.0	1993.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 17.3 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	41.8	4096	3000	283.0	566.0	411.7
	4	594		21.7			548.0	1096.0	797.1
	2	594		11.1			1072.0	2144.0	1559.3
	8	594	12 (8)	41.8			283.0	566.0	411.7
	4	594		21.7			548.0	1096.0	797.1
	2	594		11.1			1072.0	2144.0	1559.3
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	148.0	2048	1500	154.0	308.0	224.0
	4	594		80.2			290.0	580.0	421.9
	2	594		42.3			556.0	1112.0	808.8
	8	594	12 (8)	112.8			204.0	408.0	296.8
	4	594		80.2			290.0	580.0	421.9
	2	594		42.3			556.0	1112.0	808.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	283.0	566.0	411.7
	4	594		^{*2}			548.0	1096.0	797.1
	2	594		^{*2}			1072.0	2144.0	1559.3
	8	594	12 (8)	^{*2}			283.0	566.0	411.7
	4	594		^{*2}			548.0	1096.0	797.1
	2	594		^{*2}			1072.0	2144.0	1559.3

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 41.8 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	21.4	4096	3000	556.0	1112.0	808.8
	4	297		11.0			1082.5	2165.0	1574.6
	2	297		5.6			2135.5	4271.0	3106.2
	8	297	12 (8)	21.4			556.0	1112.0	808.8
	4	297		11.0			1082.5	2165.0	1574.6
	2	297		5.6			2135.5	4271.0	3106.2
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	78.1	2048	1500	298.0	596.0	433.5
	4	297		41.5			566.5	1133.0	824.0
	2	297		21.4			1103.5	2207.0	1605.1
	8	297	12 (8)	78.1			298.0	596.0	433.5
	4	297		41.5			566.5	1133.0	824.0
	2	297		21.4			1103.5	2207.0	1605.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	556.0	1112.0	808.8
	4	297		^{*2}			1082.5	2165.0	1574.6
	2	297		^{*2}			2135.5	4271.0	3106.2
	8	297	12 (8)	^{*2}			556.0	1112.0	808.8
	4	297		^{*2}			1082.5	2165.0	1574.6
	2	297		^{*2}			2135.5	4271.0	3106.2

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 21.4 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	61.1	4096	3000	192.5	385.0	280.0
	4	891		32.1			369.5	739.0	537.5
	2	891		16.6			718.5	1437.0	1045.1
	8	891	12 (8)	57.7			204.0	408.0	296.8
	4	891		32.1			369.5	739.0	537.5
	2	891		16.6			718.5	1437.0	1045.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	160.0	2048	1500	142.0	284.0	206.6
	4	891		116.6			197.5	395.0	287.3
	2	891		62.4			374.5	749.0	544.8
	8	891	12 (8)	112.8			204.0	408.0	296.8
	4	891		112.8			204.0	408.0	296.8
	2	891		62.4			374.5	749.0	544.8
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	192.5	385.0	280.0
	4	891		^{*2}			369.5	739.0	537.5
	2	891		^{*2}			718.5	1437.0	1045.1
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			369.5	739.0	537.5
	2	891		^{*2}			718.5	1437.0	1045.1

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 61.1 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	31.7	4096	3000	374.0	748.0	544.0
	4	445.5		16.4			725.5	1451.0	1055.3
	2	445.5		8.3			1427.5	2855.0	2076.4
	8	445.5	12 (8)	31.7			374.0	748.0	544.0
	4	445.5		16.4			725.5	1451.0	1055.3
	2	445.5		8.3			1427.5	2855.0	2076.4
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	114.0	2048	1500	202.0	404.0	293.9
	4	445.5		61.3			381.5	763.0	555.0
	2	445.5		31.9			739.5	1479.0	1075.7
	8	445.5	12 (8)	112.8			204.0	408.0	296.8
	4	445.5		61.3			381.5	763.0	555.0
	2	445.5		31.9			739.5	1479.0	1075.7
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	374.0	748.0	544.0
	4	445.5		^{*2}			725.5	1451.0	1055.3
	2	445.5		^{*2}			1427.5	2855.0	2076.4
	8	445.5	12 (8)	^{*2}			374.0	748.0	544.0
	4	445.5		^{*2}			725.5	1451.0	1055.3
	2	445.5		^{*2}			1427.5	2855.0	2076.4

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 31.7 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	68.3	4096	3000	172.0	344.0	250.2
	1	4752	(8)	34.5			343.0	686.0	499.0
	2	4752	12	57.7			204.0	408.0	296.8
	1	4752	(8)	34.5			343.0	686.0	499.0
Vertical / Horizontal 1/2 subsampling	2	4752	10	160.0	2048	1500	142.0	284.0	206.6
	1	4752	(8)	128.5			178.5	357.0	259.7
	2	4752	12	112.8			204.0	408.0	296.8
	1	4752	(8)	112.8			204.0	408.0	296.8
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	172.0	344.0	250.2
	1	4752	(8)	^{*2}			343.0	686.0	499.0
	2	4752	12	^{*2}			204.0	408.0	296.8
	1	4752	(8)	^{*2}			343.0	686.0	499.0

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	34.5	4096	3000	343.5	687.0	499.7
	1	2376	(8)	17.3			685.5	1371.0	997.1
	2	2376	12	34.5			343.5	687.0	499.7
	1	2376	(8)	17.3			685.5	1371.0	997.1
Vertical / Horizontal 1/2 subsampling	2	2376	10	127.4	2048	1500	180.0	360.0	261.9
	1	2376	(8)	65.4			357.0	714.0	519.3
	2	2376	12	112.8			204.0	408.0	296.8
	1	2376	(8)	65.4			357.0	714.0	519.3
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	343.5	687.0	499.7
	1	2376	(8)	^{*2}			685.5	1371.0	997.1
	2	2376	12	^{*2}			343.5	687.0	499.7
	1	2376	(8)	^{*2}			685.5	1371.0	997.1

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 68.3 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	17.3	4096	3000	687.0	1374.0	999.3
	1	1188	(8)	8.7			1370.5	2741.0	1993.5
	2	1188	12	17.3			687.0	1374.0	999.3
	1	1188	(8)	8.7			1370.5	2741.0	1993.5
Vertical / Horizontal 1/2 subsampling	2	1188	10	64.9	2048	1500	360.0	720.0	523.7
	1	1188	(8)	33.0			714.0	1428.0	1038.6
	2	1188	12	64.9			360.0	720.0	523.7
	1	1188	(8)	33.0			714.0	1428.0	1038.6
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	687.0	1374.0	999.3
	1	1188	(8)	^{*2}			1370.5	2741.0	1993.5
	2	1188	12	^{*2}			687.0	1374.0	999.3
	1	1188	(8)	^{*2}			1370.5	2741.0	1993.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 17.3 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of “ROI mode”

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	60.3	2840	2840	205.5	411.0	299.0
	4	594		31.8			393.0	786.0	571.7
	2	594		16.5			758.0	1516.0	1102.6
	8	594	12 (8)	60.3			205.5	411.0	299.0
	4	594		31.8			393.0	786.0	571.7
	2	594		16.5			758.0	1516.0	1102.6
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	201.1	1420	1420	116.5	233.0	169.5
	4	594		112.1			215.0	430.0	312.8
	2	594		61.0			402.0	804.0	584.8
	8	594	12 (8)	201.1			116.5	233.0	169.5
	4	594		112.1			215.0	430.0	312.8
	2	594		61.0			402.0	804.0	584.8
2 × 2 FD binning mode	8	594	10 (8)	201.1	1420	1420	116.5	233.0	169.5
	4	594		112.1			215.0	430.0	312.8
	2	594		61.0			402.0	804.0	584.8
	8	594	12 (8)	201.1			116.5	233.0	169.5
	4	594		112.1			215.0	430.0	312.8
	2	594		61.0			402.0	804.0	584.8
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	205.5	411.0	299.0
	4	594		^{*2}			393.0	786.0	571.7
	2	594		^{*2}			758.0	1516.0	1102.6
	8	594	12 (8)	^{*2}			205.5	411.0	299.0
	4	594		^{*2}			393.0	786.0	571.7
	2	594		^{*2}			758.0	1516.0	1102.6

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	31.3	2840	2840	399.0	798.0	580.4
	4	297		16.3			768.5	1537.0	1117.9
	2	297		8.3			1507.5	3015.0	2192.8
	8	297	12 (8)	31.3			399.0	798.0	580.4
	4	297		16.3			768.5	1537.0	1117.9
	2	297		8.3			1507.5	3015.0	2192.8
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	109.3	1420	1420	221.0	442.0	321.5
	4	297		59.5			412.5	825.0	600.0
	2	297		31.1			795.5	1591.0	1157.1
	8	297	12 (8)	109.3			221.0	442.0	321.5
	4	297		59.5			412.5	825.0	600.0
	2	297		31.1			795.5	1591.0	1157.1
2 × 2 FD binning mode	8	297	10 (8)	109.3	1420	1420	221.0	442.0	321.5
	4	297		59.5			412.5	825.0	600.0
	2	297		31.1			795.5	1591.0	1157.1
	8	297	12 (8)	109.3			221.0	442.0	321.5
	4	297		59.5			412.5	825.0	600.0
	2	297		31.1			795.5	1591.0	1157.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	399.0	798.0	580.4
	4	297		^{*2}			768.5	1537.0	1117.9
	2	297		^{*2}			1507.5	3015.0	2192.8
	8	297	12 (8)	^{*2}			399.0	798.0	580.4
	4	297		^{*2}			768.5	1537.0	1117.9
	2	297		^{*2}			1507.5	3015.0	2192.8

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 31.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	86.7	2840	2840	142.0	284.0	206.6
	4	891		46.8			266.0	532.0	387.0
	2	891		24.6			509.5	1019.0	741.1
	8	891	12 (8)	60.8			204.0	408.0	296.8
	4	891		46.8			266.0	532.0	387.0
	2	891		24.6			509.5	1019.0	741.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	281.1	1420	1420	81.5	163.0	118.6
	4	891		161.3			147.5	295.0	214.6
	2	891		89.5			272.0	544.0	395.7
	8	891	12 (8)	228.0			102.0	204.0	148.4
	4	891		161.3			147.5	295.0	214.6
	2	891		89.5			272.0	544.0	395.7
2 × 2 FD binning mode	8	891	10 (8)	281.1	1420	1420	81.5	163.0	118.6
	4	891		161.3			147.5	295.0	214.6
	2	891		89.5			272.0	544.0	395.7
	8	891	12 (8)	228.0			102.0	204.0	148.4
	4	891		161.3			147.5	295.0	214.6
	2	891		89.5			272.0	544.0	395.7
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	4	891		^{*2}			266.0	532.0	387.0
	2	891		^{*2}			509.5	1019.0	741.1
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			266.0	532.0	387.0
	2	891		^{*2}			509.5	1019.0	741.1

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 86.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	46.2	2840	2840	269.5	539.0	392.0
	4	445.5		24.3			516.0	1032.0	750.6
	2	445.5		12.4			1009.0	2018.0	1467.7
	8	445.5	12 (8)	46.2			269.5	539.0	392.0
	4	445.5		24.3			516.0	1032.0	750.6
	2	445.5		12.4			1009.0	2018.0	1467.7
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	157.6	1420	1420	151.0	302.0	219.7
	4	445.5		87.4			278.5	557.0	405.1
	2	445.5		46.0			534.5	1069.0	777.5
	8	445.5	12 (8)	157.6			151.0	302.0	219.7
	4	445.5		87.4			278.5	557.0	405.1
	2	445.5		46.0			534.5	1069.0	777.5
2 × 2 FD binning mode	8	445.5	10 (8)	157.6	1420	1420	151.0	302.0	219.7
	4	445.5		87.4			278.5	557.0	405.1
	2	445.5		46.0			534.5	1069.0	777.5
	8	445.5	12 (8)	157.6			151.0	302.0	219.7
	4	445.5		87.4			278.5	557.0	405.1
	2	445.5		46.0			534.5	1069.0	777.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	269.5	539.0	392.0
	4	445.5		^{*2}			516.0	1032.0	750.6
	2	445.5		^{*2}			1009.0	2018.0	1467.7
	8	445.5	12 (8)	^{*2}			269.5	539.0	392.0
	4	445.5		^{*2}			516.0	1032.0	750.6
	2	445.5		^{*2}			1009.0	2018.0	1467.7

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 46.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	86.7	2840	2840	142.0	284.0	206.6
	1	4752	(8)	51.7			240.5	481.0	349.9
	2	4752	12	60.8			204.0	408.0	296.8
	1	4752	(8)	51.7			240.5	481.0	349.9
Vertical / Horizontal 1/2 subsampling	2	4752	10	318.8	1420	1420	71.0	142.0	103.3
	1	4752	(8)	185.6			127.5	255.0	185.5
	2	4752	12	228.0			102.0	204.0	148.4
	1	4752	(8)	185.6			127.5	255.0	185.5
2 × 2 FD binning mode	2	4752	10	318.8	1420	1420	71.0	142.0	103.3
	1	4752	(8)	185.6			127.5	255.0	185.5
	2	4752	12	228.0			102.0	204.0	148.4
	1	4752	(8)	185.6			127.5	255.0	185.5
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	1	4752	(8)	^{*2}			240.5	481.0	349.9
	2	4752	12	^{*2}			204.0	408.0	296.8
	1	4752	(8)	^{*2}			240.5	481.0	349.9

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	51.3	2840	2840	242.5	485.0	352.8
	1	2376	(8)	26.1			480.5	961.0	699.0
	2	2376	12	51.3			242.5	485.0	352.8
	1	2376	(8)	26.1			480.5	961.0	699.0
Vertical / Horizontal 1/2 subsampling	2	2376	10	187.1	1420	1420	126.5	253.0	184.0
	1	2376	(8)	95.5			255.0	510.0	371.0
	2	2376	12	187.1			126.5	253.0	184.0
	1	2376	(8)	95.5			255.0	510.0	371.0
2 × 2 FD binning mode	2	2376	10	187.1	1420	1420	126.5	253.0	184.0
	1	2376	(8)	95.5			255.0	510.0	371.0
	2	2376	12	187.1			126.5	253.0	184.0
	1	2376	(8)	95.5			255.0	510.0	371.0
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	242.5	485.0	352.8
	1	2376	(8)	^{*2}			480.5	961.0	699.0
	2	2376	12	^{*2}			242.5	485.0	352.8
	1	2376	(8)	^{*2}			480.5	961.0	699.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 86.7 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	25.8	2840	2840	485.0	970.0	705.5
	1	1188	(8)	13.1			961.0	1922.0	1397.9
	2	1188	12	25.8			485.0	970.0	705.5
	1	1188	(8)	13.1			961.0	1922.0	1397.9
Vertical / Horizontal 1/2 subsampling	2	1188	10	96.2	1420	1420	253.0	506.0	368.0
	1	1188	(8)	48.3			509.5	1019.0	741.1
	2	1188	12	96.2			253.0	506.0	368.0
	1	1188	(8)	48.3			509.5	1019.0	741.1
2 × 2 FD binning mode	2	1188	10	96.2	1420	1420	253.0	506.0	368.0
	1	1188	(8)	48.3			509.5	1019.0	741.1
	2	1188	12	96.2			253.0	506.0	368.0
	1	1188	(8)	48.3			509.5	1019.0	741.1
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	485.0	970.0	705.5
	1	1188	(8)	^{*2}			961.0	1922.0	1397.9
	2	1188	12	^{*2}			485.0	970.0	705.5
	1	1188	(8)	^{*2}			961.0	1922.0	1397.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 25.8 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	60.3	2840	2840	205.5	411.0	299.0
	4	594		31.8			393.0	786.0	571.7
	2	594		16.5			758.0	1516.0	1102.6
	8	594	12 (8)	60.3			205.5	411.0	299.0
	4	594		31.8			393.0	786.0	571.7
	2	594		16.5			758.0	1516.0	1102.6
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	168.2	1420	1420	142.0	284.0	206.6
	4	594		112.8			215.0	430.0	312.8
	2	594		61.4			402.0	804.0	584.8
	8	594	12 (8)	118.7			204.0	408.0	296.8
	4	594		112.8			215.0	430.0	312.8
	2	594		61.4			402.0	804.0	584.8
ROI	8	594	10 (8)	*2	*3	*3	205.5	411.0	299.0
	4	594		*2			393.0	786.0	571.7
	2	594		*2			758.0	1516.0	1102.6
	8	594	12 (8)	*2			205.5	411.0	299.0
	4	594		*2			393.0	786.0	571.7
	2	594		*2			758.0	1516.0	1102.6

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 60.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	31.3	2840	2840	399.0	798.0	580.4
	4	297		16.3			768.5	1537.0	1117.9
	2	297		8.3			1507.5	3015.0	2192.8
	8	297	12 (8)	31.3			399.0	798.0	580.4
	4	297		16.3			768.5	1537.0	1117.9
	2	297		8.3			1507.5	3015.0	2192.8
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	110.0	1420	1420	221.0	442.0	321.5
	4	297		59.8			412.5	825.0	600.0
	2	297		31.3			795.5	1591.0	1157.1
	8	297	12 (8)	110.0			221.0	442.0	321.5
	4	297		59.8			412.5	825.0	600.0
	2	297		31.3			795.5	1591.0	1157.1
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	399.0	798.0	580.4
	4	297		^{*2}			768.5	1537.0	1117.9
	2	297		^{*2}			1507.5	3015.0	2192.8
	8	297	12 (8)	^{*2}			399.0	798.0	580.4
	4	297		^{*2}			768.5	1537.0	1117.9
	2	297		^{*2}			1507.5	3015.0	2192.8

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 31.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	86.7	2840	2840	142.0	284.0	206.6
	4	891		46.8			266.0	532.0	387.0
	2	891		24.6			509.5	1019.0	741.1
	8	891	12 (8)	60.8			204.0	408.0	296.8
	4	891		46.8			266.0	532.0	387.0
	2	891		24.6			509.5	1019.0	741.1
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	168.2	1420	1420	142.0	284.0	206.6
	4	891		162.1			147.5	295.0	214.6
	2	891		90.0			272.0	544.0	395.7
	8	891	12 (8)	118.7			204.0	408.0	296.8
	4	891		118.7			204.0	408.0	296.8
	2	891		90.0			272.0	544.0	395.7
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	4	891		^{*2}			266.0	532.0	387.0
	2	891		^{*2}			509.5	1019.0	741.1
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			266.0	532.0	387.0
	2	891		^{*2}			509.5	1019.0	741.1

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 86.7 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	46.2	2840	2840	269.5	539.0	392.0
	4	445.5		24.3			516.0	1032.0	750.6
	2	445.5		12.4			1009.0	2018.0	1467.7
	8	445.5	12 (8)	46.2			269.5	539.0	392.0
	4	445.5		24.3			516.0	1032.0	750.6
	2	445.5		12.4			1009.0	2018.0	1467.7
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	158.4	1420	1420	151.0	302.0	219.7
	4	445.5		87.9			278.5	557.0	405.1
	2	445.5		46.4			534.5	1069.0	777.5
	8	445.5	12 (8)	118.7			204.0	408.0	296.8
	4	445.5		87.9			278.5	557.0	405.1
	2	445.5		46.4			534.5	1069.0	777.5
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	269.5	539.0	392.0
	4	445.5		^{*2}			516.0	1032.0	750.6
	2	445.5		^{*2}			1009.0	2018.0	1467.7
	8	445.5	12 (8)	^{*2}			269.5	539.0	392.0
	4	445.5		^{*2}			516.0	1032.0	750.6
	2	445.5		^{*2}			1009.0	2018.0	1467.7

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 46.2 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	86.7	2840	2840	142.0	284.0	206.6
	1	4752	(8)	51.7			240.5	481.0	349.9
	2	4752	12	60.8			204.0	408.0	296.8
	1	4752	(8)	51.7			240.5	481.0	349.9
Vertical / Horizontal 1/2 subsampling	2	4752	10	168.2	1420	1420	142.0	284.0	206.6
	1	4752	(8)	168.2			142.0	284.0	206.6
	2	4752	12	118.7			204.0	408.0	296.8
	1	4752	(8)	118.7			204.0	408.0	296.8
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	1	4752	(8)	^{*2}			240.5	481.0	349.9
	2	4752	12	^{*2}			204.0	408.0	296.8
	1	4752	(8)	^{*2}			240.5	481.0	349.9

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	51.3	2840	2840	242.5	485.0	352.8
	1	2376	(8)	26.1			480.5	961.0	699.0
	2	2376	12	51.3			242.5	485.0	352.8
	1	2376	(8)	26.1			480.5	961.0	699.0
Vertical / Horizontal 1/2 subsampling	2	2376	10	168.2	1420	1420	142.0	284.0	206.6
	1	2376	(8)	95.9			255.0	510.0	371.0
	2	2376	12	118.7			204.0	408.0	296.8
	1	2376	(8)	95.9			255.0	510.0	371.0
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	242.5	485.0	352.8
	1	2376	(8)	^{*2}			480.5	961.0	699.0
	2	2376	12	^{*2}			242.5	485.0	352.8
	1	2376	(8)	^{*2}			480.5	961.0	699.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 86.7 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	25.8	2840	2840	485.0	970.0	705.5
	1	1188	(8)	13.1			961.0	1922.0	1397.9
	2	1188	12	25.8			485.0	970.0	705.5
	1	1188	(8)	13.1			961.0	1922.0	1397.9
Vertical / Horizontal 1/2 subsampling	2	1188	10	96.6	1420	1420	253.0	506.0	368.0
	1	1188	(8)	48.7			509.5	1019.0	741.1
	2	1188	12	96.6			253.0	506.0	368.0
	1	1188	(8)	48.7			509.5	1019.0	741.1
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	485.0	970.0	705.5
	1	1188	(8)	^{*2}			961.0	1922.0	1397.9
	2	1188	12	^{*2}			485.0	970.0	705.5
	1	1188	(8)	^{*2}			961.0	1922.0	1397.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 25.8 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	92.3	2448	2048	181.5	363.0	264.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
	8	594	12 (8)	82.4			204.0	408.0	296.8
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	296.0	1224	1024	104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
	8	594	12 (8)	296.0			104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
2 × 2 FD binning mode	8	594	10 (8)	296.0	1224	1024	104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
	8	594	12 (8)	296.0			104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	181.5	363.0	264.0
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0
	8	594	12 (8)	^{*2}			204.0	408.0	296.8
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 92.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	48.3	2448	2048	351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
	8	297	12 (8)	48.3			351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	164.1	1224	1024	197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
	8	297	12 (8)	164.1			197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
2 × 2 FD binning mode	8	297	10 (8)	164.1	1224	1024	197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
	8	297	12 (8)	164.1			197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5
	8	297	12 (8)	^{*2}			351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 48.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	117.2	2448	2048	142.0	284.0	206.6
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
	8	891	12 (8)	82.4			204.0	408.0	296.8
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	404.7	1224	1024	73.5	147.0	107.0
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
	8	891	12 (8)	302.3			102.0	204.0	148.4
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
2 × 2 FD binning mode	8	891	10 (8)	404.7	1224	1024	73.5	147.0	107.0
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
	8	891	12 (8)	302.3			102.0	204.0	148.4
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 117.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	71.0	2448	2048	237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
	8	445.5	12 (8)	71.0			237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	233.8	1224	1024	135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
	8	445.5	12 (8)	233.8			135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
2 × 2 FD binning mode	8	445.5	10 (8)	233.8	1224	1024	135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
	8	445.5	12 (8)	233.8			135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5
	8	445.5	12 (8)	^{*2}			237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 71.0 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	117.2	2448	2048	142.0	284.0	206.6
	1	4752	(8)	80.3			209.5	419.0	304.8
	2	4752	12	82.4			204.0	408.0	296.8
	1	4752	(8)	80.3			209.5	419.0	304.8
Vertical / Horizontal 1/2 subsampling	2	4752	10	418.9	1224	1024	71.0	142.0	103.3
	1	4752	(8)	278.0			112.0	224.0	163.0
	2	4752	12	302.3			102.0	204.0	148.4
	1	4752	(8)	278.0			112.0	224.0	163.0
2 × 2 FD binning mode	2	4752	10	418.9	1224	1024	71.0	142.0	103.3
	1	4752	(8)	278.0			112.0	224.0	163.0
	2	4752	12	302.3			102.0	204.0	148.4
	1	4752	(8)	278.0			112.0	224.0	163.0
ROI	2	4752	10	*2	*3	*3	142.0	284.0	206.6
	1	4752	(8)	*2			209.5	419.0	304.8
	2	4752	12	*2			204.0	408.0	296.8
	1	4752	(8)	*2			209.5	419.0	304.8

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	79.5	2448	2048	211.5	423.0	307.7
	1	2376	(8)	40.6			419.0	838.0	609.5
	2	2376	12	79.5			211.5	423.0	307.7
	1	2376	(8)	40.6			419.0	838.0	609.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	280.5	1224	1024	111.0	222.0	161.5
	1	2376	(8)	144.8			224.0	448.0	325.9
	2	2376	12	280.5			111.0	222.0	161.5
	1	2376	(8)	144.8			224.0	448.0	325.9
2 × 2 FD binning mode	2	2376	10	280.5	1224	1024	111.0	222.0	161.5
	1	2376	(8)	144.8			224.0	448.0	325.9
	2	2376	12	280.5			111.0	222.0	161.5
	1	2376	(8)	144.8			224.0	448.0	325.9
ROI	2	2376	10	*2	*3	*3	211.5	423.0	307.7
	1	2376	(8)	*2			419.0	838.0	609.5
	2	2376	12	*2			211.5	423.0	307.7
	1	2376	(8)	*2			419.0	838.0	609.5

*1 The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 117.2 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	40.2	2448	2048	423.0	846.0	615.3
	1	1188	(8)	20.4			838.0	1676.0	1219.0
	2	1188	12	40.2			423.0	846.0	615.3
	1	1188	(8)	20.4			838.0	1676.0	1219.0
Vertical / Horizontal 1/2 subsampling	2	1188	10	146.1	1224	1024	222.0	444.0	323.0
	1	1188	(8)	74.2			448.0	896.0	651.7
	2	1188	12	146.1			222.0	444.0	323.0
	1	1188	(8)	74.2			448.0	896.0	651.7
2 × 2 FD binning mode	2	1188	10	146.1	1224	1024	222.0	444.0	323.0
	1	1188	(8)	74.2			448.0	896.0	651.7
	2	1188	12	146.1			222.0	444.0	323.0
	1	1188	(8)	74.2			448.0	896.0	651.7
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	423.0	846.0	615.3
	1	1188	(8)	^{*2}			838.0	1676.0	1219.0
	2	1188	12	^{*2}			423.0	846.0	615.3
	1	1188	(8)	^{*2}			838.0	1676.0	1219.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 40.2 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	92.3	2448	2048	181.5	363.0	264.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
	8	594	12 (8)	82.4			204.0	408.0	296.8
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	224.9	1224	1024	142.0	284.0	206.6
	4	594		170.2			191.0	382.0	277.9
	2	594		93.9			354.0	708.0	515.0
	8	594	12 (8)	159.6			204.0	408.0	296.8
	4	594		159.6			204.0	408.0	296.8
	2	594		93.9			354.0	708.0	515.0
ROI	8	594	10 (8)	*2	*3	*3	181.5	363.0	264.0
	4	594		*2			345.0	690.0	501.9
	2	594		*2			662.0	1324.0	963.0
	8	594	12 (8)	*2			204.0	408.0	296.8
	4	594		*2			345.0	690.0	501.9
	2	594		*2			662.0	1324.0	963.0

*1 The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 92.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

*2 See the section of "ROI mode"

*3 Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	48.3	2448	2048	351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
	8	297	12 (8)	48.3			351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	165.3	1224	1024	197.0	394.0	286.6
	4	297		91.2			364.5	729.0	530.2
	2	297		48.1			699.5	1399.0	1017.5
	8	297	12 (8)	159.6			204.0	408.0	296.8
	4	297		91.2			364.5	729.0	530.2
	2	297		48.1			699.5	1399.0	1017.5
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5
	8	297	12 (8)	^{*2}			351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 48.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	117.2	2448	2048	142.0	284.0	206.6
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
	8	891	12 (8)	82.4			204.0	408.0	296.8
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	224.9	1224	1024	142.0	284.0	206.6
	4	891		224.9			142.0	284.0	206.6
	2	891		136.8			240.0	480.0	349.1
	8	891	12 (8)	159.6			204.0	408.0	296.8
	4	891		159.6			204.0	408.0	296.8
	2	891		136.8			240.0	480.0	349.1
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0
	8	891	12 (8)	^{*2}			204.0	408.0	296.8
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 117.2 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	71.0	2448	2048	237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
	8	445.5	12 (8)	71.0			237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	224.9	1224	1024	142.0	284.0	206.6
	4	445.5		133.2			246.5	493.0	358.6
	2	445.5		71.0			470.5	941.0	684.4
	8	445.5	12 (8)	159.6			204.0	408.0	296.8
	4	445.5		133.2			246.5	493.0	358.6
	2	445.5		71.0			470.5	941.0	684.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5
	8	445.5	12 (8)	^{*2}			237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 71.0 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	4752	10	117.2	2448	2048	142.0	284.0	206.6
	1	4752	(8)	80.3			209.5	419.0	304.8
	2	4752	12	82.4			204.0	408.0	296.8
	1	4752	(8)	80.3			209.5	419.0	304.8
Vertical / Horizontal 1/2 subsampling	2	4752	10	224.9	1224	1024	142.0	284.0	206.6
	1	4752	(8)	224.9			142.0	284.0	206.6
	2	4752	12	159.6			204.0	408.0	296.8
	1	4752	(8)	159.6			204.0	408.0	296.8
ROI	2	4752	10	^{*2}	^{*3}	^{*3}	142.0	284.0	206.6
	1	4752	(8)	^{*2}			209.5	419.0	304.8
	2	4752	12	^{*2}			204.0	408.0	296.8
	1	4752	(8)	^{*2}			209.5	419.0	304.8

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	2376	10	79.5	2448	2048	211.5	423.0	307.7
	1	2376	(8)	40.6			419.0	838.0	609.5
	2	2376	12	79.5			211.5	423.0	307.7
	1	2376	(8)	40.6			419.0	838.0	609.5
Vertical / Horizontal 1/2 subsampling	2	2376	10	224.9	1224	1024	142.0	284.0	206.6
	1	2376	(8)	146.1			224.0	448.0	325.9
	2	2376	12	159.6			204.0	408.0	296.8
	1	2376	(8)	146.1			224.0	448.0	325.9
ROI	2	2376	10	^{*2}	^{*3}	^{*3}	211.5	423.0	307.7
	1	2376	(8)	^{*2}			419.0	838.0	609.5
	2	2376	12	^{*2}			211.5	423.0	307.7
	1	2376	(8)	^{*2}			419.0	838.0	609.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 117.2 [frame/s] mode: 4752 [Mbps/Lane] × 2 = 9.504 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h

SLVS – EC output

Drive mode	SLVS – EC Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	2	1188	10	40.2	2448	2048	423.0	846.0	615.3
	1	1188	(8)	20.4			838.0	1676.0	1219.0
	2	1188	12	40.2			423.0	846.0	615.3
	1	1188	(8)	20.4			838.0	1676.0	1219.0
Vertical / Horizontal 1/2 subsampling	2	1188	10	147.4	1224	1024	222.0	444.0	323.0
	1	1188	(8)	74.6			448.0	896.0	651.7
	2	1188	12	147.4			222.0	444.0	323.0
	1	1188	(8)	74.6			448.0	896.0	651.7
ROI	2	1188	10	^{*2}	^{*3}	^{*3}	423.0	846.0	615.3
	1	1188	(8)	^{*2}			838.0	1676.0	1219.0
	2	1188	12	^{*2}			423.0	846.0	615.3
	1	1188	(8)	^{*2}			838.0	1676.0	1219.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 40.2 [frame/s] mode: 1188 [Mbps/Lane] × 2 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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FREQ (CID = 02h, Address = DCh, [1:0]) = 0h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	92.3	2448	2048	181.5	363.0	264.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
	8	594	12 (8)	84.0			200.0	400.0	291.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	296.0	1224	1024	104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
	8	594	12 (8)	296.0			104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
2 × 2 FD binning mode	8	594	10 (8)	296.0	1224	1024	104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
	8	594	12 (8)	296.0			104.5	209.0	152.0
	4	594		168.7			191.0	382.0	277.9
	2	594		93.3			354.0	708.0	515.0
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	181.5	363.0	264.0
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0
	8	594	12 (8)	^{*2}			200.0	400.0	291.0
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 92.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	48.3	2448	2048	351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
	8	297	12 (8)	48.3			351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	164.1	1224	1024	197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
	8	297	12 (8)	164.1			197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
2 × 2 FD binning mode	8	297	10 (8)	164.1	1224	1024	197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
	8	297	12 (8)	164.1			197.0	394.0	286.6
	4	297		90.6			364.5	729.0	530.2
	2	297		47.7			699.5	1399.0	1017.5
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5
	8	297	12 (8)	^{*2}			351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 48.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	93.4	2448	2048	179.5	359.0	261.1
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
	8	891	12 (8)	84.0			200.0	400.0	291.0
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	339.2	1224	1024	90.0	180.0	131.0
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
	8	891	12 (8)	308.3			100.0	200.0	145.5
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
2 × 2 FD binning mode	8	891	10 (8)	339.2	1224	1024	90.0	180.0	131.0
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
	8	891	12 (8)	308.3			100.0	200.0	145.5
	4	891		240.0			131.5	263.0	191.3
	2	891		135.6			240.0	480.0	349.1
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	179.5	359.0	261.1
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0
	8	891	12 (8)	^{*2}			200.0	400.0	291.0
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 93.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	71.0	2448	2048	237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
	8	445.5	12 (8)	71.0			237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	233.8	1224	1024	135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
	8	445.5	12 (8)	233.8			135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
2 × 2 FD binning mode	8	445.5	10 (8)	233.8	1224	1024	135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
	8	445.5	12 (8)	233.8			135.0	270.0	196.4
	4	445.5		132.1			246.5	493.0	358.6
	2	445.5		70.7			470.5	941.0	684.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5
	8	445.5	12 (8)	^{*2}			237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 71.0 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

5.26. IMX548-AAQJ-C

FREQ (CID = 02h, Address = DCh, [1:0]) = 0h

SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	594	10 (8)	92.3	2448	2048	181.5	363.0	264.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
	8	594	12 (8)	84.0			200.0	400.0	291.0
	4	594		49.2			345.0	690.0	501.9
	2	594		25.8			662.0	1324.0	963.0
Vertical / Horizontal 1/2 subsampling	8	594	10 (8)	180.4	1224	1024	179.5	359.0	261.1
	4	594		170.2			191.0	382.0	277.9
	2	594		93.9			354.0	708.0	515.0
	8	594	12 (8)	162.8			200.0	400.0	291.0
	4	594		162.8			200.0	400.0	291.0
	2	594		93.9			354.0	708.0	515.0
ROI	8	594	10 (8)	^{*2}	^{*3}	^{*3}	181.5	363.0	264.0
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0
	8	594	12 (8)	^{*2}			200.0	400.0	291.0
	4	594		^{*2}			345.0	690.0	501.9
	2	594		^{*2}			662.0	1324.0	963.0

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 92.3 [frame/s] mode: 594 [Mbps/ch] × 8 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 1h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	297	10 (8)	48.3	2448	2048	351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
	8	297	12 (8)	48.3			351.0	702.0	510.6
	4	297		25.4			672.5	1345.0	978.2
	2	297		13.0			1315.5	2631.0	1913.5
Vertical / Horizontal 1/2 subsampling	8	297	10 (8)	165.3	1224	1024	197.0	394.0	286.6
	4	297		91.2			364.5	729.0	530.2
	2	297		48.1			699.5	1399.0	1017.5
	8	297	12 (8)	162.8			200.0	400.0	291.0
	4	297		91.2			364.5	729.0	530.2
	2	297		48.1			699.5	1399.0	1017.5
ROI	8	297	10 (8)	^{*2}	^{*3}	^{*3}	351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5
	8	297	12 (8)	^{*2}			351.0	702.0	510.6
	4	297		^{*2}			672.5	1345.0	978.2
	2	297		^{*2}			1315.5	2631.0	1913.5

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 48.3 [frame/s] mode: 297 [Mbps/ch] × 8 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 2h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	891	10 (8)	93.4	2448	2048	179.5	359.0	261.1
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
	8	891	12 (8)	84.0			200.0	400.0	291.0
	4	891		72.0			234.0	468.0	340.4
	2	891		38.2			445.5	891.0	648.0
Vertical / Horizontal 1/2 subsampling	8	891	10 (8)	180.4	1224	1024	179.5	359.0	261.1
	4	891		180.4			179.5	359.0	261.1
	2	891		136.8			240.0	480.0	349.1
	8	891	12 (8)	162.8			200.0	400.0	291.0
	4	891		162.8			200.0	400.0	291.0
	2	891		136.8			240.0	480.0	349.1
ROI	8	891	10 (8)	^{*2}	^{*3}	^{*3}	179.5	359.0	261.1
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0
	8	891	12 (8)	^{*2}			200.0	400.0	291.0
	4	891		^{*2}			234.0	468.0	340.4
	2	891		^{*2}			445.5	891.0	648.0

^{*1} The total data rate is the data rate multiplied by the number of channels.
Example) In All-pixel 93.4 [frame/s] mode: 891 [Mbps/ch] × 8 = 7.128 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FREQ (CID = 02h, Address = DCh, [1:0]) = 3h
SLVS output

Drive mode	SLVS ch ^{*1}	Data rate [Mbps/ch]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	8	445.5	10 (8)	71.0	2448	2048	237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
	8	445.5	12 (8)	71.0			237.5	475.0	345.5
	4	445.5		37.7			452.0	904.0	657.5
	2	445.5		19.4			881.0	1762.0	1281.5
Vertical / Horizontal 1/2 subsampling	8	445.5	10 (8)	180.4	1224	1024	179.5	359.0	261.1
	4	445.5		133.2			246.5	493.0	358.6
	2	445.5		71.0			470.5	941.0	684.4
	8	445.5	12 (8)	162.8			200.0	400.0	291.0
	4	445.5		133.2			246.5	493.0	358.6
	2	445.5		71.0			470.5	941.0	684.4
ROI	8	445.5	10 (8)	^{*2}	^{*3}	^{*3}	237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5
	8	445.5	12 (8)	^{*2}			237.5	475.0	345.5
	4	445.5		^{*2}			452.0	904.0	657.5
	2	445.5		^{*2}			881.0	1762.0	1281.5

^{*1} The total data rate is the data rate multiplied by the number of channels.

Example) In All-pixel 71.0 [frame/s] mode: 445.5 [Mbps/ch] × 8 = 3.564 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

5.27. IMX565-AAMJ-C

Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	42.6	4096	3000	277.5	555.0	403.7
	2	1188	(8)	22.2			535.0	1070.0	778.2
	4	1188	12	42.6			277.5	555.0	403.7
	2	1188	(8)	22.2			535.0	1070.0	778.2
Vertical / Horizontal 1/2 subsampling	4	1188	10	152.4	2048	1500	148.5	297.0	216.0
	2	1188	(8)	83.5			277.0	554.0	403.0
	4	1188	12	152.4			148.5	297.0	216.0
	2	1188	(8)	83.5			277.0	554.0	403.0
2 × 2 FD binning mode	4	1188	10	152.4	2048	1500	148.5	297.0	216.0
	2	1188	(8)	83.5			277.0	554.0	403.0
	4	1188	12	152.4			148.5	297.0	216.0
	2	1188	(8)	83.5			277.0	554.0	403.0
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	277.5	555.0	403.7
	2	1188	(8)	^{*2}			535.0	1070.0	778.2
	4	1188	12	^{*2}			277.5	555.0	403.7
	2	1188	(8)	^{*2}			535.0	1070.0	778.2

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	32.2	4096	3000	368.0	736.0	535.3
	2	891	(8)	16.7			711.0	1422.0	1034.2
	4	891	12	32.2			368.0	736.0	535.3
	2	891	(8)	16.7			711.0	1422.0	1034.2
Vertical / Horizontal 1/2 subsampling	4	891	10	116.6	2048	1500	196.0	392.0	285.1
	2	891	(8)	63.3			367.0	734.0	533.9
	4	891	12	116.6			196.0	392.0	285.1
	2	891	(8)	63.3			367.0	734.0	533.9
2 × 2 FD binning mode	4	891	10	116.6	2048	1500	196.0	392.0	285.1
	2	891	(8)	63.3			367.0	734.0	533.9
	4	891	12	116.6			196.0	392.0	285.1
	2	891	(8)	63.3			367.0	734.0	533.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	368.0	736.0	535.3
	2	891	(8)	^{*2}			711.0	1422.0	1034.2
	4	891	12	^{*2}			368.0	736.0	535.3
	2	891	(8)	^{*2}			711.0	1422.0	1034.2

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.6 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	21.7	4096	3000	548.0	1096.0	797.1
	2	594	(8)	11.2			1062.5	2125.0	1545.5
	4	594	12	21.7			548.0	1096.0	797.1
	2	594	(8)	11.2			1062.5	2125.0	1545.5
Vertical / Horizontal 1/2 subsampling	4	594	10	79.8	2048	1500	290.0	580.0	421.9
	2	594	(8)	42.7			546.5	1093.0	795.0
	4	594	12	79.8			290.0	580.0	421.9
	2	594	(8)	42.7			546.5	1093.0	795.0
2 × 2 FD binning mode	4	594	10	79.8	2048	1500	290.0	580.0	421.9
	2	594	(8)	42.7			546.5	1093.0	795.0
	4	594	12	79.8			290.0	580.0	421.9
	2	594	(8)	42.7			546.5	1093.0	795.0
ROI	4	594	10	^{*2}	^{*3}	^{*3}	548.0	1096.0	797.1
	2	594	(8)	^{*2}			1062.5	2125.0	1545.5
	4	594	12	^{*2}			548.0	1096.0	797.1
	2	594	(8)	^{*2}			1062.5	2125.0	1545.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 21.7 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

5.28. IMX565-AAQJ-C

Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	42.6	4096	3000	277.5	555.0	403.7
	2	1188	(8)	22.2			535.0	1070.0	778.2
	4	1188	12	42.6			277.5	555.0	403.7
	2	1188	(8)	22.2			535.0	1070.0	778.2
Vertical / Horizontal 1/2 subsampling	4	1188	10	153.1	2048	1500	148.5	297.0	216.0
	2	1188	(8)	83.9			277.0	554.0	403.0
	4	1188	12	112.8			204.0	408.0	296.8
	2	1188	(8)	83.9			277.0	554.0	403.0
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	277.5	555.0	403.7
	2	1188	(8)	^{*2}			535.0	1070.0	778.2
	4	1188	12	^{*2}			277.5	555.0	403.7
	2	1188	(8)	^{*2}			535.0	1070.0	778.2

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	32.2	4096	3000	368.0	736.0	535.3
	2	891	(8)	16.7			711.0	1422.0	1034.2
	4	891	12	32.2			368.0	736.0	535.3
	2	891	(8)	16.7			711.0	1422.0	1034.2
Vertical / Horizontal 1/2 subsampling	4	891	10	117.3	2048	1500	196.0	392.0	285.1
	2	891	(8)	63.7			367.0	734.0	533.9
	4	891	12	112.8			204.0	408.0	296.8
	2	891	(8)	63.7			367.0	734.0	533.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	368.0	736.0	535.3
	2	891	(8)	^{*2}			711.0	1422.0	1034.2
	4	891	12	^{*2}			368.0	736.0	535.3
	2	891	(8)	^{*2}			711.0	1422.0	1034.2

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 42.6 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	21.7	4096	3000	548.0	1096.0	797.1
	2	594	(8)	11.2			1062.5	2125.0	1545.5
	4	594	12	21.7			548.0	1096.0	797.1
	2	594	(8)	11.2			1062.5	2125.0	1545.5
Vertical / Horizontal 1/2 subsampling	4	594	10	80.2	2048	1500	290.0	580.0	421.9
	2	594	(8)	43.1			546.5	1093.0	795.0
	4	594	12	80.2			290.0	580.0	421.9
	2	594	(8)	43.1			546.5	1093.0	795.0
ROI	4	594	10	^{*2}	^{*3}	^{*3}	548.0	1096.0	797.1
	2	594	(8)	^{*2}			1062.5	2125.0	1545.5
	4	594	12	^{*2}			548.0	1096.0	797.1
	2	594	(8)	^{*2}			1062.5	2125.0	1545.5

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 21.7 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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5.29. IMX566-AAMJ-C

Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	62.6	2840	2840	198.0	396.0	288.0
	2	1188	(8)	33.2			376.0	752.0	547.0
	4	1188	12	60.8			204.0	408.0	296.8
	2	1188	(8)	33.2			376.0	752.0	547.0
Vertical / Horizontal 1/2 subsampling	4	1188	10	214.4	1420	1420	109.0	218.0	158.6
	2	1188	(8)	121.7			198.0	396.0	288.0
	4	1188	12	214.4			109.0	218.0	158.6
	2	1188	(8)	121.7			198.0	396.0	288.0
2 × 2 FD binning mode	4	1188	10	214.4	1420	1420	109.0	218.0	158.6
	2	1188	(8)	121.7			198.0	396.0	288.0
	4	1188	12	214.4			109.0	218.0	158.6
	2	1188	(8)	121.7			198.0	396.0	288.0
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	198.0	396.0	288.0
	2	1188	(8)	^{*2}			376.0	752.0	547.0
	4	1188	12	^{*2}			204.0	408.0	296.8
	2	1188	(8)	^{*2}			376.0	752.0	547.0

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	47.5	2840	2840	262.0	524.0	381.1
	2	891	(8)	25.1			499.0	998.0	725.9
	4	891	12	47.5			262.0	524.0	381.1
	2	891	(8)	25.1			499.0	998.0	725.9
Vertical / Horizontal 1/2 subsampling	4	891	10	165.9	1420	1420	143.0	286.0	208.0
	2	891	(8)	93.1			261.5	523.0	380.4
	4	891	12	165.9			143.0	286.0	208.0
	2	891	(8)	93.1			261.5	523.0	380.4
2 × 2 FD binning mode	4	891	10	165.9	1420	1420	143.0	286.0	208.0
	2	891	(8)	93.1			261.5	523.0	380.4
	4	891	12	165.9			143.0	286.0	208.0
	2	891	(8)	93.1			261.5	523.0	380.4
ROI	4	891	10	^{*2}	^{*3}	^{*3}	262.0	524.0	381.1
	2	891	(8)	^{*2}			499.0	998.0	725.9
	4	891	12	^{*2}			262.0	524.0	381.1
	2	891	(8)	^{*2}			499.0	998.0	725.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 62.6 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	32.1	2840	2840	389.0	778.0	565.9
	2	594	(8)	16.8			744.5	1489.0	1083.0
	4	594	12	32.1			389.0	778.0	565.9
	2	594	(8)	16.8			744.5	1489.0	1083.0
Vertical / Horizontal 1/2 subsampling	4	594	10	114.2	1420	1420	211.0	422.0	307.0
	2	594	(8)	63.2			388.5	777.0	565.1
	4	594	12	114.2			211.0	422.0	307.0
	2	594	(8)	63.2			388.5	777.0	565.1
2 × 2 FD binning mode	4	594	10	114.2	1420	1420	211.0	422.0	307.0
	2	594	(8)	63.2			388.5	777.0	565.1
	4	594	12	114.2			211.0	422.0	307.0
	2	594	(8)	63.2			388.5	777.0	565.1
ROI	4	594	10	^{*2}	^{*3}	^{*3}	389.0	778.0	565.9
	2	594	(8)	^{*2}			744.5	1489.0	1083.0
	4	594	12	^{*2}			389.0	778.0	565.9
	2	594	(8)	^{*2}			744.5	1489.0	1083.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 32.1 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	62.6	2840	2840	198.0	396.0	288.0
	2	1188	(8)	33.2			376.0	752.0	547.0
	4	1188	12	60.8			204.0	408.0	296.8
	2	1188	(8)	33.2			376.0	752.0	547.0
Vertical / Horizontal 1/2 subsampling	4	1188	10	168.2	1420	1420	142.0	284.0	206.6
	2	1188	(8)	122.3			198.0	396.0	288.0
	4	1188	12	118.7			204.0	408.0	296.8
	2	1188	(8)	118.7			204.0	408.0	296.8
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	198.0	396.0	288.0
	2	1188	(8)	^{*2}			376.0	752.0	547.0
	4	1188	12	^{*2}			204.0	408.0	296.8
	2	1188	(8)	^{*2}			376.0	752.0	547.0

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	47.5	2840	2840	262.0	524.0	381.1
	2	891	(8)	25.1			499.0	998.0	725.9
	4	891	12	47.5			262.0	524.0	381.1
	2	891	(8)	25.1			499.0	998.0	725.9
Vertical / Horizontal 1/2 subsampling	4	891	10	167.0	1420	1420	143.0	286.0	208.0
	2	891	(8)	93.5			261.5	523.0	380.4
	4	891	12	118.7			204.0	408.0	296.8
	2	891	(8)	93.5			261.5	523.0	380.4
ROI	4	891	10	^{*2}	^{*3}	^{*3}	262.0	524.0	381.1
	2	891	(8)	^{*2}			499.0	998.0	725.9
	4	891	12	^{*2}			262.0	524.0	381.1
	2	891	(8)	^{*2}			499.0	998.0	725.9

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 62.6 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	32.1	2840	2840	389.0	778.0	565.9
	2	594	(8)	16.8			744.5	1489.0	1083.0
	4	594	12	32.1			389.0	778.0	565.9
	2	594	(8)	16.8			744.5	1489.0	1083.0
Vertical / Horizontal 1/2 subsampling	4	594	10	114.9	1420	1420	211.0	422.0	307.0
	2	594	(8)	63.4			388.5	777.0	565.1
	4	594	12	114.9			211.0	422.0	307.0
	2	594	(8)	63.4			388.5	777.0	565.1
ROI	4	594	10	^{*2}	^{*3}	^{*3}	389.0	778.0	565.9
	2	594	(8)	^{*2}			744.5	1489.0	1083.0
	4	594	12	^{*2}			389.0	778.0	565.9
	2	594	(8)	^{*2}			744.5	1489.0	1083.0

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 32.1 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	96.2	2448	2048	174.0	348.0	253.1
	2	1188	(8)	51.7			328.0	656.0	477.1
	4	1188	12	82.4			204.0	408.0	296.8
	2	1188	(8)	51.7			328.0	656.0	477.1
Vertical / Horizontal 1/2 subsampling	4	1188	10	316.8	1224	1024	97.0	194.0	141.1
	2	1188	(8)	183.9			174.0	348.0	253.1
	4	1188	12	302.3			102.0	204.0	148.4
	2	1188	(8)	183.9			174.0	348.0	253.1
2 × 2 FD binning mode	4	1188	10	316.8	1224	1024	97.0	194.0	141.1
	2	1188	(8)	183.9			174.0	348.0	253.1
	4	1188	12	302.3			102.0	204.0	148.4
	2	1188	(8)	183.9			174.0	348.0	253.1
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	174.0	348.0	253.1
	2	1188	(8)	^{*2}			328.0	656.0	477.1
	4	1188	12	^{*2}			204.0	408.0	296.8
	2	1188	(8)	^{*2}			328.0	656.0	477.1

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	73.3	2448	2048	230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
	4	891	12	73.3			230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
Vertical / Horizontal 1/2 subsampling	4	891	10	248.5	1224	1024	127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
	4	891	12	248.5			127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
2 × 2 FD binning mode	4	891	10	248.5	1224	1024	127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
	4	891	12	248.5			127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8
	4	891	12	^{*2}			230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 96.2 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	49.8	2448	2048	341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
	4	594	12	49.8			341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
Vertical / Horizontal 1/2 subsampling	4	594	10	171.7	1224	1024	187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
	4	594	12	171.7			187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
2 × 2 FD binning mode	4	594	10	171.7	1224	1024	187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
	4	594	12	171.7			187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
ROI	4	594	10	^{*2}	^{*3}	^{*3}	341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3
	4	594	12	^{*2}			341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 49.8 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	96.2	2448	2048	174.0	348.0	253.1
	2	1188	(8)	51.7			328.0	656.0	477.1
	4	1188	12	82.4			204.0	408.0	296.8
	2	1188	(8)	51.7			328.0	656.0	477.1
Vertical / Horizontal 1/2 subsampling	4	1188	10	224.9	1224	1024	142.0	284.0	206.6
	2	1188	(8)	185.8			174.0	348.0	253.1
	4	1188	12	159.6			204.0	408.0	296.8
	2	1188	(8)	159.6			204.0	408.0	296.8
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	174.0	348.0	253.1
	2	1188	(8)	^{*2}			328.0	656.0	477.1
	4	1188	12	^{*2}			204.0	408.0	296.8
	2	1188	(8)	^{*2}			328.0	656.0	477.1

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	73.3	2448	2048	230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
	4	891	12	73.3			230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
Vertical / Horizontal 1/2 subsampling	4	891	10	224.9	1224	1024	142.0	284.0	206.6
	2	891	(8)	142.6			229.5	459.0	333.9
	4	891	12	159.6			204.0	408.0	296.8
	2	891	(8)	142.6			229.5	459.0	333.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8
	4	891	12	^{*2}			230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 96.2 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	49.8	2448	2048	341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
	4	594	12	49.8			341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
Vertical / Horizontal 1/2 subsampling	4	594	10	173.5	1224	1024	187.0	374.0	272.0
	2	594	(8)	97.5			340.5	681.0	495.3
	4	594	12	159.6			204.0	408.0	296.8
	2	594	(8)	97.5			340.5	681.0	495.3
ROI	4	594	10	^{*2}	^{*3}	^{*3}	341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3
	4	594	12	^{*2}			341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3

- ^{*1} The total data rate is the data rate multiplied by the number of lanes.
Example) In All-pixel 49.8 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]
- ^{*2} See the section of "ROI mode"
- ^{*3} Designated cropping area (ROI)

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Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	93.4	2448	2048	179.5	359.0	261.1
	2	1188	(8)	51.7			328.0	656.0	477.1
	4	1188	12	84.0			200.0	400.0	291.0
	2	1188	(8)	51.7			328.0	656.0	477.1
Vertical / Horizontal 1/2 subsampling	4	1188	10	316.8	1224	1024	97.0	194.0	141.1
	2	1188	(8)	183.9			174.0	348.0	253.1
	4	1188	12	308.3			100.0	200.0	145.5
	2	1188	(8)	183.9			174.0	348.0	253.1
2 × 2 FD binning mode	4	1188	10	316.8	1224	1024	97.0	194.0	141.1
	2	1188	(8)	183.9			174.0	348.0	253.1
	4	1188	12	308.3			100.0	200.0	145.5
	2	1188	(8)	183.9			174.0	348.0	253.1
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	179.5	359.0	261.1
	2	1188	(8)	^{*2}			328.0	656.0	477.1
	4	1188	12	^{*2}			200.0	400.0	291.0
	2	1188	(8)	^{*2}			328.0	656.0	477.1

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	73.3	2448	2048	230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
	4	891	12	73.3			230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
Vertical / Horizontal 1/2 subsampling	4	891	10	248.5	1224	1024	127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
	4	891	12	248.5			127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
2 × 2 FD binning mode	4	891	10	248.5	1224	1024	127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
	4	891	12	248.5			127.0	254.0	184.8
	2	891	(8)	141.4			229.5	459.0	333.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8
	4	891	12	^{*2}			230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 93.4 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	49.8	2448	2048	341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
	4	594	12	49.8			341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
Vertical / Horizontal 1/2 subsampling	4	594	10	171.7	1224	1024	187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
	4	594	12	171.7			187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
2 × 2 FD binning mode	4	594	10	171.7	1224	1024	187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
	4	594	12	171.7			187.0	374.0	272.0
	2	594	(8)	96.6			340.5	681.0	495.3
ROI	4	594	10	^{*2}	^{*3}	^{*3}	341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3
	4	594	12	^{*2}			341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 49.8 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

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Data rate = 1188 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	1188	10	93.4	2448	2048	179.5	359.0	261.1
	2	1188	(8)	51.7			328.0	656.0	477.1
	4	1188	12	84.0			200.0	400.0	291.0
	2	1188	(8)	51.7			328.0	656.0	477.1
Vertical / Horizontal 1/2 subsampling	4	1188	10	180.4	1224	1024	179.5	359.0	261.1
	2	1188	(8)	180.4			179.5	359.0	261.1
	4	1188	12	162.8			200.0	400.0	291.0
	2	1188	(8)	162.8			200.0	400.0	291.0
ROI	4	1188	10	^{*2}	^{*3}	^{*3}	179.5	359.0	261.1
	2	1188	(8)	^{*2}			328.0	656.0	477.1
	4	1188	12	^{*2}			200.0	400.0	291.0
	2	1188	(8)	^{*2}			328.0	656.0	477.1

Data rate = 891 Mbps

CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	891	10	73.3	2448	2048	230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
	4	891	12	73.3			230.0	460.0	334.6
	2	891	(8)	39.1			435.0	870.0	632.8
Vertical / Horizontal 1/2 subsampling	4	891	10	180.4	1224	1024	179.5	359.0	261.1
	2	891	(8)	142.6			229.5	459.0	333.9
	4	891	12	162.8			200.0	400.0	291.0
	2	891	(8)	142.6			229.5	459.0	333.9
ROI	4	891	10	^{*2}	^{*3}	^{*3}	230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8
	4	891	12	^{*2}			230.0	460.0	334.6
	2	891	(8)	^{*2}			435.0	870.0	632.8

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 93.4 [frame/s] mode: 1188 [Mbps/Lane] × 4 = 4.752 [Gbps]

^{*2} See the section of "ROI mode"^{*3} Designated cropping area (ROI)

Data rate = 594 Mbps
CSI-2 output

Drive mode	CSI-2 Lane ^{*1}	Data rate [Mbps/Lane]	A/D conversion (output gradation)	Frame rate [frame/s]	Number of recording pixels		Number of INCK in 1 H		
					H	V	INCK: 37.125 MHz	INCK: 74.25 MHz	INCK: 54 MHz
All pixel	4	594	10	49.8	2448	2048	341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
	4	594	12	49.8			341.0	682.0	496.0
	2	594	(8)	26.3			648.5	1297.0	943.3
Vertical / Horizontal 1/2 subsampling	4	594	10	173.5	1224	1024	187.0	374.0	272.0
	2	594	(8)	97.5			340.5	681.0	495.3
	4	594	12	162.8			200.0	400.0	291.0
	2	594	(8)	97.5			340.5	681.0	495.3
ROI	4	594	10	^{*2}	^{*3}	^{*3}	341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3
	4	594	12	^{*2}			341.0	682.0	496.0
	2	594	(8)	^{*2}			648.5	1297.0	943.3

^{*1} The total data rate is the data rate multiplied by the number of lanes.

Example) In All-pixel 49.8 [frame/s] mode: 594 [Mbps/Lane] × 4 = 2.376 [Gbps]

^{*2} See the section of "ROI mode"

^{*3} Designated cropping area (ROI)

FRAMOS Technologies Inc.
Internal use only

6. Drive Mode Setting

Please refer to the product specification for register setup other than those lists.

6.1. IMX530-AAMJ-C

6.1.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								21.9 [frame/s]	11.2 [frame/s]	21.9 [frame/s]	11.2 [frame/s]	FREQ = 0h
								11.1 [frame/s]	5.6 [frame/s]	11.1 [frame/s]	5.6 [frame/s]	FREQ = 1h
								30.1 [frame/s]	15.8 [frame/s]	30.1 [frame/s]	15.8 [frame/s]	FREQ = 2h
								15.4 [frame/s]	8.0 [frame/s]	15.4 [frame/s]	8.0 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	127Ch	1274h	127Ch	1274h	FREQ = 0h				
D5h	[7:0]			1274h	1270h	1274h	1270h	FREQ = 1h				
D6h	[7:0]			127Ch	1278h	127Ch	1278h	FREQ = 2h				
				1274h	1270h	1274h	1270h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h				
D9h	[7:0]			584h	ACDh	584h	ACDh	FREQ = 1h				
				208h	3DEh	208h	3DEh	FREQ = 2h				
				3F8h	799h	3F8h	799h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				04h	04h	04h	04h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h				
				08h	04h	08h	04h	FREQ = 1h				
				0Ch	08h	0Ch	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				102.6 [frame/s]	67.4 [frame/s]	35.0 [frame/s]	17.8 [frame/s]	74.4 [frame/s]	67.4 [frame/s]	35.0 [frame/s]	17.8 [frame/s]	FREQ = 0h
				64.1 [frame/s]	34.4 [frame/s]	17.7 [frame/s]	8.9 [frame/s]	44.4 [frame/s]	23.7 [frame/s]	11.9 [frame/s]	6.0 [frame/s]	FREQ = 1h
				32.6 [frame/s]	17.3 [frame/s]	8.8 [frame/s]	4.4 [frame/s]	32.6 [frame/s]	17.3 [frame/s]	8.8 [frame/s]	4.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	12B8h	129Ch	1288h	1278h	12A4h	129Ch	1284h	1278h	FREQ = 0h
D5h	[7:0]			129Ch	1284h	1278h	1270h	128Ch	127Ch	1274h	1270h	FREQ = 1h
D6h	[7:0]			1284h	1278h	1270h	1270h	1284h	1278h	1270h	1270h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.1.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								79.4 [frame/s]	41.7 [frame/s]	79.4 [frame/s]	41.7 [frame/s]	FREQ = 0h
								41.1 [frame/s]	21.4 [frame/s]	41.1 [frame/s]	21.4 [frame/s]	FREQ = 1h
								100.1 [frame/s]	54.8 [frame/s]	100.1 [frame/s]	54.8 [frame/s]	FREQ = 2h
								53.1 [frame/s]	28.8 [frame/s]	53.1 [frame/s]	28.8 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	970h	968h	970h	968h	FREQ = 0h				
D5h	[7:0]			968h	960h	968h	960h	FREQ = 1h				
D6h	[7:0]			978h	968h	978h	968h	FREQ = 2h				
				968h	960h	968h	960h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
D9h	[7:0]			2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
				132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	10h	10h	10h	FREQ = 0h				
				10h	08h	10h	08h	FREQ = 1h				
				18h	10h	18h	10h	FREQ = 2h				
				10h	08h	10h	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACH	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				385.2 [frame/s]	240.6 [frame/s]	132.3 [frame/s]	68.2 [frame/s]	283.3 [frame/s]	240.6 [frame/s]	132.3 [frame/s]	68.2 [frame/s]	FREQ = 0h
				227.5 [frame/s]	126.7 [frame/s]	67.6 [frame/s]	34.3 [frame/s]	162.5 [frame/s]	90.0 [frame/s]	45.5 [frame/s]	23.0 [frame/s]	FREQ = 1h
				118.7 [frame/s]	64.7 [frame/s]	34.0 [frame/s]	17.1 [frame/s]	118.7 [frame/s]	64.7 [frame/s]	34.0 [frame/s]	17.1 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	9E8h	9B8h	988h	970h	9C0h	9B8h	988h	970h	FREQ = 0h
D5h	[7:0]			9A8h	988h	970h	960h	998h	978h	968h	960h	FREQ = 1h
D6h	[7:0]			978h	970h	960h	960h	978h	970h	960h	960h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	7Ch	E6h	1C2h	69h	7Ch	E6h	1C2h	FREQ = 0h
D9h	[7:0]			84h	F0h	1C6h	384h	BAh	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	10h	08h	08h	10h	10h	08h	08h	FREQ = 0h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				08h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				20h	18h	10h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	10h	08h	08h	10h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	38h	20h	10h	40h	38h	20h	10h	FREQ = 0h
				30h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.1.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								79.4 [frame/s]	41.7 [frame/s]	79.4 [frame/s]	41.7 [frame/s]	FREQ = 0h
								41.1 [frame/s]	21.4 [frame/s]	41.1 [frame/s]	21.4 [frame/s]	FREQ = 1h
								100.1 [frame/s]	54.8 [frame/s]	100.1 [frame/s]	54.8 [frame/s]	FREQ = 2h
								53.1 [frame/s]	28.8 [frame/s]	53.1 [frame/s]	28.8 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	970h	968h	970h	968h	FREQ = 0h				
D5h	[7:0]			968h	960h	968h	960h	FREQ = 1h				
D6h	[7:0]			978h	968h	978h	968h	FREQ = 2h				
				968h	960h	968h	960h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
D9h	[7:0]			2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
				132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	10h	10h	10h	FREQ = 0h				
				10h	08h	10h	08h	FREQ = 1h				
				18h	10h	18h	10h	FREQ = 2h				
				10h	08h	10h	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				385.2 [frame/s]	240.6 [frame/s]	132.3 [frame/s]	68.2 [frame/s]	283.3 [frame/s]	240.6 [frame/s]	132.3 [frame/s]	68.2 [frame/s]	FREQ = 0h
				227.5 [frame/s]	126.7 [frame/s]	67.6 [frame/s]	34.3 [frame/s]	162.5 [frame/s]	90.0 [frame/s]	45.5 [frame/s]	23.0 [frame/s]	FREQ = 1h
				118.7 [frame/s]	64.7 [frame/s]	34.0 [frame/s]	17.1 [frame/s]	118.7 [frame/s]	64.7 [frame/s]	34.0 [frame/s]	17.1 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	9E8h	9B8h	988h	970h	9C0h	9B8h	988h	970h	FREQ = 0h
D5h	[7:0]			9A8h	988h	970h	960h	998h	978h	968h	960h	FREQ = 1h
D6h	[7:0]			978h	970h	960h	960h	978h	970h	960h	960h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	7Ch	E6h	1C2h	69h	7Ch	E6h	1C2h	FREQ = 0h
D9h	[7:0]			84h	F0h	1C6h	384h	BAh	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	10h	08h	08h	10h	10h	08h	08h	FREQ = 0h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				08h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				20h	18h	10h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	10h	08h	08h	10h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	38h	20h	10h	40h	38h	20h	10h	FREQ = 0h
				30h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.1.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h
				584h	ACDh	584h	ACDh	FREQ = 1h
D9h	[7:0]			208h	3DEh	208h	3DEh	FREQ = 2h
				3F8h	799h	3F8h	799h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.2. IMX530-AAQJ-C

6.2.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						21.9 [frame/s]	11.2 [frame/s]	21.9 [frame/s]	11.2 [frame/s]	FREQ = 0h
						11.1 [frame/s]	5.6 [frame/s]	11.1 [frame/s]	5.6 [frame/s]	FREQ = 1h
						30.1 [frame/s]	15.8 [frame/s]	30.1 [frame/s]	15.8 [frame/s]	FREQ = 2h
						15.4 [frame/s]	8.0 [frame/s]	15.4 [frame/s]	8.0 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	127Ch	1274h	127Ch	1274h	FREQ = 0h		
D5h	[7:0]			1274h	1270h	1274h	1270h	FREQ = 1h		
D6h	[7:0]			127Ch	1278h	127Ch	1278h	FREQ = 2h		
				1274h	1270h	1274h	1270h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h		
D9h	[7:0]			584h	ACDh	584h	ACDh	FREQ = 1h		
				208h	3DEh	208h	3DEh	FREQ = 2h		
				3F8h	799h	3F8h	799h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				04h	04h	04h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				0Ch	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				102.6 [frame/s]	67.4 [frame/s]	35.0 [frame/s]	17.8 [frame/s]	74.4 [frame/s]	67.4 [frame/s]	35.0 [frame/s]	17.8 [frame/s]	FREQ = 0h
				64.1 [frame/s]	34.4 [frame/s]	17.7 [frame/s]	8.9 [frame/s]	44.4 [frame/s]	23.7 [frame/s]	11.9 [frame/s]	6.0 [frame/s]	FREQ = 1h
				32.6 [frame/s]	17.3 [frame/s]	8.8 [frame/s]	4.4 [frame/s]	32.6 [frame/s]	17.3 [frame/s]	8.8 [frame/s]	4.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	12B8h	129Ch	1288h	1278h	12A4h	129Ch	1284h	1278h	FREQ = 0h
D5h	[7:0]			129Ch	1284h	1278h	1270h	128Ch	127Ch	1274h	1270h	FREQ = 1h
D6h	[7:0]			1284h	1278h	1270h	1270h	1284h	1278h	1270h	1270h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.2.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								79.9 [frame/s]	42.1 [frame/s]	79.9 [frame/s]	42.1 [frame/s]	FREQ = 0h
								41.4 [frame/s]	21.6 [frame/s]	41.4 [frame/s]	21.6 [frame/s]	FREQ = 1h
								100.7 [frame/s]	55.3 [frame/s]	100.7 [frame/s]	55.3 [frame/s]	FREQ = 2h
								53.6 [frame/s]	29.0 [frame/s]	53.6 [frame/s]	29.0 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	960h	954h	960h	954h	FREQ = 0h				
D5h	[7:0]			954h	94Ch	954h	94Ch	FREQ = 1h				
D6h	[7:0]			968h	954h	968h	954h	FREQ = 2h				
				954h	94Ch	954h	94Ch	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
				2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
D9h	[7:0]			132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	04h	08h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				08h	04h	08h	04h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	0Ch	08h	0Ch	08h	FREQ = 0h				
				08h	04h	08h	04h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	0Ch	10h	0Ch	FREQ = 0h				
				0Ch	08h	0Ch	08h	FREQ = 1h				
				14h	0Ch	14h	0Ch	FREQ = 2h				
				0Ch	08h	0Ch	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				200.8 [frame/s]	200.8 [frame/s]	133.3 [frame/s]	68.7 [frame/s]	146.3 [frame/s]	146.3 [frame/s]	133.3 [frame/s]	68.8 [frame/s]	FREQ = 0h
				200.8 [frame/s]	127.8 [frame/s]	68.2 [frame/s]	34.6 [frame/s]	146.3 [frame/s]	90.8 [frame/s]	45.9 [frame/s]	23.2 [frame/s]	FREQ = 1h
				119.3 [frame/s]	65.2 [frame/s]	34.3 [frame/s]	17.3 [frame/s]	119.3 [frame/s]	65.2 [frame/s]	34.3 [frame/s]	17.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	990h	990h	974h	960h	97Ch	97Ch	974h	95Ch	FREQ = 0h
D5h	[7:0]			990h	974h	95Ch	950h	97Ch	964h	954h	94Ch	FREQ = 1h
D6h	[7:0]			96Ch	95Ch	950h	948h	96Ch	95Ch	950h	948h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	97h	E6h	1C2h	D1h	D1h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			97h	F0h	1C6h	384h	D1h	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	08h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	1Ch	14h	0Ch	14h	14h	14h	0Ch	FREQ = 0h
				1Ch	14h	0Ch	08h	14h	0Ch	08h	04h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	2Ch	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	1Ch	10h	08h	20h	14h	0Ch	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.2.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h
				584h	ACDh	584h	ACDh	FREQ = 1h
D9h	[7:0]			208h	3DEh	208h	3DEh	FREQ = 2h
				3F8h	799h	3F8h	799h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.3. IMX531-AAMJ-C

6.3.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								26.0 [frame/s]	13.4 [frame/s]	26.0 [frame/s]	13.4 [frame/s]	FREQ = 0h
								13.2 [frame/s]	6.8 [frame/s]	13.2 [frame/s]	6.8 [frame/s]	FREQ = 1h
								35.3 [frame/s]	18.7 [frame/s]	35.3 [frame/s]	18.7 [frame/s]	FREQ = 2h
								18.2 [frame/s]	9.5 [frame/s]	18.2 [frame/s]	9.5 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	121Ch	1214h	121Ch	1214h	FREQ = 0h				
D5h	[7:0]			1214h	1210h	1214h	1210h	FREQ = 1h				
D6h	[7:0]			1228h	1218h	1224h	1218h	FREQ = 2h				
				1218h	1210h	1218h	1210h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	266h	4A8h	266h	4A8h	FREQ = 0h				
D9h	[7:0]			4B8h	935h	4B8h	935h	FREQ = 1h				
				1C4h	356h	1C4h	356h	FREQ = 2h				
				370h	689h	370h	689h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				08h	04h	04h	04h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				0Ch	08h	0Ch	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h				
				08h	04h	08h	04h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				104.7 [frame/s]	80.1 [frame/s]	42.5 [frame/s]	21.4 [frame/s]	75.9 [frame/s]	75.9 [frame/s]	42.5 [frame/s]	21.4 [frame/s]	FREQ = 0h
				75.9 [frame/s]	40.9 [frame/s]	21.3 [frame/s]	10.7 [frame/s]	52.9 [frame/s]	28.6 [frame/s]	14.4 [frame/s]	7.2 [frame/s]	FREQ = 1h
				38.7 [frame/s]	20.7 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	38.7 [frame/s]	20.7 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	1258h	1248h	122Ch	121Ch	1244h	1244h	122Ch	121Ch	FREQ = 0h
D5h	[7:0]			1244h	1228h	121Ch	1214h	1234h	121Ch	1214h	1210h	FREQ = 1h
D6h	[7:0]			1228h	1218h	1214h	1210h	1228h	1218h	1214h	1210h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	C6h	177h	2EBh	D1h	D1h	177h	2EBh	FREQ = 0h
D9h	[7:0]			D1h	186h	2EDh	5D6h	12Dh	22Fh	45Ah	8AAh	FREQ = 1h
				19Ch	306h	5DAh	BABh	19Ch	306h	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	18h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	10h	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	20h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	10h	0Ch	08h	18h	0Ch	08h	04h	FREQ = 1h
				10h	08h	08h	04h	10h	08h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.3.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						94.1 [frame/s]	50.0 [frame/s]	94.1 [frame/s]	50.0 [frame/s]	FREQ = 0h
						48.8 [frame/s]	25.6 [frame/s]	48.8 [frame/s]	25.6 [frame/s]	FREQ = 1h
						118.3 [frame/s]	65.5 [frame/s]	118.3 [frame/s]	65.5 [frame/s]	FREQ = 2h
						62.2 [frame/s]	34.2 [frame/s]	62.2 [frame/s]	34.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	948h	938h	948h	938h	FREQ = 0h		
D5h	[7:0]			938h	930h	938h	930h	FREQ = 1h		
D6h	[7:0]			948h	940h	948h	940h	FREQ = 2h		
				940h	930h	940h	930h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	14Ch	274h	14Ch	274h	FREQ = 0h		
				284h	4CDh	284h	4CDh	FREQ = 1h		
D9h	[7:0]			108h	1DEh	108h	1DEh	FREQ = 2h		
				1F8h	399h	1F8h	399h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	10h	10h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	18h	10h	18h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				18h	10h	18h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				392.6 [frame/s]	282.5 [frame/s]	155.4 [frame/s]	80.8 [frame/s]	288.8 [frame/s]	283.4 [frame/s]	155.4 [frame/s]	80.8 [frame/s]	FREQ = 0h
				264.6 [frame/s]	148.7 [frame/s]	80.1 [frame/s]	40.6 [frame/s]	192.0 [frame/s]	106.6 [frame/s]	54.8 [frame/s]	27.7 [frame/s]	FREQ = 1h
				138.1 [frame/s]	76.6 [frame/s]	40.3 [frame/s]	20.3 [frame/s]	138.5 [frame/s]	76.6 [frame/s]	40.3 [frame/s]	20.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	9B8h	998h	960h	940h	990h	990h	960h	940h	FREQ = 0h
D5h	[7:0]			988h	960h	940h	930h	970h	948h	938h	930h	FREQ = 1h
D6h	[7:0]			960h	940h	930h	930h	958h	940h	930h	930h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	6Bh	C7h	184h	69h	6Bh	C7h	184h	FREQ = 0h
D9h	[7:0]			73h	D0h	187h	308h	A0h	125h	23Eh	471h	FREQ = 1h
				E0h	199h	30Eh	60Fh	E0h	199h	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	18h	10h	08h	10h	10h	10h	08h	FREQ = 0h
				10h	10h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				28h	18h	10h	08h	20h	10h	08h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	40h	20h	10h	40h	40h	20h	10h	FREQ = 0h
				38h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				20h	10h	08h	08h	20h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.3.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						94.1 [frame/s]	50.0 [frame/s]	94.1 [frame/s]	50.0 [frame/s]	FREQ = 0h
						48.8 [frame/s]	25.6 [frame/s]	48.8 [frame/s]	25.6 [frame/s]	FREQ = 1h
						118.3 [frame/s]	65.5 [frame/s]	118.3 [frame/s]	65.5 [frame/s]	FREQ = 2h
						62.2 [frame/s]	34.2 [frame/s]	62.2 [frame/s]	34.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	948h	938h	948h	938h	FREQ = 0h		
D5h	[7:0]			938h	930h	938h	930h	FREQ = 1h		
D6h	[7:0]			948h	940h	948h	940h	FREQ = 2h		
				940h	930h	940h	930h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	14Ch	274h	14Ch	274h	FREQ = 0h		
				284h	4CDh	284h	4CDh	FREQ = 1h		
D9h	[7:0]			108h	1DEh	108h	1DEh	FREQ = 2h		
				1F8h	399h	1F8h	399h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	10h	10h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	18h	10h	18h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				18h	10h	18h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				392.6 [frame/s]	282.5 [frame/s]	155.4 [frame/s]	80.8 [frame/s]	288.8 [frame/s]	283.4 [frame/s]	155.4 [frame/s]	80.8 [frame/s]	FREQ = 0h
				264.6 [frame/s]	148.7 [frame/s]	80.1 [frame/s]	40.6 [frame/s]	192.0 [frame/s]	106.6 [frame/s]	54.8 [frame/s]	27.7 [frame/s]	FREQ = 1h
				138.1 [frame/s]	76.6 [frame/s]	40.3 [frame/s]	20.3 [frame/s]	138.5 [frame/s]	76.6 [frame/s]	40.3 [frame/s]	20.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	9B8h	998h	960h	940h	990h	990h	960h	940h	FREQ = 0h
D5h	[7:0]			988h	960h	940h	930h	970h	948h	938h	930h	FREQ = 1h
D6h	[7:0]			960h	940h	930h	930h	958h	940h	930h	930h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	6Bh	C7h	184h	69h	6Bh	C7h	184h	FREQ = 0h
D9h	[7:0]			73h	D0h	187h	308h	A0h	125h	23Eh	471h	FREQ = 1h
				E0h	199h	30Eh	60Fh	E0h	199h	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	18h	10h	08h	10h	10h	10h	08h	FREQ = 0h
				10h	10h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				28h	18h	10h	08h	20h	10h	08h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	40h	20h	10h	40h	40h	20h	10h	FREQ = 0h
				38h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				20h	10h	08h	08h	20h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.3.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	266h	4A8h	266h	4A8h	FREQ = 0h
				4B8h	935h	4B8h	935h	FREQ = 1h
D9h	[7:0]			1C4h	356h	1C4h	356h	FREQ = 2h
				370h	689h	370h	689h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	C6h	177h	2EBh	D1h	D1h	177h	2EBh	FREQ = 0h
D9h	[7:0]			D1h	186h	2EDh	5D6h	12Dh	22Fh	45Ah	8AAh	FREQ = 1h
				19Ch	306h	5DAh	BABh	19Ch	306h	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	18h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	10h	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	20h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	10h	0Ch	08h	18h	0Ch	08h	04h	FREQ = 1h
				10h	08h	08h	04h	10h	08h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.4. IMX531-AAQJ-C

6.4.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						26.0 [frame/s]	13.4 [frame/s]	26.0 [frame/s]	13.4 [frame/s]	FREQ = 0h
						13.2 [frame/s]	6.8 [frame/s]	13.2 [frame/s]	6.8 [frame/s]	FREQ = 1h
						35.3 [frame/s]	18.7 [frame/s]	35.3 [frame/s]	18.7 [frame/s]	FREQ = 2h
						18.2 [frame/s]	9.5 [frame/s]	18.2 [frame/s]	9.5 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	121Ch	1214h	121Ch	1214h	FREQ = 0h		
D5h	[7:0]			1214h	1210h	1214h	1210h	FREQ = 1h		
D6h	[7:0]			1228h	1218h	1224h	1218h	FREQ = 2h		
				1218h	1210h	1218h	1210h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	266h	4A8h	266h	4A8h	FREQ = 0h		
D9h	[7:0]			4B8h	935h	4B8h	935h	FREQ = 1h		
				1C4h	356h	1C4h	356h	FREQ = 2h		
				370h	689h	370h	689h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	04h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				0Ch	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				104.7 [frame/s]	80.1 [frame/s]	42.5 [frame/s]	21.4 [frame/s]	75.9 [frame/s]	75.9 [frame/s]	42.5 [frame/s]	21.4 [frame/s]	FREQ = 0h
				75.9 [frame/s]	40.9 [frame/s]	21.3 [frame/s]	10.7 [frame/s]	52.9 [frame/s]	28.6 [frame/s]	14.4 [frame/s]	7.2 [frame/s]	FREQ = 1h
				38.7 [frame/s]	20.7 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	38.7 [frame/s]	20.7 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	1258h	1248h	122Ch	121Ch	1244h	1244h	122Ch	121Ch	FREQ = 0h
D5h	[7:0]			1244h	1228h	121Ch	1214h	1234h	121Ch	1214h	1210h	FREQ = 1h
D6h	[7:0]			1228h	1218h	1214h	1210h	1228h	1218h	1214h	1210h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	C6h	177h	2EBh	D1h	D1h	177h	2EBh	FREQ = 0h
D9h	[7:0]			D1h	186h	2EDh	5D6h	12Dh	22Fh	45Ah	8AAh	FREQ = 1h
				19Ch	306h	5DAh	BABh	19Ch	306h	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	18h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	10h	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	20h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	10h	0Ch	08h	18h	0Ch	08h	04h	FREQ = 1h
				10h	08h	08h	04h	10h	08h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.4.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						94.7 [frame/s]	50.5 [frame/s]	94.7 [frame/s]	50.5 [frame/s]	FREQ = 0h
						49.2 [frame/s]	25.9 [frame/s]	49.2 [frame/s]	25.9 [frame/s]	FREQ = 1h
						118.9 [frame/s]	66.1 [frame/s]	118.9 [frame/s]	66.1 [frame/s]	FREQ = 2h
						62.7 [frame/s]	34.5 [frame/s]	62.7 [frame/s]	34.5 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	938h	924h	938h	924h	FREQ = 0h		
D5h	[7:0]			924h	91Ch	924h	91Ch	FREQ = 1h		
D6h	[7:0]			93Ch	92Ch	93Ch	92Ch	FREQ = 2h		
				92Ch	920h	92Ch	920h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	14Ch	274h	14Ch	274h	FREQ = 0h		
				284h	4CDh	284h	4CDh	FREQ = 1h		
D9h	[7:0]			108h	1DEh	108h	1DEh	FREQ = 2h		
				1F8h	399h	1F8h	399h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	08h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				10h	0Ch	10h	0Ch	FREQ = 2h		
				0Ch	08h	0Ch	08h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	14h	0Ch	14h	0Ch	FREQ = 0h		
				0Ch	08h	0Ch	08h	FREQ = 1h		
				18h	10h	18h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				204.8 [frame/s]	204.8 [frame/s]	156.5 [frame/s]	81.3 [frame/s]	149.2 [frame/s]	149.2 [frame/s]	149.2 [frame/s]	81.3 [frame/s]	FREQ = 0h
				204.8 [frame/s]	149.9 [frame/s]	80.7 [frame/s]	40.9 [frame/s]	149.2 [frame/s]	107.1 [frame/s]	55.2 [frame/s]	28.0 [frame/s]	FREQ = 1h
				139.5 [frame/s]	77.1 [frame/s]	40.6 [frame/s]	20.5 [frame/s]	139.7 [frame/s]	77.1 [frame/s]	40.6 [frame/s]	20.5 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	960h	960h	950h	930h	94Ch	94Ch	94Ch	930h	FREQ = 0h
D5h	[7:0]			960h	94Ch	930h	920h	94Ch	93Ch	924h	91Ch	FREQ = 1h
D6h	[7:0]			948h	930h	920h	918h	944h	930h	920h	918h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	97h	C7h	184h	D1h	D1h	D1h	184h	FREQ = 0h
D9h	[7:0]			97h	D0h	187h	308h	D1h	125h	23Eh	471h	FREQ = 1h
				E0h	199h	30Eh	60Fh	E0h	199h	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				0Ch	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	08h	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	1Ch	18h	0Ch	14h	14h	14h	0Ch	FREQ = 0h
				1Ch	14h	0Ch	08h	14h	10h	08h	04h	FREQ = 1h
				14h	0Ch	08h	04h	14h	0Ch	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	2Ch	20h	10h	20h	20h	20h	10h	FREQ = 0h
				2Ch	20h	10h	08h	20h	18h	0Ch	08h	FREQ = 1h
				1Ch	10h	08h	04h	1Ch	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.4.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	266h	4A8h	266h	4A8h	FREQ = 0h
				4B8h	935h	4B8h	935h	FREQ = 1h
D9h	[7:0]			1C4h	356h	1C4h	356h	FREQ = 2h
				370h	689h	370h	689h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	C6h	177h	2EBh	D1h	D1h	177h	2EBh	FREQ = 0h
D9h	[7:0]			D1h	186h	2EDh	5D6h	12Dh	22Fh	45Ah	8AAh	FREQ = 1h
				19Ch	306h	5DAh	BABh	19Ch	306h	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	18h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	10h	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	20h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	10h	0Ch	08h	18h	0Ch	08h	04h	FREQ = 1h
				10h	08h	08h	04h	10h	08h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.5. IMX532-AAMJ-C

6.5.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								32.7 [frame/s]	16.8 [frame/s]	32.7 [frame/s]	16.8 [frame/s]	FREQ = 0h
								16.6 [frame/s]	8.5 [frame/s]	16.6 [frame/s]	8.5 [frame/s]	FREQ = 1h
								45.1 [frame/s]	23.7 [frame/s]	45.1 [frame/s]	23.7 [frame/s]	FREQ = 2h
								23.1 [frame/s]	12.1 [frame/s]	23.1 [frame/s]	12.1 [frame/s]	FREQ = 3h
				594 Mbps/ch								FREQ = 0h
				297 Mbps/ch								FREQ = 1h
891 Mbps/ch								FREQ = 2h				
445.5 Mbps/ch								FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	C5Ch	C54h	C5Ch	C54h	FREQ = 0h				
D5h	[7:0]			C54h	C50h	C54h	C50h	FREQ = 1h				
D6h	[7:0]			C5Ch	C58h	C5Ch	C58h	FREQ = 2h				
				C54h	C50h	C54h	C50h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h				
D9h	[7:0]			584h	ACDh	584h	ACDh	FREQ = 1h				
				208h	3DEh	208h	3DEh	FREQ = 2h				
				3F8h	799h	3F8h	799h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				04h	04h	04h	04h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h				
				08h	04h	08h	04h	FREQ = 1h				
				0Ch	08h	0Ch	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				152.5 [frame/s]	100.5 [frame/s]	52.3 [frame/s]	26.7 [frame/s]	110.8 [frame/s]	100.5 [frame/s]	52.3 [frame/s]	26.7 [frame/s]	FREQ = 0h
				95.6 [frame/s]	51.4 [frame/s]	26.4 [frame/s]	13.3 [frame/s]	66.3 [frame/s]	35.5 [frame/s]	17.9 [frame/s]	9.0 [frame/s]	FREQ = 1h
				48.7 [frame/s]	25.9 [frame/s]	13.2 [frame/s]	6.6 [frame/s]	48.7 [frame/s]	25.9 [frame/s]	13.2 [frame/s]	6.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	C98h	C7Ch	C68h	C58h	C84h	C7Ch	C64h	C58h	FREQ = 0h
D5h	[7:0]			C7Ch	C64h	C58h	C50h	C6Ch	C5Ch	C54h	C50h	FREQ = 1h
D6h	[7:0]			C64h	C58h	C50h	C50h	C64h	C58h	C50h	C50h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.5.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								117.5 [frame/s]	61.9 [frame/s]	117.5 [frame/s]	61.9 [frame/s]	FREQ = 0h
								60.9 [frame/s]	31.8 [frame/s]	60.9 [frame/s]	31.8 [frame/s]	FREQ = 1h
								147.9 [frame/s]	81.3 [frame/s]	147.9 [frame/s]	81.3 [frame/s]	FREQ = 2h
								78.8 [frame/s]	42.8 [frame/s]	78.8 [frame/s]	42.8 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	660h	658h	660h	658h	FREQ = 0h				
D5h	[7:0]			658h	650h	658h	650h	FREQ = 1h				
D6h	[7:0]			668h	658h	668h	658h	FREQ = 2h				
				658h	650h	658h	650h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
D9h	[7:0]			2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
				132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	10h	10h	10h	FREQ = 0h				
				10h	08h	10h	08h	FREQ = 1h				
				18h	10h	18h	10h	FREQ = 2h				
				10h	08h	10h	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				557.6 [frame/s]	351.4 [frame/s]	194.9 [frame/s]	101.1 [frame/s]	413.0 [frame/s]	351.4 [frame/s]	194.9 [frame/s]	101.1 [frame/s]	FREQ = 0h
				333.2 [frame/s]	186.8 [frame/s]	100.2 [frame/s]	51.0 [frame/s]	238.7 [frame/s]	133.1 [frame/s]	67.5 [frame/s]	34.1 [frame/s]	FREQ = 1h
				175.4 [frame/s]	95.7 [frame/s]	50.6 [frame/s]	25.5 [frame/s]	175.4 [frame/s]	95.7 [frame/s]	50.6 [frame/s]	25.5 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	6D8h	6A8h	678h	660h	6B0h	6A8h	678h	660h	FREQ = 0h
D5h	[7:0]			698h	678h	660h	650h	688h	668h	658h	650h	FREQ = 1h
D6h	[7:0]			668h	660h	650h	650h	668h	660h	650h	650h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	7Ch	E6h	1C2h	69h	7Ch	E6h	1C2h	FREQ = 0h
D9h	[7:0]			84h	F0h	1C6h	384h	BAh	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	10h	08h	08h	10h	10h	08h	08h	FREQ = 0h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				08h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				20h	18h	10h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	10h	08h	08h	10h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	38h	20h	10h	40h	38h	20h	10h	FREQ = 0h
				30h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.5.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								117.5 [frame/s]	61.9 [frame/s]	117.5 [frame/s]	61.9 [frame/s]	FREQ = 0h
								60.9 [frame/s]	31.8 [frame/s]	60.9 [frame/s]	31.8 [frame/s]	FREQ = 1h
								147.9 [frame/s]	81.3 [frame/s]	147.9 [frame/s]	81.3 [frame/s]	FREQ = 2h
								78.8 [frame/s]	42.8 [frame/s]	78.8 [frame/s]	42.8 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	660h	658h	660h	658h	FREQ = 0h				
D5h	[7:0]			658h	650h	658h	650h	FREQ = 1h				
D6h	[7:0]			668h	658h	668h	658h	FREQ = 2h				
				658h	650h	658h	650h	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
D9h	[7:0]			2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
				132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				08h	08h	08h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	10h	08h	10h	08h	FREQ = 0h				
				08h	08h	08h	08h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	08h	08h	08h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	10h	10h	10h	FREQ = 0h				
				10h	08h	10h	08h	FREQ = 1h				
				18h	10h	18h	10h	FREQ = 2h				
				10h	08h	10h	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACCh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	FREQ = 0h
				557.6 [frame/s]	351.4 [frame/s]	194.9 [frame/s]	101.1 [frame/s]	413.0 [frame/s]	351.4 [frame/s]	194.9 [frame/s]	101.1 [frame/s]	FREQ = 1h
				333.2 [frame/s]	186.8 [frame/s]	100.2 [frame/s]	51.0 [frame/s]	238.7 [frame/s]	133.1 [frame/s]	67.5 [frame/s]	34.1 [frame/s]	FREQ = 2h
				175.4 [frame/s]	95.7 [frame/s]	50.6 [frame/s]	25.5 [frame/s]	175.4 [frame/s]	95.7 [frame/s]	50.6 [frame/s]	25.5 [frame/s]	FREQ = 0h
				4752 Mbps/lane								FREQ = 1h
				2376 Mbps/lane								FREQ = 2h
1188 Mbps/lane												
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	6D8h	6A8h	678h	660h	6B0h	6A8h	678h	660h	FREQ = 0h
D5h	[7:0]			698h	678h	660h	650h	688h	668h	658h	650h	FREQ = 1h
D6h	[7:0]			668h	660h	650h	650h	668h	660h	650h	650h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	4Ch	7Ch	E6h	1C2h	69h	7Ch	E6h	1C2h	FREQ = 0h
D9h	[7:0]			84h	F0h	1C6h	384h	BAh	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	18h	10h	08h	08h	10h	10h	08h	08h	FREQ = 0h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 1h
				08h	08h	08h	08h	08h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	38h	28h	18h	10h	28h	28h	18h	10h	FREQ = 0h
				20h	18h	10h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	10h	08h	08h	10h	10h	08h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	50h	38h	20h	10h	40h	38h	20h	10h	FREQ = 0h
				30h	20h	10h	08h	28h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.5.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h
				584h	ACDh	584h	ACDh	FREQ = 1h
D9h	[7:0]			208h	3DEh	208h	3DEh	FREQ = 2h
				3F8h	799h	3F8h	799h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.6. IMX532-AAQJ-C

6.6.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						32.7 [frame/s]	16.8 [frame/s]	32.7 [frame/s]	16.8 [frame/s]	FREQ = 0h
						16.6 [frame/s]	8.5 [frame/s]	16.6 [frame/s]	8.5 [frame/s]	FREQ = 1h
						45.1 [frame/s]	23.7 [frame/s]	45.1 [frame/s]	23.7 [frame/s]	FREQ = 2h
						23.1 [frame/s]	12.1 [frame/s]	23.1 [frame/s]	12.1 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	C5Ch	C54h	C5Ch	C54h	FREQ = 0h		
D5h	[7:0]			C54h	C50h	C54h	C50h	FREQ = 1h		
D6h	[7:0]			C5Ch	C58h	C5Ch	C58h	FREQ = 2h		
				C54h	C50h	C54h	C50h	FREQ = 3h		
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h		
D9h	[7:0]			584h	ACDh	584h	ACDh	FREQ = 1h		
				208h	3DEh	208h	3DEh	FREQ = 2h		
				3F8h	799h	3F8h	799h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				04h	04h	04h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				0Ch	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				152.5 [frame/s]	100.5 [frame/s]	52.3 [frame/s]	26.7 [frame/s]	110.8 [frame/s]	100.5 [frame/s]	52.3 [frame/s]	26.7 [frame/s]	FREQ = 0h
				95.6 [frame/s]	51.4 [frame/s]	26.4 [frame/s]	13.3 [frame/s]	66.3 [frame/s]	35.5 [frame/s]	17.9 [frame/s]	9.0 [frame/s]	FREQ = 1h
				48.7 [frame/s]	25.9 [frame/s]	13.2 [frame/s]	6.6 [frame/s]	48.7 [frame/s]	25.9 [frame/s]	13.2 [frame/s]	6.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	C98h	C7Ch	C68h	C58h	C84h	C7Ch	C64h	C58h	FREQ = 0h
D5h	[7:0]			C7Ch	C64h	C58h	C50h	C6Ch	C5Ch	C54h	C50h	FREQ = 1h
D6h	[7:0]			C64h	C58h	C50h	C50h	C64h	C58h	C50h	C50h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.6.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				8 ch	4 ch	8 ch	4 ch					
								118.7 [frame/s]	62.7 [frame/s]	118.7 [frame/s]	62.7 [frame/s]	FREQ = 0h
								61.7 [frame/s]	32.2 [frame/s]	61.7 [frame/s]	32.2 [frame/s]	FREQ = 1h
								149.4 [frame/s]	82.3 [frame/s]	149.4 [frame/s]	82.3 [frame/s]	FREQ = 2h
								79.8 [frame/s]	43.3 [frame/s]	79.8 [frame/s]	43.3 [frame/s]	FREQ = 3h
								594 Mbps/ch				FREQ = 0h
								297 Mbps/ch				FREQ = 1h
								891 Mbps/ch				FREQ = 2h
				445.5 Mbps/ch				FREQ = 3h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	650h	644h	650h	644h	FREQ = 0h				
D5h	[7:0]			644h	63Ch	644h	63Ch	FREQ = 1h				
D6h	[7:0]			658h	644h	658h	644h	FREQ = 2h				
				644h	63Ch	644h	63Ch	FREQ = 3h				
D8h	[7:0]	HMAX	00A6h	183h	2E2h	183h	2E2h	FREQ = 0h				
				2EEh	5A1h	2EEh	5A1h	FREQ = 1h				
D9h	[7:0]			132h	232h	132h	232h	FREQ = 2h				
				244h	431h	244h	431h	FREQ = 3h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	0Ch	08h	04h	08h	04h	FREQ = 0h				
				04h	04h	04h	04h	FREQ = 1h				
				08h	04h	08h	04h	FREQ = 2h				
				04h	04h	04h	04h	FREQ = 3h				
E3h	[7:0]	GMTWT	48h	0Ch	08h	0Ch	08h	FREQ = 0h				
				08h	04h	08h	04h	FREQ = 1h				
				10h	08h	10h	08h	FREQ = 2h				
				08h	04h	08h	04h	FREQ = 3h				
E6h	[7:0]	GSDLY	34h	10h	0Ch	10h	0Ch	FREQ = 0h				
				0Ch	08h	0Ch	08h	FREQ = 1h				
				14h	0Ch	14h	0Ch	FREQ = 2h				
				0Ch	08h	0Ch	08h	FREQ = 3h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				295.5 [frame/s]	295.5 [frame/s]	197.3 [frame/s]	102.1 [frame/s]	216.0 [frame/s]	216.0 [frame/s]	197.3 [frame/s]	102.3 [frame/s]	FREQ = 0h
				295.5 [frame/s]	189.1 [frame/s]	101.4 [frame/s]	51.5 [frame/s]	216.0 [frame/s]	134.8 [frame/s]	68.3 [frame/s]	34.6 [frame/s]	FREQ = 1h
				176.7 [frame/s]	96.9 [frame/s]	51.1 [frame/s]	25.9 [frame/s]	176.7 [frame/s]	96.9 [frame/s]	51.1 [frame/s]	25.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	680h	680h	664h	650h	66Ch	66Ch	664h	64Ch	FREQ = 0h
D5h	[7:0]			680h	664h	64Ch	640h	66Ch	654h	644h	63Ch	FREQ = 1h
D6h	[7:0]			65Ch	64Ch	640h	638h	65Ch	64Ch	640h	638h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	97h	97h	E6h	1C2h	D1h	D1h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			97h	F0h	1C6h	384h	D1h	154h	2A5h	540h	FREQ = 1h
				102h	1DBh	38Ch	707h	102h	1DBh	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	0Ch	08h	08h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	1Ch	14h	0Ch	14h	14h	14h	0Ch	FREQ = 0h
				1Ch	14h	0Ch	08h	14h	0Ch	08h	04h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	2Ch	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	1Ch	10h	08h	20h	14h	0Ch	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

6.6.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	2CCh	574h	2CCh	574h	FREQ = 0h
				584h	ACDh	584h	ACDh	FREQ = 1h
D9h	[7:0]			208h	3DEh	208h	3DEh	FREQ = 2h
				3F8h	799h	3F8h	799h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	0Ch	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	0h				

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	00A6h	97h	E7h	1BFh	370h	D1h	E7h	1BFh	370h	FREQ = 0h
D9h	[7:0]			F3h	1C7h	377h	6E0h	160h	295h	51Eh	A31h	FREQ = 1h
				1E0h	389h	6EEh	DBFh	1E0h	389h	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	08h	04h	0Ch	08h	04h	04h	FREQ = 0h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	1Ch	14h	0Ch	08h	14h	14h	0Ch	08h	FREQ = 0h
				14h	0Ch	08h	04h	0Ch	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 0h
				1Ch	10h	08h	04h	14h	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	10h	08h	04h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.7 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.7. Common registers according to the setting value of HMAX (IMX530 / IMX531 / IMX532)

When HMAX is changed from the "Register list", the following registers must be set according to the setting range.

SLVS (Common for all readout driving modes)

Address	bit	Register name	Initial Value	Setting value					Remarks
				to 21Ch (540d)	21Dh (541d) to 3B4h (948d)	3B5h (949d) to 7F4h (2036d)	7F5h (2037d) to D28h (3368d)	D29h (3369d) to	Setting range of HMAX
Chip ID = 08h									
4Ch	[7:0]		00h	00h	73h				
4Eh	[7:0]		00h	00h	6Ah	7Bh	F2h	FFh	
4Fh	[7:0]		00h	00h	03h	07h	0Bh	0Fh	
74h	[7:0]		00h	00h	73h				
76h	[7:0]		00h	00h	6Ah	7Bh	F2h	FFh	
77h	[7:0]		00h	00h	03h	07h	0Bh	0Fh	
9Ch	[7:0]		00h	00h	73h				
9Eh	[7:0]		00h	00h	6Ah	7Bh	F2h	FFh	
9Fh	[7:0]		00h	00h	03h	07h	0Bh	0Fh	

SLVS – EC (Common for all readout driving modes)

Address	bit	Register name	Initial Value	Setting value					Remarks
				to 1D7h (471d)	1D8h (472d) to 3CBh (971d)	3CCh (972d) to 908h (2312d)	909h (2313d) to E3Dh (3645d)	E3Eh (3646d) to	Setting range of HMAX
Chip ID = 08h									
4Ch	[7:0]		00h	00h	73h				
4Eh	[7:0]		00h	00h	74h	D1h	19h	FFh	
4Fh	[7:0]		00h	00h	03h	07h	0Ch	0Fh	
74h	[7:0]		00h	00h	73h				
76h	[7:0]		00h	00h	74h	D1h	19h	FFh	
77h	[7:0]		00h	00h	03h	07h	0Ch	0Fh	
9Ch	[7:0]		00h	00h	73h				
9Eh	[7:0]		00h	00h	74h	D1h	19h	FFh	
9Fh	[7:0]		00h	00h	03h	07h	0Ch	0Fh	

6.8. IMX535-AAMJ-C

6.8.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						41.5 [frame/s]	21.5 [frame/s]	41.5 [frame/s]	21.5 [frame/s]	FREQ = 0h
						21.2 [frame/s]	10.9 [frame/s]	21.2 [frame/s]	10.9 [frame/s]	FREQ = 1h
						60.7 [frame/s]	31.9 [frame/s]	60.7 [frame/s]	31.9 [frame/s]	FREQ = 2h
						31.5 [frame/s]	16.3 [frame/s]	31.5 [frame/s]	16.3 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	C54h	C44h	C54h	C44h	FREQ = 0h		
D5h	[7:0]			C44h	C38h	C44h	C38h	FREQ = 1h		
D6h	[7:0]			C68h	C4Ch	C68h	C4Ch	FREQ = 2h		
				C4Ch	C40h	C4Ch	C40h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	236h	448h	236h	448h	FREQ = 0h		
				458h	875h	458h	875h	FREQ = 1h		
D9h	[7:0]			181h	2E3h	181h	2E3h	FREQ = 2h		
				2ECh	5ABh	2ECh	5ABh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	04h	04h	04h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	08h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	1Ch	10h	1Ch	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				28h	14h	28h	14h	FREQ = 2h		
				14h	0Ch	14h	0Ch	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				10h	0Ch	10h	0Ch	FREQ = 2h		
				0Ch	08h	0Ch	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				175.0 [frame/s]	125.9 [frame/s]	67.7 [frame/s]	34.3 [frame/s]	120.9 [frame/s]	120.9 [frame/s]	67.7 [frame/s]	34.3 [frame/s]	FREQ = 0h
				119.0 [frame/s]	65.4 [frame/s]	34.2 [frame/s]	17.2 [frame/s]	119.0 [frame/s]	65.4 [frame/s]	34.2 [frame/s]	17.2 [frame/s]	FREQ = 1h
				61.5 [frame/s]	33.2 [frame/s]	17.2 [frame/s]	8.6 [frame/s]	61.5 [frame/s]	33.2 [frame/s]	17.2 [frame/s]	8.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	CD8h	CA8h	C70h	C50h	CA0h	CA0h	C70h	C50h	FREQ = 0h
D5h	[7:0]			CA0h	C6Ch	C50h	C40h	CA0h	C6Ch	C50h	C40h	FREQ = 1h
D6h	[7:0]			C68h	C4Ch	C40h	C38h	C68h	C4Ch	C40h	C38h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	B6h	158h	2AEh	BEh	BEh	158h	2AEh	FREQ = 0h
D9h	[7:0]			C1h	165h	2AFh	55Bh	C1h	165h	2AFh	55Bh	FREQ = 1h
				17Ch	2C6h	55Eh	AB5h	17Ch	2C6h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	50h	2Ch	18h	4Ch	4Ch	2Ch	18h	FREQ = 0h
				4Ch	28h	18h	0Ch	4Ch	28h	18h	0Ch	FREQ = 1h
				28h	14h	0Ch	08h	28h	14h	0Ch	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	24h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	14h	0Ch	08h	20h	14h	0Ch	08h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.8.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						145.5 [frame/s]	78.4 [frame/s]	145.5 [frame/s]	78.4 [frame/s]	FREQ = 0h
						76.7 [frame/s]	40.7 [frame/s]	76.7 [frame/s]	40.7 [frame/s]	FREQ = 1h
						205.5 [frame/s]	114.6 [frame/s]	205.5 [frame/s]	114.6 [frame/s]	FREQ = 2h
						112.0 [frame/s]	60.2 [frame/s]	112.0 [frame/s]	60.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	678h	660h	678h	660h	FREQ = 0h		
D5h	[7:0]			658h	648h	658h	648h	FREQ = 1h		
D6h	[7:0]			6A0h	668h	6A0h	668h	FREQ = 2h		
				668h	650h	668h	650h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	134h	244h	134h	244h	FREQ = 0h		
				254h	46Dh	254h	46Dh	FREQ = 1h		
D9h	[7:0]			D5h	18Bh	D5h	18Bh	FREQ = 2h		
				194h	2FBh	194h	2FBh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	08h	08h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	30h	20h	30h	20h	FREQ = 0h		
				18h	10h	18h	10h	FREQ = 1h		
				48h	28h	48h	28h	FREQ = 2h		
				28h	18h	28h	18h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	18h	10h	18h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				20h	10h	20h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				592.4 [frame/s]	417.2 [frame/s]	236.9 [frame/s]	126.2 [frame/s]	430.3 [frame/s]	417.2 [frame/s]	236.9 [frame/s]	126.2 [frame/s]	FREQ = 0h
				385.5 [frame/s]	230.6 [frame/s]	125.1 [frame/s]	64.0 [frame/s]	385.5 [frame/s]	230.6 [frame/s]	125.1 [frame/s]	64.0 [frame/s]	FREQ = 1h
				210.4 [frame/s]	120.7 [frame/s]	63.5 [frame/s]	32.3 [frame/s]	210.4 [frame/s]	120.7 [frame/s]	63.5 [frame/s]	32.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	788h	718h	6B0h	670h	718h	718h	6B0h	670h	FREQ = 0h
D5h	[7:0]			708h	6B0h	670h	658h	708h	6B0h	670h	658h	FREQ = 1h
D6h	[7:0]			6A0h	670h	658h	648h	6A0h	670h	658h	648h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	62h	B7h	165h	5Fh	62h	B7h	165h	FREQ = 0h
D9h	[7:0]			6Bh	BCh	168h	2CAh	6Bh	BCh	168h	2CAh	FREQ = 1h
				D0h	175h	2D0h	594h	D0h	175h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	18h	10h	08h	18h	18h	10h	08h	FREQ = 0h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	98h	50h	28h	98h	98h	50h	28h	FREQ = 0h
				88h	50h	28h	18h	88h	50h	28h	18h	FREQ = 1h
				48h	28h	18h	10h	48h	28h	18h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	40h	28h	18h	40h	40h	28h	18h	FREQ = 0h
				40h	28h	18h	10h	40h	28h	18h	10h	FREQ = 1h
				20h	18h	10h	08h	20h	18h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.8.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						145.5 [frame/s]	78.4 [frame/s]	145.5 [frame/s]	78.4 [frame/s]	FREQ = 0h
						76.7 [frame/s]	40.7 [frame/s]	76.7 [frame/s]	40.7 [frame/s]	FREQ = 1h
						205.5 [frame/s]	114.6 [frame/s]	205.5 [frame/s]	114.6 [frame/s]	FREQ = 2h
						112.0 [frame/s]	60.2 [frame/s]	112.0 [frame/s]	60.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	678h	660h	678h	660h	FREQ = 0h		
D5h	[7:0]			658h	648h	658h	648h	FREQ = 1h		
D6h	[7:0]			6A0h	668h	6A0h	668h	FREQ = 2h		
				668h	650h	668h	650h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	134h	244h	134h	244h	FREQ = 0h		
				254h	46Dh	254h	46Dh	FREQ = 1h		
D9h	[7:0]			D5h	18Bh	D5h	18Bh	FREQ = 2h		
				194h	2FBh	194h	2FBh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	08h	08h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	30h	20h	30h	20h	FREQ = 0h		
				18h	10h	18h	10h	FREQ = 1h		
				48h	28h	48h	28h	FREQ = 2h		
				28h	18h	28h	18h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	18h	10h	18h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				20h	10h	20h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				592.4 [frame/s]	417.2 [frame/s]	236.9 [frame/s]	126.2 [frame/s]	430.3 [frame/s]	417.2 [frame/s]	236.9 [frame/s]	126.2 [frame/s]	FREQ = 0h
				385.5 [frame/s]	230.6 [frame/s]	125.1 [frame/s]	64.0 [frame/s]	385.5 [frame/s]	230.6 [frame/s]	125.1 [frame/s]	64.0 [frame/s]	FREQ = 1h
				210.4 [frame/s]	120.7 [frame/s]	63.5 [frame/s]	32.3 [frame/s]	210.4 [frame/s]	120.7 [frame/s]	63.5 [frame/s]	32.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	788h	718h	6B0h	670h	718h	718h	6B0h	670h	FREQ = 0h
D5h	[7:0]			708h	6B0h	670h	658h	708h	6B0h	670h	658h	FREQ = 1h
D6h	[7:0]			6A0h	670h	658h	648h	6A0h	670h	658h	648h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	62h	B7h	165h	5Fh	62h	B7h	165h	FREQ = 0h
D9h	[7:0]			6Bh	BCh	168h	2CAh	6Bh	BCh	168h	2CAh	FREQ = 1h
				D0h	175h	2D0h	594h	D0h	175h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	18h	10h	08h	18h	18h	10h	08h	FREQ = 0h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	98h	50h	28h	98h	98h	50h	28h	FREQ = 0h
				88h	50h	28h	18h	88h	50h	28h	18h	FREQ = 1h
				48h	28h	18h	10h	48h	28h	18h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	40h	28h	18h	40h	40h	28h	18h	FREQ = 0h
				40h	28h	18h	10h	40h	28h	18h	10h	FREQ = 1h
				20h	18h	10h	08h	20h	18h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.8.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	236h	448h	236h	448h	FREQ = 0h
				458h	875h	458h	875h	FREQ = 1h
D9h	[7:0]			181h	2E3h	181h	2E3h	FREQ = 2h
				2ECh	5ABh	2ECh	5ABh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				28h	14h	28h	14h	FREQ = 2h
				14h	0Ch	14h	0Ch	FREQ = 3h
E6h	[7:0]	GSDLY	30h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				10h	0Ch	10h	0Ch	FREQ = 2h
				0Ch	08h	0Ch	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	B6h	158h	2AEh	BEh	BEh	158h	2AEh	FREQ = 0h
D9h	[7:0]			C1h	165h	2AFh	55Bh	C1h	165h	2AFh	55Bh	FREQ = 1h
				17Ch	2C6h	55Eh	AB5h	17Ch	2C6h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	50h	2Ch	18h	4Ch	4Ch	2Ch	18h	FREQ = 0h
				4Ch	28h	18h	0Ch	4Ch	28h	18h	0Ch	FREQ = 1h
				28h	14h	0Ch	08h	28h	14h	0Ch	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	24h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	14h	0Ch	08h	20h	14h	0Ch	08h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

6.9. IMX535-AAQJ-C

6.9.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						41.6 [frame/s]	21.6 [frame/s]	41.6 [frame/s]	21.6 [frame/s]	FREQ = 0h
						21.2 [frame/s]	10.9 [frame/s]	21.2 [frame/s]	10.9 [frame/s]	FREQ = 1h
						60.7 [frame/s]	31.9 [frame/s]	60.7 [frame/s]	31.9 [frame/s]	FREQ = 2h
						31.5 [frame/s]	16.3 [frame/s]	31.5 [frame/s]	16.3 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	C50h	C40h	C50h	C40h	FREQ = 0h		
D5h	[7:0]			C40h	C34h	C40h	C34h	FREQ = 1h		
D6h	[7:0]			C64h	C48h	C64h	C48h	FREQ = 2h		
				C48h	C3Ch	C48h	C3Ch	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	236h	448h	236h	448h	FREQ = 0h		
				458h	875h	458h	875h	FREQ = 1h		
D9h	[7:0]			181h	2E3h	181h	2E3h	FREQ = 2h		
				2ECh	5ABh	2ECh	5ABh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	04h	04h	04h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	08h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	1Ch	10h	1Ch	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				28h	14h	28h	14h	FREQ = 2h		
				14h	0Ch	14h	0Ch	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				10h	0Ch	10h	0Ch	FREQ = 2h		
				0Ch	08h	0Ch	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				175.2 [frame/s]	126.0 [frame/s]	67.8 [frame/s]	34.3 [frame/s]	121.0 [frame/s]	121.0 [frame/s]	67.8 [frame/s]	34.3 [frame/s]	FREQ = 0h
				119.1 [frame/s]	65.4 [frame/s]	34.3 [frame/s]	17.2 [frame/s]	119.1 [frame/s]	65.4 [frame/s]	34.3 [frame/s]	17.2 [frame/s]	FREQ = 1h
				61.5 [frame/s]	33.2 [frame/s]	17.2 [frame/s]	8.6 [frame/s]	61.5 [frame/s]	33.2 [frame/s]	17.2 [frame/s]	8.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	CD4h	CA4h	C6Ch	C4Ch	C9Ch	C9Ch	C6Ch	C4Ch	FREQ = 0h
D5h	[7:0]			C9Ch	C68h	C4Ch	C3Ch	C9Ch	C68h	C4Ch	C3Ch	FREQ = 1h
D6h	[7:0]			C64h	C48h	C3Ch	C34h	C64h	C48h	C3Ch	C34h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	B6h	158h	2AEh	BEh	BEh	158h	2AEh	FREQ = 0h
D9h	[7:0]			C1h	165h	2AFh	55Bh	C1h	165h	2AFh	55Bh	FREQ = 1h
				17Ch	2C6h	55Eh	AB5h	17Ch	2C6h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	50h	2Ch	18h	4Ch	4Ch	2Ch	18h	FREQ = 0h
				4Ch	28h	18h	0Ch	4Ch	28h	18h	0Ch	FREQ = 1h
				28h	14h	0Ch	08h	28h	14h	0Ch	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	24h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	14h	0Ch	08h	20h	14h	0Ch	08h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.9.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						146.9 [frame/s]	79.6 [frame/s]	146.9 [frame/s]	79.6 [frame/s]	FREQ = 0h
						77.6 [frame/s]	41.1 [frame/s]	77.6 [frame/s]	41.1 [frame/s]	FREQ = 1h
						207.9 [frame/s]	115.7 [frame/s]	207.9 [frame/s]	115.7 [frame/s]	FREQ = 2h
						113.1 [frame/s]	60.9 [frame/s]	113.1 [frame/s]	60.9 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	668h	648h	668h	648h	FREQ = 0h		
D5h	[7:0]			644h	638h	644h	638h	FREQ = 1h		
D6h	[7:0]			68Ch	658h	68Ch	658h	FREQ = 2h		
				658h	63Ch	658h	63Ch	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	134h	244h	134h	244h	FREQ = 0h		
				254h	46Dh	254h	46Dh	FREQ = 1h		
D9h	[7:0]			D5h	18Bh	D5h	18Bh	FREQ = 2h		
				194h	2FBh	194h	2FBh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				0Ch	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	30h	1Ch	30h	1Ch	FREQ = 0h		
				18h	10h	18h	10h	FREQ = 1h		
				44h	24h	44h	24h	FREQ = 2h		
				24h	14h	24h	14h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	14h	0Ch	14h	0Ch	FREQ = 0h		
				0Ch	08h	0Ch	08h	FREQ = 1h		
				20h	10h	20h	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				330.7 [frame/s]	330.7 [frame/s]	239.7 [frame/s]	127.4 [frame/s]	232.0 [frame/s]	232.0 [frame/s]	232.0 [frame/s]	127.4 [frame/s]	FREQ = 0h
				330.7 [frame/s]	233.9 [frame/s]	126.3 [frame/s]	64.9 [frame/s]	232.0 [frame/s]	232.0 [frame/s]	126.3 [frame/s]	64.9 [frame/s]	FREQ = 1h
				212.9 [frame/s]	121.9 [frame/s]	64.4 [frame/s]	32.7 [frame/s]	212.9 [frame/s]	121.9 [frame/s]	64.4 [frame/s]	32.7 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	6CCh	6CCh	69Ch	660h	694h	694h	694h	660h	FREQ = 0h
D5h	[7:0]			6CCh	698h	660h	640h	694h	694h	660h	640h	FREQ = 1h
D6h	[7:0]			68Ch	660h	640h	634h	68Ch	660h	640h	634h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	81h	B7h	165h	BEh	BEh	BEh	165h	FREQ = 0h
D9h	[7:0]			81h	BCh	168h	2CAh	BEh	BEh	168h	2CAh	FREQ = 1h
				D0h	175h	2D0h	594h	D0h	175h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	50h	28h	4Ch	4Ch	4Ch	28h	FREQ = 0h
				70h	4Ch	28h	14h	4Ch	4Ch	28h	14h	FREQ = 1h
				44h	28h	14h	0Ch	44h	28h	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	24h	14h	20h	20h	20h	14h	FREQ = 0h
				30h	24h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 1h
				20h	14h	0Ch	08h	20h	14h	0Ch	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.9.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	236h	448h	236h	448h	FREQ = 0h
D9h	[7:0]			458h	875h	458h	875h	FREQ = 1h
				181h	2E3h	181h	2E3h	FREQ = 2h
				2ECh	5ABh	2ECh	5ABh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	04h	04h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				28h	14h	28h	14h	FREQ = 2h
				14h	0Ch	14h	0Ch	FREQ = 3h
E6h	[7:0]	GSDLY	30h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				10h	0Ch	10h	0Ch	FREQ = 2h
				0Ch	08h	0Ch	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	B6h	158h	2AEh	BEh	BEh	158h	2AEh	FREQ = 0h
D9h	[7:0]			C1h	165h	2AFh	55Bh	C1h	165h	2AFh	55Bh	FREQ = 1h
				17Ch	2C6h	55Eh	AB5h	17Ch	2C6h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				0Ch	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	50h	2Ch	18h	4Ch	4Ch	2Ch	18h	FREQ = 0h
				4Ch	28h	18h	0Ch	4Ch	28h	18h	0Ch	FREQ = 1h
				28h	14h	0Ch	08h	28h	14h	0Ch	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	24h	14h	0Ch	20h	20h	14h	0Ch	FREQ = 0h
				20h	14h	0Ch	08h	20h	14h	0Ch	08h	FREQ = 1h
				10h	0Ch	08h	04h	10h	0Ch	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.10. IMX536-AAMJ-C

6.10.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						59.9 [frame/s]	31.6 [frame/s]	59.9 [frame/s]	31.6 [frame/s]	FREQ = 0h
						31.1 [frame/s]	16.2 [frame/s]	31.1 [frame/s]	16.2 [frame/s]	FREQ = 1h
						87.0 [frame/s]	46.5 [frame/s]	87.0 [frame/s]	46.5 [frame/s]	FREQ = 2h
						45.9 [frame/s]	24.1 [frame/s]	45.9 [frame/s]	24.1 [frame/s]	FREQ = 3h
				594 Mbps/ch				FREQ = 0h		
				297 Mbps/ch				FREQ = 1h		
				891 Mbps/ch				FREQ = 2h		
445.5 Mbps/ch				FREQ = 3h						
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	BC4h	BA8h	BC4h	BA8h	FREQ = 0h		
D5h	[7:0]			BA8h	B9Ch	BA8h	B9Ch	FREQ = 1h		
D6h	[7:0]			BDCh	BB4h	BDCh	BB4h	FREQ = 2h		
				BB4h	BA4h	BB4h	BA4h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	19Bh	312h	19Bh	312h	FREQ = 0h		
D9h	[7:0]			31Eh	601h	31Eh	601h	FREQ = 1h		
				119h	214h	119h	214h	FREQ = 2h		
				21Bh	408h	21Bh	408h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	08h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	FREQ = 0h		
				14h	0Ch	14h	0Ch	FREQ = 1h		
				34h	1Ch	34h	1Ch	FREQ = 2h		
				1Ch	10h	1Ch	10h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	10h	08h	10h	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				18h	0Ch	18h	0Ch	FREQ = 2h		
				0Ch	08h	0Ch	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				184.0 [frame/s]	182.5 [frame/s]	99.4 [frame/s]	51.3 [frame/s]	127.2 [frame/s]	127.2 [frame/s]	99.4 [frame/s]	51.3 [frame/s]	FREQ = 0h
				170.2 [frame/s]	96.5 [frame/s]	50.9 [frame/s]	25.9 [frame/s]	127.2 [frame/s]	96.5 [frame/s]	50.9 [frame/s]	25.9 [frame/s]	FREQ = 1h
				89.2 [frame/s]	49.1 [frame/s]	25.6 [frame/s]	13.0 [frame/s]	89.2 [frame/s]	49.1 [frame/s]	25.6 [frame/s]	13.0 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	C38h	C38h	BE8h	BBCh	C00h	C00h	BE8h	BBCh	FREQ = 0h
D5h	[7:0]			C2Ch	BE0h	BBCh	BA4h	C00h	BE0h	BBCh	BA4h	FREQ = 1h
D6h	[7:0]			BDCh	BBCh	BA4h	B98h	BDCh	BBCh	BA4h	B98h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	82h	F5h	1E1h	BEh	BEh	F5h	1E1h	FREQ = 0h
D9h	[7:0]			8Ch	FDh	1E5h	3C1h	BEh	FDh	1E5h	3C1h	FREQ = 1h
				112h	1F7h	3CAh	782h	112h	1F7h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				10h	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	3Ch	20h	4Ch	4Ch	3Ch	20h	FREQ = 0h
				68h	38h	20h	10h	4Ch	38h	20h	10h	FREQ = 1h
				34h	20h	10h	08h	34h	20h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	18h	10h	08h	20h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.10.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						198.1 [frame/s]	110.6 [frame/s]	198.1 [frame/s]	110.6 [frame/s]	FREQ = 0h
						108.2 [frame/s]	58.5 [frame/s]	108.2 [frame/s]	58.5 [frame/s]	FREQ = 1h
						277.7 [frame/s]	159.7 [frame/s]	277.7 [frame/s]	159.7 [frame/s]	FREQ = 2h
						156.0 [frame/s]	85.8 [frame/s]	156.0 [frame/s]	85.8 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	648h	618h	648h	618h	FREQ = 0h		
D5h	[7:0]			610h	600h	610h	600h	FREQ = 1h		
D6h	[7:0]			668h	628h	668h	628h	FREQ = 2h		
				628h	610h	628h	610h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	E9h	1AEh	E9h	1AEh	FREQ = 0h		
D9h	[7:0]			1BAh	339h	1BAh	339h	FREQ = 1h		
				A3h	127h	A3h	127h	FREQ = 2h		
				12Eh	22Dh	12Eh	22Dh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	40h	28h	40h	28h	FREQ = 0h		
				20h	18h	20h	18h	FREQ = 1h		
				58h	30h	58h	30h	FREQ = 2h		
				30h	20h	30h	20h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	20h	10h	20h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				28h	18h	28h	18h	FREQ = 2h		
				18h	10h	18h	10h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				618.1 [frame/s]	575.8 [frame/s]	349.6 [frame/s]	183.8 [frame/s]	450.2 [frame/s]	450.2 [frame/s]	349.6 [frame/s]	183.8 [frame/s]	FREQ = 0h
				516.1 [frame/s]	328.9 [frame/s]	185.2 [frame/s]	93.8 [frame/s]	450.2 [frame/s]	328.9 [frame/s]	185.2 [frame/s]	93.8 [frame/s]	FREQ = 1h
				288.8 [frame/s]	173.6 [frame/s]	94.5 [frame/s]	47.6 [frame/s]	288.8 [frame/s]	173.6 [frame/s]	94.5 [frame/s]	47.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	738h	718h	688h	630h	6C8h	6C8h	688h	630h	FREQ = 0h
D5h	[7:0]			6F0h	688h	630h	610h	6C8h	688h	630h	610h	FREQ = 1h
D6h	[7:0]			670h	630h	610h	5F8h	670h	630h	610h	5F8h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	47h	7Fh	FFh	5Fh	5Fh	7Fh	FFh	FREQ = 0h
D9h	[7:0]			51h	87h	FDh	1FEh	5Fh	87h	FDh	1FEh	FREQ = 1h
				9Ch	10Eh	1FAh	3FBh	9Ch	10Eh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	20h	10h	08h	18h	18h	10h	08h	FREQ = 0h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	C8h	70h	38h	98h	98h	70h	38h	FREQ = 0h
				B0h	70h	38h	20h	98h	70h	38h	20h	FREQ = 1h
				60h	38h	20h	10h	60h	38h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	58h	30h	18h	40h	40h	30h	18h	FREQ = 0h
				50h	30h	18h	10h	40h	30h	18h	10h	FREQ = 1h
				28h	18h	10h	08h	28h	18h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.10.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						198.1 [frame/s]	110.6 [frame/s]	198.1 [frame/s]	110.6 [frame/s]	FREQ = 0h
						108.2 [frame/s]	58.5 [frame/s]	108.2 [frame/s]	58.5 [frame/s]	FREQ = 1h
						277.7 [frame/s]	159.7 [frame/s]	277.7 [frame/s]	159.7 [frame/s]	FREQ = 2h
						156.0 [frame/s]	85.8 [frame/s]	156.0 [frame/s]	85.8 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	648h	618h	648h	618h	FREQ = 0h		
D5h	[7:0]			610h	600h	610h	600h	FREQ = 1h		
D6h	[7:0]			668h	628h	668h	628h	FREQ = 2h		
				628h	610h	628h	610h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	E9h	1AEh	E9h	1AEh	FREQ = 0h		
				1BAh	339h	1BAh	339h	FREQ = 1h		
D9h	[7:0]			A3h	127h	A3h	127h	FREQ = 2h		
				12Eh	22Dh	12Eh	22Dh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	40h	28h	40h	28h	FREQ = 0h		
				20h	18h	20h	18h	FREQ = 1h		
				58h	30h	58h	30h	FREQ = 2h		
				30h	20h	30h	20h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	20h	10h	20h	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				28h	18h	28h	18h	FREQ = 2h		
				18h	10h	18h	10h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				618.1 [frame/s]	575.8 [frame/s]	349.6 [frame/s]	183.8 [frame/s]	450.2 [frame/s]	450.2 [frame/s]	349.6 [frame/s]	183.8 [frame/s]	FREQ = 0h
				516.1 [frame/s]	328.9 [frame/s]	185.2 [frame/s]	93.8 [frame/s]	450.2 [frame/s]	328.9 [frame/s]	185.2 [frame/s]	93.8 [frame/s]	FREQ = 1h
				288.8 [frame/s]	173.6 [frame/s]	94.5 [frame/s]	47.6 [frame/s]	288.8 [frame/s]	173.6 [frame/s]	94.5 [frame/s]	47.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	738h	718h	688h	630h	6C8h	6C8h	688h	630h	FREQ = 0h
D5h	[7:0]			6F0h	688h	630h	610h	6C8h	688h	630h	610h	FREQ = 1h
D6h	[7:0]			670h	630h	610h	5F8h	670h	630h	610h	5F8h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	47h	7Fh	FFh	5Fh	5Fh	7Fh	FFh	FREQ = 0h
D9h	[7:0]			51h	87h	FDh	1FEh	5Fh	87h	FDh	1FEh	FREQ = 1h
				9Ch	10Eh	1FAh	3FBh	9Ch	10Eh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	20h	10h	08h	18h	18h	10h	08h	FREQ = 0h
				18h	10h	08h	08h	18h	10h	08h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	C8h	70h	38h	98h	98h	70h	38h	FREQ = 0h
				B0h	70h	38h	20h	98h	70h	38h	20h	FREQ = 1h
				60h	38h	20h	10h	60h	38h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	58h	30h	18h	40h	40h	30h	18h	FREQ = 0h
				50h	30h	18h	10h	40h	30h	18h	10h	FREQ = 1h
				28h	18h	10h	08h	28h	18h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.10.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	19Bh	312h	19Bh	312h	FREQ = 0h
				31Eh	601h	31Eh	601h	FREQ = 1h
D9h	[7:0]			119h	214h	119h	214h	FREQ = 2h
				21Bh	408h	21Bh	408h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	FREQ = 0h
				14h	0Ch	14h	0Ch	FREQ = 1h
				34h	1Ch	34h	1Ch	FREQ = 2h
				1Ch	10h	1Ch	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	10h	08h	10h	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				18h	0Ch	18h	0Ch	FREQ = 2h
				0Ch	08h	0Ch	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	82h	F5h	1E1h	BEh	BEh	F5h	1E1h	FREQ = 0h
D9h	[7:0]			8Ch	FDh	1E5h	3C1h	BEh	FDh	1E5h	3C1h	FREQ = 1h
				112h	1F7h	3CAh	782h	112h	1F7h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				10h	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	3Ch	20h	4Ch	4Ch	3Ch	20h	FREQ = 0h
				68h	38h	20h	10h	4Ch	38h	20h	10h	FREQ = 1h
				34h	20h	10h	08h	34h	20h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	18h	10h	08h	20h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

6.11. IMX536-AAQJ-C

6.11.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						60.0 [frame/s]	31.6 [frame/s]	60.0 [frame/s]	31.6 [frame/s]	FREQ = 0h
						31.2 [frame/s]	16.2 [frame/s]	31.2 [frame/s]	16.2 [frame/s]	FREQ = 1h
						87.1 [frame/s]	46.6 [frame/s]	87.1 [frame/s]	46.6 [frame/s]	FREQ = 2h
						46.0 [frame/s]	24.1 [frame/s]	46.0 [frame/s]	24.1 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	BC0h	BA4h	BC0h	BA4h	FREQ = 0h		
D5h	[7:0]			BA4h	B98h	BA4h	B98h	FREQ = 1h		
D6h	[7:0]			BD8h	BB0h	BD8h	BB0h	FREQ = 2h		
				BB0h	BA0h	BB0h	BA0h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	19Bh	312h	19Bh	312h	FREQ = 0h		
D9h	[7:0]			31Eh	601h	31Eh	601h	FREQ = 1h		
				119h	214h	119h	214h	FREQ = 2h		
				21Bh	408h	21Bh	408h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	04h	08h	04h	FREQ = 2h		
				04h	04h	04h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	FREQ = 0h		
				14h	0Ch	14h	0Ch	FREQ = 1h		
				34h	1Ch	34h	1Ch	FREQ = 2h		
				1Ch	10h	1Ch	10h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	10h	08h	10h	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				18h	0Ch	18h	0Ch	FREQ = 2h		
				0Ch	08h	0Ch	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				184.2 [frame/s]	182.8 [frame/s]	99.5 [frame/s]	51.4 [frame/s]	127.3 [frame/s]	127.3 [frame/s]	99.5 [frame/s]	51.4 [frame/s]	FREQ = 0h
				170.4 [frame/s]	96.6 [frame/s]	51.0 [frame/s]	25.9 [frame/s]	127.3 [frame/s]	96.6 [frame/s]	51.0 [frame/s]	25.9 [frame/s]	FREQ = 1h
				89.3 [frame/s]	49.2 [frame/s]	25.7 [frame/s]	13.0 [frame/s]	89.3 [frame/s]	49.2 [frame/s]	25.7 [frame/s]	13.0 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	C34h	C34h	BE4h	BB8h	BFCh	BFCh	BE4h	BB8h	FREQ = 0h
D5h	[7:0]			C28h	BDCh	BB8h	BA0h	BFCh	BDCh	BB8h	BA0h	FREQ = 1h
D6h	[7:0]			BD8h	BB8h	BA0h	B94h	BD8h	BB8h	BA0h	B94h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	82h	F5h	1E1h	BEh	BEh	F5h	1E1h	FREQ = 0h
D9h	[7:0]			8Ch	FDh	1E5h	3C1h	BEh	FDh	1E5h	3C1h	FREQ = 1h
				112h	1F7h	3CAh	782h	112h	1F7h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				10h	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	3Ch	20h	4Ch	4Ch	3Ch	20h	FREQ = 0h
				68h	38h	20h	10h	4Ch	38h	20h	10h	FREQ = 1h
				34h	20h	10h	08h	34h	20h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	18h	10h	08h	20h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.11.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						200.6 [frame/s]	111.8 [frame/s]	200.6 [frame/s]	111.8 [frame/s]	FREQ = 0h
						109.0 [frame/s]	59.3 [frame/s]	109.0 [frame/s]	59.3 [frame/s]	FREQ = 1h
						280.4 [frame/s]	160.9 [frame/s]	243.6 [frame/s]	160.9 [frame/s]	FREQ = 2h
						157.2 [frame/s]	87.2 [frame/s]	157.2 [frame/s]	87.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	634h	608h	634h	608h	FREQ = 0h		
D5h	[7:0]			604h	5ECh	604h	5ECh	FREQ = 1h		
D6h	[7:0]			658h	61Ch	644h	61Ch	FREQ = 2h		
				61Ch	5F8h	61Ch	5F8h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	E9h	1AEh	E9h	1AEh	FREQ = 0h		
				1BAh	339h	1BAh	339h	FREQ = 1h		
D9h	[7:0]			A3h	127h	BEh	127h	FREQ = 2h		
				12Eh	22Dh	12Eh	22Dh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				0Ch	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	40h	24h	40h	24h	FREQ = 0h		
				20h	14h	20h	14h	FREQ = 1h		
				58h	30h	4Ch	30h	FREQ = 2h		
				30h	1Ch	30h	1Ch	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	1Ch	10h	1Ch	10h	FREQ = 0h		
				10h	08h	10h	08h	FREQ = 1h		
				28h	18h	20h	18h	FREQ = 2h		
				18h	0Ch	18h	0Ch	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				346.7 [frame/s]	346.7 [frame/s]	346.7 [frame/s]	185.2 [frame/s]	243.6 [frame/s]	243.6 [frame/s]	243.6 [frame/s]	185.2 [frame/s]	FREQ = 0h
				346.7 [frame/s]	332.1 [frame/s]	186.6 [frame/s]	95.2 [frame/s]	243.6 [frame/s]	243.6 [frame/s]	186.6 [frame/s]	95.2 [frame/s]	FREQ = 1h
				291.6 [frame/s]	174.9 [frame/s]	96.0 [frame/s]	48.1 [frame/s]	243.6 [frame/s]	174.9 [frame/s]	96.0 [frame/s]	48.1 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	67Ch	67Ch	67Ch	624h	644h	644h	644h	624h	FREQ = 0h
D5h	[7:0]			67Ch	678h	624h	5F8h	644h	644h	624h	5F8h	FREQ = 1h
D6h	[7:0]			660h	624h	5F8h	5E8h	644h	624h	5F8h	5E8h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	81h	81h	FFh	BEh	BEh	BEh	FFh	FREQ = 0h
D9h	[7:0]			81h	87h	FDh	1FEh	BEh	BEh	FDh	1FEh	FREQ = 1h
				9Ch	10Eh	1FAh	3FBh	BEh	10Eh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	10h	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	10h	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				10h	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	70h	38h	4Ch	4Ch	4Ch	38h	FREQ = 0h
				70h	6Ch	38h	1Ch	4Ch	4Ch	38h	1Ch	FREQ = 1h
				5Ch	38h	1Ch	10h	4Ch	38h	1Ch	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	30h	18h	20h	20h	20h	18h	FREQ = 0h
				30h	30h	18h	0Ch	20h	20h	18h	0Ch	FREQ = 1h
				28h	18h	0Ch	08h	20h	18h	0Ch	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.11.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	19Bh	312h	19Bh	312h	FREQ = 0h
				31Eh	601h	31Eh	601h	FREQ = 1h
D9h	[7:0]			119h	214h	119h	214h	FREQ = 2h
				21Bh	408h	21Bh	408h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
				04h	04h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	FREQ = 0h
				14h	0Ch	14h	0Ch	FREQ = 1h
				34h	1Ch	34h	1Ch	FREQ = 2h
				1Ch	10h	1Ch	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	10h	08h	10h	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				18h	0Ch	18h	0Ch	FREQ = 2h
				0Ch	08h	0Ch	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	82h	F5h	1E1h	BEh	BEh	F5h	1E1h	FREQ = 0h
D9h	[7:0]			8Ch	FDh	1E5h	3C1h	BEh	FDh	1E5h	3C1h	FREQ = 1h
				112h	1F7h	3CAh	782h	112h	1F7h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	08h	04h	0Ch	0Ch	08h	04h	FREQ = 0h
				10h	08h	04h	04h	0Ch	08h	04h	04h	FREQ = 1h
				08h	04h	04h	04h	08h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	3Ch	20h	4Ch	4Ch	3Ch	20h	FREQ = 0h
				68h	38h	20h	10h	4Ch	38h	20h	10h	FREQ = 1h
				34h	20h	10h	08h	34h	20h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	1Ch	10h	20h	20h	1Ch	10h	FREQ = 0h
				2Ch	18h	10h	08h	20h	18h	10h	08h	FREQ = 1h
				18h	10h	08h	04h	18h	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.12. IMX537-AAMJ-C

6.12.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						91.4 [frame/s]	48.7 [frame/s]	91.4 [frame/s]	48.7 [frame/s]	FREQ = 0h
						47.9 [frame/s]	25.1 [frame/s]	47.9 [frame/s]	25.1 [frame/s]	FREQ = 1h
						131.7 [frame/s]	71.3 [frame/s]	131.7 [frame/s]	71.3 [frame/s]	FREQ = 2h
						70.2 [frame/s]	37.4 [frame/s]	70.2 [frame/s]	37.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	8BCh	8A0h	8BCh	8A0h	FREQ = 0h		
D5h	[7:0]			8A0h	890h	8A0h	890h	FREQ = 1h		
D6h	[7:0]			8D8h	8B0h	8D8h	8B0h	FREQ = 2h		
				8B0h	894h	8B0h	894h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	16Bh	2B2h	FREQ = 0h		
D9h	[7:0]			2BEh	541h	2BEh	541h	FREQ = 1h		
				F9h	1D4h	F9h	1D4h	FREQ = 2h		
				1DBh	388h	1DBh	388h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	28h	18h	28h	18h	FREQ = 0h		
				18h	0Ch	18h	0Ch	FREQ = 1h		
				3Ch	20h	3Ch	20h	FREQ = 2h		
				20h	10h	20h	10h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	14h	0Ch	14h	0Ch	FREQ = 0h		
				0Ch	08h	0Ch	08h	FREQ = 1h		
				1Ch	10h	1Ch	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				245.5 [frame/s]	245.5 [frame/s]	152.1 [frame/s]	79.5 [frame/s]	170.7 [frame/s]	170.7 [frame/s]	152.1 [frame/s]	79.5 [frame/s]	FREQ = 0h
				245.5 [frame/s]	148.2 [frame/s]	78.7 [frame/s]	40.2 [frame/s]	170.7 [frame/s]	148.2 [frame/s]	78.7 [frame/s]	40.2 [frame/s]	FREQ = 1h
				135.5 [frame/s]	75.9 [frame/s]	39.8 [frame/s]	20.2 [frame/s]	135.5 [frame/s]	75.9 [frame/s]	39.8 [frame/s]	20.2 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	928h	928h	8E8h	8B4h	8F0h	8F0h	8E8h	8B4h	FREQ = 0h
D5h	[7:0]			928h	8E4h	8B4h	898h	8F0h	8E4h	8B4h	898h	FREQ = 1h
D6h	[7:0]			8D8h	8B4h	898h	88Ch	8D8h	8B4h	898h	88Ch	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	81h	D6h	1A3h	BEh	BEh	D6h	1A3h	FREQ = 0h
D9h	[7:0]			81h	DCh	1A7h	346h	BEh	DCh	1A7h	346h	FREQ = 1h
				F2h	1B7h	34Eh	68Ch	F2h	1B7h	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				08h	08h	04h	04h	08h	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	44h	24h	4Ch	4Ch	44h	24h	FREQ = 0h
				70h	44h	24h	14h	4Ch	44h	24h	14h	FREQ = 1h
				3Ch	24h	14h	0Ch	3Ch	24h	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	20h	10h	20h	20h	20h	10h	FREQ = 0h
				30h	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 1h
				1Ch	10h	08h	04h	1Ch	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.12.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						290.2 [frame/s]	166.4 [frame/s]	290.2 [frame/s]	166.4 [frame/s]	FREQ = 0h
						161.3 [frame/s]	88.4 [frame/s]	161.3 [frame/s]	88.4 [frame/s]	FREQ = 1h
						397.0 [frame/s]	236.8 [frame/s]	397.0 [frame/s]	236.8 [frame/s]	FREQ = 2h
						230.7 [frame/s]	129.8 [frame/s]	230.7 [frame/s]	129.8 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	4C8h	490h	4C8h	490h	FREQ = 0h		
D5h	[7:0]			490h	480h	490h	480h	FREQ = 1h		
D6h	[7:0]			4F8h	4A8h	4F8h	4A8h	FREQ = 2h		
				4A8h	488h	4A8h	488h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	D1h	17Eh	D1h	17Eh	FREQ = 0h		
				18Ah	2D9h	18Ah	2D9h	FREQ = 1h		
D9h	[7:0]			93h	107h	93h	107h	FREQ = 2h		
				10Eh	1EDh	10Eh	1EDh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	48h	28h	48h	28h	FREQ = 0h		
				28h	18h	28h	18h	FREQ = 1h		
				68h	38h	68h	38h	FREQ = 2h		
				38h	20h	38h	20h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	20h	10h	20h	10h	FREQ = 0h		
				10h	10h	10h	10h	FREQ = 1h		
				30h	18h	30h	18h	FREQ = 2h		
				18h	10h	18h	10h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				784.5 [frame/s]	784.5 [frame/s]	509.8 [frame/s]	272.5 [frame/s]	581.5 [frame/s]	581.5 [frame/s]	509.8 [frame/s]	272.5 [frame/s]	FREQ = 0h
				714.2 [frame/s]	465.7 [frame/s]	275.0 [frame/s]	142.8 [frame/s]	581.5 [frame/s]	465.7 [frame/s]	275.0 [frame/s]	142.8 [frame/s]	FREQ = 1h
				416.9 [frame/s]	253.9 [frame/s]	144.1 [frame/s]	72.9 [frame/s]	416.9 [frame/s]	253.9 [frame/s]	144.1 [frame/s]	72.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	5B0h	5B0h	520h	4C0h	540h	540h	520h	4C0h	FREQ = 0h
D5h	[7:0]			590h	510h	4C0h	488h	540h	510h	4C0h	488h	FREQ = 1h
D6h	[7:0]			4F8h	4B8h	488h	470h	4F8h	4B8h	488h	470h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	41h	6Fh	E0h	5Fh	5Fh	6Fh	E0h	FREQ = 0h
D9h	[7:0]			49h	7Bh	DEh	1C0h	5Fh	7Bh	DEh	1C0h	FREQ = 1h
				8Ch	F2h	1BCh	380h	8Ch	F2h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	20h	18h	10h	18h	18h	18h	10h	FREQ = 0h
				20h	10h	10h	08h	18h	10h	10h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	E0h	80h	40h	98h	98h	80h	40h	FREQ = 0h
				C8h	78h	40h	20h	98h	78h	40h	20h	FREQ = 1h
				68h	40h	20h	10h	68h	40h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	60h	38h	20h	40h	40h	38h	20h	FREQ = 0h
				58h	38h	20h	10h	40h	38h	20h	10h	FREQ = 1h
				30h	20h	10h	08h	30h	20h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.12.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						290.2 [frame/s]	166.4 [frame/s]	290.2 [frame/s]	166.4 [frame/s]	FREQ = 0h
						161.3 [frame/s]	88.4 [frame/s]	161.3 [frame/s]	88.4 [frame/s]	FREQ = 1h
						397.0 [frame/s]	236.8 [frame/s]	397.0 [frame/s]	236.8 [frame/s]	FREQ = 2h
						230.7 [frame/s]	129.8 [frame/s]	230.7 [frame/s]	129.8 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	4C8h	490h	4C8h	490h	FREQ = 0h		
D5h	[7:0]			490h	480h	490h	480h	FREQ = 1h		
D6h	[7:0]			4F8h	4A8h	4F8h	4A8h	FREQ = 2h		
				4A8h	488h	4A8h	488h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	D1h	17Eh	D1h	17Eh	FREQ = 0h		
				18Ah	2D9h	18Ah	2D9h	FREQ = 1h		
D9h	[7:0]			93h	107h	93h	107h	FREQ = 2h		
				10Eh	1EDh	10Eh	1EDh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	10h	08h	10h	08h	FREQ = 0h		
				08h	08h	08h	08h	FREQ = 1h		
				10h	08h	10h	08h	FREQ = 2h		
				08h	08h	08h	08h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	48h	28h	48h	28h	FREQ = 0h		
				28h	18h	28h	18h	FREQ = 1h		
				68h	38h	68h	38h	FREQ = 2h		
				38h	20h	38h	20h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	20h	10h	20h	10h	FREQ = 0h		
				10h	10h	10h	10h	FREQ = 1h		
				30h	18h	30h	18h	FREQ = 2h		
				18h	10h	18h	10h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				784.5 [frame/s]	784.5 [frame/s]	509.8 [frame/s]	272.5 [frame/s]	581.5 [frame/s]	581.5 [frame/s]	509.8 [frame/s]	272.5 [frame/s]	FREQ = 0h
				714.2 [frame/s]	465.7 [frame/s]	275.0 [frame/s]	142.8 [frame/s]	581.5 [frame/s]	465.7 [frame/s]	275.0 [frame/s]	142.8 [frame/s]	FREQ = 1h
				416.9 [frame/s]	253.9 [frame/s]	144.1 [frame/s]	72.9 [frame/s]	416.9 [frame/s]	253.9 [frame/s]	144.1 [frame/s]	72.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	5B0h	5B0h	520h	4C0h	540h	540h	520h	4C0h	FREQ = 0h
D5h	[7:0]			590h	510h	4C0h	488h	540h	510h	4C0h	488h	FREQ = 1h
D6h	[7:0]			4F8h	4B8h	488h	470h	4F8h	4B8h	488h	470h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	41h	41h	6Fh	E0h	5Fh	5Fh	6Fh	E0h	FREQ = 0h
D9h	[7:0]			49h	7Bh	DEh	1C0h	5Fh	7Bh	DEh	1C0h	FREQ = 1h
				8Ch	F2h	1BCh	380h	8Ch	F2h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	20h	20h	18h	10h	18h	18h	18h	10h	FREQ = 0h
				20h	10h	10h	08h	18h	10h	10h	08h	FREQ = 1h
				10h	08h	08h	08h	10h	08h	08h	08h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	E0h	E0h	80h	40h	98h	98h	80h	40h	FREQ = 0h
				C8h	78h	40h	20h	98h	78h	40h	20h	FREQ = 1h
				68h	40h	20h	10h	68h	40h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	60h	60h	38h	20h	40h	40h	38h	20h	FREQ = 0h
				58h	38h	20h	10h	40h	38h	20h	10h	FREQ = 1h
				30h	20h	10h	08h	30h	20h	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.12.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	16Bh	2B2h	FREQ = 0h
				2BEh	541h	2BEh	541h	FREQ = 1h
D9h	[7:0]			F9h	1D4h	F9h	1D4h	FREQ = 2h
				1DBh	388h	1DBh	388h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	28h	18h	28h	18h	FREQ = 0h
				18h	0Ch	18h	0Ch	FREQ = 1h
				3Ch	20h	3Ch	20h	FREQ = 2h
				20h	10h	20h	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	0Ch	08h	FREQ = 1h
				1Ch	10h	1Ch	10h	FREQ = 2h
				10h	08h	10h	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	81h	D6h	1A3h	BEh	BEh	D6h	1A3h	FREQ = 0h
D9h	[7:0]			81h	DCh	1A7h	346h	BEh	DCh	1A7h	346h	FREQ = 1h
				F2h	1B7h	34Eh	68Ch	F2h	1B7h	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				08h	08h	04h	04h	08h	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	44h	24h	4Ch	4Ch	44h	24h	FREQ = 0h
				70h	44h	24h	14h	4Ch	44h	24h	14h	FREQ = 1h
				3Ch	24h	14h	0Ch	3Ch	24h	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	20h	10h	20h	20h	20h	10h	FREQ = 0h
				30h	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 1h
				1Ch	10h	08h	04h	1Ch	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 104

6.13. IMX537-AAQJ-C

6.13.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						91.6 [frame/s]	48.8 [frame/s]	91.6 [frame/s]	48.8 [frame/s]	FREQ = 0h
						47.9 [frame/s]	25.2 [frame/s]	47.9 [frame/s]	25.2 [frame/s]	FREQ = 1h
						131.9 [frame/s]	71.4 [frame/s]	131.9 [frame/s]	71.4 [frame/s]	FREQ = 2h
						70.4 [frame/s]	37.4 [frame/s]	70.4 [frame/s]	37.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	8B8h	89Ch	8B8h	89Ch	FREQ = 0h		
D5h	[7:0]			89Ch	88Ch	89Ch	88Ch	FREQ = 1h		
D6h	[7:0]			8D4h	8ACh	8D4h	8ACh	FREQ = 2h		
				8ACh	890h	8ACh	890h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	16Bh	2B2h	FREQ = 0h		
				2BEh	541h	2BEh	541h	FREQ = 1h		
D9h	[7:0]			F9h	1D4h	F9h	1D4h	FREQ = 2h		
				1DBh	388h	1DBh	388h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h		
				04h	04h	04h	04h	FREQ = 1h		
				08h	08h	08h	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	28h	18h	28h	18h	FREQ = 0h		
				18h	0Ch	18h	0Ch	FREQ = 1h		
				3Ch	20h	3Ch	20h	FREQ = 2h		
				20h	10h	20h	10h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	14h	0Ch	14h	0Ch	FREQ = 0h		
				0Ch	08h	0Ch	08h	FREQ = 1h		
				1Ch	10h	1Ch	10h	FREQ = 2h		
				10h	08h	10h	08h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				245.9 [frame/s]	245.9 [frame/s]	152.4 [frame/s]	79.6 [frame/s]	171.0 [frame/s]	171.0 [frame/s]	152.4 [frame/s]	79.6 [frame/s]	FREQ = 0h
				245.9 [frame/s]	148.5 [frame/s]	78.9 [frame/s]	40.3 [frame/s]	171.0 [frame/s]	148.5 [frame/s]	78.9 [frame/s]	40.3 [frame/s]	FREQ = 1h
				135.7 [frame/s]	76.0 [frame/s]	39.9 [frame/s]	20.2 [frame/s]	135.7 [frame/s]	76.0 [frame/s]	39.9 [frame/s]	20.2 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	924h	924h	8E4h	8B0h	8ECh	8ECh	8E4h	8B0h	FREQ = 0h
D5h	[7:0]			924h	8E0h	8B0h	894h	8ECh	8E0h	8B0h	894h	FREQ = 1h
D6h	[7:0]			8D4h	8B0h	894h	888h	8D4h	8B0h	894h	888h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	81h	D6h	1A3h	BEh	BEh	D6h	1A3h	FREQ = 0h
D9h	[7:0]			81h	DCh	1A7h	346h	BEh	DCh	1A7h	346h	FREQ = 1h
				F2h	1B7h	34Eh	68Ch	F2h	1B7h	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				08h	08h	04h	04h	08h	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	44h	24h	4Ch	4Ch	44h	24h	FREQ = 0h
				70h	44h	24h	14h	4Ch	44h	24h	14h	FREQ = 1h
				3Ch	24h	14h	0Ch	3Ch	24h	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	20h	10h	20h	20h	20h	10h	FREQ = 0h
				30h	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 1h
				1Ch	10h	08h	04h	1Ch	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.13.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				8 ch	4 ch	8 ch	4 ch			
						295.0 [frame/s]	168.1 [frame/s]	295.0 [frame/s]	168.1 [frame/s]	FREQ = 0h
						163.5 [frame/s]	90.2 [frame/s]	163.5 [frame/s]	90.2 [frame/s]	FREQ = 1h
						403.4 [frame/s]	239.2 [frame/s]	322.4 [frame/s]	239.2 [frame/s]	FREQ = 2h
						233.0 [frame/s]	131.6 [frame/s]	233.0 [frame/s]	131.6 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000CD 4h	4B4h	484h	4B4h	484h	FREQ = 0h		
D5h	[7:0]			480h	468h	480h	468h	FREQ = 1h		
D6h	[7:0]			4E4h	49Ch	4BCh	49Ch	FREQ = 2h		
				49Ch	478h	49Ch	478h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	D1h	17Eh	D1h	17Eh	FREQ = 0h		
D9h	[7:0]			18Ah	2D9h	18Ah	2D9h	FREQ = 1h		
				93h	107h	BEh	107h	FREQ = 2h		
				10Eh	1EDh	10Eh	1EDh	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	FREQ = 0h		
				08h	04h	08h	04h	FREQ = 1h		
				10h	08h	0Ch	08h	FREQ = 2h		
				08h	04h	08h	04h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	44h	28h	44h	28h	FREQ = 0h		
				24h	14h	24h	14h	FREQ = 1h		
				64h	38h	4Ch	38h	FREQ = 2h		
				38h	20h	38h	20h	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	20h	10h	20h	10h	FREQ = 0h		
				10h	0Ch	10h	0Ch	FREQ = 1h		
				2Ch	18h	20h	18h	FREQ = 2h		
				18h	10h	18h	10h	FREQ = 3h		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
				453.9 [frame/s]	453.9 [frame/s]	453.9 [frame/s]	277.1 [frame/s]	322.4 [frame/s]	322.4 [frame/s]	322.4 [frame/s]	277.1 [frame/s]	FREQ = 0h
				453.9 [frame/s]	453.9 [frame/s]	279.6 [frame/s]	144.3 [frame/s]	322.4 [frame/s]	322.4 [frame/s]	279.6 [frame/s]	144.3 [frame/s]	FREQ = 1h
				422.2 [frame/s]	258.2 [frame/s]	145.6 [frame/s]	73.9 [frame/s]	322.4 [frame/s]	258.2 [frame/s]	145.6 [frame/s]	73.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
2376 Mbps/lane								FREQ = 1h				
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	4F4h	4F4h	4F4h	4ACh	4BCh	4BCh	4BCh	4ACh	FREQ = 0h
D5h	[7:0]			4F4h	4F4h	4ACh	47Ch	4BCh	4BCh	4ACh	47Ch	FREQ = 1h
D6h	[7:0]			4E8h	4A4h	47Ch	460h	4BCh	4A4h	47Ch	460h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	81h	81h	81h	E0h	BEh	BEh	BEh	E0h	FREQ = 0h
D9h	[7:0]			81h	81h	DEh	1C0h	BEh	BEh	DEh	1C0h	FREQ = 1h
				8Ch	F2h	1BCh	380h	BEh	F2h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	10h	0Ch	0Ch	0Ch	0Ch	0Ch	FREQ = 0h
				10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 1h
				10h	08h	08h	04h	0Ch	08h	08h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	70h	40h	4Ch	4Ch	4Ch	40h	FREQ = 0h
				70h	70h	40h	20h	4Ch	4Ch	40h	20h	FREQ = 1h
				68h	3Ch	20h	10h	4Ch	3Ch	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	30h	1Ch	20h	20h	20h	1Ch	FREQ = 0h
				30h	30h	1Ch	10h	20h	20h	1Ch	10h	FREQ = 1h
				2Ch	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

6.13.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				8 ch	4 ch	8 ch	4 ch	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	16Bh	2B2h	FREQ = 0h
				2BEh	541h	2BEh	541h	FREQ = 1h
D9h	[7:0]			F9h	1D4h	F9h	1D4h	FREQ = 2h
				1DBh	388h	1DBh	388h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h				
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				08h	08h	08h	08h	FREQ = 2h
				08h	04h	08h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	28h	18h	28h	18h	FREQ = 0h
				18h	0Ch	18h	0Ch	FREQ = 1h
				3Ch	20h	3Ch	20h	FREQ = 2h
				20h	10h	20h	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	0Ch	08h	FREQ = 1h
				1Ch	10h	1Ch	10h	FREQ = 2h
				10h	08h	10h	08h	FREQ = 3h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	0h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

SLVS – EC

Address	bit	Register name	Initial Value	Setting value								Remarks
				AD = 10 bit				AD = 12 bit				
				8 lane	4 lane	2 lane	1 lane	8 lane	4 lane	2 lane	1 lane	
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	*								
D5h	[7:0]											
D6h	[7:0]											
D8h	[7:0]	HMAX	0081h	81h	81h	D6h	1A3h	BEh	BEh	D6h	1A3h	FREQ = 0h
D9h	[7:0]			81h	DCh	1A7h	346h	BEh	DCh	1A7h	346h	FREQ = 1h
				F2h	1B7h	34Eh	68Ch	F2h	1B7h	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	10h	10h	10h	0Ch	08h	0Ch	0Ch	0Ch	08h	FREQ = 0h
				10h	0Ch	08h	04h	0Ch	0Ch	08h	04h	FREQ = 1h
				08h	08h	04h	04h	08h	08h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	70h	70h	44h	24h	4Ch	4Ch	44h	24h	FREQ = 0h
				70h	44h	24h	14h	4Ch	44h	24h	14h	FREQ = 1h
				3Ch	24h	14h	0Ch	3Ch	24h	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	30h	30h	20h	10h	20h	20h	20h	10h	FREQ = 0h
				30h	1Ch	10h	08h	20h	1Ch	10h	08h	FREQ = 1h
				1Ch	10h	08h	04h	1Ch	10h	08h	04h	FREQ = 2h
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h				1h				
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 100

6.14. IMX540-AAMJ-C

6.14.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					21.9 [frame/s]	11.2 [frame/s]	5.7 [frame/s]	21.9 [frame/s]	11.2 [frame/s]	5.7 [frame/s]	FREQ = 0h
					11.1 [frame/s]	5.7 [frame/s]	2.8 [frame/s]	11.1 [frame/s]	5.7 [frame/s]	2.8 [frame/s]	FREQ = 1h
					30.2 [frame/s]	15.9 [frame/s]	8.2 [frame/s]	30.2 [frame/s]	15.9 [frame/s]	8.2 [frame/s]	FREQ = 2h
					15.5 [frame/s]	8.1 [frame/s]	4.1 [frame/s]	15.5 [frame/s]	8.1 [frame/s]	4.1 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
				891 Mbps/ch						FREQ = 2h	
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	126Ch	1264h	1260h	126Ch	1264h	1260h	FREQ = 0h	
D5h	[7:0]			1264h	1260h	125Eh	1264h	1260h	125Eh	FREQ = 1h	
D6h	[7:0]			1270h	1268h	1262h	1270h	1268h	1262h	FREQ = 2h	
				1264h	1262h	125Eh	1264h	1262h	125Eh	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h	
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h	
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h	
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				04h	02h	02h	04h	02h	02h	FREQ = 2h	
				02h	02h	02h	02h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				08h	06h	04h	08h	06h	04h	FREQ = 2h	
				04h	04h	02h	04h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				06h	04h	02h	06h	04h	02h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				2 lane		1 lane		2 lane		1 lane		FREQ = 0h
				35.1 [frame/s]		17.8 [frame/s]		35.1 [frame/s]		17.8 [frame/s]		FREQ = 1h
				17.7 [frame/s]		8.9 [frame/s]		12.0 [frame/s]		6.0 [frame/s]		FREQ = 2h
				8.8 [frame/s]		4.4 [frame/s]		8.8 [frame/s]		4.4 [frame/s]		FREQ = 0h
				4752 Mbps/lane								FREQ = 1h
				2376 Mbps/lane								FREQ = 2h
1188 Mbps/lane												
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	1276h	126Ah	1276h	126Ah	FREQ = 0h				
D5h	[7:0]			126Ah	1262h	1264h	1260h	FREQ = 1h				
D6h	[7:0]			1262h	125Eh	1262h	125Eh	FREQ = 2h				
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h				
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h				
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h				
				04h	02h	02h	02h	FREQ = 1h				
				02h	02h	02h	02h	FREQ = 2h				
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h				
				06h	04h	04h	02h	FREQ = 1h				
				04h	02h	04h	02h	FREQ = 2h				
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h				
				08h	04h	06h	04h	FREQ = 1h				
				04h	02h	04h	02h	FREQ = 2h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.14.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							80.0 [frame/s]	42.2 [frame/s]	22.0 [frame/s]	80.0 [frame/s]	42.2 [frame/s]	22.0 [frame/s]	FREQ = 0h
							41.5 [frame/s]	21.6 [frame/s]	11.0 [frame/s]	41.5 [frame/s]	21.6 [frame/s]	11.0 [frame/s]	FREQ = 1h
							100.9 [frame/s]	55.4 [frame/s]	30.1 [frame/s]	100.9 [frame/s]	55.4 [frame/s]	30.1 [frame/s]	FREQ = 2h
							53.6 [frame/s]	29.1 [frame/s]	15.2 [frame/s]	53.6 [frame/s]	29.1 [frame/s]	15.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	95Ch	950h	948h	95Ch	950h	948h	FREQ = 0h			
D5h	[7:0]			950h	948h	944h	950h	948h	944h	FREQ = 1h			
D6h	[7:0]			964h	950h	948h	964h	950h	948h	FREQ = 2h			
				950h	948h	944h	950h	948h	944h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	183h	2E2h	58Ch	FREQ = 0h			
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h			
				132h	232h	40Eh	132h	232h	40Eh	FREQ = 2h			
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	08h	04h	10h	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	10h	0Ch	08h	10h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				14h	0Ch	08h	14h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				133.6 [frame/s]	68.8 [frame/s]	133.6 [frame/s]	68.8 [frame/s]	FREQ = 0h
				68.3 [frame/s]	34.6 [frame/s]	46.0 [frame/s]	23.2 [frame/s]	FREQ = 1h
				34.3 [frame/s]	17.4 [frame/s]	34.3 [frame/s]	17.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	970h	95Ch	970h	95Ch	FREQ = 0h
D5h	[7:0]			958h	94Ch	950h	948h	FREQ = 1h
D6h	[7:0]			94Ch	944h	94Ch	944h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	E6h	1C2h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.14.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							80.0 [frame/s]	42.2 [frame/s]	22.0 [frame/s]	80.0 [frame/s]	42.2 [frame/s]	22.0 [frame/s]	FREQ = 0h
							41.5 [frame/s]	21.6 [frame/s]	11.0 [frame/s]	41.5 [frame/s]	21.6 [frame/s]	11.0 [frame/s]	FREQ = 1h
							100.9 [frame/s]	55.4 [frame/s]	30.1 [frame/s]	100.9 [frame/s]	55.4 [frame/s]	30.1 [frame/s]	FREQ = 2h
							53.6 [frame/s]	29.1 [frame/s]	15.2 [frame/s]	53.6 [frame/s]	29.1 [frame/s]	15.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
				891 Mbps/ch						FREQ = 2h			
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	95Ch	950h	948h	95Ch	950h	948h	FREQ = 0h			
D5h	[7:0]			950h	948h	944h	950h	948h	944h	FREQ = 1h			
D6h	[7:0]			964h	950h	948h	964h	950h	948h	FREQ = 2h			
				950h	948h	944h	950h	948h	944h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	183h	2E2h	58Ch	FREQ = 0h			
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h			
				132h	232h	40Eh	132h	232h	40Eh	FREQ = 2h			
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	08h	04h	10h	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	10h	0Ch	08h	10h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				14h	0Ch	08h	14h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				133.6 [frame/s]	68.8 [frame/s]	133.6 [frame/s]	68.8 [frame/s]	FREQ = 0h
				68.3 [frame/s]	34.6 [frame/s]	46.0 [frame/s]	23.2 [frame/s]	FREQ = 1h
				34.3 [frame/s]	17.4 [frame/s]	34.3 [frame/s]	17.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	970h	95Ch	970h	95Ch	FREQ = 0h
D5h	[7:0]			958h	94Ch	950h	948h	FREQ = 1h
D6h	[7:0]			94Ch	944h	94Ch	944h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	E6h	1C2h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.14.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h
				584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h
D9h	[7:0]			208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				04h	02h	02h	04h	02h	02h	FREQ = 2h
				02h	02h	02h	02h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	08h	06h	04h	FREQ = 2h
				04h	04h	02h	04h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set “6.20 Common register” according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.15. IMX540-AAQJ-C

6.15.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					21.9 [frame/s]	11.2 [frame/s]	5.7 [frame/s]	21.9 [frame/s]	11.2 [frame/s]	5.7 [frame/s]	FREQ = 0h
					11.1 [frame/s]	5.7 [frame/s]	2.8 [frame/s]	11.1 [frame/s]	5.7 [frame/s]	2.8 [frame/s]	FREQ = 1h
					30.2 [frame/s]	15.9 [frame/s]	8.2 [frame/s]	30.2 [frame/s]	15.9 [frame/s]	8.2 [frame/s]	FREQ = 2h
					15.5 [frame/s]	8.1 [frame/s]	4.1 [frame/s]	15.5 [frame/s]	8.1 [frame/s]	4.1 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
				891 Mbps/ch						FREQ = 2h	
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	126Ch	1264h	1260h	126Ch	1264h	1260h	FREQ = 0h	
D5h	[7:0]			1264h	1260h	125Eh	1264h	1260h	125Eh	FREQ = 1h	
D6h	[7:0]			1270h	1268h	1262h	1270h	1268h	1262h	FREQ = 2h	
				1264h	1262h	125Eh	1264h	1262h	125Eh	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h	
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h	
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h	
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				04h	02h	02h	04h	02h	02h	FREQ = 2h	
				02h	02h	02h	02h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				08h	06h	04h	08h	06h	04h	FREQ = 2h	
				04h	04h	02h	04h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				06h	04h	02h	06h	04h	02h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				35.1 [frame/s]	17.8 [frame/s]	35.1 [frame/s]	17.8 [frame/s]	FREQ = 0h
				17.7 [frame/s]	8.9 [frame/s]	12.0 [frame/s]	6.0 [frame/s]	FREQ = 1h
				8.8 [frame/s]	4.4 [frame/s]	8.8 [frame/s]	4.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	1276h	126Ah	1276h	126Ah	FREQ = 0h
D5h	[7:0]			126Ah	1262h	1264h	1260h	FREQ = 1h
D6h	[7:0]			1262h	125Eh	1262h	125Eh	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.15.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							80.2 [frame/s]	42.3 [frame/s]	22.0 [frame/s]	74.3 [frame/s]	42.3 [frame/s]	22.0 [frame/s]	FREQ = 0h
							41.6 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	41.6 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	FREQ = 1h
							101.2 [frame/s]	55.5 [frame/s]	30.2 [frame/s]	74.3 [frame/s]	55.5 [frame/s]	30.2 [frame/s]	FREQ = 2h
							53.7 [frame/s]	29.2 [frame/s]	15.2 [frame/s]	53.7 [frame/s]	29.2 [frame/s]	15.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
				891 Mbps/ch						FREQ = 2h			
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	956h	948h	940h	954h	948h	940h	FREQ = 0h			
D5h	[7:0]			948h	940h	93Ch	948h	940h	93Ch	FREQ = 1h			
D6h	[7:0]			95Ch	94Ch	940h	954h	94Ch	940h	FREQ = 2h			
				94Ch	940h	93Ch	94Ch	940h	93Ch	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	1A2h	2E2h	58Ch	FREQ = 0h			
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h			
				132h	232h	40Eh	1A2h	232h	40Eh	FREQ = 2h			
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				06h	04h	02h	06h	04h	02h	FREQ = 2h			
				04h	02h	02h	04h	02h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Ch	06h	04h	0Ah	06h	04h	FREQ = 0h			
				06h	04h	02h	06h	04h	02h	FREQ = 1h			
				0Eh	08h	04h	0Ah	08h	04h	FREQ = 2h			
				08h	04h	02h	08h	04h	02h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	10h	0Ah	06h	10h	0Ah	06h	FREQ = 0h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h			
				14h	0Ch	06h	10h	0Ch	06h	FREQ = 2h			
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				102.5 [frame/s]	69.1 [frame/s]	74.3 [frame/s]	69.1 [frame/s]	FREQ = 0h
				68.6 [frame/s]	34.7 [frame/s]	46.1 [frame/s]	23.3 [frame/s]	FREQ = 1h
				34.4 [frame/s]	17.4 [frame/s]	34.4 [frame/s]	17.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	95Eh	952h	954h	952h	FREQ = 0h
D5h	[7:0]			950h	946h	948h	940h	FREQ = 1h
D6h	[7:0]			944h	93Eh	944h	93Eh	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	12Eh	1C2h	1A2h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	06h	06h	06h	FREQ = 0h
				04h	04h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Eh	0Ah	0Ah	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	16h	0Eh	10h	0Eh	FREQ = 0h
				0Eh	08h	0Ah	06h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.15.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h
				584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h
D9h	[7:0]			208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				04h	02h	02h	04h	02h	02h	FREQ = 2h
				02h	02h	02h	02h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	08h	06h	04h	FREQ = 2h
				04h	04h	02h	04h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.16. IMX541-AAMJ-C

6.16.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
				26.1 [frame/s]	13.5 [frame/s]	6.8 [frame/s]	26.1 [frame/s]	13.5 [frame/s]	6.8 [frame/s]	FREQ = 0h
				13.3 [frame/s]	6.8 [frame/s]	3.4 [frame/s]	13.3 [frame/s]	6.8 [frame/s]	3.4 [frame/s]	FREQ = 1h
				35.4 [frame/s]	18.8 [frame/s]	9.8 [frame/s]	35.4 [frame/s]	18.8 [frame/s]	9.8 [frame/s]	FREQ = 2h
				18.2 [frame/s]	9.6 [frame/s]	4.9 [frame/s]	18.2 [frame/s]	9.6 [frame/s]	4.9 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	120Eh	1204h	1200h	120Eh	1204h	1200h	FREQ = 0h
D5h	[7:0]			1204h	1200h	11FEh	1204h	1200h	11FEh	FREQ = 1h
D6h	[7:0]			1216h	120Ah	1202h	1216h	120Ah	1202h	FREQ = 2h
				120Ah	1202h	11FEh	120Ah	1202h	11FEh	FREQ = 3h
D8h	[7:0]	HMAX	00A6h	266h	4A8h	920h	266h	4A8h	920h	FREQ = 0h
D9h	[7:0]			4B8h	935h	122Fh	4B8h	935h	122Fh	FREQ = 1h
				1C4h	356h	666h	1C4h	356h	666h	FREQ = 2h
				370h	689h	CBBh	370h	689h	CBBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				06h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	02h	08h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 2h
				08h	04h	02h	08h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				42.7 [frame/s]	21.5 [frame/s]	38.3 [frame/s]	21.5 [frame/s]	FREQ = 0h
				21.4 [frame/s]	10.7 [frame/s]	14.4 [frame/s]	7.2 [frame/s]	FREQ = 1h
				10.7 [frame/s]	5.3 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	121Ch	120Ch	1218h	120Ch	FREQ = 0h
D5h	[7:0]			120Ch	1204h	1204h	1200h	FREQ = 1h
D6h	[7:0]			1204h	1200h	1204h	1200h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	177h	2EBh	1A2h	2EBh	FREQ = 0h
D9h	[7:0]			2EDh	5D6h	45Ah	8AAh	FREQ = 1h
				5DAh	BABh	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ch	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.16.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							94.9 [frame/s]	50.6 [frame/s]	26.4 [frame/s]	94.9 [frame/s]	50.6 [frame/s]	26.4 [frame/s]	FREQ = 0h
							49.3 [frame/s]	25.9 [frame/s]	13.3 [frame/s]	49.3 [frame/s]	25.9 [frame/s]	13.3 [frame/s]	FREQ = 1h
							119.1 [frame/s]	66.2 [frame/s]	35.9 [frame/s]	119.1 [frame/s]	66.2 [frame/s]	35.9 [frame/s]	FREQ = 2h
							62.8 [frame/s]	34.5 [frame/s]	18.2 [frame/s]	62.8 [frame/s]	34.5 [frame/s]	18.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	934h	920h	918h	934h	920h	918h	FREQ = 0h			
D5h	[7:0]			920h	918h	914h	920h	918h	914h	FREQ = 1h			
D6h	[7:0]			938h	928h	91Ch	938h	928h	91Ch	FREQ = 2h			
				928h	91Ch	914h	928h	91Ch	914h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	14Ch	274h	4B8h	14Ch	274h	4B8h	FREQ = 0h			
D9h	[7:0]			284h	4CDh	95Fh	284h	4CDh	95Fh	FREQ = 1h			
				108h	1DEh	376h	108h	1DEh	376h	FREQ = 2h			
				1F8h	399h	6DBh	1F8h	399h	6DBh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	10h	08h	04h	10h	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	14h	0Ch	08h	14h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				18h	10h	08h	18h	10h	08h	FREQ = 2h			
				10h	08h	04h	10h	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				156.7 [frame/s]	81.5 [frame/s]	149.5 [frame/s]	81.5 [frame/s]	FREQ = 0h
				80.8 [frame/s]	41.0 [frame/s]	55.3 [frame/s]	28.0 [frame/s]	FREQ = 1h
				40.7 [frame/s]	20.5 [frame/s]	40.7 [frame/s]	20.5 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	94Ch	92Ch	948h	92Ch	FREQ = 0h
D5h	[7:0]			92Ch	91Ch	920h	918h	FREQ = 1h
D6h	[7:0]			91Ch	914h	91Ch	914h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	C7h	184h	D1h	184h	FREQ = 0h
D9h	[7:0]			187h	308h	23Eh	471h	FREQ = 1h
				30Eh	60Fh	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	18h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	20h	10h	20h	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.16.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							94.9 [frame/s]	50.6 [frame/s]	26.4 [frame/s]	94.9 [frame/s]	50.6 [frame/s]	26.4 [frame/s]	FREQ = 0h
							49.3 [frame/s]	25.9 [frame/s]	13.3 [frame/s]	49.3 [frame/s]	25.9 [frame/s]	13.3 [frame/s]	FREQ = 1h
							119.1 [frame/s]	66.2 [frame/s]	35.9 [frame/s]	119.1 [frame/s]	66.2 [frame/s]	35.9 [frame/s]	FREQ = 2h
							62.8 [frame/s]	34.5 [frame/s]	18.2 [frame/s]	62.8 [frame/s]	34.5 [frame/s]	18.2 [frame/s]	FREQ = 3h
				594 Mbps/ch									FREQ = 0h
				297 Mbps/ch									FREQ = 1h
				891 Mbps/ch									FREQ = 2h
445.5 Mbps/ch									FREQ = 3h				
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	934h	920h	918h	934h	920h	918h	FREQ = 0h			
D5h	[7:0]			920h	918h	914h	920h	918h	914h	FREQ = 1h			
D6h	[7:0]			938h	928h	91Ch	938h	928h	91Ch	FREQ = 2h			
				928h	91Ch	914h	928h	91Ch	914h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	14Ch	274h	4B8h	14Ch	274h	4B8h	FREQ = 0h			
D9h	[7:0]			284h	4CDh	95Fh	284h	4CDh	95Fh	FREQ = 1h			
				108h	1DEh	376h	108h	1DEh	376h	FREQ = 2h			
				1F8h	399h	6DBh	1F8h	399h	6DBh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	10h	08h	04h	10h	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	14h	0Ch	08h	14h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				18h	10h	08h	18h	10h	08h	FREQ = 2h			
				10h	08h	04h	10h	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				156.7 [frame/s]	81.5 [frame/s]	149.5 [frame/s]	81.5 [frame/s]	FREQ = 0h
				80.8 [frame/s]	41.0 [frame/s]	55.3 [frame/s]	28.0 [frame/s]	FREQ = 1h
				40.7 [frame/s]	20.5 [frame/s]	40.7 [frame/s]	20.5 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	94Ch	92Ch	948h	92Ch	FREQ = 0h
D5h	[7:0]			92Ch	91Ch	920h	918h	FREQ = 1h
D6h	[7:0]			91Ch	914h	91Ch	914h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	C7h	184h	D1h	184h	FREQ = 0h
D9h	[7:0]			187h	308h	23Eh	471h	FREQ = 1h
				30Eh	60Fh	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	18h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	20h	10h	20h	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.16.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	266h	4A8h	920h	266h	4A8h	920h	FREQ = 0h
				4B8h	935h	122Fh	4B8h	935h	122Fh	FREQ = 1h
D9h	[7:0]			1C4h	356h	666h	1C4h	356h	666h	FREQ = 2h
				370h	689h	CBBh	370h	689h	CBBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				06h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	02h	08h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 2h
				08h	04h	02h	08h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	177h	2EBh	1A2h	2EBh	FREQ = 0h
D9h	[7:0]			2EDh	5D6h	45Ah	8AAh	FREQ = 1h
				5DAh	BABh	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ch	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.17. IMX541-AAQJ-C

6.17.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					26.1 [frame/s]	13.5 [frame/s]	6.8 [frame/s]	26.1 [frame/s]	13.5 [frame/s]	6.8 [frame/s]	FREQ = 0h
					13.3 [frame/s]	6.8 [frame/s]	3.4 [frame/s]	13.3 [frame/s]	6.8 [frame/s]	3.4 [frame/s]	FREQ = 1h
					35.4 [frame/s]	18.8 [frame/s]	9.8 [frame/s]	35.4 [frame/s]	18.8 [frame/s]	9.8 [frame/s]	FREQ = 2h
					18.2 [frame/s]	9.6 [frame/s]	4.9 [frame/s]	18.2 [frame/s]	9.6 [frame/s]	4.9 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	120Eh	1204h	1200h	120Eh	1204h	1200h	FREQ = 0h	
D5h	[7:0]			1204h	1200h	11FEh	1204h	1200h	11FEh	FREQ = 1h	
D6h	[7:0]			1216h	120Ah	1202h	1216h	120Ah	1202h	FREQ = 2h	
				120Ah	1202h	11FEh	120Ah	1202h	11FEh	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	266h	4A8h	920h	266h	4A8h	920h	FREQ = 0h	
D9h	[7:0]			4B8h	935h	122Fh	4B8h	935h	122Fh	FREQ = 1h	
				1C4h	356h	666h	1C4h	356h	666h	FREQ = 2h	
				370h	689h	CBBh	370h	689h	CBBh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				06h	04h	02h	06h	04h	02h	FREQ = 2h	
				04h	02h	02h	04h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	08h	04h	02h	08h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 2h	
				06h	04h	02h	06h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 2h	
				08h	04h	02h	08h	04h	02h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				42.7 [frame/s]	21.5 [frame/s]	38.3 [frame/s]	21.5 [frame/s]	FREQ = 0h
				21.4 [frame/s]	10.7 [frame/s]	14.4 [frame/s]	7.2 [frame/s]	FREQ = 1h
				10.7 [frame/s]	5.3 [frame/s]	10.7 [frame/s]	5.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	121Ch	120Ch	1218h	120Ch	FREQ = 0h
D5h	[7:0]			120Ch	1204h	1204h	1200h	FREQ = 1h
D6h	[7:0]			1204h	1200h	1204h	1200h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	177h	2EBh	1A2h	2EBh	FREQ = 0h
D9h	[7:0]			2EDh	5D6h	45Ah	8AAh	FREQ = 1h
				5DAh	BABh	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ch	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.17.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							95.2 [frame/s]	50.7 [frame/s]	26.4 [frame/s]	75.9 [frame/s]	50.7 [frame/s]	26.4 [frame/s]	FREQ = 0h
							49.4 [frame/s]	26.0 [frame/s]	13.3 [frame/s]	49.4 [frame/s]	26.0 [frame/s]	13.3 [frame/s]	FREQ = 1h
							104.6 [frame/s]	66.4 [frame/s]	36.0 [frame/s]	75.9 [frame/s]	66.4 [frame/s]	36.0 [frame/s]	FREQ = 2h
							63.0 [frame/s]	34.6 [frame/s]	18.2 [frame/s]	63.0 [frame/s]	34.6 [frame/s]	18.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	92Ch	91Ah	910h	924h	91Ah	910h	FREQ = 0h			
D5h	[7:0]			91Ah	910h	90Ch	91Ah	910h	90Ch	FREQ = 1h			
D6h	[7:0]			92Eh	920h	916h	924h	920h	916h	FREQ = 2h			
				920h	914h	90Eh	920h	914h	90Eh	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	14Ch	274h	4B8h	1A2h	274h	4B8h	FREQ = 0h			
D9h	[7:0]			284h	4CDh	95Fh	284h	4CDh	95Fh	FREQ = 1h			
				12Eh	1DEh	376h	1A2h	1DEh	376h	FREQ = 2h			
				1F8h	399h	6DBh	1F8h	399h	6DBh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				06h	04h	04h	06h	04h	04h	FREQ = 2h			
				04h	02h	02h	04h	02h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Eh	08h	04h	0Ah	08h	04h	FREQ = 0h			
				08h	04h	02h	08h	04h	02h	FREQ = 1h			
				0Eh	0Ah	06h	0Ah	0Ah	06h	FREQ = 2h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	14h	0Ah	06h	10h	0Ah	06h	FREQ = 0h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h			
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 2h			
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				104.6 [frame/s]	81.7 [frame/s]	75.9 [frame/s]	75.9 [frame/s]	FREQ = 0h
				81.0 [frame/s]	41.1 [frame/s]	55.4 [frame/s]	28.1 [frame/s]	FREQ = 1h
				40.8 [frame/s]	20.6 [frame/s]	40.8 [frame/s]	20.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	92Eh	926h	924h	924h	FREQ = 0h
D5h	[7:0]			926h	916h	91Ch	910h	FREQ = 1h
D6h	[7:0]			916h	90Eh	916h	90Eh	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	12Eh	184h	1A2h	1A2h	FREQ = 0h
D9h	[7:0]			187h	308h	23Eh	471h	FREQ = 1h
				30Eh	60Fh	30Eh	60Fh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	06h	06h	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Eh	0Ch	0Ah	0Ah	FREQ = 0h
				0Ch	06h	08h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	16h	10h	10h	10h	FREQ = 0h
				10h	08h	0Ch	06h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.17.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	266h	4A8h	920h	266h	4A8h	920h	FREQ = 0h
				4B8h	935h	122Fh	4B8h	935h	122Fh	FREQ = 1h
D9h	[7:0]			1C4h	356h	666h	1C4h	356h	666h	FREQ = 2h
				370h	689h	CBBh	370h	689h	CBBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				06h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	08h	04h	02h	08h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 2h
				08h	04h	02h	08h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	177h	2EBh	1A2h	2EBh	FREQ = 0h
D9h	[7:0]			2EDh	5D6h	45Ah	8AAh	FREQ = 1h
				5DAh	BABh	5DAh	BABh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ch	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.18. IMX542-AAMJ-C

6.18.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					32.9 [frame/s]	16.9 [frame/s]	8.6 [frame/s]	32.9 [frame/s]	16.9 [frame/s]	8.6 [frame/s]	FREQ = 0h
					16.7 [frame/s]	8.5 [frame/s]	4.3 [frame/s]	16.7 [frame/s]	8.5 [frame/s]	4.3 [frame/s]	FREQ = 1h
					45.3 [frame/s]	23.8 [frame/s]	12.3 [frame/s]	45.3 [frame/s]	23.8 [frame/s]	12.3 [frame/s]	FREQ = 2h
					23.2 [frame/s]	12.1 [frame/s]	6.2 [frame/s]	23.2 [frame/s]	12.1 [frame/s]	6.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	C4Ch	C44h	C40h	C4Ch	C44h	C40h	FREQ = 0h	
D5h	[7:0]			C44h	C40h	C3Eh	C44h	C40h	C3Eh	FREQ = 1h	
D6h	[7:0]			C50h	C48h	C42h	C50h	C48h	C42h	FREQ = 2h	
				C44h	C42h	C3Eh	C44h	C42h	C3Eh	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h	
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h	
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h	
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				04h	02h	02h	04h	02h	02h	FREQ = 2h	
				02h	02h	02h	02h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				08h	06h	04h	08h	06h	04h	FREQ = 2h	
				04h	04h	02h	04h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				06h	04h	02h	06h	04h	02h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				52.5 [frame/s]	26.8 [frame/s]	52.5 [frame/s]	26.8 [frame/s]	FREQ = 0h
				26.6 [frame/s]	13.4 [frame/s]	18.0 [frame/s]	9.0 [frame/s]	FREQ = 1h
				13.3 [frame/s]	6.7 [frame/s]	13.3 [frame/s]	6.7 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	C56h	C4Ah	C56h	C4Ah	FREQ = 0h
D5h	[7:0]			C4Ah	C42h	C44h	C40h	FREQ = 1h
D6h	[7:0]			C42h	C3Eh	C42h	C3Eh	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.18.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							119.0 [frame/s]	62.8 [frame/s]	32.8 [frame/s]	119.0 [frame/s]	62.8 [frame/s]	32.8 [frame/s]	FREQ = 0h
							61.8 [frame/s]	32.3 [frame/s]	16.5 [frame/s]	61.8 [frame/s]	32.3 [frame/s]	16.5 [frame/s]	FREQ = 1h
							149.7 [frame/s]	82.5 [frame/s]	44.9 [frame/s]	149.7 [frame/s]	82.5 [frame/s]	44.9 [frame/s]	FREQ = 2h
							80.0 [frame/s]	43.4 [frame/s]	22.7 [frame/s]	80.0 [frame/s]	43.4 [frame/s]	22.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
				891 Mbps/ch						FREQ = 2h			
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	64Ch	640h	638h	64Ch	640h	638h	FREQ = 0h			
D5h	[7:0]			640h	638h	634h	640h	638h	634h	FREQ = 1h			
D6h	[7:0]			654h	640h	638h	654h	640h	638h	FREQ = 2h			
				640h	638h	634h	640h	638h	634h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	183h	2E2h	58Ch	FREQ = 0h			
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h			
				132h	232h	40Eh	132h	232h	40Eh	FREQ = 2h			
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	08h	04h	10h	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	10h	0Ch	08h	10h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				14h	0Ch	08h	14h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				197.8 [frame/s]	102.3 [frame/s]	197.8 [frame/s]	102.3 [frame/s]	FREQ = 0h
				101.7 [frame/s]	51.6 [frame/s]	68.5 [frame/s]	34.7 [frame/s]	FREQ = 1h
				51.2 [frame/s]	25.9 [frame/s]	51.2 [frame/s]	25.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	660h	64Ch	660h	64Ch	FREQ = 0h
D5h	[7:0]			648h	63Ch	640h	638h	FREQ = 1h
D6h	[7:0]			63Ch	634h	63Ch	634h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	E6h	1C2h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.18.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							119.0 [frame/s]	62.8 [frame/s]	32.8 [frame/s]	119.0 [frame/s]	62.8 [frame/s]	32.8 [frame/s]	FREQ = 0h
							61.8 [frame/s]	32.3 [frame/s]	16.5 [frame/s]	61.8 [frame/s]	32.3 [frame/s]	16.5 [frame/s]	FREQ = 1h
							149.7 [frame/s]	82.5 [frame/s]	44.9 [frame/s]	149.7 [frame/s]	82.5 [frame/s]	44.9 [frame/s]	FREQ = 2h
							80.0 [frame/s]	43.4 [frame/s]	22.7 [frame/s]	80.0 [frame/s]	43.4 [frame/s]	22.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0012E Ch	64Ch	640h	638h	64Ch	640h	638h	FREQ = 0h			
D5h	[7:0]			640h	638h	634h	640h	638h	634h	FREQ = 1h			
D6h	[7:0]			654h	640h	638h	654h	640h	638h	FREQ = 2h			
				640h	638h	634h	640h	638h	634h	FREQ = 3h			
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	183h	2E2h	58Ch	FREQ = 0h			
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h			
				132h	232h	40Eh	132h	232h	40Eh	FREQ = 2h			
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	0Ch	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				08h	04h	04h	08h	04h	04h	FREQ = 2h			
				04h	04h	04h	04h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	48h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				10h	08h	04h	10h	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E6h	[7:0]	GSDLY	34h	10h	0Ch	08h	10h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				14h	0Ch	08h	14h	0Ch	08h	FREQ = 2h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				197.8 [frame/s]	102.3 [frame/s]	197.8 [frame/s]	102.3 [frame/s]	FREQ = 0h
				101.7 [frame/s]	51.6 [frame/s]	68.5 [frame/s]	34.7 [frame/s]	FREQ = 1h
				51.2 [frame/s]	25.9 [frame/s]	51.2 [frame/s]	25.9 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	660h	64Ch	660h	64Ch	FREQ = 0h
D5h	[7:0]			648h	63Ch	640h	638h	FREQ = 1h
D6h	[7:0]			63Ch	634h	63Ch	634h	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	E6h	1C2h	E6h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	08h	08h	08h	08h	FREQ = 0h
				04h	04h	04h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	14h	0Ch	14h	0Ch	FREQ = 0h
				0Ch	08h	08h	04h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	1Ch	10h	1Ch	10h	FREQ = 0h
				10h	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.18.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				04h	02h	02h	04h	02h	02h	FREQ = 2h
				02h	02h	02h	02h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	08h	06h	04h	FREQ = 2h
				04h	04h	02h	04h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.19. IMX542-AAQJ-C

6.19.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					32.9 [frame/s]	16.9 [frame/s]	8.6 [frame/s]	32.9 [frame/s]	16.9 [frame/s]	8.6 [frame/s]	FREQ = 0h
					16.7 [frame/s]	8.5 [frame/s]	4.3 [frame/s]	16.7 [frame/s]	8.5 [frame/s]	4.3 [frame/s]	FREQ = 1h
					45.3 [frame/s]	23.8 [frame/s]	12.3 [frame/s]	45.3 [frame/s]	23.8 [frame/s]	12.3 [frame/s]	FREQ = 2h
					23.2 [frame/s]	12.1 [frame/s]	6.2 [frame/s]	23.2 [frame/s]	12.1 [frame/s]	6.2 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	C4Ch	C44h	C40h	C4Ch	C44h	C40h	FREQ = 0h	
D5h	[7:0]			C44h	C40h	C3Eh	C44h	C40h	C3Eh	FREQ = 1h	
D6h	[7:0]			C50h	C48h	C42h	C50h	C48h	C42h	FREQ = 2h	
				C44h	C42h	C3Eh	C44h	C42h	C3Eh	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h	
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h	
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h	
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				04h	02h	02h	04h	02h	02h	FREQ = 2h	
				02h	02h	02h	02h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				08h	06h	04h	08h	06h	04h	FREQ = 2h	
				04h	04h	02h	04h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				06h	04h	02h	06h	04h	02h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks				
				AD = 10 bit		AD = 12 bit						
				2 lane		1 lane		2 lane		1 lane		
				52.5 [frame/s]		26.8 [frame/s]		52.5 [frame/s]		26.8 [frame/s]		FREQ = 0h
				26.6 [frame/s]		13.4 [frame/s]		18.0 [frame/s]		9.0 [frame/s]		FREQ = 1h
				13.3 [frame/s]		6.7 [frame/s]		13.3 [frame/s]		6.7 [frame/s]		FREQ = 2h
				4752 Mbps/lane								FREQ = 0h
				2376 Mbps/lane								FREQ = 1h
1188 Mbps/lane								FREQ = 2h				
Chip ID = 02h												
D4h	[7:0]	VMAX	0012E Ch	C56h	C4Ah	C56h	C4Ah	FREQ = 0h				
D5h	[7:0]			C4Ah	C42h	C44h	C40h	FREQ = 1h				
D6h	[7:0]			C42h	C3Eh	C42h	C3Eh	FREQ = 2h				
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h				
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h				
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h				
DCh	[1:0]	FREQ	0h	0h / 1h / 2h								
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h				
				04h	02h	02h	02h	FREQ = 1h				
				02h	02h	02h	02h	FREQ = 2h				
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h				
				06h	04h	04h	02h	FREQ = 1h				
				04h	02h	04h	02h	FREQ = 2h				
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h				
				08h	04h	06h	04h	FREQ = 1h				
				04h	02h	04h	02h	FREQ = 2h				
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h		1h						
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	1h								

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.19.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					119.4 [frame/s]	63.1 [frame/s]	33.0 [frame/s]	110.7 [frame/s]	63.1 [frame/s]	33.0 [frame/s]	FREQ = 0h
					62.1 [frame/s]	32.5 [frame/s]	16.6 [frame/s]	62.1 [frame/s]	32.5 [frame/s]	16.6 [frame/s]	FREQ = 1h
					150.5 [frame/s]	82.7 [frame/s]	45.1 [frame/s]	110.7 [frame/s]	82.7 [frame/s]	45.1 [frame/s]	FREQ = 2h
					80.2 [frame/s]	43.6 [frame/s]	22.8 [frame/s]	80.2 [frame/s]	43.6 [frame/s]	22.8 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
				891 Mbps/ch						FREQ = 2h	
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0012E Ch	646h	638h	630h	644h	638h	630h	FREQ = 0h	
D5h	[7:0]			638h	630h	62Ch	638h	630h	62Ch	FREQ = 1h	
D6h	[7:0]			64Ch	63Ch	630h	644h	63Ch	630h	FREQ = 2h	
				63Ch	630h	62Ch	63Ch	630h	62Ch	FREQ = 3h	
D8h	[7:0]	HMAX	00A6h	183h	2E2h	58Ch	1A2h	2E2h	58Ch	FREQ = 0h	
D9h	[7:0]			2EEh	5A1h	B07h	2EEh	5A1h	B07h	FREQ = 1h	
				132h	232h	40Eh	1A2h	232h	40Eh	FREQ = 2h	
				244h	431h	80Bh	244h	431h	80Bh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	0Ch	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				06h	04h	02h	06h	04h	02h	FREQ = 2h	
				04h	02h	02h	04h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	48h	0Ch	06h	04h	0Ah	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				0Eh	08h	04h	0Ah	08h	04h	FREQ = 2h	
				08h	04h	02h	08h	04h	02h	FREQ = 3h	
E6h	[7:0]	GSDLY	34h	10h	0Ah	06h	10h	0Ah	06h	FREQ = 0h	
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h	
				14h	0Ch	06h	10h	0Ch	06h	FREQ = 2h	
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				152.3 [frame/s]	102.9 [frame/s]	110.7 [frame/s]	102.9 [frame/s]	FREQ = 0h
				102.2 [frame/s]	51.8 [frame/s]	68.8 [frame/s]	34.8 [frame/s]	FREQ = 1h
				51.4 [frame/s]	26.0 [frame/s]	51.4 [frame/s]	26.0 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	64Eh	642h	644h	642h	FREQ = 0h
D5h	[7:0]			640h	636h	638h	630h	FREQ = 1h
D6h	[7:0]			634h	62Eh	634h	62Eh	FREQ = 2h
D8h	[7:0]	HMAX	00A6h	12Eh	1C2h	1A2h	1C2h	FREQ = 0h
D9h	[7:0]			1C6h	384h	2A5h	540h	FREQ = 1h
				38Ch	707h	38Ch	707h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	06h	06h	06h	FREQ = 0h
				04h	04h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Eh	0Ah	0Ah	0Ah	FREQ = 0h
				0Ah	06h	06h	04h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	16h	0Eh	10h	0Eh	FREQ = 0h
				0Eh	08h	0Ah	06h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

6.19.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0012E Ch	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	00A6h	2CCh	574h	AB8h	2CCh	574h	AB8h	FREQ = 0h
D9h	[7:0]			584h	ACDh	155Fh	584h	ACDh	155Fh	FREQ = 1h
				208h	3DEh	776h	208h	3DEh	776h	FREQ = 2h
				3F8h	799h	EDBh	3F8h	799h	EDBh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	0Ch	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				04h	02h	02h	04h	02h	02h	FREQ = 2h
				02h	02h	02h	02h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	48h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	08h	06h	04h	FREQ = 2h
				04h	04h	02h	04h	04h	02h	FREQ = 3h
E6h	[7:0]	GSDLY	34h	0Ah	06h	04h	0Ah	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* In addition to the above, set "6.20 Common register" according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0012E Ch	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	00A6h	1BFh	370h	1BFh	370h	FREQ = 0h
D9h	[7:0]			377h	6E0h	51Eh	A31h	FREQ = 1h
				6EEh	DBFh	6EEh	DBFh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	0Ch	06h	04h	06h	04h	FREQ = 0h
				04h	02h	02h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	48h	0Ah	06h	0Ah	06h	FREQ = 0h
				06h	04h	04h	02h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E6h	[7:0]	GSDLY	34h	0Eh	08h	0Eh	08h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* In addition to the above, set “6.20 Common register” according to the setting value of HMAX.

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 88

6.20. Common registers according to the setting value of HMAX (IMX540 / IMX541 / IMX542)

When HMAX is changed from the "Register list", the following registers must be set according to the setting range.

6.20.1. IMX540-AAMJ-C, IMX541-AAMJ-C, IMX542-AAMJ-C

SLVS IF (All pixel scan mode, ROI mode)

Address	bit	Register name	Initial Value	Setting value						Remarks
				to 209h (521d)	20Ah (522d) to 239h (569d)	23Ah (570d) to 23Fh (575d)	240h (576d) to 2B0h (688d)	2B1h (689d) to 331h (817d)	332h (818d) to 3B9h (953d)	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	00h	73h					
4Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
4Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	
74h	[7:0]		00h	00h	73h					
76h	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
77h	[7:0]		00h	00h	00h	01h	01h	02h	03h	
9Ch	[7:0]		00h	00h	73h					
9Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
9Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	

Address	bit	Register name	Initial Value	Setting value						Remarks
				3BAh (954d) to 4AFh (1199d)	4B0h (1200d) to 60Ah (1546d)	60Bh (1547d) to 731h (1841d)	732h (1842d) to 863h (2147d)	864h (2148d) to 9CCh (2508d)	9CDh (2509d) to	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	73h						
4Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
4Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
74h	[7:0]		00h	73h						
76h	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
77h	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
9Ch	[7:0]		00h	73h						
9Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
9Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	

SLVS IF (Vertical / Horizontal 1/2 Subsampling mode, 2 × 2 FD Binning mode)

Address	bit	Register name	Initial Value	Setting value							Remarks
				to	211h (529d)	27Ah (634d)	2B1h (689d)	302h (770d)	34Bh (843d)	3B5h (949d)	Setting range of HMAX
					210h (528d)	to 279h (633d)	to 2B0h (688d)	to 301h (769d)	to 34Ah (842d)	to 3B4h (948d)	
Chip ID = 08h											
4Ch	[7:0]		00h	00h	73h						
4Eh	[7:0]		00h	00h	40h	9Fh	D4h	36h	6Ah	0Dh	
4Fh	[7:0]		00h	00h	02h	02h	02h	03h	03h	04h	
74h	[7:0]		00h	00h	73h						
76h	[7:0]		00h	00h	40h	9Fh	D4h	36h	6Ah	0Dh	
77h	[7:0]		00h	00h	02h	02h	02h	03h	03h	04h	
9Ch	[7:0]		00h	00h	73h						
9Eh	[7:0]		00h	00h	40h	9Fh	D4h	36h	6Ah	0Dh	
9Fh	[7:0]		00h	00h	02h	02h	02h	03h	03h	04h	

Address	bit	Register name	Initial Value	Setting value							Remarks
				456h (1110d) to 4F6h (1270d)	4F7h (1271d) to 5D3h (1491d)	5D4h (1492d) to 6D3h (1747d)	6D4h (1748d) to 7F4h (2036d)	7F5h (2037d) to 98Dh (2445d)	98Eh (2446d) to BD1h (3025d)	BD2h (3026d) to	Setting range of HMAX
Chip ID = 08h											
4Ch	[7:0]		00h	73h							
4Eh	[7:0]		00h	BFh	81h	51h	7Bh	CEh	4Fh	FFh	
4Fh	[7:0]		00h	04h	05h	06h	07h	08h	0Bh	0Fh	
74h	[7:0]		00h	73h							
76h	[7:0]		00h	BFh	81h	51h	7Bh	CEh	4Fh	FFh	
77h	[7:0]		00h	04h	05h	06h	07h	08h	0Bh	0Fh	
9Ch	[7:0]		00h	73h							
9Eh	[7:0]		00h	BFh	81h	51h	7Bh	CEh	4Fh	FFh	
9Fh	[7:0]		00h	04h	05h	06h	07h	08h	0Bh	0Fh	

SLVS – EC IF (All pixel scan mode, ROI mode)

Address	bit	Register name	Initial Value	Setting value						Remarks
				to 209h (521d)	20Ah (522d) to 239h (569d)	23Ah (570d) to 23Fh (575d)	240h (576d) to 2B0h (688d)	2B1h (689d) to 331h (817d)	332h (818d) to 3B9h (953d)	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	00h	73h					
4Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
4Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	
74h	[7:0]		00h	00h	73h					
76h	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
77h	[7:0]		00h	00h	00h	01h	01h	02h	03h	
9Ch	[7:0]		00h	00h	73h					
9Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
9Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	

Address	bit	Register name	Initial Value	Setting value						Remarks
				3BAh (954d) to 4AFh (1199d)	4B0h (1200d) to 60Ah (1546d)	60Bh (1547d) to 731h (1841d)	732h (1842d) to 863h (2147d)	864h (2148d) to 9CCh (2508d)	9CDh (2509d) to	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	73h						
4Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
4Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
74h	[7:0]		00h	73h						
76h	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
77h	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
9Ch	[7:0]		00h	73h						
9Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
9Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	

SLVS – EC IF (Vertical / Horizontal 1/2 Subsampling mode, 2 × 2 FD Binning mode)

Address	bit	Register name	Initial Value	Setting value						Remarks
					20Ah (522d)	27Dh (637d)	2E8h (744d)	347h (839d)	3CCh (972d)	Setting range of HMAX
				to 209h (521d)	to 27Ch (636d)	to 2E7h (743d)	to 346h (838d)	to 3CBh (971d)	to 587h (1415d)	
Chip ID = 08h										
4Ch	[7:0]		00h	00h	73h					
4Eh	[7:0]		00h	00h	28h	AEh	EDh	5Bh	D6h	
4Fh	[7:0]		00h	00h	02h	02h	02h	03h	04h	
74h	[7:0]		00h	00h	73h					
76h	[7:0]		00h	00h	28h	AEh	EDh	5Bh	D6h	
77h	[7:0]		00h	00h	02h	02h	02h	03h	04h	
9Ch	[7:0]		00h	00h	73h					
9Eh	[7:0]		00h	00h	28h	AEh	EDh	5Bh	D6h	
9Fh	[7:0]		00h	00h	02h	02h	02h	03h	04h	

Address	bit	Register name	Initial Value	Setting value					Remarks
				588h (1416d) to 74Eh (1870d)	74Fh (1871d) to AAFh (2735d)	AB0h (2736d) to F00h (3840d)	F01h (3841d) to 14E1h (5345d)	14E2h (5346d) to	Setting range of HMAX
Chip ID = 08h									
4Ch	[7:0]		00h	73h					
4Eh	[7:0]		00h	4Eh	21h	3Eh	01h	FFh	
4Fh	[7:0]		00h	06h	09h	0Eh	0Fh	0Fh	
74h	[7:0]		00h	73h					
76h	[7:0]		00h	4Eh	21h	3Eh	01h	FFh	
77h	[7:0]		00h	06h	09h	0Eh	0Fh	0Fh	
9Ch	[7:0]		00h	73h					
9Eh	[7:0]		00h	4Eh	21h	3Eh	01h	FFh	
9Fh	[7:0]		00h	06h	09h	0Eh	0Fh	0Fh	

6.20.2. IMX540-AAQJ-C, IMX541-AAQJ-C, IMX542-AAQJ-C

SLVS IF / SLVS – EC IF (Common for all readout driving modes)

Address	bit	Register name	Initial Value	Setting value						Remarks
				to 209h (521d)	20Ah (522d) to 239h (569d)	23Ah (570d) to 23Fh (575d)	240h (576d) to 2B0h (688d)	2B1h (689d) to 331h (817d)	332h (818d) to 3B9h (953d)	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	00h	73h					
4Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
4Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	
74h	[7:0]		00h	00h	73h					
76h	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
77h	[7:0]		00h	00h	00h	01h	01h	02h	03h	
9Ch	[7:0]		00h	00h	73h					
9Eh	[7:0]		00h	00h	CBh	29h	34h	16h	18h	
9Fh	[7:0]		00h	00h	00h	01h	01h	02h	03h	

Address	bit	Register name	Initial Value	Setting value						Remarks
				3BAh (954d) to 4AFh (1199d)	4B0h (1200d) to 60Ah (1546d)	60Bh (1547d) to 731h (1841d)	732h (1842d) to 863h (2147d)	864h (2148d) to 9CCh (2508d)	9CDh (2509d) to	Setting range of HMAX
Chip ID = 08h										
4Ch	[7:0]		00h	73h						
4Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
4Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
74h	[7:0]		00h	73h						
76h	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
77h	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	
9Ch	[7:0]		00h	73h						
9Eh	[7:0]		00h	28h	14h	CAh	18h	7Ch	FFh	
9Fh	[7:0]		00h	04h	06h	08h	0Bh	0Dh	0Fh	

6.21. IMX545-AAMJ-C

6.21.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					41.8 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	41.8 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	FREQ = 0h
					21.4 [frame/s]	11.0 [frame/s]	5.6 [frame/s]	21.4 [frame/s]	11.0 [frame/s]	5.6 [frame/s]	FREQ = 1h
					61.1 [frame/s]	32.1 [frame/s]	16.6 [frame/s]	57.7 [frame/s]	32.1 [frame/s]	16.6 [frame/s]	FREQ = 2h
					31.7 [frame/s]	16.4 [frame/s]	8.3 [frame/s]	31.7 [frame/s]	16.4 [frame/s]	8.3 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	000C D4h	C40h	C2Ch	C24h	C40h	C2Ch	C24h	FREQ = 0h	
D5h	[7:0]			C2Ch	C24h	C1Eh	C2Ch	C24h	C1Eh	FREQ = 1h	
D6h	[7:0]			C52h	C38h	C28h	C50h	C38h	C28h	FREQ = 2h	
				C38h	C28h	C22h	C38h	C28h	C22h	FREQ = 3h	
D8h	[7:0]	HMAX	0081h	236h	448h	860h	236h	448h	860h	FREQ = 0h	
D9h	[7:0]			458h	875h	10AFh	458h	875h	10AFh	FREQ = 1h	
				181h	2E3h	59Dh	198h	2E3h	59Dh	FREQ = 2h	
				2ECh	5ABh	B27h	2ECh	5ABh	B27h	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	10h	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				06h	04h	02h	06h	04h	02h	FREQ = 2h	
				04h	02h	02h	04h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	08h	1Ah	0Eh	08h	FREQ = 0h	
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 1h	
				26h	14h	0Ah	24h	14h	0Ah	FREQ = 2h	
				14h	0Ah	06h	14h	0Ah	06h	FREQ = 3h	
E6h	[7:0]	GSDLY	30h	0Ch	06h	04h	0Ch	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 2h	
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				68.3 [frame/s]	34.5 [frame/s]	57.7 [frame/s]	34.5 [frame/s]	FREQ = 0h
				34.5 [frame/s]	17.3 [frame/s]	34.5 [frame/s]	17.3 [frame/s]	FREQ = 1h
				17.3 [frame/s]	8.7 [frame/s]	17.3 [frame/s]	8.7 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	C58h	C3Ah	C50h	C3Ah	FREQ = 0h
D5h	[7:0]			C3Ah	C2Ah	C3Ah	C2Ah	FREQ = 1h
D6h	[7:0]			C2Ah	C22h	C2Ah	C22h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	158h	2AEh	198h	2AEh	FREQ = 0h
D9h	[7:0]			2AFh	55Bh	2AFh	55Bh	FREQ = 1h
				55Eh	AB5h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	FREQ = 0h
				16h	0Ch	16h	0Ch	FREQ = 1h
				0Ch	06h	0Ch	06h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	0Ah	06h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.21.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
				147.3 [frame/s]	79.8 [frame/s]	42.0 [frame/s]	147.3 [frame/s]	79.8 [frame/s]	42.0 [frame/s]	FREQ = 0h
				77.8 [frame/s]	41.2 [frame/s]	21.3 [frame/s]	77.8 [frame/s]	41.2 [frame/s]	21.3 [frame/s]	FREQ = 1h
				208.4 [frame/s]	116.0 [frame/s]	62.1 [frame/s]	208.4 [frame/s]	116.0 [frame/s]	62.1 [frame/s]	FREQ = 2h
				113.4 [frame/s]	61.1 [frame/s]	31.6 [frame/s]	113.4 [frame/s]	61.1 [frame/s]	31.6 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	664h	644h	634h	664h	644h	634h	FREQ = 0h
D5h	[7:0]			640h	634h	628h	640h	634h	628h	FREQ = 1h
D6h	[7:0]			688h	654h	63Ch	688h	654h	63Ch	FREQ = 2h
				654h	638h	630h	654h	638h	630h	FREQ = 3h
D8h	[7:0]	HMAX	0081h	134h	244h	458h	134h	244h	458h	FREQ = 0h
D9h	[7:0]			254h	46Dh	89Fh	254h	46Dh	89Fh	FREQ = 1h
				D5h	18Bh	2EDh	D5h	18Bh	2EDh	FREQ = 2h
				194h	2FBh	5C7h	194h	2FBh	5C7h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	04h	08h	04h	04h	FREQ = 0h
				04h	04h	04h	04h	04h	04h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				08h	04h	04h	08h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	30h	1Ch	10h	30h	1Ch	10h	FREQ = 0h
				18h	10h	08h	18h	10h	08h	FREQ = 1h
				44h	24h	14h	44h	24h	14h	FREQ = 2h
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 3h
E6h	[7:0]	GSDLY	30h	14h	0Ch	08h	14h	0Ch	08h	FREQ = 0h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h
				20h	10h	0Ch	20h	10h	0Ch	FREQ = 2h
				10h	08h	08h	10h	08h	08h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				240.3 [frame/s]	127.7 [frame/s]	217.1 [frame/s]	127.7 [frame/s]	FREQ = 0h
				126.6 [frame/s]	65.1 [frame/s]	126.6 [frame/s]	65.1 [frame/s]	FREQ = 1h
				64.6 [frame/s]	32.8 [frame/s]	64.6 [frame/s]	32.8 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	698h	65Ch	68Ch	65Ch	FREQ = 0h
D5h	[7:0]			65Ch	63Ch	65Ch	63Ch	FREQ = 1h
D6h	[7:0]			63Ch	630h	63Ch	630h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	B7h	165h	CCh	165h	FREQ = 0h
D9h	[7:0]			168h	2CAh	168h	2CAh	FREQ = 1h
				2D0h	594h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	50h	28h	48h	28h	FREQ = 0h
				28h	14h	28h	14h	FREQ = 1h
				14h	0Ch	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	24h	14h	20h	14h	FREQ = 0h
				14h	0Ch	14h	0Ch	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.21.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							147.3 [frame/s]	79.8 [frame/s]	42.0 [frame/s]	147.3 [frame/s]	79.8 [frame/s]	42.0 [frame/s]	FREQ = 0h
							77.8 [frame/s]	41.2 [frame/s]	21.3 [frame/s]	77.8 [frame/s]	41.2 [frame/s]	21.3 [frame/s]	FREQ = 1h
							208.4 [frame/s]	116.0 [frame/s]	62.1 [frame/s]	208.4 [frame/s]	116.0 [frame/s]	62.1 [frame/s]	FREQ = 2h
							113.4 [frame/s]	61.1 [frame/s]	31.6 [frame/s]	113.4 [frame/s]	61.1 [frame/s]	31.6 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	000C D4h	664h	644h	634h	664h	644h	634h	FREQ = 0h			
D5h	[7:0]			640h	634h	628h	640h	634h	628h	FREQ = 1h			
D6h	[7:0]			688h	654h	63Ch	688h	654h	63Ch	FREQ = 2h			
				654h	638h	630h	654h	638h	630h	FREQ = 3h			
D8h	[7:0]	HMAX	0081h	134h	244h	458h	134h	244h	458h	FREQ = 0h			
D9h	[7:0]			254h	46Dh	89Fh	254h	46Dh	89Fh	FREQ = 1h			
				D5h	18Bh	2EDh	D5h	18Bh	2EDh	FREQ = 2h			
				194h	2FBh	5C7h	194h	2FBh	5C7h	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	10h	08h	04h	04h	08h	04h	04h	FREQ = 0h			
				04h	04h	04h	04h	04h	04h	FREQ = 1h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	70h	30h	1Ch	10h	30h	1Ch	10h	FREQ = 0h			
				18h	10h	08h	18h	10h	08h	FREQ = 1h			
				44h	24h	14h	44h	24h	14h	FREQ = 2h			
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 3h			
E6h	[7:0]	GSDLY	30h	14h	0Ch	08h	14h	0Ch	08h	FREQ = 0h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 1h			
				20h	10h	0Ch	20h	10h	0Ch	FREQ = 2h			
				10h	08h	08h	10h	08h	08h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				240.3 [frame/s]	127.7 [frame/s]	217.1 [frame/s]	127.7 [frame/s]	FREQ = 0h
				126.6 [frame/s]	65.1 [frame/s]	126.6 [frame/s]	65.1 [frame/s]	FREQ = 1h
				64.6 [frame/s]	32.8 [frame/s]	64.6 [frame/s]	32.8 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	698h	65Ch	68Ch	65Ch	FREQ = 0h
D5h	[7:0]			65Ch	63Ch	65Ch	63Ch	FREQ = 1h
D6h	[7:0]			63Ch	630h	63Ch	630h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	B7h	165h	CCh	165h	FREQ = 0h
D9h	[7:0]			168h	2CAh	168h	2CAh	FREQ = 1h
				2D0h	594h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	50h	28h	48h	28h	FREQ = 0h
				28h	14h	28h	14h	FREQ = 1h
				14h	0Ch	14h	0Ch	FREQ = 2h
E6h	[7:0]	GSDLY	30h	24h	14h	20h	14h	FREQ = 0h
				14h	0Ch	14h	0Ch	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.21.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	236h	448h	860h	236h	448h	860h	FREQ = 0h
				458h	875h	10AFh	458h	875h	10AFh	FREQ = 1h
D9h	[7:0]			181h	2E3h	59Dh	198h	2E3h	59Dh	FREQ = 2h
				2ECh	5ABh	B27h	2ECh	5ABh	B27h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				06h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	08h	1Ah	0Eh	08h	FREQ = 0h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 1h
				26h	14h	0Ah	24h	14h	0Ah	FREQ = 2h
				14h	0Ah	06h	14h	0Ah	06h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	0Ch	06h	04h	0Ch	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 2h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	158h	2AEh	198h	2AEh	FREQ = 0h
D9h	[7:0]			2AFh	55Bh	2AFh	55Bh	FREQ = 1h
				55Eh	AB5h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	FREQ = 0h
				16h	0Ch	16h	0Ch	FREQ = 1h
				0Ch	06h	0Ch	06h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	0Ah	06h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.22. IMX545-AAQJ-C

6.22.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					41.8 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	41.8 [frame/s]	21.7 [frame/s]	11.1 [frame/s]	FREQ = 0h
					21.4 [frame/s]	11.0 [frame/s]	5.6 [frame/s]	21.4 [frame/s]	11.0 [frame/s]	5.6 [frame/s]	FREQ = 1h
					61.1 [frame/s]	32.1 [frame/s]	16.6 [frame/s]	57.7 [frame/s]	32.1 [frame/s]	16.6 [frame/s]	FREQ = 2h
					31.7 [frame/s]	16.4 [frame/s]	8.3 [frame/s]	31.7 [frame/s]	16.4 [frame/s]	8.3 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	000C D4h	C40h	C2Ch	C24h	C40h	C2Ch	C24h	FREQ = 0h	
D5h	[7:0]			C2Ch	C24h	C1Eh	C2Ch	C24h	C1Eh	FREQ = 1h	
D6h	[7:0]			C52h	C38h	C28h	C50h	C38h	C28h	FREQ = 2h	
				C38h	C28h	C22h	C38h	C28h	C22h	FREQ = 3h	
D8h	[7:0]	HMAX	0081h	236h	448h	860h	236h	448h	860h	FREQ = 0h	
D9h	[7:0]			458h	875h	10AFh	458h	875h	10AFh	FREQ = 1h	
				181h	2E3h	59Dh	198h	2E3h	59Dh	FREQ = 2h	
				2ECh	5ABh	B27h	2ECh	5ABh	B27h	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	10h	04h	02h	02h	04h	02h	02h	FREQ = 0h	
				02h	02h	02h	02h	02h	02h	FREQ = 1h	
				06h	04h	02h	06h	04h	02h	FREQ = 2h	
				04h	02h	02h	04h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	08h	1Ah	0Eh	08h	FREQ = 0h	
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 1h	
				26h	14h	0Ah	24h	14h	0Ah	FREQ = 2h	
				14h	0Ah	06h	14h	0Ah	06h	FREQ = 3h	
E6h	[7:0]	GSDLY	30h	0Ch	06h	04h	0Ch	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 2h	
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				68.3 [frame/s]	34.5 [frame/s]	57.7 [frame/s]	34.5 [frame/s]	FREQ = 0h
				34.5 [frame/s]	17.3 [frame/s]	34.5 [frame/s]	17.3 [frame/s]	FREQ = 1h
				17.3 [frame/s]	8.7 [frame/s]	17.3 [frame/s]	8.7 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	C58h	C3Ah	C50h	C3Ah	FREQ = 0h
D5h	[7:0]			C3Ah	C2Ah	C3Ah	C2Ah	FREQ = 1h
D6h	[7:0]			C2Ah	C22h	C2Ah	C22h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	158h	2AEh	198h	2AEh	FREQ = 0h
D9h	[7:0]			2AFh	55Bh	2AFh	55Bh	FREQ = 1h
				55Eh	AB5h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	FREQ = 0h
				16h	0Ch	16h	0Ch	FREQ = 1h
				0Ch	06h	0Ch	06h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	0Ah	06h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.22.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
				148.0 [frame/s]	80.2 [frame/s]	42.3 [frame/s]	112.8 [frame/s]	80.2 [frame/s]	42.3 [frame/s]	FREQ = 0h
				78.1 [frame/s]	41.5 [frame/s]	21.4 [frame/s]	78.1 [frame/s]	41.5 [frame/s]	21.4 [frame/s]	FREQ = 1h
				160.0 [frame/s]	116.6 [frame/s]	62.4 [frame/s]	112.8 [frame/s]	112.8 [frame/s]	62.4 [frame/s]	FREQ = 2h
				114.0 [frame/s]	61.3 [frame/s]	31.9 [frame/s]	112.8 [frame/s]	61.3 [frame/s]	31.9 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	65Ch	63Ch	628h	64Ch	63Ch	628h	FREQ = 0h
D5h	[7:0]			63Ah	628h	620h	63Ah	628h	620h	FREQ = 1h
D6h	[7:0]			662h	64Ch	634h	64Ch	64Ch	634h	FREQ = 2h
				64Ch	632h	624h	64Ch	632h	624h	FREQ = 3h
D8h	[7:0]	HMAX	0081h	134h	244h	458h	198h	244h	458h	FREQ = 0h
D9h	[7:0]			254h	46Dh	89Fh	254h	46Dh	89Fh	FREQ = 1h
				11Ch	18Bh	2EDh	198h	198h	2EDh	FREQ = 2h
				194h	2FBh	5C7h	198h	2FBh	5C7h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	08h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	06h	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	2Eh	1Ah	0Eh	24h	1Ah	0Eh	FREQ = 0h
				18h	0Eh	08h	18h	0Eh	08h	FREQ = 1h
				32h	24h	14h	24h	24h	14h	FREQ = 2h
				24h	14h	0Ah	24h	14h	0Ah	FREQ = 3h
E6h	[7:0]	GSDLY	30h	14h	0Ch	06h	10h	0Ch	06h	FREQ = 0h
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 1h
				16h	10h	0Ah	10h	10h	0Ah	FREQ = 2h
				10h	08h	06h	10h	08h	06h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				160.0 [frame/s]	128.5 [frame/s]	112.8 [frame/s]	112.8 [frame/s]	FREQ = 0h
				127.4 [frame/s]	65.4 [frame/s]	112.8 [frame/s]	65.4 [frame/s]	FREQ = 1h
				64.9 [frame/s]	33.0 [frame/s]	64.9 [frame/s]	33.0 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	662h	652h	64Ch	64Ch	FREQ = 0h
D5h	[7:0]			652h	634h	64Ch	634h	FREQ = 1h
D6h	[7:0]			634h	624h	634h	624h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	165h	198h	198h	FREQ = 0h
D9h	[7:0]			168h	2CAh	198h	2CAh	FREQ = 1h
				2D0h	594h	2D0h	594h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	FREQ = 0h
				06h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	28h	24h	24h	FREQ = 0h
				28h	14h	24h	14h	FREQ = 1h
				14h	0Ah	14h	0Ah	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	12h	10h	10h	FREQ = 0h
				12h	0Ah	10h	0Ah	FREQ = 1h
				0Ah	06h	0Ah	06h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	1h				

6.22.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	236h	448h	860h	236h	448h	860h	FREQ = 0h
				458h	875h	10AFh	458h	875h	10AFh	FREQ = 1h
D9h	[7:0]			181h	2E3h	59Dh	198h	2E3h	59Dh	FREQ = 2h
				2ECh	5ABh	B27h	2ECh	5ABh	B27h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	04h	02h	02h	04h	02h	02h	FREQ = 0h
				02h	02h	02h	02h	02h	02h	FREQ = 1h
				06h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	08h	1Ah	0Eh	08h	FREQ = 0h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 1h
				26h	14h	0Ah	24h	14h	0Ah	FREQ = 2h
				14h	0Ah	06h	14h	0Ah	06h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	0Ch	06h	04h	0Ch	06h	04h	FREQ = 0h
				06h	04h	02h	06h	04h	02h	FREQ = 1h
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 2h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	158h	2AEh	198h	2AEh	FREQ = 0h
D9h	[7:0]			2AFh	55Bh	2AFh	55Bh	FREQ = 1h
				55Eh	AB5h	55Eh	AB5h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	FREQ = 0h
				16h	0Ch	16h	0Ch	FREQ = 1h
				0Ch	06h	0Ch	06h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	FREQ = 0h
				0Ah	06h	0Ah	06h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.23. IMX546-AAMJ-C

6.23.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							60.3 [frame/s]	31.8 [frame/s]	16.5 [frame/s]	60.3 [frame/s]	31.8 [frame/s]	16.5 [frame/s]	FREQ = 0h
							31.3 [frame/s]	16.3 [frame/s]	8.3 [frame/s]	31.3 [frame/s]	16.3 [frame/s]	8.3 [frame/s]	FREQ = 1h
							86.7 [frame/s]	46.8 [frame/s]	24.6 [frame/s]	60.8 [frame/s]	46.8 [frame/s]	24.6 [frame/s]	FREQ = 2h
							46.2 [frame/s]	24.3 [frame/s]	12.4 [frame/s]	46.2 [frame/s]	24.3 [frame/s]	12.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	000C D4h	BB0h	B94h	B88h	BB0h	B94h	B88h	FREQ = 0h			
D5h	[7:0]			B94h	B86h	B82h	B94h	B86h	B82h	FREQ = 1h			
D6h	[7:0]			BC6h	BA2h	B8Ch	BB0h	BA2h	B8Ch	FREQ = 2h			
				BA2h	B8Ch	B84h	BA2h	B8Ch	B84h	FREQ = 3h			
D8h	[7:0]	HMAX	0081h	19Bh	312h	5ECh	19Bh	312h	5ECh	FREQ = 0h			
				31Eh	601h	BC7h	31Eh	601h	BC7h	FREQ = 1h			
D9h	[7:0]			11Ch	214h	3FBh	198h	214h	3FBh	FREQ = 2h			
				21Bh	408h	7E2h	21Bh	408h	7E2h	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				08h	04h	02h	06h	04h	02h	FREQ = 2h			
				04h	02h	02h	04h	02h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	70h	24h	12h	0Ah	24h	12h	0Ah	FREQ = 0h			
				12h	0Ah	06h	12h	0Ah	06h	FREQ = 1h			
				32h	1Ch	0Eh	24h	1Ch	0Eh	FREQ = 2h			
				1Ch	0Eh	08h	1Ch	0Eh	08h	FREQ = 3h			
E6h	[7:0]	GSDLY	30h	10h	08h	06h	10h	08h	06h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				16h	0Ch	06h	10h	0Ch	06h	FREQ = 2h			
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				86.7 [frame/s]	51.7 [frame/s]	60.8 [frame/s]	51.7 [frame/s]	FREQ = 0h
				51.3 [frame/s]	26.1 [frame/s]	51.3 [frame/s]	26.1 [frame/s]	FREQ = 1h
				25.8 [frame/s]	13.1 [frame/s]	25.8 [frame/s]	13.1 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	BC6h	BA6h	BB0h	BA6h	FREQ = 0h
D5h	[7:0]			BA6h	B90h	BA6h	B90h	FREQ = 1h
D6h	[7:0]			B90h	B84h	B90h	B84h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	1E1h	198h	1E1h	FREQ = 0h
D9h	[7:0]			1E5h	3C1h	1E5h	3C1h	FREQ = 1h
				3CAh	782h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	1Eh	24h	1Eh	FREQ = 0h
				1Eh	10h	1Eh	10h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	0Eh	10h	0Eh	FREQ = 0h
				0Eh	08h	0Eh	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.23.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							201.1 [frame/s]	112.1 [frame/s]	61.0 [frame/s]	201.1 [frame/s]	112.1 [frame/s]	61.0 [frame/s]	FREQ = 0h
							109.3 [frame/s]	59.5 [frame/s]	31.1 [frame/s]	109.3 [frame/s]	59.5 [frame/s]	31.1 [frame/s]	FREQ = 1h
							281.1 [frame/s]	161.3 [frame/s]	89.5 [frame/s]	228.0 [frame/s]	161.3 [frame/s]	89.5 [frame/s]	FREQ = 2h
							157.6 [frame/s]	87.4 [frame/s]	46.0 [frame/s]	157.6 [frame/s]	87.4 [frame/s]	46.0 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
				891 Mbps/ch						FREQ = 2h			
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	000C D4h	630h	604h	5E8h	630h	604h	5E8h	FREQ = 0h			
D5h	[7:0]			600h	5E8h	5DCh	600h	5E8h	5DCh	FREQ = 1h			
D6h	[7:0]			654h	618h	5F4h	63Ch	618h	5F4h	FREQ = 2h			
				618h	5F4h	5E4h	618h	5F4h	5E4h	FREQ = 3h			
D8h	[7:0]	HMAX	0081h	E9h	1AEh	324h	E9h	1AEh	324h	FREQ = 0h			
D9h	[7:0]			1BAh	339h	637h	1BAh	339h	637h	FREQ = 1h			
				A3h	127h	220h	CCh	127h	220h	FREQ = 2h			
				12Eh	22Dh	42Dh	12Eh	22Dh	42Dh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	10h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h			
				08h	04h	04h	08h	04h	04h	FREQ = 1h			
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h			
				08h	04h	04h	08h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	70h	40h	24h	14h	40h	24h	14h	FREQ = 0h			
				20h	14h	0Ch	20h	14h	0Ch	FREQ = 1h			
				58h	30h	1Ch	48h	30h	1Ch	FREQ = 2h			
				30h	1Ch	10h	30h	1Ch	10h	FREQ = 3h			
E6h	[7:0]	GSDLY	30h	1Ch	10h	08h	1Ch	10h	08h	FREQ = 0h			
				10h	08h	04h	10h	08h	04h	FREQ = 1h			
				28h	18h	0Ch	20h	18h	0Ch	FREQ = 2h			
				18h	0Ch	08h	18h	0Ch	08h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				318.8 [frame/s]	185.6 [frame/s]	228.0 [frame/s]	185.6 [frame/s]	FREQ = 0h
				187.1 [frame/s]	95.5 [frame/s]	187.1 [frame/s]	95.5 [frame/s]	FREQ = 1h
				96.2 [frame/s]	48.3 [frame/s]	96.2 [frame/s]	48.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	668h	620h	63Ch	620h	FREQ = 0h
D5h	[7:0]			620h	5F4h	620h	5F4h	FREQ = 1h
D6h	[7:0]			5F4h	5E4h	5F4h	5E4h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	8Eh	FFh	CCh	FFh	FREQ = 0h
D9h	[7:0]			FDh	1FEh	FDh	1FEh	FREQ = 1h
				1FAh	3FBh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	10h	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	64h	38h	48h	38h	FREQ = 0h
				38h	1Ch	38h	1Ch	FREQ = 1h
				1Ch	10h	1Ch	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	2Ch	18h	20h	18h	FREQ = 0h
				18h	0Ch	18h	0Ch	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.23.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
				201.1 [frame/s]	112.1 [frame/s]	61.0 [frame/s]	201.1 [frame/s]	112.1 [frame/s]	61.0 [frame/s]	FREQ = 0h
				109.3 [frame/s]	59.5 [frame/s]	31.1 [frame/s]	109.3 [frame/s]	59.5 [frame/s]	31.1 [frame/s]	FREQ = 1h
				281.1 [frame/s]	161.3 [frame/s]	89.5 [frame/s]	228.0 [frame/s]	161.3 [frame/s]	89.5 [frame/s]	FREQ = 2h
				157.6 [frame/s]	87.4 [frame/s]	46.0 [frame/s]	157.6 [frame/s]	87.4 [frame/s]	46.0 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
891 Mbps/ch						FREQ = 2h				
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	630h	604h	5E8h	630h	604h	5E8h	FREQ = 0h
D5h	[7:0]			600h	5E8h	5DCh	600h	5E8h	5DCh	FREQ = 1h
D6h	[7:0]			654h	618h	5F4h	63Ch	618h	5F4h	FREQ = 2h
				618h	5F4h	5E4h	618h	5F4h	5E4h	FREQ = 3h
D8h	[7:0]	HMAX	0081h	E9h	1AEh	324h	E9h	1AEh	324h	FREQ = 0h
D9h	[7:0]			1BAh	339h	637h	1BAh	339h	637h	FREQ = 1h
				A3h	127h	220h	CCh	127h	220h	FREQ = 2h
				12Eh	22Dh	42Dh	12Eh	22Dh	42Dh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h
				08h	04h	04h	08h	04h	04h	FREQ = 1h
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h
				08h	04h	04h	08h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	40h	24h	14h	40h	24h	14h	FREQ = 0h
				20h	14h	0Ch	20h	14h	0Ch	FREQ = 1h
				58h	30h	1Ch	48h	30h	1Ch	FREQ = 2h
				30h	1Ch	10h	30h	1Ch	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	1Ch	10h	08h	1Ch	10h	08h	FREQ = 0h
				10h	08h	04h	10h	08h	04h	FREQ = 1h
				28h	18h	0Ch	20h	18h	0Ch	FREQ = 2h
				18h	0Ch	08h	18h	0Ch	08h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				318.8 [frame/s]	185.6 [frame/s]	228.0 [frame/s]	185.6 [frame/s]	FREQ = 0h
				187.1 [frame/s]	95.5 [frame/s]	187.1 [frame/s]	95.5 [frame/s]	FREQ = 1h
				96.2 [frame/s]	48.3 [frame/s]	96.2 [frame/s]	48.3 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	668h	620h	63Ch	620h	FREQ = 0h
D5h	[7:0]			620h	5F4h	620h	5F4h	FREQ = 1h
D6h	[7:0]			5F4h	5E4h	5F4h	5E4h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	8Eh	FFh	CCh	FFh	FREQ = 0h
D9h	[7:0]			FDh	1FEh	FDh	1FEh	FREQ = 1h
				1FAh	3FBh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	10h	08h	0Ch	08h	FREQ = 0h
				08h	04h	08h	04h	FREQ = 1h
				04h	04h	04h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	64h	38h	48h	38h	FREQ = 0h
				38h	1Ch	38h	1Ch	FREQ = 1h
				1Ch	10h	1Ch	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	2Ch	18h	20h	18h	FREQ = 0h
				18h	0Ch	18h	0Ch	FREQ = 1h
				0Ch	08h	0Ch	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.23.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	19Bh	312h	5ECh	19Bh	312h	5ECh	FREQ = 0h
				31Eh	601h	BC7h	31Eh	601h	BC7h	FREQ = 1h
D9h	[7:0]			11Ch	214h	3FBh	198h	214h	3FBh	FREQ = 2h
				21Bh	408h	7E2h	21Bh	408h	7E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	24h	12h	0Ah	24h	12h	0Ah	FREQ = 0h
				12h	0Ah	06h	12h	0Ah	06h	FREQ = 1h
				32h	1Ch	0Eh	24h	1Ch	0Eh	FREQ = 2h
				1Ch	0Eh	08h	1Ch	0Eh	08h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	10h	08h	06h	10h	08h	06h	FREQ = 0h
				08h	04h	04h	08h	04h	04h	FREQ = 1h
				16h	0Ch	06h	10h	0Ch	06h	FREQ = 2h
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	11Ch	1E1h	198h	1E1h	FREQ = 0h
D9h	[7:0]			1E5h	3C1h	1E5h	3C1h	FREQ = 1h
				3CAh	782h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	1Eh	24h	1Eh	FREQ = 0h
				1Eh	10h	1Eh	10h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	0Eh	10h	0Eh	FREQ = 0h
				0Eh	08h	0Eh	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.24. IMX546-AAQJ-C

6.24.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					60.3 [frame/s]	31.8 [frame/s]	16.5 [frame/s]	60.3 [frame/s]	31.8 [frame/s]	16.5 [frame/s]	FREQ = 0h
					31.3 [frame/s]	16.3 [frame/s]	8.3 [frame/s]	31.3 [frame/s]	16.3 [frame/s]	8.3 [frame/s]	FREQ = 1h
					86.7 [frame/s]	46.8 [frame/s]	24.6 [frame/s]	60.8 [frame/s]	46.8 [frame/s]	24.6 [frame/s]	FREQ = 2h
					46.2 [frame/s]	24.3 [frame/s]	12.4 [frame/s]	46.2 [frame/s]	24.3 [frame/s]	12.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	000C D4h	BB0h	B94h	B88h	BB0h	B94h	B88h	FREQ = 0h	
D5h	[7:0]			B94h	B86h	B82h	B94h	B86h	B82h	FREQ = 1h	
D6h	[7:0]			BC6h	BA2h	B8Ch	BB0h	BA2h	B8Ch	FREQ = 2h	
				BA2h	B8Ch	B84h	BA2h	B8Ch	B84h	FREQ = 3h	
D8h	[7:0]	HMAX	0081h	19Bh	312h	5ECh	19Bh	312h	5ECh	FREQ = 0h	
				31Eh	601h	BC7h	31Eh	601h	BC7h	FREQ = 1h	
D9h	[7:0]			11Ch	214h	3FBh	198h	214h	3FBh	FREQ = 2h	
				21Bh	408h	7E2h	21Bh	408h	7E2h	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h	
				04h	02h	02h	04h	02h	02h	FREQ = 1h	
				08h	04h	02h	06h	04h	02h	FREQ = 2h	
				04h	02h	02h	04h	02h	02h	FREQ = 3h	
E3h	[7:0]	GMTWT	70h	24h	12h	0Ah	24h	12h	0Ah	FREQ = 0h	
				12h	0Ah	06h	12h	0Ah	06h	FREQ = 1h	
				32h	1Ch	0Eh	24h	1Ch	0Eh	FREQ = 2h	
				1Ch	0Eh	08h	1Ch	0Eh	08h	FREQ = 3h	
E6h	[7:0]	GSDLY	30h	10h	08h	06h	10h	08h	06h	FREQ = 0h	
				08h	04h	04h	08h	04h	04h	FREQ = 1h	
				16h	0Ch	06h	10h	0Ch	06h	FREQ = 2h	
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks	
				AD = 10 bit		AD = 12 bit			
				2 lane	1 lane	2 lane	1 lane		
				86.7 [frame/s]	51.7 [frame/s]	60.8 [frame/s]	51.7 [frame/s]		
				51.3 [frame/s]	26.1 [frame/s]	51.3 [frame/s]	26.1 [frame/s]		
				25.8 [frame/s]	13.1 [frame/s]	25.8 [frame/s]	13.1 [frame/s]		
				4752 Mbps/lane					FREQ = 0h
				2376 Mbps/lane					FREQ = 1h
				1188 Mbps/lane					FREQ = 2h
Chip ID = 02h									
D4h	[7:0]	VMAX	000C D4h	BC6h	BA6h	BB0h	BA6h	FREQ = 0h	
D5h	[7:0]			BA6h	B90h	BA6h	B90h	FREQ = 1h	
D6h	[7:0]			B90h	B84h	B90h	B84h	FREQ = 2h	
D8h	[7:0]	HMAX	0081h	11Ch	1E1h	198h	1E1h	FREQ = 0h	
D9h	[7:0]			1E5h	3C1h	1E5h	3C1h	FREQ = 1h	
				3CAh	782h	3CAh	782h	FREQ = 2h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h					
E2h	[7:0]	GMRWT	10h	08h	04h	06h	04h	FREQ = 0h	
				04h	02h	04h	02h	FREQ = 1h	
				02h	02h	02h	02h	FREQ = 2h	
E3h	[7:0]	GMTWT	70h	32h	1Eh	24h	1Eh	FREQ = 0h	
				1Eh	10h	1Eh	10h	FREQ = 1h	
				10h	08h	10h	08h	FREQ = 2h	
E6h	[7:0]	GSDLY	30h	16h	0Eh	10h	0Eh	FREQ = 0h	
				0Eh	08h	0Eh	08h	FREQ = 1h	
				08h	04h	08h	04h	FREQ = 2h	
Chip ID = 04h									
00h	[5:4]	ADBIT	0h	0h		1h			
Chip ID = 06h									
30h	[1:0]	ODBIT	0h	2h					
Chip ID = 07h									
ACH	[5]		1h	1h					

6.24.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							168.2 [frame/s]	112.8 [frame/s]	61.4 [frame/s]	118.7 [frame/s]	112.8 [frame/s]	61.4 [frame/s]	FREQ = 0h
							110.0 [frame/s]	59.8 [frame/s]	31.3 [frame/s]	110.0 [frame/s]	59.8 [frame/s]	31.3 [frame/s]	FREQ = 1h
							168.2 [frame/s]	162.1 [frame/s]	90.0 [frame/s]	118.7 [frame/s]	118.7 [frame/s]	90.0 [frame/s]	FREQ = 2h
							158.4 [frame/s]	87.9 [frame/s]	46.4 [frame/s]	118.7 [frame/s]	87.9 [frame/s]	46.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	000C D4h	612h	5FAh	5E0h	5FCh	5FAh	5E0h	FREQ = 0h			
D5h	[7:0]			5F6h	5E0h	5D2h	5F6h	5E0h	5D2h	FREQ = 1h			
D6h	[7:0]			612h	610h	5ECh	5FCh	5FCh	5ECh	FREQ = 2h			
				610h	5ECh	5D8h	5FCh	5ECh	5D8h	FREQ = 3h			
D8h	[7:0]	HMAX	0081h	11Ch	1AEh	324h	198h	1AEh	324h	FREQ = 0h			
D9h	[7:0]			1BAh	339h	637h	1BAh	339h	637h	FREQ = 1h			
				11Ch	127h	220h	198h	198h	220h	FREQ = 2h			
				12Eh	22Dh	42Dh	198h	22Dh	42Dh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	10h	08h	06h	04h	06h	06h	04h	FREQ = 0h			
				06h	04h	02h	06h	04h	02h	FREQ = 1h			
				08h	08h	04h	06h	06h	04h	FREQ = 2h			
				08h	04h	02h	06h	04h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	70h	32h	22h	12h	24h	22h	12h	FREQ = 0h			
				20h	12h	0Ah	20h	12h	0Ah	FREQ = 1h			
				32h	30h	1Ah	24h	24h	1Ah	FREQ = 2h			
				30h	1Ah	0Eh	24h	1Ah	0Eh	FREQ = 3h			
E6h	[7:0]	GSDLY	30h	16h	10h	08h	10h	10h	08h	FREQ = 0h			
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 1h			
				16h	16h	0Ch	10h	10h	0Ch	FREQ = 2h			
				16h	0Ch	06h	10h	0Ch	06h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				168.2 [frame/s]	168.2 [frame/s]	118.7 [frame/s]	118.7 [frame/s]	FREQ = 0h
				168.2 [frame/s]	95.9 [frame/s]	118.7 [frame/s]	95.9 [frame/s]	FREQ = 1h
				96.6 [frame/s]	48.7 [frame/s]	96.6 [frame/s]	48.7 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	612h	612h	5FCh	5FCh	FREQ = 0h
D5h	[7:0]			612h	5EEh	5FCh	5EEh	FREQ = 1h
D6h	[7:0]			5EEh	5D8h	5EEh	5D8h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	11Ch	198h	198h	FREQ = 0h
D9h	[7:0]			11Ch	1FEh	198h	1FEh	FREQ = 1h
				1FAh	3FBh	1FAh	3FBh	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	08h	06h	06h	FREQ = 0h
				08h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	32h	24h	24h	FREQ = 0h
				32h	1Ch	24h	1Ch	FREQ = 1h
				1Ch	0Eh	1Ch	0Eh	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	16h	10h	10h	FREQ = 0h
				16h	0Ch	10h	0Ch	FREQ = 1h
				0Ch	06h	0Ch	06h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.24.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	19Bh	312h	5ECh	19Bh	312h	5ECh	FREQ = 0h
				31Eh	601h	BC7h	31Eh	601h	BC7h	FREQ = 1h
D9h	[7:0]			11Ch	214h	3FBh	198h	214h	3FBh	FREQ = 2h
				21Bh	408h	7E2h	21Bh	408h	7E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	04h	02h	06h	04h	02h	FREQ = 2h
				04h	02h	02h	04h	02h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	24h	12h	0Ah	24h	12h	0Ah	FREQ = 0h
				12h	0Ah	06h	12h	0Ah	06h	FREQ = 1h
				32h	1Ch	0Eh	24h	1Ch	0Eh	FREQ = 2h
				1Ch	0Eh	08h	1Ch	0Eh	08h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	10h	08h	06h	10h	08h	06h	FREQ = 0h
				08h	04h	04h	08h	04h	04h	FREQ = 1h
				16h	0Ch	06h	10h	0Ch	06h	FREQ = 2h
				0Ch	06h	04h	0Ch	06h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	11Ch	1E1h	198h	1E1h	FREQ = 0h
D9h	[7:0]			1E5h	3C1h	1E5h	3C1h	FREQ = 1h
				3CAh	782h	3CAh	782h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	04h	06h	04h	FREQ = 0h
				04h	02h	04h	02h	FREQ = 1h
				02h	02h	02h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	1Eh	24h	1Eh	FREQ = 0h
				1Eh	10h	1Eh	10h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	0Eh	10h	0Eh	FREQ = 0h
				0Eh	08h	0Eh	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.25. IMX547-AAMJ-C

6.25.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							92.3 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	82.4 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	FREQ = 0h
							48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	FREQ = 1h
							117.2 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	82.4 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	FREQ = 2h
							71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	000C D4h	8A6h	88Ah	87Ah	8A0h	88Ah	87Ah	FREQ = 0h			
D5h	[7:0]			88Ah	87Ah	872h	88Ah	87Ah	872h	FREQ = 1h			
D6h	[7:0]			8B6h	89Ah	882h	8A0h	89Ah	882h	FREQ = 2h			
				898h	882h	876h	898h	882h	876h	FREQ = 3h			
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	52Ch	198h	2B2h	52Ch	FREQ = 0h			
D9h	[7:0]			2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h			
				11Ch	1D4h	37Bh	198h	1D4h	37Bh	FREQ = 2h			
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				08h	06h	04h	06h	06h	04h	FREQ = 2h			
				06h	04h	02h	06h	04h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	70h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h			
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h			
				32h	20h	10h	24h	20h	10h	FREQ = 2h			
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h			
E6h	[7:0]	GSDLY	30h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h			
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 2h			
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				117.2 [frame/s]	80.3 [frame/s]	82.4 [frame/s]	80.3 [frame/s]	FREQ = 0h
				79.5 [frame/s]	40.6 [frame/s]	79.5 [frame/s]	40.6 [frame/s]	FREQ = 1h
				40.2 [frame/s]	20.4 [frame/s]	40.2 [frame/s]	20.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	8B6h	89Eh	8A0h	89Eh	FREQ = 0h
D5h	[7:0]			89Eh	884h	89Eh	884h	FREQ = 1h
D6h	[7:0]			884h	876h	884h	876h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	1A3h	198h	1A3h	FREQ = 0h
D9h	[7:0]			1A7h	346h	1A7h	346h	FREQ = 1h
				34Eh	68Ch	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	FREQ = 0h
				06h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	22h	24h	22h	FREQ = 0h
				22h	12h	22h	12h	FREQ = 1h
				12h	0Ah	12h	0Ah	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	10h	10h	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h	1h			
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	1h				

6.25.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	FREQ = 0h
					164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	FREQ = 1h
					404.7 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	302.3 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	FREQ = 2h
					233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	000C D4h	4B0h	480h	464h	4B0h	480h	464h	FREQ = 0h	
D5h	[7:0]			47Ch	464h	458h	47Ch	464h	458h	FREQ = 1h	
D6h	[7:0]			4E0h	498h	474h	4B4h	498h	474h	FREQ = 2h	
				498h	474h	45Ch	498h	474h	45Ch	FREQ = 3h	
D8h	[7:0]	HMAX	0081h	D1h	17Eh	2C4h	D1h	17Eh	2C4h	FREQ = 0h	
D9h	[7:0]			18Ah	2D9h	577h	18Ah	2D9h	577h	FREQ = 1h	
				93h	107h	1E0h	CCh	107h	1E0h	FREQ = 2h	
				10Eh	1EDh	3ADh	10Eh	1EDh	3ADh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	10h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h	
				08h	04h	04h	08h	04h	04h	FREQ = 1h	
				10h	08h	04h	0Ch	08h	04h	FREQ = 2h	
				08h	04h	04h	08h	04h	04h	FREQ = 3h	
E3h	[7:0]	GMTWT	70h	44h	28h	14h	44h	28h	14h	FREQ = 0h	
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h	
				64h	38h	20h	48h	38h	20h	FREQ = 2h	
				38h	20h	10h	38h	20h	10h	FREQ = 3h	
E6h	[7:0]	GSDLY	30h	20h	10h	0Ch	20h	10h	0Ch	FREQ = 0h	
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 1h	
				2Ch	18h	10h	20h	18h	10h	FREQ = 2h	
				18h	10h	08h	18h	10h	08h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				418.9 [frame/s]	278.0 [frame/s]	302.3 [frame/s]	278.0 [frame/s]	FREQ = 0h
				280.5 [frame/s]	144.8 [frame/s]	280.5 [frame/s]	144.8 [frame/s]	FREQ = 1h
				146.1 [frame/s]	74.2 [frame/s]	146.1 [frame/s]	74.2 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	4E0h	4A8h	4B4h	4A8h	FREQ = 0h
D5h	[7:0]			4A8h	478h	4A8h	478h	FREQ = 1h
D6h	[7:0]			478h	45Ch	478h	45Ch	FREQ = 2h
D8h	[7:0]	HMAX	0081h	8Eh	E0h	CCh	E0h	FREQ = 0h
D9h	[7:0]			DEh	1C0h	DEh	1C0h	FREQ = 1h
				1BCh	380h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	10h	0Ch	0Ch	0Ch	FREQ = 0h
				0Ch	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	64h	40h	48h	40h	FREQ = 0h
				40h	20h	40h	20h	FREQ = 1h
				20h	10h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	2Ch	1Ch	20h	1Ch	FREQ = 0h
				1Ch	10h	1Ch	10h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.25.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
				296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	FREQ = 0h
				164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	FREQ = 1h
				404.7 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	302.3 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	FREQ = 2h
				233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h
				297 Mbps/ch						FREQ = 1h
				891 Mbps/ch						FREQ = 2h
445.5 Mbps/ch						FREQ = 3h				
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	4B0h	480h	464h	4B0h	480h	464h	FREQ = 0h
D5h	[7:0]			47Ch	464h	458h	47Ch	464h	458h	FREQ = 1h
D6h	[7:0]			4E0h	498h	474h	4B4h	498h	474h	FREQ = 2h
				498h	474h	45Ch	498h	474h	45Ch	FREQ = 3h
D8h	[7:0]	HMAX	0081h	D1h	17Eh	2C4h	D1h	17Eh	2C4h	FREQ = 0h
D9h	[7:0]			18Ah	2D9h	577h	18Ah	2D9h	577h	FREQ = 1h
				93h	107h	1E0h	CCh	107h	1E0h	FREQ = 2h
				10Eh	1EDh	3ADh	10Eh	1EDh	3ADh	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h
				08h	04h	04h	08h	04h	04h	FREQ = 1h
				10h	08h	04h	0Ch	08h	04h	FREQ = 2h
				08h	04h	04h	08h	04h	04h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	44h	28h	14h	44h	28h	14h	FREQ = 0h
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h
				64h	38h	20h	48h	38h	20h	FREQ = 2h
				38h	20h	10h	38h	20h	10h	FREQ = 3h
E6h	[7:0]	GSDLY	30h	20h	10h	0Ch	20h	10h	0Ch	FREQ = 0h
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 1h
				2Ch	18h	10h	20h	18h	10h	FREQ = 2h
				18h	10h	08h	18h	10h	08h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				418.9 [frame/s]	278.0 [frame/s]	302.3 [frame/s]	278.0 [frame/s]	FREQ = 0h
				280.5 [frame/s]	144.8 [frame/s]	280.5 [frame/s]	144.8 [frame/s]	FREQ = 1h
				146.1 [frame/s]	74.2 [frame/s]	146.1 [frame/s]	74.2 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	4E0h	4A8h	4B4h	4A8h	FREQ = 0h
D5h	[7:0]			4A8h	478h	4A8h	478h	FREQ = 1h
D6h	[7:0]			478h	45Ch	478h	45Ch	FREQ = 2h
D8h	[7:0]	HMAX	0081h	8Eh	E0h	CCh	E0h	FREQ = 0h
D9h	[7:0]			DEh	1C0h	DEh	1C0h	FREQ = 1h
				1BCh	380h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	10h	0Ch	0Ch	0Ch	FREQ = 0h
				0Ch	08h	0Ch	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	64h	40h	48h	40h	FREQ = 0h
				40h	20h	40h	20h	FREQ = 1h
				20h	10h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	2Ch	1Ch	20h	1Ch	FREQ = 0h
				1Ch	10h	1Ch	10h	FREQ = 1h
				10h	08h	10h	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.25.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	52Ch	198h	2B2h	52Ch	FREQ = 0h
				2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h
D9h	[7:0]			11Ch	1D4h	37Bh	198h	1D4h	37Bh	FREQ = 2h
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	06h	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h
				32h	20h	10h	24h	20h	10h	FREQ = 2h
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h
E6h	[7:0]	GSDLY	30h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 2h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	11Ch	1A3h	198h	1A3h	FREQ = 0h
D9h	[7:0]			1A7h	346h	1A7h	346h	FREQ = 1h
				34Eh	68Ch	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	FREQ = 0h
				06h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	22h	24h	22h	FREQ = 0h
				22h	12h	22h	12h	FREQ = 1h
				12h	0Ah	12h	0Ah	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	10h	10h	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.26. IMX547-AAQJ-C

6.26.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks		
				AD = 10 bit			AD = 12 bit					
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch			
				92.3 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	82.4 [frame/s]	49.2 [frame/s]	25.8 [frame/s]			
												FREQ = 0h
												FREQ = 1h
												FREQ = 2h
												FREQ = 3h
				594 Mbps/ch						FREQ = 0h		
				297 Mbps/ch						FREQ = 1h		
891 Mbps/ch						FREQ = 2h						
445.5 Mbps/ch						FREQ = 3h						
Chip ID = 02h												
D4h	[7:0]	VMAX	000C D4h	8A6h	88Ah	87Ah	8A0h	88Ah	87Ah	FREQ = 0h		
D5h	[7:0]			88Ah	87Ah	872h	88Ah	87Ah	872h	FREQ = 1h		
D6h	[7:0]			8B6h	89Ah	882h	8A0h	89Ah	882h	FREQ = 2h		
				898h	882h	876h	898h	882h	876h	FREQ = 3h		
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	52Ch	198h	2B2h	52Ch	FREQ = 0h		
D9h	[7:0]			2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h		
				11Ch	1D4h	37Bh	198h	1D4h	37Bh	FREQ = 2h		
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h		
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h								
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h		
				04h	02h	02h	04h	02h	02h	FREQ = 1h		
				08h	06h	04h	06h	06h	04h	FREQ = 2h		
				06h	04h	02h	06h	04h	02h	FREQ = 3h		
E3h	[7:0]	GMTWT	70h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h		
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h		
				32h	20h	10h	24h	20h	10h	FREQ = 2h		
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h		
E6h	[7:0]	GSDLY	30h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h		
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h		
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 2h		
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h		
Chip ID = 04h												
00h	[5:4]	ADBIT	0h	0h			1h					
Chip ID = 06h												
30h	[1:0]	ODBIT	0h	2h								
Chip ID = 07h												
ACh	[5]		1h	0h								

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				117.2 [frame/s]	80.3 [frame/s]	82.4 [frame/s]	80.3 [frame/s]	FREQ = 0h
				79.5 [frame/s]	40.6 [frame/s]	79.5 [frame/s]	40.6 [frame/s]	FREQ = 1h
				40.2 [frame/s]	20.4 [frame/s]	40.2 [frame/s]	20.4 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
				1188 Mbps/lane				FREQ = 2h
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	8B6h	89Eh	8A0h	89Eh	FREQ = 0h
D5h	[7:0]			89Eh	884h	89Eh	884h	FREQ = 1h
D6h	[7:0]			884h	876h	884h	876h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	1A3h	198h	1A3h	FREQ = 0h
D9h	[7:0]			1A7h	346h	1A7h	346h	FREQ = 1h
				34Eh	68Ch	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	FREQ = 0h
				06h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	22h	24h	22h	FREQ = 0h
				22h	12h	22h	12h	FREQ = 1h
				12h	0Ah	12h	0Ah	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	10h	10h	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	1h				

6.26.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					224.9 [frame/s]	170.2 [frame/s]	93.9 [frame/s]	159.6 [frame/s]	159.6 [frame/s]	93.9 [frame/s]	FREQ = 0h
					165.3 [frame/s]	91.2 [frame/s]	48.1 [frame/s]	159.6 [frame/s]	91.2 [frame/s]	48.1 [frame/s]	FREQ = 1h
					224.9 [frame/s]	224.9 [frame/s]	136.8 [frame/s]	159.6 [frame/s]	159.6 [frame/s]	136.8 [frame/s]	FREQ = 2h
					224.9 [frame/s]	133.2 [frame/s]	71.0 [frame/s]	159.6 [frame/s]	133.2 [frame/s]	71.0 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
891 Mbps/ch						FREQ = 2h					
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	000C D4h	48Ah	476h	45Ch	474h	474h	45Ch	FREQ = 0h	
D5h	[7:0]			474h	45Ch	44Eh	474h	45Ch	44Eh	FREQ = 1h	
D6h	[7:0]			48Ah	48Ah	46Ah	474h	474h	46Ah	FREQ = 2h	
				48Ah	46Ah	456h	474h	46Ah	456h	FREQ = 3h	
D8h	[7:0]	HMAX	0081h	11Ch	17Eh	2C4h	198h	198h	2C4h	FREQ = 0h	
D9h	[7:0]			18Ah	2D9h	577h	198h	2D9h	577h	FREQ = 1h	
				11Ch	11Ch	1E0h	198h	198h	1E0h	FREQ = 2h	
				11Ch	1EDh	3ADh	198h	1EDh	3ADh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	10h	08h	06h	04h	06h	06h	04h	FREQ = 0h	
				06h	04h	02h	06h	04h	02h	FREQ = 1h	
				08h	08h	04h	06h	06h	04h	FREQ = 2h	
				08h	04h	04h	06h	04h	04h	FREQ = 3h	
E3h	[7:0]	GMTWT	70h	32h	26h	14h	24h	24h	14h	FREQ = 0h	
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h	
				32h	32h	1Eh	24h	24h	1Eh	FREQ = 2h	
				32h	1Eh	10h	24h	1Eh	10h	FREQ = 3h	
E6h	[7:0]	GSDLY	30h	16h	10h	0Ah	10h	10h	0Ah	FREQ = 0h	
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 1h	
				16h	16h	0Eh	10h	10h	0Eh	FREQ = 2h	
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

SLVS - EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
				224.9 [frame/s]	224.9 [frame/s]	159.6 [frame/s]	159.6 [frame/s]	FREQ = 0h
				224.9 [frame/s]	146.1 [frame/s]	159.6 [frame/s]	146.1 [frame/s]	FREQ = 1h
				147.4 [frame/s]	74.6 [frame/s]	147.4 [frame/s]	74.6 [frame/s]	FREQ = 2h
				4752 Mbps/lane				FREQ = 0h
				2376 Mbps/lane				FREQ = 1h
1188 Mbps/lane				FREQ = 2h				
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	48Ah	48Ah	474h	474h	FREQ = 0h
D5h	[7:0]			48Ah	46Eh	474h	46Eh	FREQ = 1h
D6h	[7:0]			46Eh	456h	46Eh	456h	FREQ = 2h
D8h	[7:0]	HMAX	0081h	11Ch	11Ch	198h	198h	FREQ = 0h
D9h	[7:0]			11Ch	1C0h	198h	1C0h	FREQ = 1h
				1BCh	380h	1BCh	380h	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	08h	06h	06h	FREQ = 0h
				08h	06h	06h	06h	FREQ = 1h
				06h	04h	06h	04h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	32h	24h	24h	FREQ = 0h
				32h	20h	24h	20h	FREQ = 1h
				20h	10h	20h	10h	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	16h	10h	10h	FREQ = 0h
				16h	0Eh	10h	0Eh	FREQ = 1h
				0Eh	08h	0Eh	08h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACH	[5]		1h	1h				

6.26.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0081h	16Bh	2B2h	52Ch	198h	2B2h	52Ch	FREQ = 0h
				2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h
D9h	[7:0]			11Ch	1D4h	37Bh	198h	1D4h	37Bh	FREQ = 2h
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	10h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				08h	06h	04h	06h	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	70h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h
				32h	20h	10h	24h	20h	10h	FREQ = 2h
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h
E6h	[7:0]	GSDLY	30h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h
				16h	0Eh	08h	10h	0Eh	08h	FREQ = 2h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

SLVS – EC

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				2 lane	1 lane	2 lane	1 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	11Ch	1A3h	198h	1A3h	FREQ = 0h
D9h	[7:0]			1A7h	346h	1A7h	346h	FREQ = 1h
				34Eh	68Ch	34Eh	68Ch	FREQ = 2h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h				
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	FREQ = 0h
				06h	04h	06h	04h	FREQ = 1h
				04h	02h	04h	02h	FREQ = 2h
E3h	[7:0]	GMTWT	70h	32h	22h	24h	22h	FREQ = 0h
				22h	12h	22h	12h	FREQ = 1h
				12h	0Ah	12h	0Ah	FREQ = 2h
E6h	[7:0]	GSDLY	30h	16h	10h	10h	10h	FREQ = 0h
				10h	08h	10h	08h	FREQ = 1h
				08h	04h	08h	04h	FREQ = 2h
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.27. IMX548-AAMJ-C

6.27.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							92.3 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	84.0 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	FREQ = 0h
							48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	FREQ = 1h
							93.4 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	84.0 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	FREQ = 2h
							71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0008A Ah	8A6h	88Ah	87Ah	8A0h	88Ah	87Ah	FREQ = 0h			
D5h	[7:0]			88Ah	87Ah	872h	88Ah	87Ah	872h	FREQ = 1h			
D6h	[7:0]			8A6h	89Ah	882h	8A0h	89Ah	882h	FREQ = 2h			
				898h	882h	876h	898h	882h	876h	FREQ = 3h			
D8h	[7:0]	HMAX	0167h	16Bh	2B2h	52Ch	190h	2B2h	52Ch	FREQ = 0h			
D9h	[7:0]			2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h			
				167h	1D4h	37Bh	190h	1D4h	37Bh	FREQ = 2h			
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	06h	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				06h	06h	04h	06h	06h	04h	FREQ = 2h			
				06h	04h	02h	06h	04h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	28h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h			
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h			
				28h	20h	10h	24h	20h	10h	FREQ = 2h			
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h			
E6h	[7:0]	GSDLY	12h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h			
				12h	0Eh	08h	10h	0Eh	08h	FREQ = 2h			
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

6.27.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	FREQ = 0h
					164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	FREQ = 1h
					339.2 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	308.3 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	FREQ = 2h
					233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
				891 Mbps/ch						FREQ = 2h	
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0008A Ah	4B0h	480h	464h	4B0h	480h	464h	FREQ = 0h	
D5h	[7:0]			47Ch	464h	458h	47Ch	464h	458h	FREQ = 1h	
D6h	[7:0]			4C0h	498h	474h	4B4h	498h	474h	FREQ = 2h	
				498h	474h	45Ch	498h	474h	45Ch	FREQ = 3h	
D8h	[7:0]	HMAX	0167h	D1h	17Eh	2C4h	D1h	17Eh	2C4h	FREQ = 0h	
D9h	[7:0]			18Ah	2D9h	577h	18Ah	2D9h	577h	FREQ = 1h	
				B4h	107h	1E0h	C8h	107h	1E0h	FREQ = 2h	
				10Eh	1EDh	3ADh	10Eh	1EDh	3ADh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	06h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h	
				08h	04h	04h	08h	04h	04h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				08h	04h	04h	08h	04h	04h	FREQ = 3h	
E3h	[7:0]	GMTWT	28h	44h	28h	14h	44h	28h	14h	FREQ = 0h	
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h	
				50h	38h	20h	48h	38h	20h	FREQ = 2h	
				38h	20h	10h	38h	20h	10h	FREQ = 3h	
E6h	[7:0]	GSDLY	12h	20h	10h	0Ch	20h	10h	0Ch	FREQ = 0h	
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 1h	
				24h	18h	10h	20h	18h	10h	FREQ = 2h	
				18h	10h	08h	18h	10h	08h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

6.27.3. 2 × 2 FD Binning mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks	
				AD = 10 bit			AD = 12 bit				
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch		
					296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	296.0 [frame/s]	168.7 [frame/s]	93.3 [frame/s]	FREQ = 0h
					164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	164.1 [frame/s]	90.6 [frame/s]	47.7 [frame/s]	FREQ = 1h
					339.2 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	308.3 [frame/s]	240.0 [frame/s]	135.6 [frame/s]	FREQ = 2h
					233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	233.8 [frame/s]	132.1 [frame/s]	70.7 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h	
				297 Mbps/ch						FREQ = 1h	
				891 Mbps/ch						FREQ = 2h	
445.5 Mbps/ch						FREQ = 3h					
Chip ID = 02h											
D4h	[7:0]	VMAX	0008A Ah	4B0h	480h	464h	4B0h	480h	464h	FREQ = 0h	
D5h	[7:0]			47Ch	464h	458h	47Ch	464h	458h	FREQ = 1h	
D6h	[7:0]			4C0h	498h	474h	4B4h	498h	474h	FREQ = 2h	
				498h	474h	45Ch	498h	474h	45Ch	FREQ = 3h	
D8h	[7:0]	HMAX	0167h	D1h	17Eh	2C4h	D1h	17Eh	2C4h	FREQ = 0h	
D9h	[7:0]			18Ah	2D9h	577h	18Ah	2D9h	577h	FREQ = 1h	
				B4h	107h	1E0h	C8h	107h	1E0h	FREQ = 2h	
				10Eh	1EDh	3ADh	10Eh	1EDh	3ADh	FREQ = 3h	
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h							
E2h	[7:0]	GMRWT	06h	0Ch	08h	04h	0Ch	08h	04h	FREQ = 0h	
				08h	04h	04h	08h	04h	04h	FREQ = 1h	
				0Ch	08h	04h	0Ch	08h	04h	FREQ = 2h	
				08h	04h	04h	08h	04h	04h	FREQ = 3h	
E3h	[7:0]	GMTWT	28h	44h	28h	14h	44h	28h	14h	FREQ = 0h	
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h	
				50h	38h	20h	48h	38h	20h	FREQ = 2h	
				38h	20h	10h	38h	20h	10h	FREQ = 3h	
E6h	[7:0]	GSDLY	12h	20h	10h	0Ch	20h	10h	0Ch	FREQ = 0h	
				10h	0Ch	08h	10h	0Ch	08h	FREQ = 1h	
				24h	18h	10h	20h	18h	10h	FREQ = 2h	
				18h	10h	08h	18h	10h	08h	FREQ = 3h	
Chip ID = 04h											
00h	[5:4]	ADBIT	0h	0h			1h				
Chip ID = 06h											
30h	[1:0]	ODBIT	0h	2h							
Chip ID = 07h											
ACh	[5]		1h	0h							

6.27.4. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0008A Ah	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0167h	16Bh	2B2h	52Ch	190h	2B2h	52Ch	FREQ = 0h
				2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h
D9h	[7:0]			167h	1D4h	37Bh	190h	1D4h	37Bh	FREQ = 2h
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	06h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				06h	06h	04h	06h	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	28h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h
				28h	20h	10h	24h	20h	10h	FREQ = 2h
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h
E6h	[7:0]	GSDLY	12h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h
				12h	0Eh	08h	10h	0Eh	08h	FREQ = 2h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.28. IMX548-AAQJ-C

6.28.1. All pixel scan mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							92.3 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	84.0 [frame/s]	49.2 [frame/s]	25.8 [frame/s]	FREQ = 0h
							48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	48.3 [frame/s]	25.4 [frame/s]	13.0 [frame/s]	FREQ = 1h
							93.4 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	84.0 [frame/s]	72.0 [frame/s]	38.2 [frame/s]	FREQ = 2h
							71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	71.0 [frame/s]	37.7 [frame/s]	19.4 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
891 Mbps/ch						FREQ = 2h							
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0008A Ah	8A6h	88Ah	87Ah	8A0h	88Ah	87Ah	FREQ = 0h			
D5h	[7:0]			88Ah	87Ah	872h	88Ah	87Ah	872h	FREQ = 1h			
D6h	[7:0]			8A6h	89Ah	882h	8A0h	89Ah	882h	FREQ = 2h			
				898h	882h	876h	898h	882h	876h	FREQ = 3h			
D8h	[7:0]	HMAX	0167h	16Bh	2B2h	52Ch	190h	2B2h	52Ch	FREQ = 0h			
D9h	[7:0]			2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h			
				167h	1D4h	37Bh	190h	1D4h	37Bh	FREQ = 2h			
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	06h	06h	04h	02h	06h	04h	02h	FREQ = 0h			
				04h	02h	02h	04h	02h	02h	FREQ = 1h			
				06h	06h	04h	06h	06h	04h	FREQ = 2h			
				06h	04h	02h	06h	04h	02h	FREQ = 3h			
E3h	[7:0]	GMTWT	28h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h			
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h			
				28h	20h	10h	24h	20h	10h	FREQ = 2h			
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h			
E6h	[7:0]	GSDLY	12h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h			
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h			
				12h	0Eh	08h	10h	0Eh	08h	FREQ = 2h			
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

6.28.2. Vertical / Horizontal 1/2 Subsampling mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks			
				AD = 10 bit			AD = 12 bit						
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch				
							180.4 [frame/s]	170.2 [frame/s]	93.9 [frame/s]	162.8 [frame/s]	162.8 [frame/s]	93.9 [frame/s]	FREQ = 0h
							165.3 [frame/s]	91.2 [frame/s]	48.1 [frame/s]	162.8 [frame/s]	91.2 [frame/s]	48.1 [frame/s]	FREQ = 1h
							180.4 [frame/s]	180.4 [frame/s]	136.8 [frame/s]	162.8 [frame/s]	162.8 [frame/s]	136.8 [frame/s]	FREQ = 2h
							180.4 [frame/s]	133.2 [frame/s]	71.0 [frame/s]	162.8 [frame/s]	133.2 [frame/s]	71.0 [frame/s]	FREQ = 3h
				594 Mbps/ch						FREQ = 0h			
				297 Mbps/ch						FREQ = 1h			
				891 Mbps/ch						FREQ = 2h			
445.5 Mbps/ch						FREQ = 3h							
Chip ID = 02h													
D4h	[7:0]	VMAX	0008A Ah	47Ah	476h	45Ch	474h	474h	45Ch	FREQ = 0h			
D5h	[7:0]			474h	45Ch	44Eh	474h	45Ch	44Eh	FREQ = 1h			
D6h	[7:0]			47Ah	47Ah	46Ah	474h	474h	46Ah	FREQ = 2h			
				47Ah	46Ah	456h	474h	46Ah	456h	FREQ = 3h			
D8h	[7:0]	HMAX	0167h	167h	17Eh	2C4h	190h	190h	2C4h	FREQ = 0h			
D9h	[7:0]			18Ah	2D9h	577h	190h	2D9h	577h	FREQ = 1h			
				167h	167h	1E0h	190h	190h	1E0h	FREQ = 2h			
				167h	1EDh	3ADh	190h	1EDh	3ADh	FREQ = 3h			
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h									
E2h	[7:0]	GMRWT	06h	06h	06h	04h	06h	06h	04h	FREQ = 0h			
				06h	04h	02h	06h	04h	02h	FREQ = 1h			
				06h	06h	04h	06h	06h	04h	FREQ = 2h			
				06h	04h	04h	06h	04h	04h	FREQ = 3h			
E3h	[7:0]	GMTWT	28h	28h	26h	14h	24h	24h	14h	FREQ = 0h			
				24h	14h	0Ch	24h	14h	0Ch	FREQ = 1h			
				28h	28h	1Eh	24h	24h	1Eh	FREQ = 2h			
				28h	1Eh	10h	24h	1Eh	10h	FREQ = 3h			
E6h	[7:0]	GSDLY	12h	12h	10h	0Ah	10h	10h	0Ah	FREQ = 0h			
				10h	0Ah	06h	10h	0Ah	06h	FREQ = 1h			
				12h	12h	0Eh	10h	10h	0Eh	FREQ = 2h			
				12h	0Eh	08h	10h	0Eh	08h	FREQ = 3h			
Chip ID = 04h													
00h	[5:4]	ADBIT	0h	0h			1h						
Chip ID = 06h													
30h	[1:0]	ODBIT	0h	2h									
Chip ID = 07h													
ACh	[5]		1h	0h									

6.28.3. ROI mode

SLVS

Address	bit	Register name	Initial Value	Setting value						Remarks
				AD = 10 bit			AD = 12 bit			
				8 ch	4 ch	2 ch	8 ch	4 ch	2 ch	
Chip ID = 02h										
D4h	[7:0]	VMAX	0008A Ah	*						
D5h	[7:0]									
D6h	[7:0]									
D8h	[7:0]	HMAX	0167h	16Bh	2B2h	52Ch	190h	2B2h	52Ch	FREQ = 0h
				2BEh	541h	A47h	2BEh	541h	A47h	FREQ = 1h
D9h	[7:0]			167h	1D4h	37Bh	190h	1D4h	37Bh	FREQ = 2h
				1DBh	388h	6E2h	1DBh	388h	6E2h	FREQ = 3h
DCh	[1:0]	FREQ	0h	0h / 1h / 2h / 3h						
E2h	[7:0]	GMRWT	06h	06h	04h	02h	06h	04h	02h	FREQ = 0h
				04h	02h	02h	04h	02h	02h	FREQ = 1h
				06h	06h	04h	06h	06h	04h	FREQ = 2h
				06h	04h	02h	06h	04h	02h	FREQ = 3h
E3h	[7:0]	GMTWT	28h	28h	16h	0Ch	24h	16h	0Ch	FREQ = 0h
				16h	0Ch	06h	16h	0Ch	06h	FREQ = 1h
				28h	20h	10h	24h	20h	10h	FREQ = 2h
				1Eh	10h	0Ah	1Eh	10h	0Ah	FREQ = 3h
E6h	[7:0]	GSDLY	12h	12h	0Ah	06h	10h	0Ah	06h	FREQ = 0h
				0Ah	06h	04h	0Ah	06h	04h	FREQ = 1h
				12h	0Eh	08h	10h	0Eh	08h	FREQ = 2h
				0Eh	08h	04h	0Eh	08h	04h	FREQ = 3h
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h			1h			
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	0h						

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.29. IMX565-AAMJ-C

6.29.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				42.6 [frame/s]	22.2 [frame/s]	42.6 [frame/s]	22.2 [frame/s]	
				32.2 [frame/s]	16.7 [frame/s]	32.2 [frame/s]	16.7 [frame/s]	
				21.7 [frame/s]	11.2 [frame/s]	21.7 [frame/s]	11.2 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	C40h	C2Ch	C40h	C2Ch	1188 Mbps
D5h	[7:0]			C38h	C28h	C38h	C28h	891 Mbps
D6h	[7:0]			C2Ch	C24h	C2Ch	C24h	594 Mbps
D8h	[7:0]	HMAX	0081h	22Bh	42Eh	22Bh	42Eh	1188 Mbps
D9h	[7:0]			2E0h	58Eh	2E0h	58Eh	891 Mbps
				448h	84Dh	448h	84Dh	594 Mbps
E2h	[7:0]	GMRWT	10h	04h	02h	04h	02h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				02h	02h	02h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	1Ah	0Eh	1188 Mbps
				14h	0Ah	14h	0Ah	891 Mbps
				0Eh	08h	0Eh	08h	594 Mbps
E6h	[7:0]	GSDLY	30h	0Ch	06h	0Ch	06h	1188 Mbps
				0Ah	06h	0Ah	06h	891 Mbps
				06h	04h	06h	04h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
AC'h	[5]		1h			1h		

6.29.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				152.4 [frame/s]	83.5 [frame/s]	152.4 [frame/s]	83.5 [frame/s]			
				116.6 [frame/s]	63.3 [frame/s]	116.6 [frame/s]	63.3 [frame/s]	891 Mbps		
				79.8 [frame/s]	42.7 [frame/s]	79.8 [frame/s]	42.7 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	668h	644h	668h	644h	1188 Mbps		
D5h	[7:0]			658h	63Ch	658h	63Ch	891 Mbps		
D6h	[7:0]			644h	634h	644h	634h	594 Mbps		
D8h	[7:0]	HMAX	0081h	129h	22Ah	129h	22Ah	1188 Mbps		
D9h	[7:0]			188h	2DEh	188h	2DEh	891 Mbps		
				244h	445h	244h	445h	594 Mbps		
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	1188 Mbps		
				08h	04h	08h	04h	891 Mbps		
				04h	04h	04h	04h	594 Mbps		
E3h	[7:0]	GMTWT	70h	30h	1Ch	30h	1Ch	1188 Mbps		
				28h	14h	28h	14h	891 Mbps		
				1Ch	10h	1Ch	10h	594 Mbps		
E6h	[7:0]	GSDLY	30h	18h	0Ch	18h	0Ch	1188 Mbps		
				10h	0Ch	10h	0Ch	891 Mbps		
				0Ch	08h	0Ch	08h	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	1h						

6.29.3. 2 × 2 FD Binning mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks	
				AD = 10 bit		AD = 12 bit			
				4 lane	2 lane	4 lane	2 lane		
				152.4 [frame/s]	83.5 [frame/s]	152.4 [frame/s]	83.5 [frame/s]		1188 Mbps
				116.6 [frame/s]	63.3 [frame/s]	116.6 [frame/s]	63.3 [frame/s]		891 Mbps
				79.8 [frame/s]	42.7 [frame/s]	79.8 [frame/s]	42.7 [frame/s]	594 Mbps	
Chip ID = 02h									
D4h	[7:0]	VMAX	000C D4h	668h	644h	668h	644h	1188 Mbps	
D5h	[7:0]			658h	63Ch	658h	63Ch	891 Mbps	
D6h	[7:0]			644h	634h	644h	634h	594 Mbps	
D8h	[7:0]	HMAX	0081h	129h	22Ah	129h	22Ah	1188 Mbps	
D9h	[7:0]			188h	2DEh	188h	2DEh	891 Mbps	
				244h	445h	244h	445h	594 Mbps	
E2h	[7:0]	GMRWT	10h	08h	04h	08h	04h	1188 Mbps	
				08h	04h	08h	04h	891 Mbps	
				04h	04h	04h	04h	594 Mbps	
E3h	[7:0]	GMTWT	70h	30h	1Ch	30h	1Ch	1188 Mbps	
				28h	14h	28h	14h	891 Mbps	
				1Ch	10h	1Ch	10h	594 Mbps	
E6h	[7:0]	GSDLY	30h	18h	0Ch	18h	0Ch	1188 Mbps	
				10h	0Ch	10h	0Ch	891 Mbps	
				0Ch	08h	0Ch	08h	594 Mbps	
Chip ID = 04h									
00h	[5:4]	ADBIT	0h	0h		1h			
Chip ID = 06h									
30h	[1:0]	ODBIT	0h	2h					
Chip ID = 07h									
ACH	[5]		1h	1h					

6.29.4. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	22Bh	42Eh	22Bh	42Eh	1188 Mbps
D9h	[7:0]			2E0h	58Eh	2E0h	58Eh	891 Mbps
				448h	84Dh	448h	84Dh	594 Mbps
E2h	[7:0]	GMRWT	10h	04h	02h	04h	02h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				02h	02h	02h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	1Ah	0Eh	1188 Mbps
				14h	0Ah	14h	0Ah	891 Mbps
				0Eh	08h	0Eh	08h	594 Mbps
E6h	[7:0]	GSDLY	30h	0Ch	06h	0Ch	06h	1188 Mbps
				0Ah	06h	0Ah	06h	891 Mbps
				06h	04h	06h	04h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.30. IMX565-AAQJ-C

6.30.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				42.6 [frame/s]	22.2 [frame/s]	42.6 [frame/s]	22.2 [frame/s]	
				32.2 [frame/s]	16.7 [frame/s]	32.2 [frame/s]	16.7 [frame/s]	
			21.7 [frame/s]	11.2 [frame/s]	21.7 [frame/s]	11.2 [frame/s]	594 Mbps	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	C40h	C2Ch	C40h	C2Ch	1188 Mbps
D5h	[7:0]			C38h	C28h	C38h	C28h	891 Mbps
D6h	[7:0]			C2Ch	C24h	C2Ch	C24h	594 Mbps
D8h	[7:0]	HMAX	0081h	22Bh	42Eh	22Bh	42Eh	1188 Mbps
D9h	[7:0]			2E0h	58Eh	2E0h	58Eh	891 Mbps
				448h	84Dh	448h	84Dh	594 Mbps
E2h	[7:0]	GMRWT	10h	04h	02h	04h	02h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				02h	02h	02h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	1Ah	0Eh	1188 Mbps
				14h	0Ah	14h	0Ah	891 Mbps
				0Eh	08h	0Eh	08h	594 Mbps
E6h	[7:0]	GSDLY	30h	0Ch	06h	0Ch	06h	1188 Mbps
				0Ah	06h	0Ah	06h	891 Mbps
				06h	04h	06h	04h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
AC _h	[5]		1h			1h		

6.30.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				153.1 [frame/s]	83.9 [frame/s]	112.8 [frame/s]	83.9 [frame/s]			
				117.3 [frame/s]	63.7 [frame/s]	112.8 [frame/s]	63.7 [frame/s]			
				80.2 [frame/s]	43.1 [frame/s]	80.2 [frame/s]	43.1 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	660h	63Ch	64Ch	63Ch	1188 Mbps		
D5h	[7:0]			64Eh	634h	64Ch	634h	891 Mbps		
D6h	[7:0]			63Ch	628h	63Ch	628h	594 Mbps		
D8h	[7:0]	HMAX	0081h	129h	22Ah	198h	22Ah	1188 Mbps		
D9h	[7:0]			188h	2DEh	198h	2DEh	891 Mbps		
				244h	445h	244h	445h	594 Mbps		
E2h	[7:0]	GMRWT	10h	08h	04h	06h	04h	1188 Mbps		
				06h	04h	06h	04h	891 Mbps		
				04h	02h	04h	02h	594 Mbps		
E3h	[7:0]	GMTWT	70h	30h	1Ah	24h	1Ah	1188 Mbps		
				26h	14h	24h	14h	891 Mbps		
				1Ah	0Eh	1Ah	0Eh	594 Mbps		
E6h	[7:0]	GSDLY	30h	16h	0Ch	10h	0Ch	1188 Mbps		
				10h	0Ah	10h	0Ah	891 Mbps		
				0Ch	06h	0Ch	06h	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACH	[5]		1h			1h				

6.30.3. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	22Bh	42Eh	22Bh	42Eh	1188 Mbps
D9h	[7:0]			2E0h	58Eh	2E0h	58Eh	891 Mbps
				448h	84Dh	448h	84Dh	594 Mbps
E2h	[7:0]	GMRWT	10h	04h	02h	04h	02h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				02h	02h	02h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	1Ah	0Eh	1Ah	0Eh	1188 Mbps
				14h	0Ah	14h	0Ah	891 Mbps
				0Eh	08h	0Eh	08h	594 Mbps
E6h	[7:0]	GSDLY	30h	0Ch	06h	0Ch	06h	1188 Mbps
				0Ah	06h	0Ah	06h	891 Mbps
				06h	04h	06h	04h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.31. IMX566-AAMJ-C

6.31.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				62.6 [frame/s]	33.2 [frame/s]	60.8 [frame/s]	33.2 [frame/s]	
				47.5 [frame/s]	25.1 [frame/s]	47.5 [frame/s]	25.1 [frame/s]	
				32.1 [frame/s]	16.8 [frame/s]	32.1 [frame/s]	16.8 [frame/s]	1188 Mbps
								891 Mbps
								594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	BB0h	B98h	BB0h	B98h	1188 Mbps
D5h	[7:0]			BA2h	B90h	BA2h	B90h	891 Mbps
D6h	[7:0]			B96h	B88h	B96h	B88h	594 Mbps
D8h	[7:0]	HMAX	0081h	18Ch	2F0h	198h	2F0h	1188 Mbps
D9h	[7:0]			20Ch	3E6h	20Ch	3E6h	891 Mbps
				30Ah	5D1h	30Ah	5D1h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	1188 Mbps
				1Ch	10h	1Ch	10h	891 Mbps
				14h	0Ah	14h	0Ah	594 Mbps
E6h	[7:0]	GSDLY	30h	10h	0Ah	10h	0Ah	1188 Mbps
				0Ch	08h	0Ch	08h	891 Mbps
				08h	06h	08h	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
AC _h	[5]		1h			1h		

6.31.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				214.4 [frame/s]	121.7 [frame/s]	214.4 [frame/s]	121.7 [frame/s]	
				165.9 [frame/s]	93.1 [frame/s]	165.9 [frame/s]	93.1 [frame/s]	
				114.2 [frame/s]	63.2 [frame/s]	114.2 [frame/s]	63.2 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	634h	604h	634h	604h	1188 Mbps
D5h	[7:0]			61Ch	5F4h	61Ch	5F4h	891 Mbps
D6h	[7:0]			604h	5E8h	604h	5E8h	594 Mbps
D8h	[7:0]	HMAX	0081h	DAh	18Ch	DAh	18Ch	1188 Mbps
D9h	[7:0]			11Eh	20Bh	11Eh	20Bh	891 Mbps
				1A6h	309h	1A6h	309h	594 Mbps
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	1188 Mbps
				08h	04h	08h	04h	891 Mbps
				08h	04h	08h	04h	594 Mbps
E3h	[7:0]	GMTWT	70h	44h	24h	44h	24h	1188 Mbps
				34h	1Ch	34h	1Ch	891 Mbps
				24h	14h	24h	14h	594 Mbps
E6h	[7:0]	GSDLY	30h	1Ch	10h	1Ch	10h	1188 Mbps
				18h	0Ch	18h	0Ch	891 Mbps
				10h	08h	10h	08h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
ACH	[5]		1h			1h		

6.31.3. 2 × 2 FD Binning mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				214.4 [frame/s]	121.7 [frame/s]	214.4 [frame/s]	121.7 [frame/s]	
				165.9 [frame/s]	93.1 [frame/s]	165.9 [frame/s]	93.1 [frame/s]	
				114.2 [frame/s]	63.2 [frame/s]	114.2 [frame/s]	63.2 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	634h	604h	634h	604h	1188 Mbps
D5h	[7:0]			61Ch	5F4h	61Ch	5F4h	891 Mbps
D6h	[7:0]			604h	5E8h	604h	5E8h	594 Mbps
D8h	[7:0]	HMAX	0081h	DAh	18Ch	DAh	18Ch	1188 Mbps
D9h	[7:0]			11Eh	20Bh	11Eh	20Bh	891 Mbps
				1A6h	309h	1A6h	309h	594 Mbps
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	1188 Mbps
				08h	04h	08h	04h	891 Mbps
				08h	04h	08h	04h	594 Mbps
E3h	[7:0]	GMTWT	70h	44h	24h	44h	24h	1188 Mbps
				34h	1Ch	34h	1Ch	891 Mbps
				24h	14h	24h	14h	594 Mbps
E6h	[7:0]	GSDLY	30h	1Ch	10h	1Ch	10h	1188 Mbps
				18h	0Ch	18h	0Ch	891 Mbps
				10h	08h	10h	08h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
ACH	[5]		1h			1h		

6.31.4. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	18Ch	2F0h	198h	2F0h	1188 Mbps
D9h	[7:0]			20Ch	3E6h	20Ch	3E6h	891 Mbps
				30Ah	5D1h	30Ah	5D1h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	1188 Mbps
				1Ch	10h	1Ch	10h	891 Mbps
				14h	0Ah	14h	0Ah	594 Mbps
E6h	[7:0]	GSDLY	30h	10h	0Ah	10h	0Ah	1188 Mbps
				0Ch	08h	0Ch	08h	891 Mbps
				08h	06h	08h	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.32. IMX566-AAQJ-C

6.32.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				62.6 [frame/s]	33.2 [frame/s]	60.8 [frame/s]	33.2 [frame/s]	
				47.5 [frame/s]	25.1 [frame/s]	47.5 [frame/s]	25.1 [frame/s]	
				32.1 [frame/s]	16.8 [frame/s]	32.1 [frame/s]	16.8 [frame/s]	1188 Mbps
								891 Mbps
								594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	BB0h	B98h	BB0h	B98h	1188 Mbps
D5h	[7:0]			BA2h	B90h	BA2h	B90h	891 Mbps
D6h	[7:0]			B96h	B88h	B96h	B88h	594 Mbps
D8h	[7:0]	HMAX	0081h	18Ch	2F0h	198h	2F0h	1188 Mbps
D9h	[7:0]			20Ch	3E6h	20Ch	3E6h	891 Mbps
				30Ah	5D1h	30Ah	5D1h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	1188 Mbps
				1Ch	10h	1Ch	10h	891 Mbps
				14h	0Ah	14h	0Ah	594 Mbps
E6h	[7:0]	GSDLY	30h	10h	0Ah	10h	0Ah	1188 Mbps
				0Ch	08h	0Ch	08h	891 Mbps
				08h	06h	08h	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
AC _h	[5]		1h	1h				

6.32.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				168.2 [frame/s]	122.3 [frame/s]	118.7 [frame/s]	118.7 [frame/s]	
				167.0 [frame/s]	93.5 [frame/s]	118.7 [frame/s]	93.5 [frame/s]	
				114.9 [frame/s]	63.4 [frame/s]	114.9 [frame/s]	63.4 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	612h	5FCh	5FCh	5FCh	1188 Mbps
D5h	[7:0]			612h	5EEh	5FCh	5EEh	891 Mbps
D6h	[7:0]			5FAh	5E2h	5FAh	5E2h	594 Mbps
D8h	[7:0]	HMAX	0081h	11Ch	18Ch	198h	198h	1188 Mbps
D9h	[7:0]			11Eh	20Bh	198h	20Bh	891 Mbps
				1A6h	309h	1A6h	309h	594 Mbps
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	1188 Mbps
				08h	04h	06h	04h	891 Mbps
				06h	04h	06h	04h	594 Mbps
E3h	[7:0]	GMTWT	70h	32h	24h	24h	24h	1188 Mbps
				32h	1Ch	24h	1Ch	891 Mbps
				22h	14h	22h	14h	594 Mbps
E6h	[7:0]	GSDLY	30h	16h	10h	10h	10h	1188 Mbps
				16h	0Ch	10h	0Ch	891 Mbps
				10h	08h	10h	08h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
ACH	[5]		1h			1h		

6.32.3. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	18Ch	2F0h	198h	2F0h	1188 Mbps
D9h	[7:0]			20Ch	3E6h	20Ch	3E6h	891 Mbps
				30Ah	5D1h	30Ah	5D1h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				04h	02h	04h	02h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	24h	14h	24h	14h	1188 Mbps
				1Ch	10h	1Ch	10h	891 Mbps
				14h	0Ah	14h	0Ah	594 Mbps
E6h	[7:0]	GSDLY	30h	10h	0Ah	10h	0Ah	1188 Mbps
				0Ch	08h	0Ch	08h	891 Mbps
				08h	06h	08h	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.33. IMX567-AAMJ-C

6.33.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				96.2 [frame/s]	51.7 [frame/s]	82.4 [frame/s]	51.7 [frame/s]			
				73.3 [frame/s]	39.1 [frame/s]	73.3 [frame/s]	39.1 [frame/s]			
				49.8 [frame/s]	26.3 [frame/s]	49.8 [frame/s]	26.3 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	8A8h	88Ah	8A0h	88Ah	1188 Mbps		
D5h	[7:0]			89Ah	884h	89Ah	884h	891 Mbps		
D6h	[7:0]			88Ah	87Ah	88Ah	87Ah	594 Mbps		
D8h	[7:0]	HMAX	0081h	15Ch	290h	198h	290h	1188 Mbps		
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps		
				2AAh	511h	2AAh	511h	594 Mbps		
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps		
				06h	04h	06h	04h	891 Mbps		
				04h	02h	04h	02h	594 Mbps		
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	1188 Mbps		
				20h	12h	20h	12h	891 Mbps		
				16h	0Ch	16h	0Ch	594 Mbps		
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	1188 Mbps		
				0Eh	08h	0Eh	08h	891 Mbps		
				0Ah	06h	0Ah	06h	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACh	[5]		1h	1h						

6.33.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				316.8 [frame/s]	183.9 [frame/s]	302.3 [frame/s]	183.9 [frame/s]			
				248.5 [frame/s]	141.4 [frame/s]	248.5 [frame/s]	141.4 [frame/s]	891 Mbps		
				171.7 [frame/s]	96.6 [frame/s]	171.7 [frame/s]	96.6 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	4B8h	488h	4B4h	488h	1188 Mbps		
D5h	[7:0]			498h	478h	498h	478h	891 Mbps		
D6h	[7:0]			484h	468h	484h	468h	594 Mbps		
D8h	[7:0]	HMAX	0081h	C2h	15Ch	CCh	15Ch	1188 Mbps		
D9h	[7:0]			FEh	1CBh	FEh	1CBh	891 Mbps		
				176h	2A9h	176h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	1188 Mbps		
				08h	08h	08h	08h	891 Mbps		
				08h	04h	08h	04h	594 Mbps		
E3h	[7:0]	GMTWT	70h	4Ch	2Ch	48h	2Ch	1188 Mbps		
				38h	20h	38h	20h	891 Mbps		
				28h	18h	28h	18h	594 Mbps		
E6h	[7:0]	GSDLY	30h	20h	14h	20h	14h	1188 Mbps		
				18h	10h	18h	10h	891 Mbps		
				14h	0Ch	14h	0Ch	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACH	[5]		1h			1h				

6.33.3. 2 × 2 FD Binning mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				316.8 [frame/s]	183.9 [frame/s]	302.3 [frame/s]	183.9 [frame/s]			
				248.5 [frame/s]	141.4 [frame/s]	248.5 [frame/s]	141.4 [frame/s]	891 Mbps		
				171.7 [frame/s]	96.6 [frame/s]	171.7 [frame/s]	96.6 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	4B8h	488h	4B4h	488h	1188 Mbps		
D5h	[7:0]			498h	478h	498h	478h	891 Mbps		
D6h	[7:0]			484h	468h	484h	468h	594 Mbps		
D8h	[7:0]	HMAX	0081h	C2h	15Ch	CCh	15Ch	1188 Mbps		
D9h	[7:0]			FEh	1CBh	FEh	1CBh	891 Mbps		
				176h	2A9h	176h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	10h	0Ch	08h	0Ch	08h	1188 Mbps		
				08h	08h	08h	08h	891 Mbps		
				08h	04h	08h	04h	594 Mbps		
E3h	[7:0]	GMTWT	70h	4Ch	2Ch	48h	2Ch	1188 Mbps		
				38h	20h	38h	20h	891 Mbps		
				28h	18h	28h	18h	594 Mbps		
E6h	[7:0]	GSDLY	30h	20h	14h	20h	14h	1188 Mbps		
				18h	10h	18h	10h	891 Mbps		
				14h	0Ch	14h	0Ch	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACH	[5]		1h			1h				

6.33.4. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	15Ch	290h	198h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.34. IMX567-AAQJ-C

6.34.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				96.2 [frame/s]	51.7 [frame/s]	82.4 [frame/s]	51.7 [frame/s]	
				73.3 [frame/s]	39.1 [frame/s]	73.3 [frame/s]	39.1 [frame/s]	
				49.8 [frame/s]	26.3 [frame/s]	49.8 [frame/s]	26.3 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	8A8h	88Ah	8A0h	88Ah	1188 Mbps
D5h	[7:0]			89Ah	884h	89Ah	884h	891 Mbps
D6h	[7:0]			88Ah	87Ah	88Ah	87Ah	594 Mbps
D8h	[7:0]	HMAX	0081h	15Ch	290h	198h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

6.34.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				224.9 [frame/s]	185.8 [frame/s]	159.6 [frame/s]	159.6 [frame/s]		1188 Mbps	
				224.9 [frame/s]	142.6 [frame/s]	159.6 [frame/s]	142.6 [frame/s]		891 Mbps	
				173.5 [frame/s]	97.5 [frame/s]	159.6 [frame/s]	97.5 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	000C D4h	48Ah	47Ch	474h	474h	1188 Mbps		
D5h	[7:0]			48Ah	46Eh	474h	46Eh	891 Mbps		
D6h	[7:0]			478h	45Eh	474h	45Eh	594 Mbps		
D8h	[7:0]	HMAX	0081h	11Ch	15Ch	198h	198h	1188 Mbps		
D9h	[7:0]			11Ch	1CBh	198h	1CBh	891 Mbps		
				176h	2A9h	198h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	10h	08h	06h	06h	06h	1188 Mbps		
				08h	06h	06h	06h	891 Mbps		
				06h	04h	06h	04h	594 Mbps		
E3h	[7:0]	GMTWT	70h	32h	2Ah	24h	24h	1188 Mbps		
				32h	20h	24h	20h	891 Mbps		
				26h	16h	24h	16h	594 Mbps		
E6h	[7:0]	GSDLY	30h	16h	12h	10h	10h	1188 Mbps		
				16h	0Eh	10h	0Eh	891 Mbps		
				12h	0Ah	10h	0Ah	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACH	[5]		1h	1h						

6.34.3. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	000C D4h	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0081h	15Ch	290h	198h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	10h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	70h	2Ah	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	30h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.35. IMX568-AAMJ-C

6.35.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				93.4 [frame/s]	51.7 [frame/s]	84.0 [frame/s]	51.7 [frame/s]	
				73.3 [frame/s]	39.1 [frame/s]	73.3 [frame/s]	39.1 [frame/s]	
				49.8 [frame/s]	26.3 [frame/s]	49.8 [frame/s]	26.3 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	0008A Ah	8A6h	88Ah	8A0h	88Ah	1188 Mbps
D5h	[7:0]			89Ah	884h	89Ah	884h	891 Mbps
D6h	[7:0]			88Ah	87Ah	88Ah	87Ah	594 Mbps
D8h	[7:0]	HMAX	0167h	167h	290h	190h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	06h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	28h	28h	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	12h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
ACh	[5]		1h			1h		

6.35.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				316.8 [frame/s]	183.9 [frame/s]	308.3 [frame/s]	183.9 [frame/s]			
				248.5 [frame/s]	141.4 [frame/s]	248.5 [frame/s]	141.4 [frame/s]	891 Mbps		
				171.7 [frame/s]	96.6 [frame/s]	171.7 [frame/s]	96.6 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	0008A Ah	4B8h	488h	4B4h	488h	1188 Mbps		
D5h	[7:0]			498h	478h	498h	478h	891 Mbps		
D6h	[7:0]			484h	468h	484h	468h	594 Mbps		
D8h	[7:0]	HMAX	0167h	C2h	15Ch	C8h	15Ch	1188 Mbps		
D9h	[7:0]			FEh	1CBh	FEh	1CBh	891 Mbps		
				176h	2A9h	176h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	06h	0Ch	08h	0Ch	08h	1188 Mbps		
				08h	08h	08h	08h	891 Mbps		
				08h	04h	08h	04h	594 Mbps		
E3h	[7:0]	GMTWT	28h	4Ch	2Ch	48h	2Ch	1188 Mbps		
				38h	20h	38h	20h	891 Mbps		
				28h	18h	28h	18h	594 Mbps		
E6h	[7:0]	GSDLY	12h	20h	14h	20h	14h	1188 Mbps		
				18h	10h	18h	10h	891 Mbps		
				14h	0Ch	14h	0Ch	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACh	[5]		1h			1h				

6.35.3. 2 × 2 FD Binning mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				316.8 [frame/s]	183.9 [frame/s]	308.3 [frame/s]	183.9 [frame/s]			
				248.5 [frame/s]	141.4 [frame/s]	248.5 [frame/s]	141.4 [frame/s]	891 Mbps		
				171.7 [frame/s]	96.6 [frame/s]	171.7 [frame/s]	96.6 [frame/s]	594 Mbps		
Chip ID = 02h										
D4h	[7:0]	VMAX	0008A Ah	4B8h	488h	4B4h	488h	1188 Mbps		
D5h	[7:0]			498h	478h	498h	478h	891 Mbps		
D6h	[7:0]			484h	468h	484h	468h	594 Mbps		
D8h	[7:0]	HMAX	0167h	C2h	15Ch	C8h	15Ch	1188 Mbps		
D9h	[7:0]			FEh	1CBh	FEh	1CBh	891 Mbps		
				176h	2A9h	176h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	06h	0Ch	08h	0Ch	08h	1188 Mbps		
				08h	08h	08h	08h	891 Mbps		
				08h	04h	08h	04h	594 Mbps		
E3h	[7:0]	GMTWT	28h	4Ch	2Ch	48h	2Ch	1188 Mbps		
				38h	20h	38h	20h	891 Mbps		
				28h	18h	28h	18h	594 Mbps		
E6h	[7:0]	GSDLY	12h	20h	14h	20h	14h	1188 Mbps		
				18h	10h	18h	10h	891 Mbps		
				14h	0Ch	14h	0Ch	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h			2h				
Chip ID = 07h										
ACH	[5]		1h			1h				

6.35.4. ROI mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0008A Ah	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0167h	167h	290h	190h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	06h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	28h	28h	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	12h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

6.36. IMX568-AAQJ-C

6.36.1. All pixel scan mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
				93.4 [frame/s]	51.7 [frame/s]	84.0 [frame/s]	51.7 [frame/s]	
				73.3 [frame/s]	39.1 [frame/s]	73.3 [frame/s]	39.1 [frame/s]	
				49.8 [frame/s]	26.3 [frame/s]	49.8 [frame/s]	26.3 [frame/s]	594 Mbps
Chip ID = 02h								
D4h	[7:0]	VMAX	0008A Ah	8A6h	88Ah	8A0h	88Ah	1188 Mbps
D5h	[7:0]			89Ah	884h	89Ah	884h	891 Mbps
D6h	[7:0]			88Ah	87Ah	88Ah	87Ah	594 Mbps
D8h	[7:0]	HMAX	0167h	167h	290h	190h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	06h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	28h	28h	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	12h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h			2h		
Chip ID = 07h								
ACh	[5]		1h			1h		

6.36.2. Vertical / Horizontal 1/2 Subsampling mode

CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks		
				AD = 10 bit		AD = 12 bit				
				4 lane	2 lane	4 lane	2 lane			
				180.4 [frame/s]	180.4 [frame/s]	162.8 [frame/s]	162.8 [frame/s]		1188 Mbps	
				180.4 [frame/s]	142.6 [frame/s]	162.8 [frame/s]	142.6 [frame/s]		891 Mbps	
			173.5 [frame/s]	97.5 [frame/s]	162.8 [frame/s]	97.5 [frame/s]	594 Mbps			
Chip ID = 02h										
D4h	[7:0]	VMAX	0008A Ah	47Ah	47Ah	474h	474h	1188 Mbps		
D5h	[7:0]			47Ah	46Eh	474h	46Eh	891 Mbps		
D6h	[7:0]			478h	45Eh	474h	45Eh	594 Mbps		
D8h	[7:0]	HMAX	0167h	167h	167h	190h	190h	1188 Mbps		
D9h	[7:0]			167h	1CBh	190h	1CBh	891 Mbps		
				176h	2A9h	190h	2A9h	594 Mbps		
E2h	[7:0]	GMRWT	06h	06h	06h	06h	06h	1188 Mbps		
				06h	06h	06h	06h	891 Mbps		
				06h	04h	06h	04h	594 Mbps		
E3h	[7:0]	GMTWT	28h	28h	28h	24h	24h	1188 Mbps		
				28h	20h	24h	20h	891 Mbps		
				26h	16h	24h	16h	594 Mbps		
E6h	[7:0]	GSDLY	12h	12h	12h	10h	10h	1188 Mbps		
				12h	0Eh	10h	0Eh	891 Mbps		
				12h	0Ah	10h	0Ah	594 Mbps		
Chip ID = 04h										
00h	[5:4]	ADBIT	0h	0h		1h				
Chip ID = 06h										
30h	[1:0]	ODBIT	0h	2h						
Chip ID = 07h										
ACH	[5]		1h	1h						

6.36.3. ROI mode

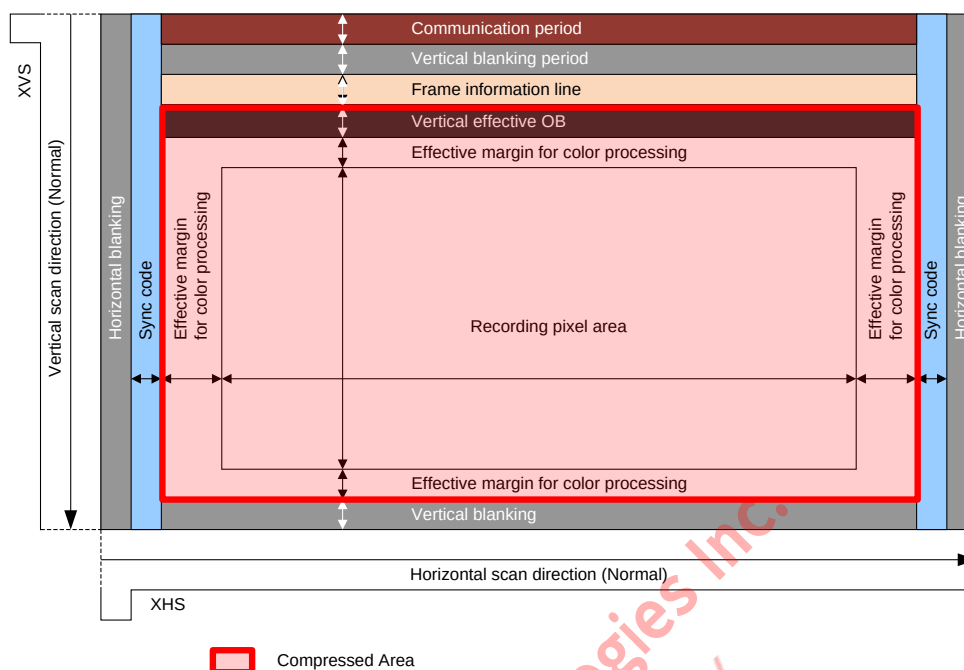
CSI-2

Address	bit	Register name	Initial Value	Setting value				Remarks
				AD = 10 bit		AD = 12 bit		
				4 lane	2 lane	4 lane	2 lane	
Chip ID = 02h								
D4h	[7:0]	VMAX	0008A Ah	*				
D5h	[7:0]							
D6h	[7:0]							
D8h	[7:0]	HMAX	0167h	167h	290h	190h	290h	1188 Mbps
D9h	[7:0]			1CCh	366h	1CCh	366h	891 Mbps
				2AAh	511h	2AAh	511h	594 Mbps
E2h	[7:0]	GMRWT	06h	06h	04h	06h	04h	1188 Mbps
				06h	04h	06h	04h	891 Mbps
				04h	02h	04h	02h	594 Mbps
E3h	[7:0]	GMTWT	28h	28h	16h	24h	16h	1188 Mbps
				20h	12h	20h	12h	891 Mbps
				16h	0Ch	16h	0Ch	594 Mbps
E6h	[7:0]	GSDLY	12h	12h	0Ah	10h	0Ah	1188 Mbps
				0Eh	08h	0Eh	08h	891 Mbps
				0Ah	06h	0Ah	06h	594 Mbps
Chip ID = 04h								
00h	[5:4]	ADBIT	0h	0h		1h		
Chip ID = 06h								
30h	[1:0]	ODBIT	0h	2h				
Chip ID = 07h								
ACh	[5]		1h	1h				

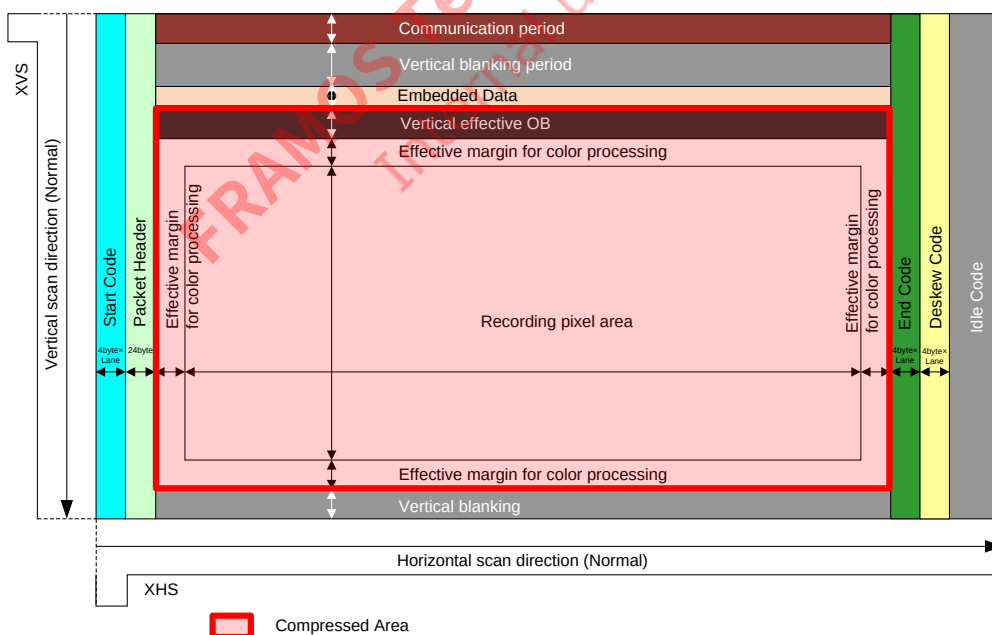
* VMAX = ROIWV1 + ROIWV2 + ROIWV3 + ... + ROIWV8 + GMRWT + GMTWT + GSDLY + 86

7. Area of Gradation Compression

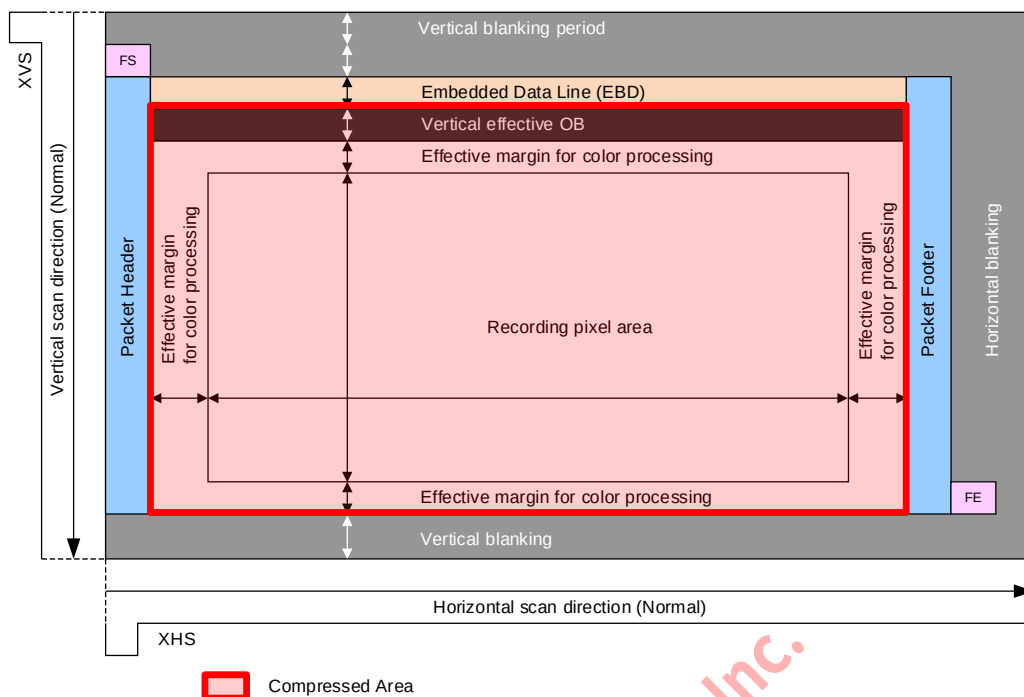
The area of Gradation Compression is as below.



Area of Gradation Compression (SLVS)



Area of Gradation Compression (SLVS – EC)



Area of Gradation Compression (CSI-2)

FRAMOS Technologies Inc.
Internal use only

8. Description of Gradation Compression Function

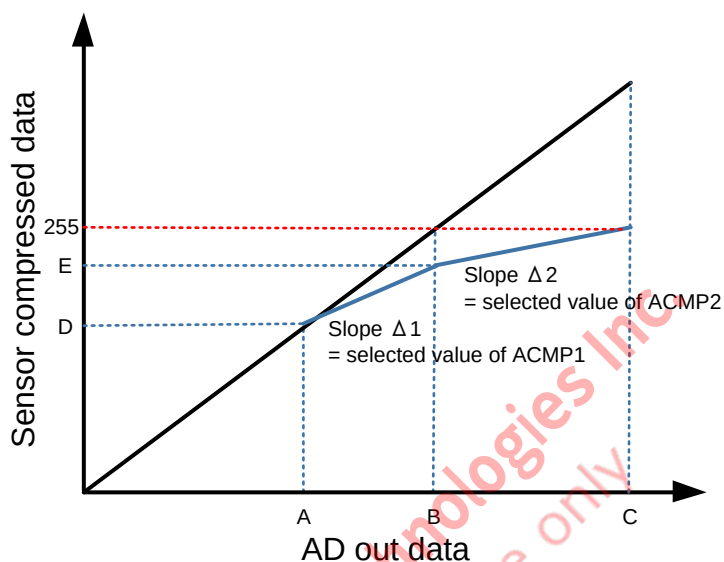
This is the process which makes the output data be 8 bit.

Gradation Compression compress as below.

In case of the setting which the result of compressed AD data is over 8 bit, the value of output data is clipped at 255.

In case of SLVS output, 0 and 255 which are compressed data output as 1 and 254.

(Refer to section "Output Signal Range" on Datasheet.)



$D = A =$ selected value of CCMP1

$B =$ selected value of CCMP2

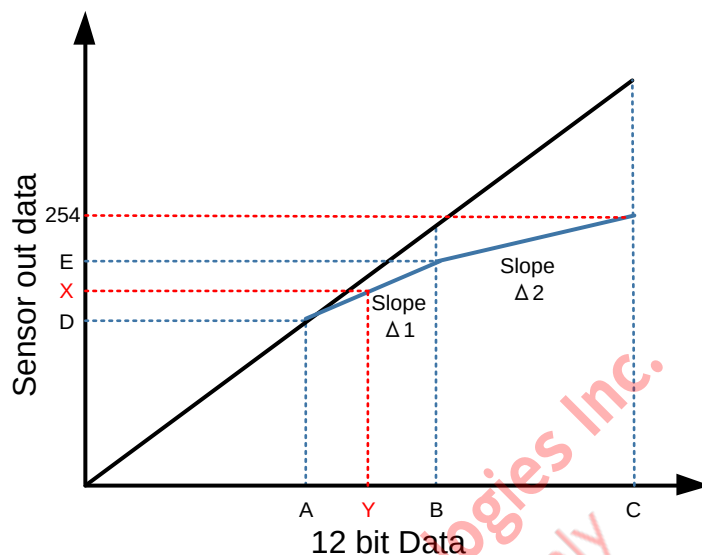
$E = (B - A) * \Delta 1 + D$

$255 = (C - B) * \Delta 2 + E$

The image of Gradation Compression

When the sensor outputs image data in Gradation Compression, it may be necessary for the following DSP to decompress the input data to process them properly. An example of inverse formula is shown below. In this case, division operations in the formula can be simply done with shift operations since $\Delta 1$ and $\Delta 2$ are power of 2.

In case of SLVS output, 0 and 255 which are compressed data output as 1 and 254. Because of it, these data can't be decompressed.



$$\begin{aligned}
 1 \leq X < D & : Y = X \\
 D \leq X < E & : Y = (X - D) / \Delta 1 + A \\
 E \leq X \leq 254 & : Y = (X - E) / \Delta 2 + B
 \end{aligned}$$

The image of Gradation Decompression

Revision History

Version	Date	Page	Remarks
Rev.0.1	2019 / 04 / 26	–	First Edition
Rev.1.0	2020 / 06 / 20	–	First Edition (Official version of IMX530 series / IMX540 series)
		All pages	Added sensor types Update register setting values
		1	Added Chapter “Product type” Change to Ultra short exposure from Super short exposure Added SLVS 2 ch output in support interface Added the restriction in support interface
Rev.2.0	2020 / 10 / 27	7 to 72	Correction: the description of ¹
		73, 75, 77, 79, 81, 83, 85, 87, 89, 91, 93, 95, 97, 99, 101, 103, 105, 107, 109, 111, 113, 116, 122, 124, 128, 144, 150, 152, 156	Delete Chip ID=08h, address F0h and F1h
		74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 96, 98, 100, 102, 104, 106, 108, 110, 112, 114, 117 to 121, 123, 125 to 127, 129 to 143, 145 to 149, 151, 153 to 155, 157	Delete Chip ID=07h, address CAh and CEh Delete Chip ID=08h, address F0h and F1h
		115, 158 to 162	Change Common registers according to HMAX
Rev.3.0	2021 / 05 / 25	1	Added IMX535 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series
		2	Change Interface expression
		9	Delete Chip ID=07h, address ACh
		12 to 167	Added Types of IMX535 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series to Section 5
		168 to 376	Added Types of IMX535 series / IMX545 series / IMX548 series / IMX565 series / IMX568 series to Section 6
		168 to 294	Correction: Initial Value of VMAX, HMAX for IMX530 series and IMX540 series VMAX : 12ECh to 0012ECh, HMAX : 0A6h to 00A6h
		168 to 376	Correction: ADBIT (Chip ID=04h, Address 00h), bit [6:5] to [5:4]
		169 to 339	Correction: Data rate unit of SLVS-EC, Mbps/ch to Mbps/lane
		378	Added CSI-2 to “Area of Gradation Compression”